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A Crèche Course in Model Theory

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This book was written to answer one question "Does a recursion theorist dare to write a book on model theory?"
Gerald E. Sacks
Saturated Model Theory (1972)

Contents

Chapter 1

Preliminaries and notation

This chapter introduces the syntax and semantic of first order logic.

The definitions of terms and formulas we give in Section 1.3 and 1.5 are more formal than required here. Our main objective is to convince the reader that a rigorous definition of language and truth is possible. However, the actual details of such a definition are not relevant for our purposes.

1.1 Tuples

A sequence is a function $a : I \to A$ whose domain is a linear order $I, <_I$. We may use the notation $a = \langle a_i : i \in I \rangle$ for sequences. A tuple is a sequence whose domain is an ordinal, say α, then we write $a = \langle a_i : i < \alpha \rangle$. When α is finite, we may also write $a = a_0, \ldots, a_{\alpha-1}$. The domain of the tuple a, the ordinal α, is denoted by $|a|$ and is called the length of a. In some contexts it may be convenient to confuse $\text{rng}(a)$, the range of a, with a. If $A = \text{rng}(a)$ we say that a is an enumeration of A.

If $J \subseteq I$ is a subset of the domain of the sequence $a = \langle a_i : i \in I \rangle$, we write $a_{\restriction J}$ for the restriction of a to J. When J is well ordered by $<_I$ we identify $a_{\restriction J}$ with a tuple. Sometimes (i.e. not always) we may overline tuples or sequences as mnemonic, as in $\bar{a}, \bar{c}$, etc. Still, restrictions are written as $a_{\restriction J}$, $c_{\restriction J}$ without the bar.

The set of tuples of elements of length α is denoted by A^α. The set of tuples of length $< \alpha$ is denoted by $A^{<\alpha}$. For instance, $A^{<\omega}$ is the set of all finite tuples of elements of A. When α is finite we do not distinguish between A^α and the α-th Cartesian power of A. In particular, we do not distinguish between A^1 and A.

When x is a tuple of variables, see Section 1.3, we will write write A^x for $A^{|x|}$. In fact, this notation is convenient when dealing with many-sorted structures, see Section 13.1.

If $a, b \in A^\alpha$ and h is a function defined on A, we write $h(a) = b$ for $h(a_i) = b_i$. We often do not distinguish between the pair $\langle a, b \rangle$ and the tuple of pairs $\langle a_i, b_i \rangle$. The context will resolve the ambiguity.

Note that there is a unique tuple of length 0, the empty set $\varnothing$, which in this context is called empty tuple. Recall that by definition $A^0 = \{\varnothing\}$ for every set A. Therefore, even when A is empty, A^0 contains the empty string.

We often concatenate tuples. If a and b are tuples, we write $a\,b$ for the concatenation or, when emphasis is required, $a^\frown b$.

1.2 Structures

Finally, we are really going to get started.

1.1 Definition A language L (also called signature) is a triple that consists of

1. a set $L_{\rm fun}$ whose elements are called function symbols
2. a set $L_{\rm rel}$ whose elements are called relation symbols
3. a function that assigns to every $f \in L_{\rm fun}$, respectively $r \in L_{\rm rel}$, non-negative integers n_f and n_r that we call arity of the function, respectively relation, symbol. We say that f is an n_f-ary function symbol, and similarly for r. A 0-ary function symbol is also called a constant.

Warning: it is customary to use the symbol L to denote both the language and the set of formulas associated to it (to be defined below). We denote by $|L|$ the cardinality of $L_{\rm fun} \cup L_{\rm rel} \cup \omega$. Note that, by definition, $|L|$ is always infinite.

1.2 Definition A structure M of signature L (for short L-structure) consists of

1. a set called the domain or support of the structure and is denoteed by the same symbol M used for the whole structure
2. a function that assigns to every $f \in L_{\rm fun}$ a total map $f^M : M^{n_f} \to M$
3. a function that assigns to every $r \in L_{\rm rel}$ a relation $r^M \subseteq M^{n_r}$.

We call f^M and r^M, the interpretation of f, respectively r, in M.

Recall that, by definition, $M^0 = \{\varnothing\}$. Therefore the interpretation of a constant c is a function that maps the unique element of M^0 to an element of M. We identify c^M with $c^M(\varnothing)$.

We may use the word model as a synonym for structure. But beware that, in some contexts, the word model is used to denote a particular kind of structure.

If M is an L-structure and $A \subseteq M$ is any subset, we write $L(A)$ for the language obtained by adding to $L_{\rm fun}$ the elements of A as constants. In this context, the elements of A are called parameters. There is a canonical expansion of M to an $L(A)$-structure that is obtained by setting $a^M = a$ for every $a \in A$.

1.3 Example The language of additive groups consists of the following function symbols:

1. a constant (that is, a function symbol of arity 0) 0
2. a unary function symbol (that is, of arity 1) $-$
3. a binary function symbol (that is, of arity 2) $+$.

In the language of multiplicative groups the three symbols above are replaced by 1, ${}^{-1}$, and $\cdot$ respectively. Any group is a structure in either of these two signatures with the obvious interpretation. Needless to say, not all structures with these signatures are groups.

The language of (unitary) rings contains all the symbols above except ${}^{-1}$. The language of ordered rings also contains the binary relation symbol $<$.

The following example is less straightforward. The reason for the choice of the language of vector spaces will become clear in Example 1.11 below.

1.4 Example Let F be a field. The language of vector spaces over F, which we denote by L_F, extends that of additive groups by a unary function symbol k for every $k \in F$.

Recall that a vector space over F is an abelian group M together with a function $\mu : F \times M \to M$ satisfying some well-known properties. To view a vector space over F as an L_F-structure, we interpret the group symbols in the obvious way and each $k \in F$ as the function $\mu(k, \text{-})$. See Example 2.6.

The languages in Examples 1.3 and 1.4, with the exception of that of ordered rings, are functional languages; that is, $L_{\text{rel}} = \varnothing$. The following example mention two important examples of relational languages, that is, languages where $L_{\text{fun}} = \varnothing$.

1.5 Example The language of strict orders contains a binary relation symbol, which is usually denoted by $<$. The language of graphs, contains a binary relation symbol $r(\text{-}, \text{-})$. See Chapter 6.

1.3 Terms

Let V be an infinite set whose elements we call variables. We use the letters x, y, z, etc. to denote variables or tuples of variables. We rarely refer to V explicitly, and we always assume that V is large enough for our needs.

We fix a signature L for the whole section.

1.6 Definition A term is a finite sequence of elements of $L_{\text{fun}} \cup V$ that are obtained inductively as follows

0. every variable, intended as a tuple of length 1, is a term
i. if $f \in L_{\text{fun}}$ and $t_1, \ldots, t_{n_f}$ are terms, then $f t_1, \ldots, t_{n_f}$ is a term.

We say L-term when we need to specify the language L.

Note that any constant f, intended as a tuple of length 1, is a term (by i, the term f is obtained concatenating $n_f = 0$ terms and prefixing by f). Terms that do not contain variables are called closed terms.

The intended meaning of, for instance, the term $+ + x\, y\, z$ is $(x + y) + z$. The first expression uses prefix notation; the second uses infix notation. When convenient, we informally use infix notation and add parentheses to improve legibility and avoid ambiguity.

The following lemma shows that prefix notation allows to write terms unambiguously without using parentheses.

1.7 Lemma (Unique legibility of terms) Let a be a sequence of terms. Suppose a can be obtained both by concatenating the terms $t_1, \ldots, t_n$ and by concatenating the terms $s_1, \ldots, s_m$. Then $n = m$ and $s_i = t_i$.

Proof. By induction on $|a|$. If $|a| = 0$ than $n = m = 0$ and there is nothing to prove. Suppose the claim holds for tuples of length k and let $a = a_1, \ldots, a_{k+1}$. Then a_1 is the first element of both t_1 and s_1. If a_1 is a variable, say x, then t_1 and s_1 are the term x and $n = m = 1$. Otherwise a_1 is a function symbol, say f. Then $t_1 = f\,\bar{t}$ and $s_1 = f\,\bar{s}$, where $\bar{t}$ and $\bar{s}$ are obtained by concatenating the terms $t'_1, \ldots, t'_p$ and $s'_1, \ldots, s'_p$. Now apply the induction hypothesis to $a_2, \ldots, a_{k+1}$ and to the terms

$\bar{t}\, t_2 \ldots t_n$ and $\bar{s}\, s_2 \ldots s_m$. □

If $x = x_1, \ldots, x_n$ is a tuple of distinct variables and $s = s_1, \ldots, s_n$ is a tuple of terms, we write $t[x/s]$ for the sequence obtained by replacing x by s coordinatewise. Proving that $t[x/s]$ is indeed a term is a tedious task that can be safely skipped.

If t is a term and $x_1, \ldots, x_n$ are (tuples of) variables, we write $t(x_1, \ldots, x_n)$ to declare that the variables occurring in t are among those that occur in $x_1, \ldots, x_n$. When a term has been presented as $t(x, y)$, we write $t(s, y)$ for $t[x/s]$.

Finally, we define the interpretation of a term in a structure M. We begin with closed terms. These are interpreted as 0-ary functions, i.e. as elements of the structure.

1.8 Definition Let t be a closed $L(M)$ term. The interpretation of t, denoted by t^M, is defined by induction on the syntax of t as follows

i. if $t = f\, t_1 \ldots t_{n_f}$, where $f \in L_{\text{fun}}$, then $t^M = f^M(t_1^M, \ldots, t_{n_f}^M)$.

Note that in i we have used Lemma 1.7 in an essential way. In fact this ensures that the sequence $t_1, \ldots, t_{n_f}$ uniquely determines the terms $t_1^M, \ldots, t_{n_f}^M$.

The inductive definition above is based on the case $n_f = 0$, that is, the case where f a constant, or a parameter. When $t = c$, a constant, $t_1 \ldots t_{n_f}$ is the empty tuple, and so $t^M = c^M(\varnothing)$, which we abbreviate as c^M. In particular, if $t = a$, a parameter, then $t^M = a^M = a$.

Now we generalize the interpretation to all (not necessarily closed) terms. If $t(x)$ is a term, we define $t^M(x) : M^x \to M$ to be the function that maps a to $t(a)^M$.

1.4 Substructures

In the working practice, a *substructure* is a subset of a structure that is closed under the interpretation of the functions in the language. But there are a few cases when we need the following formal definition.

1.9 Definition Fix a signature L and let M and N be two L-structures. We say that M is a substructure of N, and write $M \subseteq N$, if

1. the domain of M is a subset of the domain of N
2. $f^M = f^N \restriction M^{n_f}$ for every $f \in L_{\text{fun}}$
3. $r^M = r^N \cap M^{n_r}$ for every $f \in L_{\text{rel}}$.

Note that when f is a constant 2 becomes $f^M = f^N$, in particular the substructures of N contains at least all the constants of N.

If a set $A \subseteq N$ is such that

1. $f^N[A^{n_f}] \subseteq A$ for every $f \in L_{\text{fun}}$

then there is a unique substructure $M \subseteq N$ with domain A, namely, the structure with the following interpretation

2. $f^M \;=\; f^N \restriction A^{n_f}$ (which is a good definition by assumption 1)
3. $r^M \;=\; r^N \cap A^{n_r}$.

It is usual to confuse subsets of N that satisfy 1 with the unique substructure they support.

It is immediate to verify that the intersection of an arbitrary family of substructures of N is a substructure of N. Therefore, for any given $A \subseteq N$ we may define the substructure of N generated A as the intersection of all substructures of N that contain A. We write $\langle A \rangle_N$. The following easy proposition gives more concrete representation of $\langle A \rangle_N$

1.10 Lemma The following hold for every $A \subseteq N$

1. $\langle A \rangle_N \;=\; \Big\{ t^N \;:\; t \text{ a closed } L(A)\text{-term} \Big\}$
2. $\langle A \rangle_N \;=\; \Big\{ t^N(a) \;:\; t(x) \text{ an } L\text{-term and } a \in A^x \Big\}$
3. $\langle A \rangle_N \;=\; \bigcup\limits_{n \in \omega} A_n,$ where $A_0 = A$

$$A_{n+1} \;=\; A_n \;\cup\; \Big\{ f^N(a) \;:\; f \in L_{\text{fun}},\; a \in A_n^{n_f} \Big\}.$$

1.11 Example Let L be the language of groups. Let N be a group, which we consider as an L-structure in the natural way. Then the substructures of N are exactly the subgroups of N and $\langle A \rangle_N$ is the group generated by $A \subseteq N$. A similar claim is true when L_F is the signature of vector spaces over some fixed field F. The choice of the language is more or less fixed if we want that the algebraic and the model theoretic notion of substructure coincide.

1.5 Formulas

Fix a language L and a set of variables V as in Section 1.3. A formula is a finite sequence of symbols in $L_{\text{fun}} \cup L_{\text{rel}} \cup V \cup \{\doteq, \perp, \neg, \vee, \exists\}$. The last set contains the logical symbols that are called respectively

$\doteq$	equality	$\perp$	contradiction	$\neg$	negation
$\vee$	disjunction	$\exists$	existential quantifier.		

Syntactically, $\doteq$ behaves like a binary relation symbol. So, for convenience set $n_{\doteq} = 2$. However $\doteq$ is considered as a logic symbol because its semantic is fixed (it is always interpreted in the diagonal).

The definition below uses the prefix notation which simplifies the proof of the unique legibility lemma. However, in practice we always we use the infix notation: $t \doteq s$, $\varphi \vee \psi$, etc.

1.12 Definition A formula is any finite sequence is obtained with the following inductive procedure

0. if $r \in L_{\text{rel}} \cup \{\doteq\}$ and t is a tuple obtained concatenating n_r terms then $r\,t$ is a formula. Formulas of this form are called atomic

i. if φ e ψ are formulas then the following are formulas: $\perp$, $\neg\,\varphi$, $\vee\,\varphi\,\psi$, and $\exists x\,\varphi$, for any $x \in V$.

We use L to denote both the language and the set of formulas. We write L_{at} for the set of atomic formulas and L_{qf} for the set of quantifier-free formulas i.e. formulas where $\exists$ does not occur.

The proof of the following is similar to the analogous lemma for terms.

1.13 Lemma (Unique legibility of formulas) Let a be a sequence of formulas. Suppose a can be obtained both by the concatenation of the formulas $\varphi_1, \dots, \varphi_n$ or by the concatenation of the formulas $\psi_1, \dots, \psi_m$. Then $n = m$ and $\varphi_i = \psi_i$.

A formula is closed if all its variables occur under the scope of a quantifier. Closed formulas are also called sentences. We will do without a formal definition of *occurs under the scope of a quantifier* which is too lengthy. An example suffices: all occurrences of x are under the scope a quantifiers in the formula $\exists x\, \varphi$. These occurrences are called bounded. The formula $x \doteq y \wedge \exists x\, \varphi$ has free (i.e., not bound) occurrences of x and y.

Let x is a tuple of variables and t is a tuple of terms such that $|x| = |t|$. We write $\varphi[x/t]$ for the formula obtained substituting t for all free occurrences of x, coordinatewise.

We write $\varphi(x)$ to declare that the free variables in the formula φ are all among those of the tuple x. In this case we write $\varphi(t)$ for $\varphi[x/t]$.

We often use without explicit mention the following useful syntactic decomposition of formulas with parameters which we state without proof.

1.14 Lemma For every formula $\varphi(x) \in L(A)$ there is a formula $\psi(x\,;z) \in L$ and a tuple of parameters $a \in A^z$ such that $\varphi(x) = \psi(x\,;a)$.

Just as a term $t(x)$ is a name for a function $t(x)^M : M^x \to M$, a formula $\varphi(x)$ is a name for a subset $\varphi(x)^M \subseteq M^x$ which we call the subset of M defined by $\varphi(x)$. It is also very common to write $\varphi(M^x)$ for the set defined by $\varphi(x)$. In general sets of the form $\varphi(M^x)$ for some $\varphi(x) \in M$ are called definable.

1.15 Definition of truth For every formula φ with variables among those of the tuple x we define $\varphi(x)^M$ by induction as follows

o1. $$(\doteq t\, s)(x)^M = \Big\{a \in M^x \;:\; t^M(a) = s^M(a)\Big\}$$

o2. $$(r\, t_1 \dots t_n)(x)^M = \Big\{a \in M^x \;:\; \langle t_1^M(a), \dots, t_n^M(a)\rangle \in r^M\Big\}$$

i0. $$\bot(x)^M = \varnothing$$

i1. $$(\neg\xi)(x)^M = M^x \smallsetminus \xi(x)^M$$

i2. $$(\vee\, \xi\, \psi)(x)^M = \xi(x)^M \cup \psi(x)^M$$

i3. $$(\exists y\, \varphi)(x)^M = \bigcup_{a \in M} (\varphi[y/a])(x)^M.$$

Condition i2 assumes that ξ and ψ are uniquely determined by $\vee\, \xi\, \psi$. This is a guaranteed by the unique legibility o formulas, Lemma 1.13. Analogously, o1 e o2 assume Lemma 1.7.

The case when x is the empty tuple is far from trivial. Note that $\varphi(\varnothing)^M$ is a subset of $M^0 = \{\varnothing\}$. Then there are two possibilities either $\{\varnothing\}$ or $\varnothing$. We will read them as two truth values: True and False, respectively. If $\varphi^M = \{\varnothing\}$ we say that φ is true in M, if $\varphi^M = \varnothing$, we say that φ is false M and we write $M \vDash \varphi$, respectively $M \nvDash \varphi$. In words we may say that M models φ, respectively M does not model φ. It is immediate to verify that

$$\varphi(M^x) = \big\{a \in M^x : M \vDash \varphi(a)\big\}.$$

Note that usually, we say *formula* when, strictly speaking, we mean *pair* that consists of a formula and a tuple of variables. Such pairs are interpreted in definable sets (cf. Definition 1.15). In fact, if the tuple of variables were not given, the arity of the corresponding set is not determined.

In some contexts we also want to distinguish between two sorts of variables that play different roles. Some are placeholder for parameters, some are used to define a set. In the the first chapters this distinction is only a clue for the reader, in the last chapters it is an essential part of the definitions.

1.16 Definition A partitioned formula is a triple $\varphi(x\,;z)$ consisting of a formula and two tuples of variables such that the variables occurring in φ are all among x, z.

We use a semicolon to separate the two tuples of variables. Typically, z is the placeholder for parameters and x runs over the model.

1.6 Yet more notation

Now we abandon the prefix notation in favor of the infix notation. We also use the following logical connectives as abbreviations

$\top$	stands for	$\neg\bot$	tautology
$\varphi \wedge \psi$	stands for	$\neg[\neg\varphi \vee \neg\psi]$	conjunction

$\varphi \to \psi$	stands for	$\neg\varphi \vee \psi$	implication
$\varphi \leftrightarrow \psi$	stands for	$[\varphi \to \psi] \wedge [\psi \to \varphi]$	bi-implication
$\varphi \not\leftrightarrow \psi$	stands for	$\neg[\varphi \leftrightarrow \psi]$	exclusive disjunction
$\forall x\, \varphi$	stands for	$\neg\exists x \neg\, \varphi$	universal quantifier

We agree that $\to$ e $\leftrightarrow$ bind less than $\wedge$ e $\vee$. Unary connnectives (quantifiers and negation) bind stronger then binary connectives. For example

$$\exists x\, \varphi \wedge \psi \;\to\; \neg\xi \vee \vartheta \quad \text{reads as} \quad \Big[[\exists x\, \varphi] \wedge \psi\Big] \;\to\; \Big[[\neg\xi] \vee \vartheta\Big]$$

We say that $\forall x\, \varphi(x)$ and $\exists x\, \varphi(x)$ are the universal, respectively, existential closure of $\varphi(x)$. We say that $\varphi(x)$ holds in M when its universal closure is true in M. We say that $\varphi(x)$ is consistent in M when its existential closure is true in M.

The semantic of conjunction and disjunction is associative. Then for any finite set of formulas $\{\varphi_i : i \in I\}$ we can write without ambiguities

$$\bigwedge_{i\in I} \varphi_i \qquad\qquad \bigvee_{i\in I} \varphi_i$$

When $x = x_1, \ldots, x_n$ is a tuple of variables we write $\exists x\, \varphi$ or $\exists x_1, \ldots, x_n\, \varphi$ for $\exists x_1 \ldots \exists x_n\, \varphi$. There is a first order sentences that say that $\varphi(M^x)$ has at least n elements (also, no more than, or exactly n). It is convenient to use the following abbreviations.

$\exists^{\geq n} x\, \varphi(x)$	stands for	$\exists x_1, \ldots, x_n \Big[\bigwedge\limits_{1\leq i\leq n} \varphi(x_i) \;\wedge\; \bigwedge\limits_{1\leq i<j\leq n} x_i \dot{\neq} x_j \Big].$
$\exists^{\leq n} x\, \varphi(x)$	stands for	$\neg\exists^{\geq n+1} x\, \varphi(x)$
$\exists^{= n} x\, \varphi(x)$	stands for	$\exists^{\geq n} x\, \varphi(x) \;\wedge\; \exists^{\leq n} x\, \varphi(x)$

1.17 Exercise Let M be an L-structure and let $\psi(y), \varphi(x, y) \in L$. For each of the following conditions, write a sentence true in M exactly when

a. $\psi(M^y) \;\in\; \big\{\varphi(a, M^y) : a \in M^x\big\}$

b. $\big\{\varphi(a, M^y) : a \in M^x\big\}$ contains at least two sets

c. $\big\{\varphi(a, M^y) : a \in M^x\big\}$ contains only sets that are pairwise disjoint.

1.18 Exercise Let M be a structure in a signature that contains a symbol r for a binary relation. Write a sentence φ such that

a. $M \vDash \varphi$ if and only if there is $A \subseteq M$ such that $r^M \;\subseteq\; A \times \neg A$.

Remark: φ assert an asymmetric version of the property below

b. $M \vDash \psi$ if and only if there is $A \subseteq M$ such that $r^M \;\subseteq\; (A \times \neg A) \;\cup\; (\neg A \times A)$.

Assume M is a graph, what required in b is equivalent to saying that M is a *bipartite graph*, or equivalently that it has *chromatic number* 2 i.e., we can color the vertices with 2 colors so that no two adjacent vertices share the same color.

Chapter 2

Theories and elementarity

2.1 Logical consequences

A theory is a set $T \subseteq L$ of sentences. We write $M \vDash T$ when $M \vDash \varphi$ for every $\varphi \in T$. If $\varphi \in L$ is a sentence we write $T \vdash \varphi$ when

$$M \vDash T \quad \Rightarrow \quad M \vDash \varphi \qquad \text{for every } M.$$

In words, we say that φ is a logical consequence of T or that φ follows from T. If S is a theory $T \vdash S$ has a similar meaning. If $T \vdash S$ and $S \vdash T$ we say that T and S are logically equivalent. We may say that T axiomatizes S (or vice versa).

We say that a theory is consistent if it has a model. With the notation above, T is consistent if and only if $T \nvdash \perp$.

The closure of T under logical consequence is the set $\mathrm{ccl}(T)$ which is defined as follows:

$$\mathrm{ccl}(T) \quad = \quad \Big\{\varphi \in L \ : \ \text{sentence such that } T \vdash \varphi\Big\}$$

If T is a finite set, say $T = \{\varphi_1, \ldots, \varphi_n\}$ we write $\mathrm{ccl}(\varphi_1, \ldots, \varphi_n)$ for $\mathrm{ccl}(T)$. If $T = \mathrm{ccl}(T)$ we say that T is closed under logical consequences.

The theory of M is the set of sentences that hold in M and is denoted by $\mathrm{Th}(M)$. More generally, if $\mathcal{K}$ is a class of structures, $\mathrm{Th}(\mathcal{K})$ is the set of sentences that hold in every model in $\mathcal{K}$. That is

$$\mathrm{Th}(\mathcal{K}) \quad = \quad \bigcap_{M \in \mathcal{K}} \mathrm{Th}(M)$$

The class of all models of T is denoted by $\mathrm{Mod}(T)$. We say that $\mathcal{K}$ is axiomatizable if $\mathrm{Mod}(T) = \mathcal{K}$ for some theory T. If T is finite we say that $\mathcal{K}$ is finitely axiomatizable. To sum up

$$\begin{aligned}
\mathrm{Th}(M) \quad &= \quad \Big\{\varphi \ : \ M \vDash \varphi\Big\} \\
\mathrm{Th}(\mathcal{K}) \quad &= \quad \Big\{\varphi \ : \ M \vDash \varphi \text{ for all } M \in \mathcal{K}\Big\} \\
\mathrm{Mod}(T) \quad &= \quad \Big\{M \ : \ M \vDash T\Big\}
\end{aligned}$$

2.1 Example Let L be the language of multiplicative groups. Let T_g be the set containing the universal closure of following three formulas

1. $(x \cdot y) \cdot z \ = \ x \cdot (y \cdot z)$
2. $x \cdot x^{-1} \ = \ x^{-1} \cdot x \ = \ 1$
3. $x \cdot 1 \ = \ 1 \cdot x \ = x.$

Then T_g axiomatizes the theory of groups, i.e. $\mathrm{Th}(\mathcal{K})$ for $\mathcal{K}$ the class of all groups. Let φ be the universal closure of the following formula

$$z \cdot x \ = \ z \cdot y \ \rightarrow \ x \ = \ y.$$

As φ formalizes the cancellation property then $T_g \vdash \varphi$, that is, φ is a logical consequence of T_g. Now consider the sentence ψ which is the universal closure of

4. $x \cdot y \ = \ y \cdot x.$

So, commutative groups model ψ and non commutative groups model $\neg\psi$. Hence neither $T_g \vdash \psi$ nor $T_g \vdash \neg\psi$. We say that T_g does not decide ψ.

Note that even when T is a very concrete set, $\mathrm{ccl}(T)$ may be more difficult to grasp. In the example above T_g contains three sentences but $\mathrm{ccl}(T_g)$ is an infinite set containing sentences that code theorems of group theory yet to be proved.

2.2 Remark The following properties say that ccl is a finitary closure operator.

1. $T \subseteq \mathrm{ccl}(T)$ (extensive)
2. $\mathrm{ccl}(T) \ = \ \mathrm{ccl}\big(\mathrm{ccl}(T)\big)$ (idempotent)
3. $T \subseteq S \ \Rightarrow \ \mathrm{ccl}(T) \subseteq \mathrm{ccl}(S)$ (increasing)
4. $\mathrm{ccl}(T) \ = \ \bigcup\big\{\mathrm{ccl}(S) \ : \ S \text{ finite subset of } T\big\}.$ (finitary)

Properties 1-3 are easy to verify while 4 requires the compactness theorem.

In the next example we list a few algebraic theories with straightforward axiomatization.

2.3 Example We write T_{ag} for the theory of abelian groups which contains the universal closure of following

a1. $(x+y)+z \ = \ y+(x+z)$

a2. $x+(-x) \ = \ 0$

a3. $x+0 \ = \ x$

a4. $x+y \ = \ y+x.$

2.4 Example The theory T_{r} of (unitary) rings extends T_{ag} with

a5. $(x\cdot y)\cdot z \ = \ x\cdot(y\cdot z)$

a6. $1\cdot x \ = \ x\cdot 1 \ = \ x$

a7. $(x+y)\cdot z \ = \ x\cdot z+y\cdot z$

a8. $z\cdot(x+y) \ = \ z\cdot x+z\cdot y.$

The theory of commutative rings T_{cg} contains also com of examples 2.1.

2.5 Example The theory of ordered rings T_{or} extends T_{cr} with

o1. $x<z \ \rightarrow \ x+y<z+y$

o2. $0<x \wedge 0<z \ \rightarrow \ 0<x\cdot z.$

2.6 Example The axiomatization of the theory of vector spaces is less straightforward. Fix a field F. The language L_F extends the language of additive groups with a unary function for every element of F. The theory of vector fields over F extends T_{ag} with the following axioms (for all $h,k,l \in F$)

m1. $h\,(x+y) \ = \ h\,x+h\,y$

m2. $l\,x \ = \ h\,x+k\,x,$ where $l \ = \ h +_F k$

m3. $l\,x \ = \ h\,(k\,x),$ where $l \ = \ h \cdot_F k$

m4. $0_F\,x \ = \ 0$

m5. $1_F\,x \ = \ x$

The symbols 0_F and 1_F denote the zero and the unit of F. The symbols $+_F$ and $\cdot_F$ denote the sum and the product in F. These are not part of L_F, they are symbols we use in the metalanguage.

2.7 Example Recall from Example 1.5 that we represent a graph with a symmetric irreflexive relation. Therefore theory of graphs contains the following two axioms

1. $\neg r(x,x)$
2. $r(x,y) \rightarrow r(y,x)$.

Our last example is a trivial one.

2.8 Example Let L be the empty language The theory of infinite sets is axiomatized by the sentences $\exists^{\geq n} x\ (x = x)$ for all positive integer n.

2.9 Exercise Prove that $\operatorname{ccl}(\varphi \vee \psi) = \operatorname{ccl}(\varphi) \cap \operatorname{ccl}(\psi)$.

2.10 Exercise Prove that $\operatorname{Th}(\operatorname{Mod}(T)) = \operatorname{ccl}(T)$.

2.2 Elementary equivalence

The following is a fundamental notion in model theory.

2.11 Definition We say that M and N are elementarily equivalent if

ee. $N \vDash \varphi \quad \Leftrightarrow \quad M \vDash \varphi, \qquad$ for every sentence $\varphi \in L$.

In this case we write $M \equiv N$. More generally, we write $M \equiv_A N$ and say that M and N are elementarily equivalent over A if the following hold

a. $A \subseteq M \cap N$

ee'. equivalence ee above holds for every sentence $\varphi \in L(A)$.

The case when A is the whole domain of M is particularly important.

2.12 Definition When $M \equiv_M N$ we say that M is an elementary substructure of N and write $M \preceq N$.

The following lemma shows that the use of the term *substructure* in the definition above is appropriate.

2.13 Lemma If M and N are such that $M \equiv_A N$ and A is the domain of a substructure of M then A is also the domain a substructure of N and the two substructures coincide.

Proof. Let f be a function symbol and ler r be a relation symbol. It suffices to prove that $f^M(a) = f^N(a)$ for every $a \in A^{n_f}$ and that $r^M \cap A^{n_r} = r^N \cap A^{n_r}$.

If $b \in A$ is such that $b = f^M a$ then $M \vDash fa = b$. So, from $M \equiv_A N$, we obtain $N \vDash fa = b$, hence $f^N a = b$. This proves $f^M(a) = f^N(a)$.

Now let $a \in A^{n_r}$ and suppose $a \in r^M$. Then $M \vDash ra$ and, by elementarity, $N \vDash ra$, hence $a \in r^N$. By symmetry $r^M \cap A^{n_r} = r^N \cap A^{n_r}$ follows. $\square$

It is not easy to prove that two structures are elementary equivalent. A direct verification is unfeasible even for the most simple structures. It will take a few chapters before we are able to discuss concrete examples.

We generalize the definition of $\mathrm{Th}(M)$ to include parameters

$$\mathrm{Th}(M/A) = \Big\{\varphi : \text{ sentence in } L(A) \text{ such that } M \vDash \varphi\Big\}.$$

The following proposition is immediate

2.14 Proposition For every pair of structures M and N and every $A \subseteq M \cap N$ the following are equivalent

a. $M \equiv_A N$

b. $\mathrm{Th}(M/A) = \mathrm{Th}(N/A)$

c. $M \vDash \varphi(a) \quad\Leftrightarrow\quad N \vDash \varphi(a) \qquad$ for every $\varphi(x) \in L$ and every $a \in A^x$.

d. $\varphi(M^x) \cap A^x \;=\; \varphi(N^x) \cap A^x$ for every $\varphi(x) \in L$.

If we restate a and c of the proposition above when $A = M$ we obtain that the following are equivalent

a′. $M \preceq N$

b′. $N \vDash \mathrm{Th}(M/M)$

d′. $\varphi(M^x) \;=\; \varphi(N^x) \cap M^x \qquad$ for every $\varphi(x) \in L$.

Note that c' extends to all definable sets what Definition 1.9 requires for a few basic definable sets.

2.15 Example Let G be a group which we consider as a structure in the multiplicative language of groups. We show that if G is simple and $H \preceq G$ then also H is simple. Recall that G is simple if all its normal subgroups are trivial, equivalently, if for every $a \in G \smallsetminus \{1\}$ the set $\{gag^{-1} : g \in G\}$ generates the whole group G.

Assume H is not simple. Then there are $a, b \in H$ such that b is not the product of elements of $\{hah^{-1} : h \in H\}$. Then for every n

$$H \vDash \neg\exists x_1, \ldots, x_n \;\big(b \;=\; x_1 a x_1^{-1} \cdots\; x_n a x_n^{-1}\big)$$

By elementarity the same hold in G. Hence G is not simple.

2.16 Exercise Let $A \subseteq M \cap N$. Prove that $M \equiv_A N$ if and only if $M \equiv_B N$ for every finite $B \subseteq A$.

2.17 Exercise Let $M \preceq N$ and let $\varphi(x) \in L(M)$. Prove that $\varphi(M^x)$ is finite if and only if $\varphi(N)$ is finite and in this case $\varphi(N) = \varphi(M^x)$.

2.18 Exercise Let $M \preceq N$ and let $\varphi(x, z) \in L$. Suppose there are finitely many sets of the form $\varphi(a, N)$ for some $a \in N^x$. Prove that all these sets are definable over M.

2.19 Exercise Consider $\mathbb{Z}^n$ as a structure in the additive language of groups with the natural interpretation. Prove that $\mathbb{Z}^n \not\equiv \mathbb{Z}^m$ for every positive integers $n \neq m$. Hint: in $\mathbb{Z}^n$ there are at most 2^n elements that are not congruent modulo 2.

2.3 A nonstandard example

This section concerns an example that is useful to look at in some detail to get some familiarity with the notion of elementary substructure. The example is culturally interesting, because it formalises rigorously the notions of infinity and infinitesimal, which were used in Newton and Leibniz's time to develop real analysis. These notions were not well defined - in fact, they were inconsistent.

It was only in the mid 19th century that mathematicians of Weiertsrass' generation developed the notion of limit, thus providing rigorous grounds for the development of analysis.

Nonstandard analysis was developed by Abraham Robinson in the 1950s. Robinson found a way to formalise the ideas of infinity and infinitesimals through the concept of elementary extension.

The notation that follows will be used throughout this section.

The language we use contains

1. X, a relation symbol of arity n, for every $n \in \omega$ and every $X \subseteq \mathbb{R}^n$
2. f, a function symbol of arity n, for every $n \in \omega$ and every $f : \mathbb{R}^n \to \mathbb{R}$.

The standard model of real analysis is $\mathbb{R}$ with the natural interpretation of the symbols in our language, so we use the same symbol for an element of the language and its interpretation in $\mathbb{R}$. This is not an abuse of notation: in this case, elements and interpretations coincide.

Suppose $\mathbb{R}$ has a proper elementary extension ${}^*\mathbb{R}$. The existence of such an extension will be proved later. The interpretations of the symbols f and X in ${}^*\mathbb{R}$ will be denoted by ${}^*f$ and ${}^*X$, respectively. The elements of ${}^*\mathbb{R}$ are called hyperreals, the elements of $\mathbb{R}$ are called standard (hyper)reals, those in ${}^*\mathbb{R} \smallsetminus \mathbb{R}$ are called nonstandard (hyper)reals.

It is easy to verify that ${}^*\mathbb{R}$ is an ordered field. This is because the operations sum and product are in the language, and so is the order relation. The property of being an ordered field can be expressed via a set of sentences that are true in $\mathbb{R}$, and hence also in ${}^*\mathbb{R}$.

A hyperreal c is said to be infinitesimal if ${}^*\mathbb{R} \vDash |c| < \varepsilon$ for every positive standard ε. A hyperreal c is infinite if ${}^*\mathbb{R} \vDash k < |c|$ for every standard k; it is finite otherwise. Hence if c is infinite, then c^{-1} is infinitesimal. Clearly, all standard reals are finite, and 0 is the only standard infinitesimal.

The proof of the following lemma is left to the reader.

2.20 Fact Infinitesimals are closed under sum, product and multiplication by standard reals.

We begin with an existence proof.

2.21 Lemma There are infinite hyperreals and nonzero infinitesimals. Moreover, for every finite hyperreal c there is a unique standard b such that $b - c$ is infinitesimal.

Proof. If $c \in {}^*\mathbb{R}$ in infinite then c^{-1} is infinitesimal and if c is a nonzero infinitesimal, then c^{-1} is infinite. Then the first claim follows from the second one. Let c be finite. The set $\{a \in \mathbb{R} : c < a\}$ is a nonempty set of (standard) reals that is bounded from below. Let $b \in \mathbb{R}$ the least upper bound. We show that $b - c$ is an infinitesimal. Assume for a contradiction that $\varepsilon < |b - c|$ for some standard positive ε. Then $c < b - \varepsilon$, or $b + \varepsilon < c$, depending on whether $c < b$ or $b < c$. Both cases contradict that b is an infimum. This proves the existence of b.

To prove uniqueness, suppose that $b' \in \mathbb{R}$ and $b' - c$ is also infinitesimal. Then $b - b'$ is also infinitesimal. As 0 is the only standard infinitesimal $b = b'$. □

We define the following equivalence relation on ${}^*\mathbb{R}$: we write $a \approx b$ if $|a - b|$ is infinitesimal. The fact that this is an equivalence relation follows from Fact 2.20. The equivalence class of c is called the monad of c.'' By Lemma 2.21 if c is a finite hyperreal, there is a unique real in the monad of c. If c is finite, the unique standard real in the monad of c is called standard part of c and is denoted by $\mathrm{st}(c)$.

The existence of infinite nonstandard hyperreals shows that ${}^*\mathbb{R}$ is not Archimedean: standard integers are not cofinal in ${}^*\mathbb{R}$. However, by elementarity, the nonstandard integers ${}^*\mathbb{N}$ are cofinal in ${}^*\mathbb{R}$: from the perspective of an inhabitant of ${}^*\mathbb{R}$, the latter is a normal Archimedean field.

Dedekind's completeness is another fundamental property of $\mathbb{R}$ that does not hold in ${}^*\mathbb{R}$. An ordered set is Dedekind complete if every subset that is bounded above has a least upper bound. Then ${}^*\mathbb{R}$ is not Dedekind complete, e.g. the set of infinitesimals does not have a least upper bound. However, it is easy to verify that all bounded definable subsets of ${}^*\mathbb{R}$ have a least upper bound. It follows that Dedekind completeness is *not* a first-order property.

In the following lemma, the expressions on the left can be formalized as first-order sentences, and therefore they hold in $\mathbb{R}$ if and only if they hold in ${}^*\mathbb{R}$.

2.22 Proposition For every $f : \mathbb{R} \to \mathbb{R}$, for every $a, l \in \mathbb{R}$ the following equivalences hold.

a. $\lim\limits_{x \to +\infty} fx = +\infty \quad \Leftrightarrow \quad {}^*f(c)$ is positive and infinite for every infinite $c > 0$

b. $\lim\limits_{x \to +\infty} fx = l \quad \Leftrightarrow \quad {}^*f(c) \approx l$ for every infinite $c > 0$

c. $\lim\limits_{x \to a} fx = +\infty \quad \Leftrightarrow \quad {}^*f(c)$ is positive and infinite for every $c \approx a \neq c$

d. $\lim\limits_{x \to a} fx = l \quad \Leftrightarrow \quad {}^*f(c) \approx l$ for every $c \approx a \neq c$.

Proof. We prove part d and leave the rest as an exercise for the reader.

$\Rightarrow$. The left-hand side of the equivalence d asserts

1 $$\mathbb{R} \quad \vDash \quad \forall \varepsilon > 0\, \exists \delta > 0\, \forall x \Big[0 < |x - a| < \delta \quad \to \quad |fx - l| < \varepsilon \Big].$$

For consistency with standard notation in analysis, we use the Greek letters ε and δ as variables. The symbols $\dot{\varepsilon}$ e $\dot{\delta}$ denote parameters either in $\mathbb{R}$ or in ${}^*\mathbb{R}$.

Pick $c \neq a \approx c$ arbitrarily. We now check that ${}^*f(c) \approx l$, that is, $|{}^*fc - l| < \dot{\varepsilon}$ for every positive standard $\dot{\varepsilon}$. Pick $\dot{\varepsilon} \in \mathbb{R}^+$ arbitrarily and let $\dot{\delta} \in \mathbb{R}$ be given by 1. By elementarity, we have

$${}^*\mathbb{R} \quad \vDash \quad \forall x \Big[0 < |x - a| < \dot{\delta} \quad \to \quad |fx - l| < \dot{\varepsilon} \Big].$$

In particular

$$^*\mathbb{R} \quad \vDash \qquad 0<|c-a|<\dot\delta \ \ \rightarrow\ \ |fc-l|<\dot\varepsilon.$$

As $0<|c-a|<\dot\delta$ certainly holds (because $\dot\delta$ is standard) $|^*fc-l|<\dot\varepsilon$ follows.

$\Leftarrow$. Assume that 1 is false, that is,

2 $$\mathbb{R} \quad \vDash \quad \exists\,\varepsilon>0\,\forall\,\delta>0\,\exists x\,\Big[0<|x-a|<\delta \ \ \wedge\ \ \varepsilon\le|fx-l|\Big].$$

We want to show that $^*f(c)\not\approx l$ for some $c\neq a$. Let $\dot\varepsilon\in\mathbb{R}$ witness 2. By elementarity we have

$$^*\mathbb{R} \quad \vDash \qquad \forall\,\delta>0\,\exists x\,\Big[0<|x-a|<\delta \ \ \wedge\ \ \dot\varepsilon\le|fx-l|\Big].$$

Let $\dot\delta$ be an arbitrary infinitesimal. Then

$$^*\mathbb{R} \quad \vDash \qquad \exists x\,\Big[0<|x-a|<\dot\delta \ \ \wedge\ \ \dot\varepsilon\le|fx-l|\Big].$$

Any c that witnesses the above formula is such that $c\approx a\neq c$ and, simultaneously, $\varepsilon\le|^*fc-l|$. But $\dot\varepsilon$ is standard, hence $^*fc\not\approx l$. □

The following corollary is immediate.

2.23 Corollary For every $f:\mathbb{R}\to\mathbb{R}$ the following are equivalent

a. f is continuous

b. $^*f(a)\approx{}^*f(c)$ for every pair of finite hyperreals such that $c\approx a$.

In Corollary 2.23, it is important to restrict c to *finite* hyperreals, otherwise we get a stronger property.

2.24 Proposition For every $f:\mathbb{R}\to\mathbb{R}$, the following are equivalent

a. f is uniformly continuous

b. $^*f(a)\approx{}^*f(b)$ for every pair of hyperreals such that $a\approx b$.

Proof. a$\Rightarrow$b. Assume a, then

1 $$\mathbb{R} \quad \vDash \quad \forall\,\varepsilon>0\,\exists\,\delta>0\,\forall x,y\,\Big[|x-y|<\delta \ \ \rightarrow\ \ |fx-fy|<\varepsilon\Big].$$

Let $a\approx b$. We want to show that $|^*f(a)-{}^*f(b)|<\dot\varepsilon$ for every positive standard $\dot\varepsilon$. Given a standard positive $\dot\varepsilon$, let $\dot\delta$ be a standard real obtained from 1. Now elementarity gives

2 $$\mathbb{R} \quad \vDash \qquad \forall x,y\,\Big[|x-y|<\dot\delta \ \ \rightarrow\ \ |fx-fy|<\dot\varepsilon\Big].$$

Hence, in particular,

$$^*\mathbb{R} \quad \vDash \qquad |a-b|<\dot\delta \ \ \rightarrow\ \ |fa-fb|<\dot\varepsilon.$$

Since $a\approx b$ and $\dot\delta$ is standard, $|a-b|<\dot\delta$ is true. Therefore $|^*f(a)-{}^*f(b)|<\dot\varepsilon$.

b$\Rightarrow$a. Negate a

3 $$\mathbb{R} \quad \vDash \quad \exists\,\varepsilon>0\,\forall\,\delta>0\,\exists x,y\,\Big[|x-y|<\delta \ \ \wedge\ \ \varepsilon\le|fx-fy|\Big]$$

We want $a\approx b$ such that $\dot\varepsilon\le|^*f(a)-{}^*f(b)|$ for some positive standard $\dot\varepsilon$. Let $\dot\varepsilon$ be a standard real that witnesses 3. Elementarity gives

$$^*\mathbb{R} \quad \vDash \qquad \forall \delta > 0\, \exists x, y\, \Big[|x-y| < \delta \;\; \wedge \;\; \dot{\varepsilon} \leq |fx - fy|\Big].$$

Pick an arbitrary infinitesimal $\dot{\delta} > 0$ and let $a, b \in {}^*\mathbb{R}$ be given by the formula above

$$^*\mathbb{R} \quad \vDash \qquad |a-b| < \delta \;\; \wedge \;\; \dot{\varepsilon} \leq |fa - fb|.$$

Since $\dot{\delta}$ is infinitesimal, we have $a \approx b$, as required. □

The following proposition is an immediate consequence of Proposition 2.22.

2.25 Proposition For every unary function f and every standard a, the following are equivalent.

a. f is differentiable in a

b. $\mathrm{st}\left(\dfrac{f(a) - f(a+h)}{h}\right)$ exists and is constant for all infinitesimal $h \neq 0$.

2.26 Exercise Every definable (possibly with parameters) subset of $^*\mathbb{R}$ that is bounded from above has a least upper bound.

2.27 Exercise Prove that if the function $f : \mathbb{R} \to \mathbb{R}$ is injective, then $^*fa$ is nonstandard whenever a is nonstandard.

2.28 Exercise Prove that the following are equivalent for every subset $X \subseteq \mathbb{R}$

1. X is a finite set
2. $^*X = X$.

2.29 Exercise Prove that the following are equivalent for every subset $X \subseteq \mathbb{R}$

1. X is an open set in the usual topology on $\mathbb{R}$
2. $b \approx a \in X \quad \Rightarrow \quad b \in {}^*X$ for every $b \in {}^*\mathbb{R}$.

2.30 Exercise Prove that the following are equivalent for every subset $X \subseteq \mathbb{R}$

1. X is closed in the usual topology on $\mathbb{R}$
2. $a \in {}^*X \quad \Rightarrow \quad \mathrm{st}\, a \in X$ for every finite $a \in {}^*\mathbb{R}$.

2.31 Exercise Prove that the following are equivalent for every subset $X \subseteq \mathbb{R}$

1. X is bounded and closed in the usual topology on $\mathbb{R}$
2. for every $b \in {}^*X$ there is an $a \in X$ such that $a \approx b$.

2.32 Exercise For what sets $X \subseteq \mathbb{R}$ does the following hold for every $a, b \in {}^*\mathbb{R}$?

$b \approx a \in {}^*X \;\Rightarrow\; b \in {}^*X$.

2.33 Exercise Prove that $|\mathbb{R}| \leq |{}^*\mathbb{Q}|$, that is, that the cardinality of $^*\mathbb{Q}$ is at least that of the continuum. (Hint: define an injective function $f : \mathbb{R} \to {}^*\mathbb{Q}$ by choosing a nonstandard rational in the monad of every standard real.)

2.34 Exercise Prove that every proper elementary extension of $\mathbb{Q}$ in the language of rings plus a relation symbol for every subset of $\mathbb{Q}$ has infinities and infinitesimals.

2.4 Embeddings and isomorphisms

Here we prove that isomorphic structures are elementarily equivalent and a few related results. The following definition generalizes the notion of substructure (and will be further generalized in Section 3.4).

2.35 Definition An embedding of M into N is an injective total map $h : M \hookrightarrow N$ such that

1. $a \in r^M \quad \Leftrightarrow \quad ha \in r^N$ for every $r \in L_{\text{rel}}$ and $a \in M^{n_r}$;
2. $h\, f^M(a) \quad = \quad f^N(h\, a)$ for every $f \in L_{\text{fun}}$ and $a \in M^{n_f}$.

Note that when $c \in L_{\text{fun}}$ is a constant 2 reads $h\, c^M = c^N$. Therefore $M \subseteq N$ if and only if $\text{id}_M : M \to N$ is an embedding.

A surjective embedding is an isomorphism or, when domain and codomain coincide, an automorphism.

Condition 1 above and the assumption that h is injective can be summarized in the following

1′. $M \vDash r(a) \quad \Leftrightarrow \quad N \vDash r(ha)$ for every $r \in L_{\text{rel}} \cup \{\doteq\}$ and every $a \in M^{n_r}$.

Note also that, by straightforward induction on syntax, from 2 we obtain

2′ $h\, t^M(a) \quad = \quad t^N(h\, a)$ for every term $t(x)$ and every $a \in M^x$.

Combining 1′ and 2′ we obtain by straightforward induction on the syntax

3. $M \vDash \varphi(a) \quad \Leftrightarrow \quad N \vDash \varphi(ha)$ for every $\varphi(x) \in L_{\text{qf}}$ and every $a \in M^x$.

Recall that we write L_{qf} for the set of quantifier-free formulas. It is worth noting that when $M \subseteq N$ and $h = \text{id}_M$ then 3 becomes

3′ $M \vDash \varphi(a) \quad \Leftrightarrow \quad N \vDash \varphi(a)$ for every $\varphi(x) \in L_{\text{qf}}$ and for every $a \in M^x$.

In words this is summarized by saying that the truth of quantifier-free formulas is preserved under sub- and superstructure.

Finally, we prove that first order truth is preserved under isomorphism. We say that a map $h : M \to N$ fixes $A \subseteq M$ (pointwise) if $\text{id}_A \subseteq h$. An isomorphism that fixes A is also called an A-isomorphism.

2.36 Theorem If $h : M \to N$ is an isomorphism then for every $\varphi(x) \in L$

$$M \vDash \varphi(a) \quad \Leftrightarrow \quad N \vDash \varphi(ha) \text{ for every } a \in M^x$$

In particular, if h is an A-isomorphism then $M \equiv_A N$.

Proof. We proceed by induction on the syntax of $\varphi(x)$. When $\varphi(x)$ is atomic # holds by 3 above. Induction for the Boolean connectives is straightforward, so we only need to consider the existential quantifier. Assume as induction hypothesis that

$$M \vDash \varphi(a, b) \quad \Leftrightarrow \quad N \vDash \varphi(ha, hb) \text{ for every } a \in M^x \text{ and } b \in M.$$

We prove that the equivalence in the theorem holds for the formula $\exists y\, \varphi(x, y)$.

$$\begin{aligned} M \vDash \exists y\, \varphi(a, y) \quad &\Leftrightarrow \quad M \vDash \varphi(a, b) \text{ for some } b \in M \\ &\Leftrightarrow \quad N \vDash \varphi(ha, hb) \text{ for some } b \in M \quad \text{(by induction hypothesis)} \\ &\Leftrightarrow \quad N \vDash \varphi(ha, c) \text{for some } c \in N \quad (\Leftarrow \text{ by surjectivity}) \\ &\Leftrightarrow \quad N \vDash \exists y\, \varphi(ha, y). \end{aligned}$$

□

2.37 Corollary If $h : M \to N$ is an isomorphism then for every $\varphi(x) \in L$.

$$h\big[\varphi(M^x)\big] \quad = \quad \varphi(N^x)$$

We can now give a few very simple examples of elementarily equivalent structures.

2.38 Example Let L be the language of strict orders. Consider the intervals of $\mathbb{R}$ as structures in the natural way. The intervals $[0,1]$ and $[0,2]$ are isomorphic, hence $[0,1] \equiv [0,2]$ follows from Theorem 2.36. Clearly, $[0,1]$ is a substructure of $[0,2]$. However $[0,1] \not\preceq [0,2]$, in fact the formula $\forall x\,(x \leq 1)$ holds in $[0,1]$ but is false in $[0,2]$. This shows that $M \subseteq N$ and $M \equiv N$ does not imply $M \preceq M$.

Now we prove that $(0,1) \preceq (0,2)$. By Exercise 2.16 it suffices to verify that for every finite $B \subseteq (0,1)$ we have $(0,1) \equiv_B (0,2)$. This follows again from Theorem 2.36 as $(0,1)$ and $(0,2)$ are B-isomorphic for every finite $B \subseteq (0,1)$.

For the sake of completeness we also give the definition of homomorphism.

2.39 Definition A homomorphism is a total map $h : M \to N$ such that

1. $a \in r^M \quad \Rightarrow \quad ha \in r^N$ for every $r \in L_{\text{rel}}$ and $a \in M^{n_r}$;
2. $h\,f^M(a) \quad = \quad f^N(h\,a)$ for every $f \in L_{\text{fun}}$ and $a \in M^{n_f}$.

Note that only one implication is required in 1.

2.40 Exercise Prove that if $h : N \to N$ is an automorphism and $M \preceq N$ then $h[M] \preceq N$.

2.41 Exercise Let L be the empty language. Let $A, D \subseteq M$. Prove that the following are equivalent

1. D is definable over A
2. either D is finite and $D \subseteq A$, or $\neg D$ is finite and $\neg D \subseteq A$.

Hint: as structures are plain sets, every bijection $f : M \to M$ is an automorphism.

2.42 Exercise Prove that if $\varphi(x)$ is an existential formula and $h : M \hookrightarrow N$ is an embedding then

$$M \vDash \varphi(a) \quad \Rightarrow \quad N \vDash \varphi(ha) \qquad\qquad \text{for every } a \in M^x.$$

Recall that existential formulas as those of the form $\exists y\, \psi(x,y)$ for $\psi(x,y) \in L_{\text{qf}}$. Note that Theorem 10.7 proves that the property above characterizes existential formulas.

2.43 Exercise Let M be the model with domain $\mathbb{Z}$ in the language of additive groups which is interpreted in the natural way. Prove that there is no existential formula $\varphi(x)$ such that $\varphi(M^x)$ is the set of odd integers. Hint: use Exercise 2.42.

2.44 Exercise Let $\mathbb{Q}^+$ be the multiplicative group of positive rationals. Let M be the subgroup of the numbers of the form n/m for some odd integers m and n. Prove that $M \preceq \mathbb{Q}^+$. Hint: use the fundamental theorem of arithmetic and reason as in Example 2.38.

2.45 Exercise Prove that, in the language of strict orders, $\mathbb{R} \smallsetminus \{0\} \preceq \mathbb{R}$ and $\mathbb{R} \smallsetminus \{0\} \not\preceq \mathbb{R}$.

2.5 Quotient structure

The content of this section is mainly technical and only required later in the course. Its reading may be postponed.

If E is an equivalence relation on N we write $[c]_E$ for the equivalence class of $c \in N$. We use the same symbol for the equivalence relation on N^n defined as follow: if $a = a_1, \ldots, a_n$ and $b = b_1, \ldots, b_n$ are n-tuples of elements of N then $a\ E\ b$ means that $a_i\ E\ b_i$ holds for all i. It is easy to see that $b_1, \ldots, b_n \in [a, ..., a_n]_E$ if and only if $b_i \in [a_i]_E$ for all i. Therefore we use the notation $[a]_E$ for both the equivalence class of $a \in N^n$ and the tuple of equivalence classes $[a_1]_E, \ldots, [a_n]_E$.

2.46 Definition We say that the equivalence relation E on a structure N is a congruence if for every $f \in L_{\text{fun}}$

c1. $\qquad a\ E\ b \quad \Rightarrow \quad f^N a\ E\ f^N b;$

When E is a congruence on N we write N/E for the a structure that has as domain the set of E-equivalence classes in N and the following interpretation of $f \in L_{\text{fun}}$ and $r \in L_{\text{rel}}$:

c2. $\qquad f^{N/E}[a]_E \quad = \quad \big[f^N a\big]_E\,;$

c3. $\qquad [a]_E \ \in\ r^{N/E} \quad \Leftrightarrow \quad [a]_E \cap r^N \neq \varnothing.$

We call N/E the quotient structure.

By c1 the quotient structure is well defined. The reader will recognize it as a familiar notion by the following proposition (which is not required in the following and requires the notion of homomorphism, see Definition 2.39. Recall that the kernel of a total map $h : N \to M$ is the equivalence relation E such that

$$a\ E\ b \quad \Leftrightarrow \quad ha = hb$$

for every $a, b \in N$.

2.47 Proposition Let $h : N \to M$ be a surjective homomorphism and let E be the kernel of h. Then there is an isomorphism k that makes the following diagram commute

$$\begin{tikzcd} N \arrow[r, "h"] \arrow[d, "\pi"'] & M \\ N/E \arrow[ur, "k"'] & \end{tikzcd}$$

where $\pi : a \mapsto [a]_E$ is the projection map.

⚠ Quotients clutter the notation with brackets. To avoid the mess, we prefer to reason in N and tweak the satisfaction relation. Warning: though this is what everyone informally does, it is rarely spelled out so, feel free to reguard the following as a pedantry. In a nutshell, we are going to define the satisfaction relation $N/E \overset{*}{\vDash} \varphi(a)$ as an abbreviation of $N/E \vDash \varphi([a]_E)$.

Recall that in model theory, equality is not treated as a all other predicates. In fact, the interpretation of equality is fixed to always be the identity relation. In a few contexts is convinient to allow any congruence to interprete equality. This allows to work in N while thinking of N/E.

We define $N/E \overset{*}{\vDash}$ to be $N \vDash$ but with equality interpreted with E. The proposition below shows that this is the same thing as the regular truth in the quotient structure, $N/E \vDash$.

2.48 Definition For t_2, t_2 closed terms of $L(N)$ define

1* $\quad N/E \overset{*}{\vDash} t_1 = t_2 \quad\Leftrightarrow\quad t_1^N \, E \, t_2^N$

For t a tuple of closed terms of $L(N)$ and $r \in L_{\rm rel}$ a relation symbol

2* $\quad N/E \overset{*}{\vDash} r\,t \quad\Leftrightarrow\quad t^N \, E \, a \quad$ for some $a \in r^N$

Finally the definition is extended to all sentences $\varphi \in L(N)$ by induction in the usual way

3* $\quad N/E \overset{*}{\vDash} \neg\varphi \quad\Leftrightarrow\quad$ not $N/E \overset{*}{\vDash} \varphi$

4* $\quad N/E \overset{*}{\vDash} \varphi \wedge \psi \quad\Leftrightarrow\quad N/E \overset{*}{\vDash} \varphi$ and $N/E \overset{*}{\vDash} \psi$

5* $\quad N/E \overset{*}{\vDash} \exists x\, \varphi(x) \quad\Leftrightarrow\quad N/E \overset{*}{\vDash} \varphi(a) \quad$ for some $a \in N$.

Now, by induction on the syntax of formulas one can prove $\overset{*}{\vDash}$ does what required. In particular, $N/E \overset{*}{\vDash} \varphi(a) \leftrightarrow \varphi(b)$ for every $a \, E \, b$.

2.49 Proposition Let E be a congruence relation of N. Then the following are equivalent for every $\varphi(x) \in L$

1. $N/E \overset{*}{\vDash} \varphi(a)$;
2. $N/E \vDash \varphi([a]_E)$.

2.6 Completeness

A theory T is maximally consistent if it is consistent and there is no consistent theory S such that $T \subset S$. Equivalently, T contains every sentence φ consistent with T, that is, such that $T \cup \{\varphi\}$ is consistent. Clearly a maximally consistent theory is closed under logical consequences.

A theory T is complete if ccl$T$ is maximally consistent. Concrete examples will be given in the next chapters as it is not easy to prove that a theory is complete.

2.50 Proposition The following are equivalent

a. T is maximally consistent

b. $T = \mathrm{Th}(M)$ for some structure M

c. T is consistent and $\varphi \in T$ or $\neg\varphi \in T$ for every sentence φ.

Proof. To prove a⇒b, assume that T is consistent. Then there is $M \vDash T$. Therefore $T \subseteq \mathrm{Th}(M)$. As T is maximally consistent $T = \mathrm{Th}(M)$. Implication b⇒c is immediate. As for c⇒a note that if $T \cup \{\varphi\}$ is consistent then $\neg\varphi \notin T$ therefore $\varphi \in T$ follows from c. □

The proof of the proposition below is left as an exercise for the reader.

2.51 Proposition The following are equivalent

a. T is complete
b. there is a unique maximally consistent theory S such that $T \subseteq S$
c. T is consistent and $T \vdash \mathrm{Th}(M)$ for every $M \vDash T$
d. T is consistent and either $T \vdash \varphi$ or $T \vdash \neg\varphi$ for every sentence φ
e. T is consistent and $M \equiv N$ for every pair of models of T.

2.52 Exercise Prove that the following are equivalent

a. T is complete
b. for every sentence φ, o $T \vdash \varphi$ o $T \vdash \neg\varphi$ but not both.

By contrast prove that the following are *not* equivalent

a. T is maximally consistent
b. for every sentence φ, o $\varphi \in T$ o $\neg\varphi \in T$ but not both.

2.53 Exercise Prove that if T has exactly 2 maximally consistent extension T_1 and T_2 then there is a sentence φ such that $T, \varphi \vdash T_1$ and $T, \neg\varphi \vdash T_2$. State and prove the generalization to finitely many maximally consistent extensions.

2.7 The Tarski-Vaught test

There is no natural notion of *smallest* elementary substructure containing a set of parameters A. The downward Löwenheim-Skolem, which we prove in the next section, is the best result that holds in full generality. Given an arbitrary $A \subseteq N$ we shall construct a model $M \preceq N$ containing A that is small in the sense of cardinality. The construction selects one by one the elements of M that are required to realise the condition $M \preceq N$. Unfortunately, Definition 2.12 supposes full knowledge of the truth in M and it may not be applied during the construction. The following lemma comes to our rescue with a property equivalent to $M \preceq N$ that only mention the truth in N.

2.54 Lemma (Tarski-Vaught test) For every $A \subseteq N$ the following are equivalent

1. A is the domain of a structure $M \preceq N$
2. for every formula $\varphi(x) \in L(A)$, with $|x| = 1$,
   $N \vDash \exists x\, \varphi(x) \quad \Rightarrow \quad N \vDash \varphi(b)$ for some $b \in A$.

Proof. 1⇒2.

$$\begin{aligned} N \vDash \exists x\, \varphi(x) \ &\Rightarrow\ M \vDash \exists x\, \varphi(x) \\ &\Rightarrow\ M \vDash \varphi(b) \qquad \text{for some } b \in M \\ &\Rightarrow\ N \vDash \varphi(b) \qquad \text{for some } b \in A. \end{aligned}$$

2⇒1. Firstly, note that A is the domain of a substructure of N, that is, $f^N a \in A$ for every $f \in L_{\mathrm{fun}}$ and every $a \in A^{n_f}$. In fact, this follows from 2 with $fa = x$ for $\varphi(x)$.

Write M for the substructure of N with domain A. By induction on the syntax we prove that for every $\xi(x) \in L$

$$M \vDash \xi(a) \ \Leftrightarrow\ N \vDash \xi(a) \qquad \text{for every } a \in M^x.$$

If $\xi(x)$ is atomic the claim follows from $M \subseteq N$ and the remarks underneath Definition 2.35. The case of Boolean connectives is straightforward, so only the existential quantifier requires a proof. So, let $\xi(x)$ be the formula $\exists y\, \psi(x, y)$ and assume the induction hypothesis holds for $\psi(x, y)$

$$\begin{aligned} M \vDash \exists y\, \psi(a, y) \ &\Leftrightarrow\ M \vDash \psi(a, b) && \text{for some } b \in M \\ &\Leftrightarrow\ N \vDash \psi(a, b) && \text{for some } b \in M \\ &\Leftrightarrow\ N \vDash \exists y\, \psi(a, y). \end{aligned}$$

The second equivalence holds by induction hypothesis, in the last equivalence we use 2 for the implication $\Leftarrow$. □

2.8 Downward Löwenheim-Skolem

The main theorem of this section was proved by Löwenheim at the beginning of the last century. Skolem gave a simpler proof immediately afterwards. At the time, the result was perceived as paradoxical.

A few years earlier, Zermelo and Fraenkel provided a formalization of set theory in a first order language. The downward Löwenheim-Skolem theorem implies the existence of an infinite countable model M of set theory: this is the so-called Skolem paradox. The existence of M seems paradoxical because, in particular, a sentence that formalises the axiom of power set holds in M. Therefore M contains an element b which, in M, is the set of subsets of the natural numbers. But the set of elements of b is a subset of M, and therefore it is countable.

In fact, this is not a contradiction, because the expression *all subsets of the natural numbers* does not have the same meaning in M as it has in the real world. The notion of cardinality, too, acquires a different meaning. In the language of set theory, there is a first order sentence that formalises the fact that b is uncountable: the sentence says that there is no bijection between b and the natural numbers. Therefore the bijection between the elements of b and the natural numbers (which exists in the real world) does not belong to M. The notion of equinumerosity has a different meaning in M and in the real world, but those who live in M cannot realise this.

2.55 Theorem (Downward Löwenheim-Skolem) Let N be an infinite structure and fix some set $A \subseteq N$. Then there is a structure M of cardinality $\leq |L(A)|$ such that $A \subseteq M \preceq N$.

Proof. Set $\lambda = |L(A)|$. Below we construct a chain $\langle A_i : i < \omega \rangle$ of subsets of N. The chain begins at $A_0 = A$. Finally we set $M = \bigcup_{i<\omega} A_i$. All A_i will have cardinality $\leq \lambda$ so $|M| \leq \lambda$ follows.

Now we construct A_{i+1} given A_i. Assume as induction hypothesis that $|A_i| \leq \lambda$. Then $|L(A_i)| \leq \lambda$. For some fixed variable x let $\langle \varphi_k(x) \ : \ k < \lambda \rangle$ be an enumeration of the formulas in $L(A_i)$ that are consistent in N. For every k pick $a_k \in N$ such that $N \vDash \varphi_k(a_k)$. Define $A_{i+1} = A_i \cup \{a_k \ : \ k < \lambda\}$. Then $|A_{i+1}| \leq \lambda$ is clear.

We use the Tarski-Vaught test to prove $M \preceq N$. Suppose $\varphi(x) \in L(M)$ is consistent in N. As finitely many parameters occur in formulas, $\varphi(x) \in L(A_i)$ for some i. Then $\varphi(x)$ is among the formulas we enumerated at stage i and $A_{i+1} \subseteq M$ contains a solution of $\varphi(x)$. □

We will need to adapt the construction above to meet more requirements on the model M. To better control the elements that end up in M it is convenient to add one element at the time (above we add λ elements at each stage). We need to enumerate formulas with care if we want to complete the construction by stage λ.

Second proof of the downward Löwenheim-Skolem Theorem. Let $\pi : \lambda^2 \to \lambda$ be a bijection such that $j, k \leq \pi(j,k)$ for all $j, k < \lambda$. Suppose we have defined the sets A_j for every $j \leq i$ and let $\langle \varphi_{j,k}(x) \;:\; k < \lambda \rangle$ be an enumeration of the consistent formulas of $L(A_j)$. Let $j, k \leq i$ be such that $\pi(j,k) = i$. Let b be a solution of the formula $\varphi_{j,k}(x)$ and define $A_{i+1} = A_i \cup \{b\}$.

We use Tarski-Vaught test to prove $M \preceq N$. Let $\varphi(x) \in L(M)$ be consistent in N. Then $\varphi(x) \in L(A_j)$ for some j. Then $\varphi(x) = \varphi_{j,k}$ for some k. Hence a witness of $\varphi(x)$ is enumerated in M at stage $\pi(j,k) + 1$. □

2.56 Exercise Assume L is countable and let $M \preceq N$ have arbitrary (large) cardinality. Let $A \subseteq N$ be countable. Prove there is a countable model K such that $A \subseteq K \preceq N$ and $K \cap M \preceq N$ (in particular, $K \cap M$ is a model). Hint: adapt the construction used to prove the downward Löwenheim-Skolem Theorem.

2.57 Exercise The language contains only $<$, the relation of strict order. Prove that there is a countable ordinal α that is an elementary substructure of ω_1, the first uncountable ordinal.

2.9 Elementary chains

An elementary chain is a chain $\langle M_i : i < \lambda \rangle$ of structures such that $M_i \preceq M_j$ for every $i < j < \lambda$. The union (or limit) of the chain is the structure with as domain the set $\bigcup_{i<\lambda} M_i$ and as relations and functions the union of the relations and functions of M_i. It is plain that all structures in the chain are substructures of the limit.

2.58 Lemma Let $\langle M_i : i \in \lambda \rangle$ be an elementary chain of structures. Let N be the union of the chain. Then $M_i \preceq N$ for every i.

Proof. By induction on the syntax of $\varphi(x) \in L$ we prove

$$M_i \vDash \varphi(a) \;\Leftrightarrow\; N \vDash \varphi(a) \qquad \text{for every } i < \lambda \text{ and every } a \in M_i^x$$

As remarked in 3' of Section 2.4, the claim holds for quantifier-free formulas. Induction for Boolean connectives is straightforward so we only need to consider the existential quantifier

$$\begin{aligned} M_i \vDash \exists y\, \varphi(a,y) \;&\Rightarrow\; M_i \vDash \varphi(a,b) && \text{for some } b \in M_i. \\ &\Rightarrow\; N \vDash \varphi(a,b) && \text{for some } b \in M_i \subseteq N \end{aligned}$$

where the second implication follows from the induction hypothesis. Vice versa

$$N \vDash \exists y\, \varphi(a,y) \;\Rightarrow\; N \vDash \varphi(a,b) \qquad \text{for some } b \in N$$

Without loss of generality we can assume that $b \in M_j$ for some $j \geq i$ and obtain

$$\Rightarrow\; M_j \vDash \varphi(a,b) \qquad \text{for some } b \in M_j$$

Now apply the induction hypothesis to $\varphi(x,y)$ and M_j

$$\Rightarrow \quad M_j \vDash \exists y\, \varphi(a, y)$$
$$\Rightarrow \quad M_i \vDash \exists y\, \varphi(a, y)$$

where the last implication holds because $M_i \preceq M_j$. □

2.59 Exercise Let $\langle M_i : i \in \lambda \rangle$ be an chain of elementary substructures of N. Let M be the union of the chain. Prove that $M \preceq N$ and note that Lemma 2.58 is not required.

2.60 Exercise Give an alternative proof of Exercise 2.56 using the downward Löwenheim-Skolem Theorem (instead of its proof). Hint: construct two countable chains of countable models such that $K_i \cap M \subseteq M_i \preceq N$ and $A \cup M_i \subseteq K_{i+1} \preceq N$. The required model is $K = \bigcup_{i \in \omega} K_i$. In fact it is easy to check that $K \cap M = \bigcup_{i \in \omega} M_i$.

Chapter 3

Types and morphisms

There is a lot of notation in this chapter. The reader may skim through and return to it when necessary. In Section 3.1 and 3.2 we introduce distributive lattices and prime filters and prove Stone's representation theorem for distributive lattices. In Section 3.3 we discuss lattices that arise from sets of formulas and their prime filters (prime types). These sections are only required for the discussion of Hilbert's Nullstellensatz, see Section 8.7 below.

3.1 Semilattices and filters

A preorder is a set $\mathbb{P}$ with a reflexive and transitive relation. We usually denote the preorder by $\leq$. If $A, B \subseteq \mathbb{P}$ we write $A \leq B$ if $a \leq b$ for every $a \in A$ and $b \in B$. We write $a \leq B$ and $A \leq b$ for $\{a\} \leq B$ and $A \leq \{b\}$ respectively.

Quotienting a preorder by the equivalence relation

$$a \sim b \quad \Leftrightarrow \quad a \leq b \quad \text{and} \quad b \leq a$$

gives a (partial) order. We often do not distinguish between a preorder and the partial order associated to it. Preorders are very common; here we are mainly interested in preorders induced by the relation of logical consequence

$$\varphi \leq \psi \quad \Leftrightarrow \quad \varphi \vdash \psi$$

or, more generally, induced by the relation of logical consequence modulo a theory T, that is,

$$\varphi \leq \psi \quad \Leftrightarrow \quad T \cup \{\varphi\} \vdash \psi.$$

A partial order $\mathbb{P}$ is a lower semilattice if for each pair $a, b \in \mathbb{P}$ there is a maximal element c such that $c \leq \{a, b\}$. We call c the meet of a and b. The meet is unique and is denoted by $a \wedge b$. Dually, a partial order is an upper semilattice if for each pair of elements a and b there is a minimal element c such that $\{a, b\} \leq c$. This c is called the join of a and b. The join is unique and is denoted by $a \vee b$. A lattice is simultaneously a lower and an upper semilattice.

For instance, the family of subsets of $\mathbb{R}$ that are finite union of left-open, right-closed intervals is an upper-semilattice w.r.t. the relation of inclusion. However, it is not a lattice.

An element c such that $c \leq \mathbb{P}$ is called a lower bound or a bottom. An element such that $\mathbb{P} \leq c$ is called an upper bound or a top. Lower and upper bounds are unique and will be denoted by 0, respectively 1. Other symbols common in the literature are $\bot$, respectively $\top$. A semilattice is bounded if it has both an upper and a lower bound.

For the rest of this section we assume that $\mathbb{P}$ is a bounded lower semilattice.

The meet is associative and commutative

$$\begin{aligned}(a \wedge b) \wedge c &= a \wedge (b \wedge c)\\ a \wedge b &= a \wedge b.\end{aligned}$$

Hence we may unambiguously write $a_1 \wedge \cdots \wedge a_n$. When $C \subseteq \mathbb{P}$ is finite, we write $\wedge C$ for the meet of all the elements of C. We agree that $\wedge\varnothing = 1$.

In an upper semilattice, the dual properties hold for the join. We write $\vee C$ for the join of all elements in C and we agree that $\vee\varnothing = 0$.

A filter of $\mathbb{P}$ is a non-empty set $F \subseteq \mathbb{P}$ that satisfies the following for all $a, b \in \mathbb{P}$

f1. $a \in F$ and $a \leq b \;\Rightarrow\; b \in F$

f2. $a, b \in F \;\Rightarrow\; a \wedge b \in F$.

We say that F is a proper filter if $F \neq \mathbb{P}$, equivalently if $0 \notin F$. We say that F is principal if $F = \{b \;:\; a \leq b\}$ for some $a \in \mathbb{P}$. More precisely, we say that F is the principal filter generated by a. A proper filter F is maximal if there is no filter H such that $F \subset H \subset \mathbb{P}$.

A set $B \subseteq \mathbb{P}$ has the finite intersection property if $\wedge C \neq \varnothing$ for every finite $C \subseteq B$. For $B \subseteq \mathbb{P}$ we define the filter generated by B to be the intersection of all the filters containing B. It is easy to verify that this is indeed a filter. When B is a finite set, the filter generated by B is the principal filter generated by $\wedge B$. In general we have the following.

3.1 Proposition For every $B \subseteq \mathbb{P}$, the filter generated by B is the set

$$\{a \;:\; \wedge C \leq a \text{ for some finite non-empty } C \subseteq B\}.$$

In particular B is contained in a proper filter if and only if it has the finite intersection property.

We say that a filter F is maximal relative to c if $c \notin F$ and $c \in H$ for every $H \supset F$. So, a filter F is maximal if it is maximal relative to 0. We say that F is relatively maximal when it is maximal relative to some c.

3.2 Proposition Let $B \subseteq \mathbb{P}$ and let $c \in \mathbb{P}$. If $\wedge C \nleq c$ for every finite non-empty $C \subseteq B$, then B is contained in a maximal filter relative to c.

Proof. Let $\mathcal{F}$ be the set of filters F such that $B \subseteq F$ and $c \notin F$. By Proposition 3.1, $\mathcal{F}$ is non-empty. It is immediate that $\mathcal{F}$ is closed under unions of arbitrary chains. Then, by Zorn's lemma, $\mathcal{F}$ has a maximal element. □

3.3 Exercise Let $B \subseteq \mathbb{P}$. Let $c \in \mathbb{P}$ be such that $\wedge C \nleq c$ for every finite non-empty $C \subseteq B$. Prove that the following are equivalent

1. B is a maximal filter relative to c
2. $a \notin B \;\Rightarrow\; b \wedge a \leq c$ for some $b \in B$.

3.4 Exercise Let $F \subseteq \mathbb{P}$ be a non-principal filter. Is F always contained in a maximal non-principal filter ?

3.2 Distributive lattices and prime filters

Let $\mathbb{P}$ be a lattice. We say that $\mathbb{P}$ is distributive if for every $a, b, c \in \mathbb{P}$

$$
\begin{aligned}
a \wedge (b \vee c) &= (a \wedge b) \vee (a \wedge c) \\
a \vee (b \wedge c) &= (a \vee b) \wedge (a \vee c)
\end{aligned}
$$

Throughout this section we assume that $\mathbb{P}$ is a bounded distributive lattice.

A proper filter F is prime if for every $a, b \in \mathbb{P}$

$$a \vee b \in F \quad \Rightarrow \quad a \in F \quad \text{or} \quad b \in F.$$

3.5 Proposition Every relatively maximal filter of $\mathbb{P}$ is prime.

Proof. Let F be maximal relative to c and assume that $a \notin F$ and $b \notin F$. Then, by Exercise 3.3, there are $d_1, d_2 \in F$ such that $d_1 \wedge a \leq c$ and $d_2 \wedge b \leq c$. Let $d = d_1 \wedge d_2$. Then $d \wedge a \leq c$ and $d \wedge b \leq c$ and therefore $(d \wedge a) \vee (d \wedge b) \leq c$. Hence, by distributivity, $d \wedge (a \vee b) \leq c$. Then $a \vee b \notin F$. □

The Stone space of $\mathbb{P}$ is a topological space that we denote by $S(\mathbb{P})$. The points of $S(\mathbb{P})$ are the prime filters of $\mathbb{P}$. The closed sets of the Stone topology are arbitrary intersections of sets of the form

$$[a]_{\mathbb{P}} \quad = \quad \Big\{ F \; : \; \text{prime filter such that } a \in F \Big\}.$$

for $a \in \mathbb{P}$. In other words, the sets above form a base of closed sets of the Stone topology. Using 1 and 3 in the following proposition the reader can easily check that this is indeed a base for a topology.

3.6 Proposition For every $a, b \in \mathbb{P}$ we have

1. $[0]_{\mathbb{P}} = \varnothing$
2. $[1]_{\mathbb{P}} = S(\mathbb{P})$
3. $[a]_{\mathbb{P}} \cup [b]_{\mathbb{P}} = [a \vee b]_{\mathbb{P}}$
4. $[a]_{\mathbb{P}} \cap [b]_{\mathbb{P}} = [a \wedge b]_{\mathbb{P}}$.

Proof. The verification is immediate. Only 3 requires that the filters in $S(\mathbb{P})$ are prime. □

The closed subsets of $S(\mathbb{P})$ ordered by inclusion form a distributive lattice. The following is a representation theorem for distributive lattices.

3.7 Theorem The map $a \mapsto [a]_{\mathbb{P}}$ is an embedding of $\mathbb{P}$ in the lattice of the closed subsets of $S(\mathbb{P})$. In particular

1. $0 \mapsto \varnothing$
2. $1 \mapsto S(\mathbb{P})$
3. $a \vee b \mapsto [a]_{\mathbb{P}} \cup [b]_{\mathbb{P}}$
4. $a \wedge b \mapsto [a]_{\mathbb{P}} \cap [b]_{\mathbb{P}}$.

Proof. It is immediate that the map above preserves the order and Proposition 3.6 shows that it preserves the lattice operations. We prove that the map is injective. Let $a \neq b$, say $a \nleq b$. We claim that $[a]_{\mathbb{P}} \nsubseteq [b]_{\mathbb{P}}$. There is a filter F that contains a and is maximal relative to b. By Proposition 3.5 such an F is prime. Then $F \in [a]_{\mathbb{P}} \smallsetminus [b]_{\mathbb{P}}$. □

3.8 Theorem With the Stone topology, $S(\mathbb{P})$ is a compact space.

Proof. Let $\langle [a_i]_{\mathbb{P}} : i \in I \rangle$ be basic closed sets such that for every finite $J \subseteq I$

a. $$\bigcap_{i \in J} [a_i]_{\mathbb{P}} \neq \varnothing.$$

We claim that

b. $$\bigcap_{i \in I} [a_i]_{\mathbb{P}} \neq \varnothing.$$

By 4 of Proposition 3.6 and a we obtain that $\wedge C \nleq 0$ for every finite $C \subseteq \{a_i : i \in I\}$. By Proposition 3.2, there is a maximal (relative to 0) filter containing $\{a_i : i \in I\}$. By Proposition 3.5, such filter is prime and it belongs to the intersection in b. □

Let $a, b \in \mathbb{P}$. If $a \wedge b = 0$ and $a \vee b = 1$, we say that b is the complement of a (and vice versa). The complement of an element need not exist. If the complement exists it is unique, the complement of a is denoted by $\neg a$.

3.9 Lemma Let $U \subseteq S(\mathbb{P})$ be a clopen set. Then $U = [a]_{\mathbb{P}}$ for some $a \in \mathbb{P}$, and $\neg a$ exists.

Proof. As both U and $S(\mathbb{P}) \smallsetminus U$ are closed, for some sets $A, B \subseteq \mathbb{P}$

$$\bigcap_{x \in A} [x]_{\mathbb{P}} = U$$

$$\bigcap_{y \in B} [y]_{\mathbb{P}} = S(\mathbb{P}) \smallsetminus U.$$

By compactness, that is Theorem 3.8, there are some finite $A_0 \subseteq A$ and $B_0 \subseteq B$ such that

$$\bigcap_{x \in A_0} [x]_{\mathbb{P}} \cap \bigcap_{y \in B_0} [y]_{\mathbb{P}} = \varnothing$$

Let $a = \wedge A_0$ and $b = \wedge B_0$. From claim 4 of Proposition 3.6 we obtain $[a]_{\mathbb{P}} \cap [b]_{\mathbb{P}} = \varnothing$. Therefore $U \subseteq [a]_{\mathbb{P}} \subseteq S(\mathbb{P}) \smallsetminus [b]_{\mathbb{P}} \subseteq U$. Hence $U = [a]_{\mathbb{P}}$ and $b = \neg a$. □

A Boolean algebra is a bounded distributive lattice where every element has a complement. In a Boolean algebra, the sets $[a]_{\mathbb{P}}$ are clopen and they form also a base of open set of the topology of $S(\mathbb{P})$. A topology that has a base of clopen sets is called zero-dimensional. By the following proposition the Stone topology of a Boolean algebra is Hausdorff.

A proper filter of a Boolean algebra $\mathbb{P}$ is an ultrafilter if either $a \in F$ or $\neg a \in F$ for every $a \in \mathbb{P}$.

3.10 Proposition Let $\mathbb{P}$ be a Boolean algebra. Then the following are equivalent

1. F is maximal
2. F is prime
3. F is an ultrafilter.

Proof. For 2⇒3 observe that $a \vee \neg a \in F$. The rest is immediate. □

The most natural example of Boolean algebra is $\mathcal{P}(I)$ where I is any set. Union and intersection are the join, respectively meet of the algebra. We say filter on I for filter of $\mathcal{P}(I)$.

3.11 Exercise Let I be infinite. Let $F = \{a \subseteq I \ : \ |I \smallsetminus a| < \omega\}$. Then F is a filter on I which is called he Fréchet's filter. Prove that every non-principal ultrafilter on I contains Fréchet's filter.

3.12 Exercise Prove that the stone topology on $S(\mathbb{P})$ has a base of open compact sets.

3.13 Exercise Suppose we had defined $S(\mathbb{P})$ as the set of relatively maximal filters. What could possibly go wrong ?

3.3 Types as filters

In this section we work with a fixed set of formulas Δ all with free variables among those of some fixed tuple x. The tuple x is not displayed in the notation. Subsets of Δ are called Δ-types.

We associate to Δ a bounded lattice $\mathbb{P}(\Delta)$. This is the closure under conjunction and disjunction of the formulas in $\Delta \cup \{\bot, \top\}$. The (pre)order relation in $\mathbb{P}(\Delta)$ is given by

$$\psi \leq \varphi \quad \Leftrightarrow \quad T \cup \{\psi\} \ \vdash \ \varphi,$$

for some fixed theory T. In this section, to lighten notation, we absorb T in the symbol $\vdash$. If p is a Δ-type we denote by $\langle p \rangle_{\mathbb{P}}$ the filter in $\mathbb{P}(\Delta)$ generated by p.

A few propositions in this section requires the compactness theorem (for types), Theorem 5.7 which is only proved below.

3.14 Lemma For every Δ-type p

$$\langle p \rangle_{\mathbb{P}} \quad = \quad \Big\{ \varphi \ \in \ \mathbb{P}(\Delta) \quad : \quad p \ \vdash \ \varphi \Big\}.$$

In particular p is consistent if and only if $\langle p \rangle_{\mathbb{P}}$ is a proper filter.

Proof. Inclusion $\subseteq$ is clear. If $p \vdash \varphi$, by compactness, $\psi \vdash \varphi$ for some formula ψ that is conjunction of formulas in p. Then $\varphi \in \langle p \rangle_{\mathbb{P}}$ follows from $\psi \in \langle p \rangle_{\mathbb{P}}$ and $\psi \leq \varphi$. $\square$

We say that $p \subseteq \Delta$ is a principal Δ-type if $\langle p \rangle_{\mathbb{P}}$ is a principal. The following lemma is an immediate consequence of the Compactness Theorem. Note that in 3 the formula φ is arbitrary, possibly not even in $\mathbb{P}(\Delta)$.

3.15 Lemma For every Δ-type p the following are equivalent

1. p is principal
2. $\psi \ \vdash \ p \ \vdash \ \psi$ for some formula ψ (here ψ is any formula, it need not be in Δ)
3. $\varphi \ \vdash \ p$ where φ is conjunction of formulas in p.

Proof. Implications 1$\Rightarrow$2 is immediate by Lemma 3.14. To prove 2$\Rightarrow$3 suppose $\psi \vdash p \vdash \psi$. Apply compactness to obtain a formula φ, conjunction of formulas in p, such that $\varphi \vdash \psi$. Implications 3$\Rightarrow$1is trivial. $\square$

3.16 Definition We say that $p \subseteq \Delta$ is a prime Δ-type if $\langle p \rangle_{\mathbb{P}}$ is a prime filter. We say that p is a maximal Δ-type if $\langle p \rangle_{\mathbb{P}}$ is a maximal filter.

Though in general neither Δ nor $\mathbb{P}(\Delta)$ are closed under negation, Lemma 3.14 has the following consequence.

3.17 Proposition For every consistent Δ-type p the following are equivalent

1. p is maximal
2. p is consistent and either $p \vdash \varphi$ or $p \vdash \neg\varphi$ for every formula $\varphi \in \Delta$.

Proof. 2⇒1. Assume 2. As p is consistent, $\langle p \rangle_{\mathbb{P}}$ is a proper filter. To prove that it is maximal suppose $\varphi \notin \langle p \rangle_{\mathbb{P}}$. Then $p \nvdash \varphi$ and from 2 it follows that $p \vdash \neg\varphi$. Hence no proper filter contains $p \cup \{\varphi\}$.

1⇒2. As $\langle p \rangle_{\mathbb{P}}$ is proper, p is consistent. Suppose $p \nvdash \varphi$. Then $\varphi \notin \langle p \rangle_{\mathbb{P}}$ and, as $\langle p \rangle_{\mathbb{P}}$ is maximal, $p \cup \{\varphi\}$ generates the improper filter. Then $p \cup \{\varphi\}$ is inconsistent. Hence $p \vdash \neg\varphi$. □

Given a model M and a tuple $c \in M^x$ the Δ-type of c in M is the sets

$$\Delta\text{-tp}_M(c) \quad = \quad \Big\{\varphi(x) \in \Delta \ : \ M \vDash \varphi(c)\Big\}$$

When the model M is clear from the context we omit the subscript. When x and c are the empty tuple, we write $\text{Th}_\Delta(M)$ for $\Delta\text{-tp}_M(c)$.

3.18 Lemma For every Δ-type p the following are equivalent

1. p is prime
2. $p \ \cup \ \big\{\neg\varphi \ : \ \varphi \ \in \ \Delta \ \text{ such that } \ p \nvdash \varphi\big\}$ is consistent
3. $\big\{\varphi \in \Delta : p \vdash \varphi\big\} \ = \ \Delta\text{-tp}_M(c)$ for some model M and some tuple $c \in M^x$.

Proof. Implications 2⇒3⇒1 are clear, we prove 1⇒2. By compactness if the type in 2 is inconsistent then there are finitely many formulas $\varphi_1, \dots, \varphi_n \in \Delta$ such that $p \nvdash \varphi_i$ and

$$p \ \vdash \ \bigvee_{i=1}^{n} \varphi_i.$$

hence p is not prime. □

The following corollary is immediate. It simplifies the task of verifying that a given type is prime.

3.19 Corollary For every Δ-type p the following are equivalent

1. p is prime
2. $p \ \vdash \ \bigvee_{i=1}^{n} \varphi_i \quad \Rightarrow \quad p \vdash \varphi_i$ for some $i \leq n$, $\quad$ for every n and every $\varphi_1, \dots, \varphi_n \in \Delta$.

The set Δ above contains only formulas with variables among those of the tuple x. In the following it is convenient to consider sets Δ that are closed under substitution of variables with any other variable. The set of prime Δ-types is denoted by $S(\Delta)$, and we write $S_x(\Delta)$ when we restrict to types in with variables among those of the tuple x.

The most common Δ used in the sequel is the set of all formulas in $L(A)$. Then the underlying theory is $\mathrm{Th}(M/A)$ for some given model M containing A. In this case we write $\mathrm{tp}_M(c/A)$ for Δ-$\mathrm{tp}_M(c)$ or, when A is empty, $\mathrm{tp}_M(c)$. The set $S_x(\Delta)$ is denoted by $S_x(A)$. The topology on $S_x(A)$ is generated by the clopen

$$[\varphi(x)] \quad = \quad \Big\{p \in S_x(A) \;:\; \varphi(x) \in p\Big\}.$$

Frequently Δ is the set of all formulas of a given syntactic form. Then we use a more suggestive notation as summarized below.

3.20 Notation The following are some of the most common Δ-types and Δ-theories

1. at-tp(c), $\mathrm{Th}_{\mathrm{at}}(M)$ when $\Delta = L_{\mathrm{at}}$
2. at$^{\pm}$-tp(c), $\mathrm{Th}_{\mathrm{at}^{\pm}}(M)$ when $\Delta = L_{\mathrm{at}^{\pm}}$
3. qf-tp(c), $\mathrm{Th}_{\mathrm{qf}}(M)$ when $\Delta = L_{\mathrm{qf}}$.

The types/theories in 2 and 3 are logically equivalent. Most used is the theory $\mathrm{Th}_{\mathrm{at}^{\pm}}(M/M)$ which is called the diagram of M and has a dedicated symbol: $\mathrm{Diag}(M)$. The theory $\mathrm{Th}(M/M)$ is called the elementary diagram of M.

3.21 Remark Let $A \subseteq M \cap N$. The following are equivalent

1. $N \vDash \mathrm{Diag}\langle A\rangle_M$
2. $\langle A\rangle_M$ is a substructure of N.

3.4 Morphisms

First we set the meaning of the word map.

3.22 Definition A map consists a triple $f : M \to N$ where

1. M is a set (usually a structure) called the domain of the map
2. N is a set (usually a structure) called the codomain the map
3. f is a function with domain of definition $\mathrm{dom}\, f \subseteq M$ and image $\mathrm{rng}\, f \subseteq N$.

By cardinality of $f : M \to N$ we understand the cardinality of the function f.

If $\mathrm{dom}\, f = M$ we say that the map is total; if $\mathrm{rng}\, f = N$ we say that it is surjective. The composition of two maps and the inverse of a map are defined in the obvious way.

3.23 Definition Let Δ be a set of formulas. The map $h : M \to N$ is a Δ-morphism if it preserves the truth of all formulas in Δ. By this we mean that

p. $M \vDash \varphi(a) \quad \Rightarrow \quad N \vDash \varphi(ha)$ for every $\varphi(x) \in \Delta$ and every $a \in (\mathrm{dom}\, h)^x$.

When h is total we we say it is a Δ-embedding.

The following is easy remark is useful to simplify notation.

3.24 Remark When Δ is closed under substitution of free variables it is convenient to rewrite p using a fixed tuple a that enumerates $\operatorname{dom} h$ and a fixed tuple a valiables x of the same length. Then condition p can be rephrased as

p′. $M \vDash \varphi(a) \quad \Rightarrow \quad N \vDash \varphi(ha)$ for every $\varphi(x) \in \Delta$.

When $\Delta = L$ we say elementary map and elementary embedding for Δ-morphism, respectively Δ-embedding. When $\Delta = L_{\mathrm{at}}$ we say partial homomorphism and when $\Delta = L_{\mathrm{at}^{\pm}}$ we say either partial embedding or partial isomorphism. The reason for the latter name is explained in Remark 3.25. It is immediate to verify that a partial embedding which is total is an embedding as in Definition 2.35. Similarly, a partial embedding which is total and surjective is an isomorphism. The precise connection between partial and total homo/iso-morphisms is discussed in following remark.

3.25 Remark For every map $h : M \to N$ the following are equivalent

1. $h : M \to N$ is a partial isomorphism
2. there is an isomorphism $k : \langle \operatorname{dom} h\rangle_M \to \langle \operatorname{rng} h\rangle_N$ that extends h.

Moreover, this extension is unique. The equivalence holds replacing isomorphism by homomorphism (in this case, in 2 we obtain a surjection). The extension k is obtained defining $k\big(t(a)\big) = t(ha)$ for every term $t(x)$.
A similar fact holds for partial homomorphims if we replace isomorphism by epimorphism, i.e. surjective homomorphism.

We use Δ-morphisms to compare, locally, two structures. There are different ways to do this, in the proposition below we list a few synonymous expressions. But first some more notation. When x is a fixed tuple of variables, $a \in M^x$ and $b \in N^x$ we write

$M, a \Rrightarrow_{\Delta} N, b$ if $M \vDash \varphi(a) \Rightarrow N \vDash \varphi(b)$ for every $\varphi(x) \in \Delta$.

$M, a \equiv_{\Delta} N, b$ if $M \vDash \varphi(a) \Leftrightarrow N \vDash \varphi(b)$ for every $\varphi(x) \in \Delta$.

The following equivalences are immediate and will be used without explicit reference.

3.26 Proposition For every given set of formulas Δ and every map $h : M \to N$ the following are equivalent

1. $h : M \to N$ is a Δ-morphism
2. $M, a \Rrightarrow_{\Delta} N, ha$ for every $a \in (\operatorname{dom} h)^x$
3. $\Delta\text{-tp}_M(a) \subseteq \Delta\text{-tp}_N(ha)$ for every $a \in (\operatorname{dom} h)^x$
4. $N, ha \vDash p(x)$ for every $a \in (\operatorname{dom} h)^x$ and $p(x) = \Delta\text{-tp}_M(a)$.

As in Remark 3.24, one can equivalently require 2, 3, and 4 for some/any fixed tuple a that enumerates $\operatorname{dom} h$.

3.27 Remark Condition p in Definition 3.23 applies to tuples x of any length, in particular to the empty tuple. In this case $\varphi(x)$ is a sentence, $a \in (\operatorname{dom} h)^0 = \{\varnothing\}$ is the empty tuple, and p asserts that $\mathrm{Th}_\Delta(M) \subseteq \mathrm{Th}_\Delta(N)$. When $h = \varnothing$ this is actually all that p says. In fact $\varnothing^x = \varnothing$ unless $|x| = 0$. Still, $\mathrm{Th}_\Delta(M) \subseteq \mathrm{Th}_\Delta(N)$ may be a non trivial requirement.

3.28 Definition We call $\mathrm{Th}_\Delta(M)$ the Δ-theory of M. We say that the theory T is Δ-complete if $\mathrm{Th}_\Delta(M) = \mathrm{Th}_\Delta(N)$ for all $M, N \vDash T$. In other words, for any pair of models of T, the empty map $\varnothing : M \to N$ is a Δ-morphism. When T is $L_{\mathrm{at}^\pm}$-complete we say that T decides the characteristic of its models (by analogy with rings and fields). Note that when T decides the characteristic of its models, we have $\langle\varnothing\rangle_M \simeq \langle\varnothing\rangle_N$ for all $M, N \vDash T$.

We conclude this section with a couple of propositions that break Theorem 2.36 into parts. More interestingly, in Chapter 10 we shall prove a sort of converse of Propositions 3.30 and 3.31.

It is interesting to note that there is a relation between certain properties of Δ-morphisms and the closure of Δ under logical connectives. When $\mathsf{C} \subseteq \{\forall, \exists, \neg, \vee, \wedge\}$ is a set of connectives, we write $\mathsf{C}\Delta$ for the closure of Δ with respect to the connectives in C. We write $\neg\Delta$ for the set containing the negation of the formulas in Δ. Warning: do not confuse $\neg\Delta$ with $\{\neg\}\Delta$. Up to logical equivalence $\{\neg\}\Delta$ coincides with $\Delta \cup \neg\Delta$.

It is clear that Δ-morphisms are $\{\wedge, \vee\}\Delta$-morphisms.

3.29 Proposition For every given set of formulas Δ and every injective map $h : M \to N$ the following are equivalent

a. $h : M \to N$ is a $\neg\Delta$-morphism

b. $h^{-1} : M \to N$ is a Δ-morphism.

3.30 Proposition For every set of formulas Δ, every total Δ-morphism $h : M \to N$ is a $\{\exists\}\Delta$-morphism.

Proof. Formulas in $\{\exists\}\Delta$ have the form $\exists y\, \varphi(x, y)$ where y is a finite tuples of variables and $\varphi(x, y) \in \Delta$. For every tuple $a \in (\operatorname{dom} h)^x$ we have:

$$\begin{aligned}
M \vDash \exists y\, \varphi(a, y) \quad &\Rightarrow \quad M \vDash \varphi(a, b) \text{ for } b \in M^y \\
&\Rightarrow \quad N \vDash \varphi(ha, hb) \\
&\Rightarrow \quad N \vDash \varphi(ha, c) \text{ for } c \in N^y \\
&\Rightarrow \quad N \vDash \exists y\, \varphi(ha, y).
\end{aligned}$$

Note that the second implication requires the totality of $h : M \to N$ which guarantees that $\varphi(a, y)$ has a solution in $\operatorname{dom} h$. □

When $h : M \to N$ is injective the following proposition is a corollary of Propositions 3.29 and 3.30. The general proof is the dual version of the proof of Propositions 3.30.

3.31 Proposition For every set of formulas Δ, every surjective Δ-morphism $h : M \to N$ is a $\{\forall\}\Delta$-morphism.

3.32 Exercise Prove that if $h : M \to N$ is an elementary embedding $h[M]$ is an ememen-tary substructure of N and $h : M \to h[M]$ is an isomorphism.

Chapter 4

Ultraproducts

In these notes we only use ultraproducts to prove the compactness theorem. Since a syntactic proof of the compactness theorem is also given, this chapter is, strictly speaking, not required. However, the importance of ultraproducts transcends their application to model theory.

4.1 Direct products

In this and in the next section $\langle M_i : i \in I \rangle$ is a sequence of L-structures. (We are slightly abusing of the word sequence, since I is a just naked set.) The direct product of this sequence is a structure denoted by

$$N \quad = \quad \prod_{i \in I} M_i$$

and defined by conditions 1-3 below. If $M_i = M$ for all $i \in I$, we say that N is a direct power of M and denote it by M^I.

The domain of N is the set containing all functions

1. $$\hat{a} \quad : \quad I \ \rightarrow \ \bigcup_{i \in I} M_i$$ $$\hat{a} \quad : \quad i \ \mapsto \ \hat{a}i \ \in \ M_i$$

We do not distinguish between tuples of elements of N and tuple-valued functions. For instance, the tuple $\hat{a} = \langle \hat{a}_1 \ldots \hat{a}_n \rangle$ is identified with the function $\hat{a} : i \mapsto \hat{a}i = \langle \hat{a}_1 i, \ldots, \hat{a}_n i \rangle$. On a first reading of what follows, it may help to pretend that all functions and relations are unary.

The interpretation of $f \in L_{\text{fun}}$ is defined as follows:

2. $$(f^N \hat{a})i \quad = \quad f^{M_i}(\hat{a}i) \qquad \text{for all } i \in I.$$

The interpretation of $r \in L_{\text{rel}}$ is the product of the relations r^{M_i}, that is, we define

3. $$\hat{a} \in r^N \quad \Leftrightarrow \quad \hat{a}i \in r^{M_i} \qquad \text{for all } i \in I.$$

The following proposition is immediate.

4.1 Proposition If $\doteq, \wedge, \forall, \exists$ are the only logical symbol that occur in $\varphi(x) \in L$, then for every $\hat{a} \in N^x$

$\sharp$ $$N \vDash \varphi(\hat{a}) \quad \Leftrightarrow \quad M_i \vDash \varphi(\hat{a}i) \qquad \text{for all } i \in I.$$

Proof. By induction on syntax. First note that we can extend 2 to all terms $t(x)$ as follows:

2′. $$(t^N \hat{a})i \quad = \quad t^{M_i}(\hat{a}i) \qquad \text{for all } i \in I.$$

Combining 3 and 2′ gives that for every $r \in L_{\text{rel}} \cup \{=\}$ and every L-term $t(x)$

$$N \vDash r t \hat{a} \quad \Leftrightarrow \quad M_i \vDash r t \hat{a} i \qquad \text{for all } i \in I.$$

This shows that $\sharp$ holds for $\varphi(x)$ atomic. Induction for the connectives $\wedge$, $\forall$ ed $\exists$ is immediate. □

A consequence of Proposition 4.1 is that a direct product of groups, rings or vector spaces is a structure of the same sort. However, a product of fields is not a field.

4.2 Łoś's Theorem

Let I be any set. By filter on I we mean a filter of the Boolean algebra $\mathcal{P}(I)$.

We (temporary) denote by N' the following structure. The domain and the interpretation of the function symbols is as in the structure N defined in 1, respectively 2, of the previous section. As interpretation of $r \in L_{\text{rel}}$ we take

3′. $$\hat{a} \in r^{N'} \quad \Leftrightarrow \quad \big\{ i \in I \ : \ \hat{a} i \in r^{M_i} \big\} \in F\,.$$

Let F be a filter on I. We define the following congruence on N' (see Definition 2.46)

$$\hat{a} \sim_F \hat{c} \quad \Leftrightarrow \quad \big\{ i \in I \ : \ \hat{a} i = \hat{c} i \big\} \in F\,.$$

To check that $\sim_F$ is indeed a congruence, we first need to check that it is an equivalence relation. Reflexivity and symmetry are immediate, and transitivity follows from f2 in Section 3.1. Then we check that $\sim_F$ is compatible with the functions of L, that is, that c1 of Definition 2.46 is satisfied. This follows from f1 in Section 3.1.

For brevity, we write N'/F for $N'/{\sim_F}$ and $[\hat{a}]_F$ for $[\hat{a}]_{\sim_F}$. We write $N'/F \overset{*}{\vDash} \varphi(\hat{a})$ for $N'/F \vDash \varphi([\hat{a}]_F)$. A more formal interpretation of $\overset{*}{\vDash}$ is given in Definition 2.48.

The structure N'/F is called the reduced product of the structures $\langle M_i : i \in I \rangle$ or, when $M_i = M$ for all $i \in I$, the reduced power of M. When F is an ultrafilter we say ultraproduct, respectively ultrapower.

The difference between N and N' is highlighted in Exercise 4.6. For neater notation, below we write N for N'.

The following proposition is almost tautological. Note that it is a special case of Łoš's Theorem below (but it is does not require F to be an *ultra*filter).

4.2 Proposition Let $r \in L_{\text{rel}}$. Let $t(x)$ be tuple of terms of length n_r, the arity of r. Then for every $\hat{a} \in N^x$ and every filter F the following are equivalent

1. $N/F \overset{*}{\vDash} r\, t(\hat{a})$
2. $\big\{ i \ : \ M_i \vDash r\, t(\hat{a} i) \big\} \in F$.

Proof. 1⇒2. Assume 1. Then, by 2* of Definition 2.48, there is a $\hat{b} \sim t^N(\hat{a})$ such that $N \vDash r\,\hat{b}$. Therefore $\{i : M_i \vDash r\,\hat{b} i\}$ is in F. But $\hat{b} \sim t(\hat{a})$ means that $\big\{ i \ : \ \hat{b} i = t(\hat{a} i) \big\}$ is in F, hence 2 follows.

2⇒1. Is clear. □

4.3 Łoś's Theorem Let $\varphi(x) \in L$ and let F be an ultrafilter on I. Then for every $\hat{a} \in N^x$ the following are equivalent:

1. $N/F \overset{*}{\vDash} \varphi(\hat{a})$ (see Definition 2.48)
2. $\{\, i : M_i \vDash \varphi(\hat{a}i) \,\} \in F$.

Proof. We proceed by induction on the syntax of $\varphi(x)$. If $\varphi(x)$ is equality, then equivalence holds by definition of $\sim$. If $\varphi(x)$ is of the form $r\,t(x)$ for some tuple of terms $t(x)$ and $r \in L_{\mathrm{rel}}$ then 1$\Leftrightarrow$2 is Proposition 4.2.

We prove the inductive step for the connectives $\neg$, $\wedge$, and the quantifier $\exists$. We begin with $\neg$. This is the only place in the proof where the assumption that F is an *ultra* filter is required. By the inductive hypothesis,

$$N/F \overset{*}{\vDash} \neg\varphi(\hat{a}) \quad\Leftrightarrow\quad \{\, i : M_i \vDash \varphi(\hat{a}i) \,\} \notin F$$

So, as F is an *ultra* filter

$$\Leftrightarrow\quad \{\, i : M_i \vDash \neg\varphi(\hat{a}i) \,\} \in F\,.$$

Now consider $\wedge$. Assume inductively that the equivalence 1$\Leftrightarrow$2 holds for $\varphi(x)$ and $\psi(x)$. Then

$$N/F \overset{*}{\vDash} \varphi(\hat{a}) \wedge \psi(\hat{a}) \quad\Leftrightarrow\quad \{i : M_i \vDash \varphi(\hat{a}i)\} \in F \text{ and } \{i : M_i \vDash \psi(\hat{a}i)\} \in F\,.$$

As filters are closed under intersection, we obtain

$$\Leftrightarrow\quad \{\, i : M_i \vDash \varphi(\hat{a}i) \wedge \psi(\hat{a}i) \,\} \in F\,.$$

Finally, consider $\exists y$. Assume inductively that the equivalence 1$\Leftrightarrow$2 holds for $\varphi(x, y)$. Then

$$\begin{aligned} N/F \overset{*}{\vDash} \exists y\, \varphi(\hat{a}, y) \quad&\Leftrightarrow\quad N/F \overset{*}{\vDash} \varphi(\hat{a}, \hat{b}) && \text{for some } \hat{b} \in N \\ &\Leftrightarrow\quad \{\, i : M_i \vDash \varphi(\hat{a}i, \hat{b}i) \,\} \in F && \text{for some } \hat{b} \in N. \end{aligned}$$

We claim this is equivalent to

$$\Leftrightarrow\quad \{\, i : M_i \vDash \exists y\, \varphi(\hat{a}i, y) \,\} \in F.$$

The $\Rightarrow$ direction is trivial. For $\Leftarrow$, we choose as $\hat{b}$ a sequence that picks a witness of $M_i \vDash \exists y\, \varphi(\hat{a}i, y)$ if it exists, and some arbitrary element of M_i otherwise. □

Let a^I denote the element of M^I that has constant value a. The following is an immediate consequence of Łoś's theorem.

4.4 Corollary Let $N = M^I$. For every $a \in M$

$$N/F \overset{*}{\vDash} \varphi(a^I) \quad\Leftrightarrow\quad M \vDash \varphi(a)\,.$$

We often identify M with its image under the embedding $h : a \mapsto [a^I]_F$, and say that M^I/F is an elementary extension of M.

The following corollary is an immediate consequence of the compactness theorem that we prove in the next chapter but here we give a direct proof.

4.5 Corollary Every infinite structure has a proper elementary extension.

Proof. Let M be an infinite structure and let F be a *non-principal* ultrafilter on ω. It suffices to show that $h[M]$, the image of the embedding defined above, is a *proper* substructure of M^ω/F. As M is infinite, there is an injective function $\hat{d} \in M^\omega$. Then for every $a \in M$ the set $\{i : \hat{d}\,i = a\}$ is either empty or a singleton and, as F is non principal, it does not belong to F. So, by Łoš Theorem, we have $M^\omega/F \overset{*}{\vDash} \hat{d} \neq a^I$ for every $a \in M$, that is, $[\hat{d}]_F \notin h[M]$. $\square$

4.6 Exercise Let N be the structure defined in the previous section. Prove that if $M_i = M$ for all $i \in I$ then $N'/F = N/F$. More generally, prove that if for every $r \in L_{\mathrm{rel}}$, either $r^{M_i} \neq \varnothing$ for all i or $r^{M_i} = \varnothing$ for all i then $N'/F = N/F$.

4.7 Exercise Consider $\mathbb{N}$ as a structure in the language of strict orders. Let F be a non-principal ultrafilter on ω. Prove that in $\mathbb{N}^\omega$ there is a sequence $\langle \hat{a}_i : i \in \omega \rangle$ such that $\mathbb{N}^\omega/F \overset{*}{\vDash} \hat{a}_{i+1} < \hat{a}_i$.

4.8 Exercise Let I be the set of integers $i > 1$. For $i \in I$, let $\mathbb{Z}_i$ denote the additive group of integers modulo i, and let N denote the product $\prod_{i \in I} \mathbb{Z}_i$. Prove that, if F is a non-principal ultrafilter on I,

1. for some F, N/F does not contain any element of finite order
2. for some F, N/F has some elements of order 2
3. for all F, N/F contains an element of infinite order
4. for some F, $N/F \vDash \forall x\, \exists y\; my = x$ for every integer $m > 0$
5. for some F, N/F contains an element $\hat{a}$ such that $N/F \vDash \forall x\; mx \neq \hat{a}$ for every positive integer m.

Chapter 5

Compactness theorem(s)

We present two proofs of the compactness theorem. The first uses ultrapowers the second is syntactic. Finally, we generalize the theorem so that it applies to types.

5.1 Compactness via ultraproducts

A theory is finitely consistent if all its finite subsets are consistent. The following theorem is the *fiat lux* of model theory.

5.1 Compactness Theorem Every finitely consistent theory is consistent.

Proof. Let T be a finitely consistent theory. We claim that the structure N/F which we define below is a model of T. Let I be the set of consistent sentences $\xi \in L$. For every $\xi \in I$ pick some $M_\xi \vDash \xi$. For any sentence $\varphi \in L$ we define

$$X_\varphi \quad = \quad \Big\{\xi \in I \;:\; \xi \vdash \varphi\Big\}.$$

Clearly φ is consistent if and only if $X_\varphi \neq \varnothing$. Moreover $X_{\varphi\wedge\psi} \;=\; X_\varphi \cap X_\psi$. Hence, as T is finitely consistent, the set $B = \big\{X_\varphi \;:\; \varphi \in T\big\}$ has the finite intersection property. Therefore B extends to an ultrafilter F on I. Define

$$N \quad = \quad \prod_{\xi\in I} M_\xi.$$

We claim that $N/F \vDash T$. By Łoš Theorem, for every sentence $\varphi \in L$

$$N/F \vDash \varphi \;\;\Leftrightarrow\;\; \Big\{\xi \;:\; M_\xi \vDash \varphi\Big\} \;\in\; F\,.$$

By the definition of F, for every $\varphi \in T$, the set $X_\varphi \subseteq \{\xi \;:\; M_\xi \vDash \varphi\}$ belongs to F. Therefore $N/F \vDash T$, *et lux fuit*. □

The compactness theorem can be formulated in the following apparently stronger way.

5.2 Corollary If $T \vdash \varphi$ then there is some finite $S \subseteq T$ such that $S \vdash \varphi$.

Proof. Suppose $S \nvdash \varphi$ for every finite $S \subseteq T$. Then for every finite $S \subseteq T$ there is a model $M \vDash S \cup \{\neg\varphi\}$. In other words, $T \cup \{\neg\varphi\}$ is finitely consistent. By compactness $T \cup \{\neg\varphi\}$ is consistent, hence $T \nvdash \varphi$. □

5.3 Exercise Let $\Phi \subseteq L$ be a set of sentences and suppose that $\vdash \psi \leftrightarrow \bigvee\Phi$ for some sentence ψ. Prove that there is a finite $\Phi_0 \subseteq \Phi$ such that $\vdash \psi \leftrightarrow \bigvee\Phi_0$.

5.4 Exercise Let $\mathcal{C}$ be a class of structures. Let $\mathrm{Th}(\mathcal{C}) = \{\varphi \mid M \vDash \varphi \text{ for all } M \in \mathcal{C}\}$ be the theory of $\mathcal{C}$. Prove the following are equivalent

1. $N \vDash \mathrm{Th}(\mathcal{C})$
2. N is elementarily equivalent to an ultraproduct of elements of $\mathcal{C}$.

5.2 Compactness for types

Recall that a type is a set of formulas. When we present types we usually declare the variables that may occur in it – we write $p(x)$, $q(x)$, etc. where x is a tuple of variables. When x is the empty tuple, $p(x)$ is just a theory. Often we identify a finite types with the (infinite) conjunction of the formulas contained in it.

We write $M \vDash p(a)$ if $M \vDash \varphi(a)$ for every $\varphi(x) \in p$. We say that a is a solution or a realization of $p(x)$. An equivalent notation is $M, a \vDash p(x)$ or, when M is clear from the context, $a \vDash p(x)$. We say that $p(x)$ is consistent in M if it has a solution in M. In this case we may write $M \vDash \exists x\, p(x)$. We say that $p(x)$ is consistent if it is realized in some model.

We say that a type $p(x) \subseteq L$ is finitely consistent if all its finite subsets are consistent. The following fact is an immediate consequence of the compactness theorem.

5.5 Fact Every finitely consistent type $p(x) \subseteq L$ is consistent.

Proof. Let L' be the expansion of L obtained by adding the fresh symbols c, a tuple of constants of the same length as x. Then $p(c)$ is a finitely consistent theory in the language L'. By the compactness theorem there is an L'-structure $N' \vDash p(c)$. Let N be the reduct of N' to L, that is, the L-structure with the same domain and the same interpretation as N' on the symbols of L. Note that, though the constants c are not in L, the elements of the tuple $c^{N'}$ remain in N. Then $N, c^{N'} \vDash p(x)$. □

If all finite subsets of a type $p(x) \subseteq L(M)$ are consistent in the very same model M, we say that $p(x)$ is finitely consistent in M. The following theorem shows that the notion of finite consistency *in a model*, which is trivial for theories, is instead very interesting for types. Before that, we recall an important particular case of Proposition 3.26.

5.6 Lemma (Elementary Diagram Method) Let M be an infinite model. Let c be an enumeration of M. Define $q(z) = \mathrm{tp}_M(c)$. Then, for every model N realizing $q(z)$, there is an elementary embedding $h : M \hookrightarrow N$.

We call $q(z)$ the elementary diagram of M though, strictly speaking, the elementary diagram of M is theory $\mathrm{Th}(M/M) = q(c)$.

Proof. Let $N, a \vDash q(z)$. Let $h : M \hookrightarrow N$ map $c \mapsto a$. By Remark 3.24, this is an elementary embedding. □

5.7 Compactness Theorem for types Let M be an infinite model. If $p(x) \subseteq L(M)$ is finitely consistent in M then it is realized in some elementary extension of M.

Proof. Let a be an enumeration of M. It is convenient to write $p(x)$ as $p(x\,;a)$ where $p(x\,;z) \subseteq L$. Let $q(z) = \mathrm{tp}_M(a)$. Clearly, $p(x\,;z) \cup q(z)$ is finitely consistent. Then, by Fact 5.5, it is realized in some model N' by some $c', a' \in (N')^{x,z}$. Let $h : M \to N'$ map $a \mapsto a'$. By Lemma 5.6, this is an elementary embedding. Note that $hc = c'$. By Exercise 3.32, $h[M]$ is an isomorphic copy of M and is an elementary substructure of N'. By construction $N', hc \vDash p(x\,;ha)$.

We may conclude that there is a structure N (an isomorphic copy of N') that extends elementarily M and realizes $p(x\,;a)$. □

The following corollary is historically important.

5.8 Upward Löwenheim-Skolem Theorem Every infinite structure has elementary extensions of arbitrarily large cardinality.

Proof. Let $x = \langle x_i : i < \lambda \rangle$ be a tuple of distinct variables, where λ is an arbitrary cardinal. The type $p(x) = \{x_i \neq x_j : i < j < \lambda\}$ is finitely consistent in every infinite structure and every structure that realises p has cardinality $\geq \lambda$. Hence the claim follows from Theorem 5.7. □

5.9 Exercise Prove that for every type $p(x) \subseteq L(M)$ the following are equivalent

1 $p(x)$ is finitely consistent in M

2 $p(x) \cup \mathrm{Th}(M/M)$ is finitely consistent.

5.3 Compactness via syntax

Here we prove the compactness theorem using the so-called Henkin method. We divide the proof in two steps. Firstly, we observe that when the language is rich enough to name witnesses of all existential statements of the theory, these witnesses (*Henkin constants*) form a canonical model. Secondly, we show that we can add the required Henkin constants to any finitely consistent theory.

5.10 Definition Fix a language L. Assume for simplicity that formulas use only the connectives $\wedge$, $\neg$ and $\exists$. We say that T is a Henkin theory if for all formulas φ and ψ

0. $\varphi \in T \Rightarrow \neg\varphi \notin T$
1. $\neg\neg\varphi \in T \Rightarrow \varphi \in T$
2. $\varphi \wedge \psi \in T \Rightarrow \varphi \in T$ and $\psi \in T$
3. $\neg(\varphi \wedge \psi) \in T \Rightarrow \neg\varphi \in T$ or $\neg\psi \in T$
4. $\exists x\, \varphi \in T \Rightarrow \varphi[x/a] \in T$ for some closed term a
5. $\neg\exists x\, \varphi \in T \Rightarrow \neg\varphi[x/a] \in T$ for all closed terms a.

Moreover, the following holds for all closed terms a, b, c

a. $a \doteq a \in T$

b. $a \doteq b \in T \Rightarrow b \doteq a \in T$

c. $a \doteq b,\ b \doteq c \in T \Rightarrow a \doteq c \in T$

d. $a \doteq b,\ \varphi[x/a] \in T \Rightarrow \varphi[x/b] \in T$.

Fix a theory T and let M be the structure that has as domain the set of closed terms. Define for every relation symbol r

$$r^M = \{\langle a_1, \ldots, a_n \rangle \in M^n : r(a_1, \ldots, a_n) \in T\},$$

where n is the arity of r. Define for every function symbol f

$$f^M = \{\langle t, a_1, \ldots, a_n \rangle \in M^{n+1} : t = fa_1 \ldots a_n\}.$$

where n is the arity of r. An easy proof by induction shows that $t^M = t$ for all closed terms t.

Finally, let E be the relation on M that holds when $a \doteq b \in T$.

5.11 Lemma The relation E is a congruence on M (as defined in Section 2.5).

Proof. Axioms a-c ensure that E is an equivalence. We claim that that E is a congruence. This is immediate for unary functions: apply e to the formula $fx \doteq fa$. In general the claim is easily proved by induction on the arity of f. □

Condition 0 is the only negative requirement of Definition 5.10 (it requires that T does *not* contain some formula). By condition 0, Henkin theories do not contain any blatant inconsistency. Surprisingly, this is all what is needed for the existence of a model.

5.12 Theorem If T is a Henkin theory then $M/E \vDash T$.

Proof. By induction on the complexity of the formula φ in T we prove that

1. $\varphi \in T \Rightarrow M/E \overset{*}{\vDash} \varphi$ — for the notation cf. Definition 2.48
2. $\neg\varphi \in T \Rightarrow M/E \overset{*}{\nvDash} \neg\varphi$

Induction is immediate by 1-5 of Definition 5.10. Hence we only need to verify the claim for atomic formulas. Consider first the formula $\varphi = (t_1 \doteq t_2)$ where the t_i are closed terms. By the definition of M, for every closed term t we have $t^M = t$ so claim 1 is clear. As for 2, suppose $M/E \overset{*}{\vDash} t_1 \doteq t_2$, that is $t_1 E t_2$. Then $t_1 \doteq t_2 \in T$ and $\neg t_1 \doteq t_2 \notin T$ follows from axiom 0.

Now assume $\varphi = rt$ for a relation r and a tuple of closed terms t. The argument is similar: 1 is immediate; to prove 2 suppose that $M/E \overset{*}{\vDash} rt$. Then $tEs \in r^M$ for some tuple of closed terms s. Then $rs \in T$, and by d $rt \in T$. Finally from 0 we obtain $\neg rt \notin T$. □

5.13 Proposition If every finite subset of T has a model then there is a Henkin theory T' containing T. The theory T' may be in an expanded language L'.

Proof. Set $\lambda = |L|$. Let $\langle c_i : i < \lambda \rangle$ be some constants not in L. Let L_i be the language with constants among $c_{\restriction i}$. Fix a variable x and an enumeration $\langle \varphi_i(x) : i < \lambda \rangle$ of the formulas in L_λ. Suppose that the enumeration is such that $\varphi_i(x) \in L_i$.

We now construct a sequence of finitely consistent L_i-theories T_i. If α is 0 or a limit ordinal we define

$$T_\alpha = T \cup \bigcup_{i<\alpha} T_i.$$

As for successor ordinals, let S_i be a maximally finitely consistent set of L_i-formulas containing T_i. (Here we use Zorn's lemma, but see the remark below.) It is immediate that S_i satisfies all requirements in Definition 5.10 but possibly for 4.

Now, if $\exists x\, \varphi_i(x) \in S_i$ set $T_{i+1} = S_i \cup \{\varphi[x/c_i]\}$. As c_i does not occur in S_i, it is evident that T_{i+1} is finitely consistent.

Recall that we assumed $\exists x\, \varphi_i(x) \in L_i$. Then, either $\exists x\, \varphi_i(x) \in T_{i+1}$ or it is not finitely consistent with T_{i+1}. Hence stage i settle requirement 4 in Definition 5.10 as far as $\varphi_i(x)$ is concerned.

At stage λ all possible counterexamples to 4 have been ruled out, then $T' = T_\lambda$ is the required Henkin theory. □

The following theorem is an immediate corollary of the proposition above.

5.14 Compactness Theorem If T is finitely consistent then T is consistent.

To keep the proof of Proposition 5.13 short, we applied Zorn's lemma. This is not strictly necessary. In fact, if we are given a finitely consistent theory T. We can extend T to a theory S that meets Definition 5.10, up to condition 4, by adding systematically all required formulas. The procedure is effective, hence Zorn's lemma is not required.

It is interesting to consider the case when T is finite. Assume also (though this is not really necessary) that the language contains finitely many symbols and no functions other then constants. Then the construction in Proposition 5.13 is an effective procedure that produces in ω steps a model of T. At each step T_n is finite and contains only subformulas of formulas in T or variant on these obtained by substituting constants for variables.

Now suppose instead that we start with an inconsistent T. The procedure above has to come to a halt at same (finite) stage because a model of T does not exist. When the procedure halts, we end up with a finite sequence of finite theories $T_0, \ldots, T_n$ where $T_0 = T$ and T_n contains some blatant inconsistency (i.e. φ and $\neg\varphi$). Many have interpreted $T_0, \ldots, T_n$ as a formal *proof* of the inconsistency of T.

All this has little or no interest to model theory. But it highlights a fascinating phenomenon. When we say that T is inconsistent, we say that no structure models T. This expression uses a (meta linguistic) universal quantifier that ranges over the class of all structures. Yet this is equivalent to an expression that merely asserts the existence of a finite sequence of finite theories.

Chapter 6

Some relational structures

In the first section we prove that theory of dense linear orders without endpoints is ω-categorical. That is, any two such countable orders are isomorphic. This is an easy classical result of Cantor. In this chapter we examine Cantor's construction (a so-called *back-and-forth* construction) in great detail. In the second section we apply the same technique to prove that the theory of the random graph is ω-categorical.

6.1 Dense linear orders

The language of strict orders, which in this section we denote by L, contains only a binary relation symbol $<$. A structure M of signature L is a strict order if it models (the universal closure of) the following formulas

1. $x \not< x$ irreflexive
2. $x < z < y \ \rightarrow \ x < y$ transitive.

Note that the following is an immediate consequence of 1 and 2.

$x < y \ \rightarrow \ y \not< x$ antisymmetric.

We say that the order is total or linear if

li. $x < y \ \vee \ y < x \ \vee \ x = y$ linear or total.

An order is dense if

nt. $\exists x, y \ (x < y)$ nontrivial

d. $x < y \ \rightarrow \ \exists z \ (x < z < y)$ dense.

We need to require the existence of two comparable elements, then d implies that dense orders are in fact infinite. We say that the ordering has no endpoints if

e. $\exists y \ (x < y) \ \wedge \ \exists y \ (y < x)$ without endpoints.

We denote by $T_{\rm lo}$ the theory strict linear orders and by $T_{\rm dlo}$ the theory of dense linear orders without endpoints. Clearly, these are consistent theories: $\mathbb{Q}$ with the usual ordering is a model of $T_{\rm dlo}$.

We introduce some notation to improve readability of the proof of the following theorem. Let A and B be subsets of an ordered set. We write $A < B$ if $a < b$ for every $a \in A$ and $b \in B$. We write $a < B$ e $A < b$ for $\{a\} < B$, respectively $A < \{b\}$. Let $M \models T_{\rm lo}$. Then $M \models T_{\rm dlo}$ if and only if for every finite $A, B \subseteq M$ such that $A < B$ there is a c such that $A < c < B$. In fact axiom d is evident and axioms nt and e are obtained taking replacing A and/or B by the empty set.

Now we prove the first of a series of lemmas that we call extension lemmas. Recall that in the language of strict orders an injective map $k : M \to N$ is a partial isomorphism if

$$M \models a < b \ \Leftrightarrow \ N \models ka < kb \qquad \text{for every } a, b \in \operatorname{dom} k.$$

When $M, N \models T_{\rm lo}$ the direction $\Rightarrow$ suffices.

6.1 Lemma Fix $M \vDash T_{\rm lo}$ and $N \vDash T_{\rm dlo}$. Let $k : M \to N$ be a finite partial isomorphism and let $b \in M$. Then there is a partial isomorphism $h : M \to N$ that extends k and is defined in b.

Proof. Given a finite partial isomorphism $k : M \to N$ define

$$A^- \;=\; \big\{a \in \mathrm{dom}(k) \;:\; a < b\big\};$$
$$A^+ \;=\; \big\{a \in \mathrm{dom}(k) \;:\; b < a\big\}.$$

The sets A^- and A^+ are finite and partition $\mathrm{dom}\, k$, and $A^- < A^+$. As $k : M \to N$ is a partial isomorphism, $k[A^-] < k[A^+]$. Then in N there is an element c such that $k[A^-] < c < k[A^+]$. It is easy to check that setting $h = k \cup \big\{\langle b, c\rangle\big\}$ gives the required extension. □

The following is an equivalent version of Lemma 6.1.

6.2 Corollary Let $M \vDash T_{\rm lo}$ be countable and let $N \vDash T_{\rm dlo}$. Let $k : M \to N$ be a finite partial isomorphism. Then there is a (total) embedding $h : M \hookrightarrow N$ that extends k.

Proof. Let $\langle a_i : i < \omega\rangle$ be an enumeration of M. Define by induction a chain of finite partial isomorphisms $h_i : M \to N$ such that $a_i \in \mathrm{dom}\, h_{i+1}$. The construction starts with $h_0 = k$. At stage $i+1$ we chose any finite partial isomorphism $h_{i+1} : M \to N$ that extends h_i and is defined in a_i. This is possible by Lemma 6.1. In the end we set

$$h \;=\; \bigcup_{i \in \omega} h_i.$$

It is immediate to verify that $h : M \hookrightarrow N$ is the required embedding. □

We are now ready to prove that any two countable models of $T_{\rm dlo}$ are isomorphic which is a classical result of Cantor's. Actually what we prove is slightly more general than that. In fact Cantor's theorem is obtained from the theorem below by setting $k = \varnothing$, which we are allowed to, because all models have the same empty characteristic (cf. Remark 3.27).

6.3 Theorem Every finite partial isomorphism $k : M \to N$ between countable models of $T_{\rm dlo}$ extends to an isomorphism $g : M \xrightarrow{\sim} N$.

The following is the archetypal back-and-forth construction. It is important to note that it does not mention linear orders at all. It only uses the extension Lemma 6.1. The same construction can be applied in many other contexts where an extension lemma holds (cf. Theorem 7.6).

Proof. Let $\langle a_i : i < \omega\rangle$ and $\langle b_i : i < \omega\rangle$ be enumerations of M and N respectively. We define by induction a chain of finite partial isomorphisms $g_i : M \to N$ such that $a_i \in \mathrm{dom}\, g_{i+1}$ and $b_i \in \mathrm{rng}\, g_{i+1}$. In the end we set

$$g \;=\; \bigcup_{i \in \omega} g_i$$

We begin by letting $g_0 = k$. The inductive step consists of two half-steps that we call the *forth step* and *back step*. In the forth step we define $g_{i+1/2}$ such that $a_i \in \operatorname{dom} g_{i+1/2}$. In the back step to define g_{i+1} such that $b_i \in \operatorname{rng} g_{i+1}$.

By the extension lemma 6.1 there is a finite partial isomorphism $g_{i+1/2} : M \to N$ that extends g_i and is defined in a_i. Now apply the same lemma to extend $\left(g_{i+1/2}\right)^{-1}$: $N \to M$ to a finite partial isomorphism $(g_{i+1})^{-1} : N \to M$ defined in b_i. □

Let λ be an infinite cardinal. We say that a theory is λ-categorical if any two models of T of cardinality λ are isomorphic. From Theorem 6.3, taking $k = \varnothing$, we obtain the following.

6.4 Corollary The theory $T_{\rm dlo}$ is ω-categorical.

We also obtain that $T_{\rm dlo}$ is a complete theory. This is consequence of the following general fact.

6.5 Proposition If T has no finite models and is λ-categorical for some $\lambda \geq |L|$, then T is complete.

Proof. Let M and N be any two models of T. Applying the upward and/or downward Löwenheim-Skolem theorem, we may assume they both have cardinality λ (here we use that M and N are both infinite and that $\lambda \geq |L|$). Hence $M \simeq N$ and in particular $M \equiv N$. □

The following corollary is also very important and will be siscusse3d in detail in Chapter 7.1.

6.6 Corollary Every partial isomorphism $k : M \to N$ between countable models of $T_{\rm dlo}$ is elementary.

Proof. We can assume $k : M \to N$ is finite. Then there are $M' \preceq M$ and $N' \preceq N$ such that $\operatorname{dom} k \subseteq M'$ and $\operatorname{rng} k \subseteq N'$. By Theorem 6.3, k extends to an isomorphism between M' and N'. Then k is an elementary map between M' and N', and therefore also between M and N. □

6.7 Exercise Prove that the extension Lemma 6.1 characterizes models of $T_{\rm dlo}$ among models of $T_{\rm lo}$. That is, if N is a model of $T_{\rm lo}$ such that the conclusion of Lemma 6.1 holds, then $M \vDash T_{\rm dlo}$.

6.8 Exercise Prove that $T_{\rm dlo}$ is not λ-categorical for any uncountable λ.

6.9 Exercise Prove that, in the language of strict orders, $\mathbb{Q} \preceq \mathbb{R}$.

6.10 Exercise Let L be the language of strict orders expanded with the constants $\{c_i : i \in \omega\}$. Let T be the theory that extends $T_{\rm dlo}$ with the axioms $c_i < c_{i+1}$ for all i. Prove that T is complete. Find three non-isomorphic countable models of this theory. For a suitably chosen model N of T, prove the statement in Lemma 6.1, where M any model of T.

6.11 Exercise Show that in Theorem 6.5 the assumption $\lambda \geq |L|$ is necessary. Hint: let ν be an uncountable cardinal. The language contains only the ordinals $i < \nu$ as constants. The theory T says that there are infinitely many elements and either $i = 0$ for every $i < \nu$, or $i \neq j$ for every $i < j < \nu$. Prove that T is incomplete. Prove that T has countable models and that these are all isomorphic.

6.2 Random graphs

Recall that the language of graphs, which in this section we denote by L, contains only a binary relation r. A graph structure of signature L such that

1. $\neg r(x, x)$ irreflexive
2. $r(x, y) \rightarrow r(y, x)$ symmetric.

An element of a graph M is called a vertex or a node. An edge is an unordered pair of vertices $\{a, b\} \subseteq M$ such that $M \vDash r(a, b)$. In words we may say that a is adjacent to b.

A random graph is a graph that also satisfies the following axioms for every n

nt. $\exists x, y \quad (x \neq y)$ nontrivial

r_n. $\displaystyle\bigwedge_{i,j=1}^{n} x_i \neq y_j \quad \rightarrow \quad \exists z \bigwedge_{i=1}^{n} \big[r(x_i, z) \wedge \neg r(z, y_i) \wedge z \neq y_i\big]$ for every $n \in \mathbb{Z}^+$.

The theory of graphs is denoted by T_{gph} and the theory of random graphs is denoted by T_{rg}. The scheme of axioms r_n plays the same role as density in the previous section. It says that given two disjoint sets A^+ and A^- of cardinality $\leq n$ there is a vertex z that is adjacent to all vertices in A^+ and to no vertex in A^-. We explicitly required that $z \notin A^-$, by 1 it is clear that $z \notin A^+$.

Strictly speaking, the axioms r_n do not mention the cases when A^+ or A^- are empty. But as it is evident that random graphs are infinite, we can deal with them by adding redundant elements.

The following is analogous to Lemma 6.1. Recall that in the language of graphs a map $k : M \rightarrow N$ is a partial isomorphism if it is injective and

$$M \vDash r(a, b) \Leftrightarrow N \vDash r(ka, kb) \qquad \text{for every } a, b \in \operatorname{dom} k.$$

6.12 Lemma Fix $M \vDash T_{\text{gph}}$ and $N \vDash T_{\text{rg}}$. Let $k : M \rightarrow N$ be a finite partial isomorphism and let $b \in M$. Then there is a partial isomorphism $h : M \rightarrow N$ that extends k and is defined in b.

Proof. The structure of the proof is the same as in Lemma 6.1, so we use the same notation. Assume $b \notin \operatorname{dom} k$ and define

$$A^+ = \big\{x \in \operatorname{dom} k \ : \ M \vDash r(x, b)\big\} \quad \text{e} \quad A^- = \big\{y \in \operatorname{dom} k \ : \ M \vDash \neg r(y, b)\big\}.$$

These two sets are finite and disjoint, then so are $k[A^+]$ and $k[A^-]$. Then there is a $c \notin \operatorname{rng} k$ such that

$$\bigwedge_{a \in A^+} r(ka, c) \ \wedge \ \bigwedge_{a \in A^-} \neg r(ka, c).$$

As $k[A^+] \cup k[A^-] = \operatorname{rng} k$, it is immediate to verify that $h = k \cup \{\langle b, c\rangle\}$ is the required extension. □

Some readers may doubt that $T_{\rm rg}$ is consistent.

6.13 Proposition There exists a random graph.

Proof. The domain of is the set of natural numbers. Let $r(n, m)$ hold if the n-th prime number divides m or, conversely, the m-th prime number divides n. □

The same argument as in the proof of Corollary 6.2 gives the following.

6.14 Corollary Let $M \vDash T_{\rm gph}$ be countable and let $N \vDash T_{\rm rg}$. Let $k : M \to N$ be a finite partial isomorphism. Then there is a (total) embedding $h : M \hookrightarrow N$ that extends k.

The proof of Theorem 6.3 gives the following theorem and its corollary.

6.15 Theorem Every finite partial isomorphism $k : M \to N$ between countable models of $T_{\rm rg}$ extends to an isomorphism $g : M \xrightarrow{\sim} N$.

6.16 Corollary The theory $T_{\rm rg}$ is ω-categorical (and therefore complete).

The same argument as in the proof of Corollary 6.6 yields the following.

6.17 Corollary Every partial isomorphism $k : M \to N$ between countable models of $T_{\rm rg}$ is elementary.

6.18 Exercise Let $A \subseteq N \vDash T_{\rm rg}$. Let $\varphi(x) \in L(A)$. Prove that if $\varphi(N)$ nonempty and disjoint from A then $\varphi(N)$ is a random graph.

6.19 Exercise Let $N \vDash T_{\rm rg}$. Is there a formula $\varphi(x) \in L(N)$ such that both $\varphi(N)$ and $\neg\varphi(N)$ are random graphs?

6.20 Exercise Let N be free union of two random graphs N_1 and N_2. That is, $N = N_1 \sqcup N_2$ and $r^N = r^{N_1} \sqcup r^{N_2}$, where $\sqcup$ denotes the disjoint union. Prove that N is not a random graph. Show that N_1 is not definable without parameters. Write a first order formula $\psi(x, y)$ true if x and y belong to the same connected component of N. Axiomatize the class of graphs that are free union of two random graphs.

6.21 Exercise Let M be the graph that has, as verteces, the hereditarily finite sets. There is an edge between two sets a and b if either $a \in b$ or $b \in a$. Prove that M is a random graph. Modify this example to prove that $T_{\rm rg}$ is not finitely axiomatizable.

6.22 Exercise Prove that $T_{\rm rg}$ is not λ-categorical for any uncountable λ. Hint: construct a sequence $\langle a_i : i < \lambda\rangle$ that enumerates a random graph N and is such that every a_i is adiacent to only finitely many a_j with $j < i$. We can also require that N contains an uncountable anticlique (that is, a set of vertices mutually nonadjacent) but note that the enumeration forbids uncountable cliques. Compare N with its complement graph (the graph that has edges between pairs that are non-adjacent in N).

6.23 Exercise (Peter J. Cameron) Prove that for every infinite countable graph M the following are equivalent

1. M is either random, complete, or empty (i.e. $r^M = M^2$ or $r^M = \varnothing$)
2. if $M_1, M_2 \subseteq M$ are such that $M_1 \sqcup M_2 = M$, then $M_1 \simeq M$ or $M_2 \simeq M$.

Hint: 2⇒1. Assume 2 and show that if $r(a, M) = \varnothing$ for some $a \in M$ then the graph is null; if $\{b\} \cup r(b, M) = M$ for some b, then the graph is complete. Clearly, 2 implies that any finite partition of M contains an element isomorphic to M. Then the claim above generalizes as follows: if there is a finite A such that $\bigcap_{a \in A} r(a, M) = \varnothing$ then M is the empty graph and if there is a finite B such that $B \cup \bigcup_{b \in B} r(b, M) = M$ then M is the complete graph.

Suppose M is not a random graph. Fix some finite, disjoint A and B such that no c satisfies both $r(A, c) = A$ and $r(B, c) = \varnothing$. Let $M_1 = \{c : r(A, c) \neq A\}$ and $M_2 = M \smallsetminus M_1$. Now note that

$$\bigcap_{a \in A} r(a, M_1) = \varnothing \qquad \text{and} \qquad \bigcup_{b \in B} r(b, M_2) = M_2.$$

6.3 Notes and references

We refer the reader to [1] for a well-written accessible survey on the amazing model theoretic properties of the random graph.

[1] Peter J. Cameron, *The random graph* (2013), arXiv:1301.7544. t.a. in The Mathematics of Paul Erdös III.

Chapter 7

Rich models

We introduce *Fraïssé limits*, also known as *homogeneous-universal* or *generic* structures, which here we call *rich* models, after Poizat. Rich models generalize the examples in Chapters 6 and the many more to come.

Elimination of quantifiers is briefly discussed at the end of Section 7.1. For the time being we identify quantifier elimination with the property that says that all partial embeddings are elementary maps. Proofs are easier with this notion in mind. The equivalence of this property with its syntactic counterpart is only proved in Chapter 10, when the reader is more familiar with arguments of compactness.

7.1 Models and morphisms

We now define *categories of models and partial morphisms*. These are example of concrete categories as intended in category theory. However, apart from the name, in what follows we dispense with all notions of category theory as they would make the exposition less basic than intended (without providing additional technical tools).

A category (of models and partial morphisms) is a class $\mathcal{M}$ which is disjoint union of two classes: $\mathcal{M}_{\rm ob}$ and $\mathcal{M}_{\rm hom}$. The first is the class of objects and contains structures with a common signature L which we call models. The second is the class of morphisms and contains (partial) maps between models. We require that the identity maps are morphisms and that composition of two morphism is again morphism. This makes $\mathcal{M}$ a well-defined category.

For example, $\mathcal{M}$ could consist of all models of some theory T_0 and of all partial embeddings between these. Alternatively, as morphisms we could take elementary maps between models. On a first reading the reader may assume $\mathcal{M}$ is as in one of the two examples above. In the general case we need to make some assumptions on $\mathcal{M}$.

7.1 Definition For ease of reference we list together all properties required below

c1. the (partial) identity map ${\rm id}_A : M \to M$ is a morphism, for any $A \subseteq M$

c2. if $k' : M \to N$ is a morphism for all finite $k' \subseteq k$, then $k : M \to N$ is a morphism

c3. morphisms are invertible maps and the inverse of a morphism is a morphism

c4. morphisms preserve the truth of $L_{\rm at}$-formulas

c5. if M is a model and $N \equiv M$, then also N is a model

c6. every elementary map between models is a morphism.

The connected component of a model M is the subclass of models N such that there is any morphism with domain M and codomain N (or vice versa, by c3). By axiom c1 the restriction of a morphism is a morphism, therefore M and N are in the same connected component if and only if the empty map $\varnothing : M \to N$ is a morphism. If the whole category $\mathcal{M}$ consists of one connected component we say that $\mathcal{M}$ is connected.

We call c2 the finite character of morphisms. Note that it implies the following

c7. if $k_i : M \to N$ is a chain of morphisms, then $\bigcup_{i<\lambda} k_i : M \to N$ is a morphism.

The following two definitions require c3. The generalization to non-injective morphisms is not straightforward (in fact, there are two generalizations: *projective* and *inductive*). These generalizations are not very common and will not be considered here.

7.2 Definition Assume that $\mathcal{M}$ satisfies c1-c3 of Definition 7.1. We say that a model N is λ-rich if for every model M, every $b \in M$ and every morphism $k : M \to N$ of cardinality $< \lambda$ there is a $c \in N$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is a morphism. We say that N is rich if it is λ-rich for $\lambda = |N|$. When $\mathcal{M}_{\rm ob} = \mathrm{Mod}(T_0)$ for some theory T_0 and $\mathcal{M}_{\rm hom}$ is clear from the context, we say rich model of T_0.

Rich models are also called *Fraïssé limits* or *homogeneous-universal* for a reason that will soon be clear; they are also called *syndetic*. Unfortunately these names are either too long or too syndetic, so we opt for the less common term *rich* that was proposed by Poizat.

The following two notions are closely connected with richness.

7.3 Definition Assume that $\mathcal{M}$ satisfies c1-c3 of Definition 7.1. We say that a model N is λ-universal if for every model M of cardinality $\leq \lambda$ in the same connected component as N there is an embedding $k : M \hookrightarrow N$. We say that a model N is λ-homogeneous if every $k : N \to N$ of cardinality $< \lambda$ extends to a bijective morphism $h : N \rightarrow N$ (an automorphism when c4 holds).
Note that the larger $\mathcal{M}_{\rm hom}$, the stronger notion of homogeneity. When $\mathcal{M}_{\rm hom}$ contains all partial embeddings between models (the largest class of morphisms considered here), it is common to say λ-ultrahomogeneous for λ-homogeneous.
As above, when $\lambda = |N|$ we say universal, homogeneous and ultrahomogeneous.

In Section 6.1 we implicitly used $\mathcal{M}_{\rm ob} = \mathrm{Mod}(T_{\rm lo})$ and partial embeddings as $\mathcal{M}_{\rm hom}$. In Section 6.2 we used $\mathcal{M}_{\rm ob} = \mathrm{Mod}(T_{\rm gph})$ and again partial embeddings as $\mathcal{M}_{\rm hom}$. Corollary 6.2 proves that every model of $T_{\rm dlo}$ is ω-rich. Corollary 6.14 claims the analogous fact for $T_{\rm rg}$.

In the following we frequently work under the following assumption (even when not all properties are strictly necessary).

7.4 Assumption Assume $|L| \leq \lambda$ and suppose that $\mathcal{M}$ satisfies c1-c6 of Definition 7.1

The assumption on the cardinality of L is necessary to apply the downward Löwenheim-Skolem Theorem when required.

7.5 Proposition (Assume 7.4) The following are equivalent

1. N is a λ-rich model
2. for every model M of cardinality $\leq \lambda$ and every morphism $k : M \to N$ of cardinality, say $< \lambda$ there is a embedding $h : M \hookrightarrow N$ that extends k.

Proof. Closure under union of chains of morphisms, which is ensured by c7, immediately yields 1⇒2. For implication 2⇒1 consider a morphism $k : M \to N$ of cardinality $< \lambda$ and $b \in M$. By the downward Löwenheim-Skolem theorem there is an $M' \preceq M$ of cardinality λ containing $\operatorname{dom} k \cup \{b\}$. Let $h : M' \hookrightarrow N$ be the embedding obtained from 2. By c4, the map $h : M \to N$ is a composition of morphisms, hence a morphism. □

The following theorem subsumes both Theorem 6.3 and Theorem 6.15.

7.6 Theorem (Assume 7.4) Let M and N be two rich models of the same cardinality λ. Then every morphism $k : M \to N$ of cardinality $< \lambda$ extends to an isomorphism.

Proof. When $\lambda = \omega$, we can take the proof of Theorem 6.3 and replace *partial embedding* by *morphism* and the references to Lemma 6.1 by references to Definition 7.2. As for uncountable λ, we only need to extend the construction through limit stages. By c7 we can simply take the union. □

7.7 Corollary (Assume 7.4) All rich models of cardinality λ in the same connected component are isomorphic.

It is obvious that rich models are universal. By Theorem 7.6, rich models they are homogeneous. These two notions are weaker than richness. For instance, when $\mathcal{M}$ is as in Section 6.2, the countable graph with no edge is trivially ultrahomogeneous but it is not universal and a fortiori not rich. On the other hand if we add to a countable random graph an isolated point we obtain a universal graph which is not ultrahomogeneous. However, when taken together, these two properties are equivalent to richness.

7.8 Theorem (Assume 7.4) The following are equivalent:

1. N is rich
2. N is homogeneous and universal.

Proof. Implication 1⇒2 is clear as noted above, so we prove 2⇒1. We use the characterization of richness given in Proposition 7.5. Let $k : M \to N$ be a morphism such that $|k| < |N|$ and $|M| \leq |N|$. As N is universal, there is a total morphism $f : M \hookrightarrow N$. By c3 the map $k \circ f^{-1} : N \to N$ is a morphism of cardinality $< |N|$. By homogeneity, it has an extension to an automorphism $h : N \xrightarrow{\sim} N$. It is immediate that $h \circ f : M \hookrightarrow N$ is the required extension of k. □

A consequence of Theorem 7.6 is that morphisms between rich models of the same cardinality are elementary maps. However, the theorem gives no information when the models have different cardinality nor when they are merely λ-rich. This case is dealt with in Theorem 7.11, arguably the main result of this section.

7.9 Exercise Let N be the structure obtained by adding to a countable random graph an isolated point. Show that N is homogeneous if morphisms are elementary maps, but that this is not true if morphisms are simply partial embeddings.

7.10 Exercise Check that the following can be added to Proposition 7.5

3. for every model M of cardinality $< \lambda$ and every morphism $k : M \to N$ of cardinality, say $< |M|$ there is an embedding $h : M \hookrightarrow N$ that extends k.

7.2 The theory of rich models, and quantifier elimination

The theory of the rich models of $\mathcal{M}$ is the set T_1 of sentences that hold in all models that are λ-rich for some $\lambda \geq |L|$. The theorem below proves that T_1 is complete as soon as $\mathcal{M}$ is connected. Note that the following theorem generalizes Corollary 6.6 and 6.17.

7.11 Theorem (Assume 7.4) Every morphism between λ-rich models is elementary. In particular, λ-rich models in the same connected component are elementarily equivalent.

Proof. Let $k : M \to N$ be a morphism between rich models. It suffices to prove that every finite restriction of k is elementary. By c2, we may as well assume that k itself is finite. It suffices to construct $M' \preceq M$ and $N' \preceq N$ together with a morphism $h : M \to N$ that extends k and maps M' bijectively to N'. Then by c6 the map $h : M' \xrightarrow{\sim} N'$ is the composition of morphisms, hence it is a morphism. Finally, by c4, it is an isomorphism, in particular an elementary map.

In general, richness is not preserved under elementary equivalence. Therefore M' and N' need to be constructed simultaneously with h. We define a chain of functions $\langle h_i : i<\lambda \rangle$ such that $h_i : M \to N$ are morphisms and in the end we set

$$h \;=\; \bigcup_{i<\lambda} h_i, \qquad M' \;=\; \operatorname{dom} h, \qquad N' \;=\; \operatorname{rng} h.$$

We interweave the usual back-and-forth-argument with the construction in the second proof of the Löwenheim-Skolem Theorem 2.55 in order to obtain $M' \preceq M$ and $N' \preceq N$.

The chains start with $h_0 = k$. At limit stages we take the union. Now assume we have h_i. Let $\varphi(x) \in L(\operatorname{dom} h_i)$ be some formula consistent in M and pick a solution $b \in M$. By λ-richness there is a $c \in N$ such that $h_i \cup \{\langle b, c \rangle\} : M \to N$ is a morphism. Let $h_{i+1/2} = h_i \cup \{\langle b, c \rangle\}$.

Finally, as in the proof of Theorem 6.3, we extend $h_{i+1/2}$ to obtain h_{i+1} by applying the same procedure with the roles of M and N inverted and $h^{-1}_{i+1/2}$ for h_i.

In the end we obtain $M' \preceq M$ if all formulas $\varphi(x) \in L(M')$ are considered. A similar consideration holds for N'. This is achieved using the same dovetail enumeration as in our second proof the downward Löwenheim-Skolem Theorem 2.55. □

7.12 Corollary (Assume 7.4) Assume $\mathcal{M}$ is connected. Let T_1 be the theory of the rich models of $\mathcal{M}$. Then T_1 is complete.

Proof. If for a contradictiopn $\varphi, \neg\varphi \notin T_1$, then there are two λ-rich models (say, $\lambda = |L|$) such that $M \vDash \varphi$ and $N \vDash \neg\varphi$. But, as $\varnothing : M \to \mathcal{N}$ is a morphism, this contradicts the above theorem. □

Assume for simplicity that $\mathcal{M}$ is connected. Then all λ-rich models belong to $\mathrm{Mod}(T_1)$ for some complete theory T_1. It is interesting to ask if the converse is true: if T_1 is the theory of the λ-rich models of $\mathcal{M}$, do all models in $\mathrm{Mod}(T_1)$ are rich?

7.13 Remark (Assume 7.4) Assume $\mathcal{M}$ is connected. Let T_1 be the theory of the rich models of $\mathcal{M}$. If every model of T_1 is λ-rich then T_1 is λ-categorical. (This is a consequence of Corollary 7.7)

A very interesting variant of the question asked above is considered in Theorem 9.8 where it is related to an important phenomenon that we now introduce.

When partial embeddings between models of a given theory T coincide with elementary maps, it is always by a fundamental reason. Let us introduce some terminology. Let T be a consistent theory. We say that T has (or admits) elimination of quantifiers if for every $\varphi(x) \in L$ there is a quantifier-free formula $\psi(x) \in L_{\mathrm{qf}}$ such that

$$T \vdash \psi(x) \leftrightarrow \varphi(x).$$

We will discuss general criteria for elimination of quantifiers in Chapter 10. Here we report without proof the following theorem.

7.14 Theorem The following are equivalent

1. T has elimination of quantifiers
2. every partial embedding between models of T is an elementary map.

⚠ This theorem will be proved only in Chapter 10, see Remark 10.4 or Corollary 10.12. For the time being we do not need the syntactic version of elimination of quantifiers, so when saying that T *has quantifier elimination* we mean 2 of Theorem 7.14. For instance, we rephrase Corollary 6.6 and 6.17 by saying that T_{dlo} and T_{rg} have elimination of quantifiers.

In the next chapter we introduce important examples of ω-rich models that do not have an ω-categorical theory. These are algebraic structures (groups, fields etc.) which are more complex than pure relational structures. So, we conclude this section with an example of this phenomenon in an almost trivial context.

7.15 Example Let L contain a unary predicate r_n for every positive integer n. The theory T_0 contains the axioms $\neg\exists x\, [r_n(x) \wedge r_m(x)]$ for $n \neq m$ and $\exists^{\leq n} x\, r_n(x)$ for every n. Work in the category of models of T_0 and partial embeddings. Let T_1 be the theory that extends T_0 with the axioms $\exists^{=n} x\, r_n(x)$ for every n. Let $q(x)$ be the type $\{\neg r_n(x) : n \in \omega\}$. There are models of T_1 that do not realize $q(x)$, hence T_1 is not ω-categorical. It is easy to verify that the following are equivalent.

1. N is an ω-rich model
2. $N \vDash T_1$ and $q(N)$ is infinite.

The reader may use Theorem 7.11 and Compactness Theorem for Types 5.7 to prove that T_1 is complete and has elimination of quantifiers. It is also easy to verify that every uncountable model of T_1 is rich and consequently that T_1 is uncountably categorical.

7.16 Exercise The language contains only the binary relations $<$ and e. The theory T_0 says that $<$ is a strict linear order and that e is an equivalence relation. Let $\mathcal{M}$ consists of models of T_0 and partial embeddings as morphisms. Do rich models exist? Can we axiomatize their theory? If so, does it have elimination of quantifiers? Is it λ-categorical for some λ?

7.17 Exercise The language contains only two binary relations. The theory T_0 says that they are equivalence relations. Let $\mathcal{M}$ consists of models of T_0 and partial embeddings as morphisms. Do rich models exist? Can we axiomatize their theory? If so, does it have elimination of quantifiers? Is it λ-categorical for some λ?

7.18 Exercise The language contains a binary relation r and countably many unary relation symbols r_i. The theory T_0 says that r is a graph and that the r_i are mutually exclusive. Let $\mathcal{M}$ consists of models of T_0 and partial embeddings as morphisms. Axiomatize the theory T_1 of the rich models. Are all countable models of T_1 rich? Does T_1 have quantifier elimination?

7.19 Exercise Let T_0 be the theory axiomatized by $T_{\rm lo}$ and the axiom that says that every point has an immediate successor and an immediate predecessor. Let $\mathcal{M}$ consists of models of $T_{\rm lo}$ and, as morphisms, maps that preserve the distance between points. Describe a countable rich model and the theory of rich models.

7.20 Exercise A back-and-forth system between to models M and N is a nonempty set $\mathcal{P}$ of finite functions k such that

0. $k : M \to N$ is a partial embedding

1a. for every $b \in M$ there is $h \in \mathcal{P}$ such that $k \subseteq h$ and $b \in \operatorname{dom} h$

1b. for every $c \in N$ there is $h \in \mathcal{P}$ such that $k \subseteq h$ and $c \in \operatorname{rng} h$.

Prove that if L is countable and there is a back-and-forth system between M and N then there are $M' \preceq M$ and $N' \preceq N$ such that $M' \simeq N'$.

7.21 Exercise Let $\mathcal{M}$ consists of models of $T_{\rm lo}$ and, as morphisms, partial embeddings. Assume there exists a rich model N of uncountable cardinality κ. Prove that N has 2^κ Dedekind cuts. A *Dedekind cut* is a set $C \subseteq N$ that is downward closed, i.e. such that $a < b \in C$ implies $a \in C$.

7.3 Weaker notions of universality and homogeneity

We want to extend the equivalence in Theorem 7.8 to λ-rich models. For that we need to weaken the notions of λ-homogeneity. This section is more technical and could be skipped at a first reading.

7.22 Definition We say that a structure N is weakly λ-homogeneous if for every $b \in N$ every morphism $k : N \to N$ of cardinality $< \lambda$ extends to one defined in b. The term back-and-forth λ-homogeneous is also used.

7.23 Lemma (Assume 7.4) Let N be a weakly λ-homogeneous model. Let $A \subseteq N$ have cardinality $\leq \lambda$ and let $k : N \to N$ be a morphism of cardinality $< \lambda$. Then there is a model $M \preceq N$ containing A and an automorphism $h : M \xrightarrow{\sim} M$ that extends k.

Proof. Similar to the proof of Theorem 7.11. We shall construct simultaneously a chain $\langle A_i : i < \lambda \rangle$ of subsets of N and a chain functions $\langle h_i : i < \lambda \rangle$, such that $h_i : N \to N$ are morphisms. In the end we will set

$$M = \bigcup_{i<\lambda} A_i \qquad \text{and} \qquad h = \bigcup_{i<\lambda} h_i$$

The chains start with $A_0 = A \cup \operatorname{dom} k \cup \operatorname{rng} k$ and $h_0 = k$. As usual, at limit stages we take the union. Now we consider successor stages. At stage i we fix some enumerations of A_i and of $L_x(A_i)$, where $|x| = 1$. Let $\langle i_1, i_2 \rangle$ be the i-th pair of ordinals $< \lambda$. If the i_2-th formula in $L_x(A_{i_1})$ is consistent in N, let a be any of its solutions. Also let b be the i_2-th element of A_{i_1}. Let $h_{i+1} : N \to N$ be a minimal morphism that extends h_i and is such that $b \in \operatorname{dom} h_{i+1} \cap \operatorname{rng} h_{i+1}$. Define $A_{i+1} = A_i \cup \{a,\ h_{i+1}b,\ h_{i+1}^{-1}b\}$. □

7.24 Theorem (Assume 7.4) For every model N the following are equivalent

1. N is λ-rich
2. N is λ-universal and weakly λ-homogeneous.

Proof. Implication 1⇒2 is clear. To prove 2⇒1 we generalize the proof of Theorem 7.8. We assume 2 fix some morphism $k : M \to N$ of cardinality $< \lambda$ and let $b \in M$. By λ-universality and the downward Löwenheim-Skolem theorem, there is a morphism $f : M \to N$ with domain of definition $\operatorname{dom} k \cup \{b\}$. The map $f \circ k^{-1} : N \to N$ has cardinality $< \lambda$ and, by Lemma 7.23, it has an extension to an automorphism $h : N' \xrightarrow{\sim} N'$ for some $N' \preceq N$ containing $\operatorname{rng} k \cup \operatorname{rng} f$. Then $h \circ f : M \to N$ extends k and is defined on b. □

7.25 Exercise (Assume 7.4) Prove that any weakly λ-homogeneous structure of cardinality λ is homogeneous.

7.4 The amalgamation property

In this section we discuss conditions that ensure the existence of rich models.

We say that $\mathcal{M}$ has the amalgamation property if for every pair of morphisms $f_1 : M \to M_1$ and $f_2 : M \to M_2$ there are two of embeddings $g_1 : M_1 \hookrightarrow N$ and $g_2 : M_2 \hookrightarrow N$ such that $g_1 \circ f_1(a) = g_2 \circ f_2(a)$ for every a in the common domain of definition, $\operatorname{dom}(f_1) \cap \operatorname{dom}(f_2)$.

$$\begin{tikzcd}[column sep=small] & M_1 \arrow[dr, hook, "g_1"] & \\ M \arrow[ur, "f_1"] \arrow[dr, "f_1"'] & & N \\ & M_2 \arrow[ur, hook, "g_2"'] & \end{tikzcd}$$

As we assume that morphisms are invertible, we may express the amalgamation property in a more concise form. It is convenient to use the following notation. We write $M \leq N$ if $M \subseteq N$ and $\mathrm{id}_M : M \hookrightarrow N$ is a morphism. We say that $k : M \to N$ extends to $g : M \to N'$ if $k \subseteq h$ and $N \leq N'$ namely, if the following diagram commutes

$$\begin{tikzcd}[column sep=small] & N \arrow[dr, hook, "\mathrm{id}_N"] & \\ M \arrow[ur, "k"] \arrow[rr, hook, "g"'] & & N' \end{tikzcd}$$

7.26 Proposition Assume c3, then the following are equivalent

1. $\mathcal{M}$ has the amalgamation property
2. every morphism $k : M \to N$ extends to an embedding $g : M \hookrightarrow N'$.

Proof. 1⇒2 Given $k : M \to N$, the amalgamation property yields the following commutative diagram which can be simplified to the diagram at the right

$$\begin{tikzcd}[column sep=small] & N \arrow[dr, hook, "g_1"] & \\ M \arrow[ur, "k"] \arrow[dr, "\mathrm{id}_M"'] & & N' \\ & M \arrow[ur, hook, "g_2"'] & \end{tikzcd} \qquad \begin{tikzcd}[column sep=small] & N \arrow[dr, hook, "g_1"] & \\ M \arrow[ur, "k"] \arrow[rr, hook, "g_2"'] & & N' \end{tikzcd}$$

Up to isomorphism we can assume $g_1 = \mathrm{id}_N$, i.e. that $N \leq N'$. Hence $g_2 : M \hookrightarrow N'$ is the required extension of $k : M \to N$.

2⇒1 Let $f_1 : M \to M_1$ and $f_2 : M \to M_2$ be given. Let $k = f_2 \circ f_1^{-1} : M_1 \to M_2$ and let $g : M_1 \hookrightarrow N$ be the extension ensured by 2. Then we obtain

$$\begin{tikzcd} & M_1 \arrow[dd, "f_2 \circ f_1^{-1}"] \arrow[dr, hook, "g"] & \\ M \arrow[ur, "f_1"] \arrow[dr, "f_2"'] & & N \\ & M_2 \arrow[ur, hook, "\mathrm{id}_{M_2}"'] & \end{tikzcd}$$

as required. □

We say that $\langle M_i : i < \lambda \rangle$ is a $\leq$-chain if $M_i \leq M_j$ for all $i < j < \lambda$. For the next theorem to hold we need the following property

7.27 Definition We say that $\mathcal{M}$ is closed under union of $\leq$-chains if

c8. if $\langle M_i : i < \lambda \rangle$ is a $\leq$-chain, then $M_i \leq \bigcup_{j<\lambda} M_j$ for all $i < \lambda$.

The following is a general existence theorem for rich models. This general form requires large cardinalities. We leave to the reader to verify that if the number if finite morphisms is countable (up to isomorphism) then countable rich models exit.

7.28 Theorem Assume 7.4. Assume further c8 and that $\mathcal{M}$ has the amalgamation property. Let λ be such that $|L| < \lambda = \lambda^{<\lambda}$. Then there is a rich model N of cardinality λ.

Proof. We construct N as union of a $\leq$-chain of models $\langle N_i : i < \lambda \rangle$ such that $|N_i| = \lambda$. Let N_0 be any model of cardinality λ. At stage $i+1$, let $f : M \to N_i$ be the least morphism (in a well-ordering that we specify below) such that $|f| \leq |M| < \lambda$ and f has no extension to an embedding $f' : M \hookrightarrow N_i$. Apply the amalgamation property to obtain a total morphism $f' : M \hookrightarrow N'$ that extends $f : M \to N_i$. By the downward Löwenheim-Skolem Theorem we may assume $|N'| = \lambda$. Let $N_{i+1} = N'$. At limit stages take the union.

The well-ordering mentioned needs to be chosen so that in the end we forget nobody. So, first at each stage we well-order the isomorphism-type of the morphisms $f : M \to N_i$ such that $f \leq |M| < \lambda$. Then the required well-ordering is obtained by dovetailing all these well-orderings. The length of this enumeration is at most $\lambda^{<\lambda}$, which is λ by hypothesis.

We check that N is rich. Let $f : M \to N$ be a morphism and $|f| < |M| \leq \lambda$. As $|L| < \lambda$ we can approximate M with an elementary chain of structures of cardinality $< \lambda$. Hence we may as well assume that $|f| \leq |M| < \lambda$. The cofinality of λ is larger than $|f|$, hence $\operatorname{rng} f \subseteq N_i$ for some $i < \lambda$. So $f : M \to N_i$ is a morphism and at some stage j we have ensured the existence of an embedding of $f' : M \hookrightarrow N_{j+1}$ that extends f. □

7.29 Proposition Let $\mathcal{M}$ consist of all structures of some fixed signature and the elementary maps between these. Then $\mathcal{M}$ has the amalgamation property.

Proof. Let $k : M \to N$ be an elementary map. Let a enumerate $\operatorname{dom} k$ and let b enumerate M. Set $p(x\,;z) = \operatorname{tp}_M(b\,;a)$. The type $p(x\,;a)$ is consistent in M, in particular, it is finitely consistent and, by elementarity, $p(x\,;ka)$ is finitely consistent in N. By the compactness theorem, there is $N' \succeq N$ such that $N' \vDash p(c\,;ka)$ for some $c \in N'^{x}$. Hence $g = \{\langle b, c \rangle\} : M \to N'$ is the required elementary map that extends $k : M \to N$. □

Chapter 8

Some algebraic structures

The main result in this chapter is the elimination of quantifiers in algebraically closed fields, Corollary 8.22. In commutative algebra this is called Chevalley's Theorem on constructible sets. From tghis we derive Hilbert's Nullstellensatz. First we state the theorem the model theoretic language, Theorem 8.27, then we translate it in the traditional algebraic setting, Theorem 8.31. To obtain the latter theorem we introduce the model theoretic objects that correspond to prime and radical ideals of polynomials.

The first sections of this chapter are not a prerequisite for Sections 8.4–8.7, at the cost of a few repetitions in the latter.

8.1 Notation Recall that when $A \subseteq M$ we denote by $\langle A \rangle_M$ the substructure of M generated by A. Then $\langle A \rangle_M \subseteq N$ is equivalent to $N \vDash \text{Diag}\,\langle A \rangle_M$. The diagram of a structure has been defined in Notation 3.20.
In this chapter, whenever some $A \subseteq M \vDash T$ are fixed, by *model* we mean superstructures of $\langle A \rangle_M$ that models T. The notions of *logical consequence*, *consistency*, *completeness*, etc. are modified accordingly, and we write $\vdash$ for $T \cup \text{Diag}\langle A \rangle_M \vdash$.
We say that a type $p(x)$ is trivial if $\vdash p(x)$.

8.1 Abelian groups

The language L is that of additive groups. The theory T_{ag} of abelian groups is axiomatized by the universal closure of the usual axioms

a1 $(x+y)+z \;=\; y+(x+z)$

a2 $x+(-x) \;=\; (-x)+x \;=\; 0$

a3 $x+0 \;=\; 0+x \;=\; x$

a4 $x+y \;=\; y+x.$

Let x be a tuple of variables of length α, an ordinal. We write $L_{\text{ter},x}$ for the set of terms $t(x)$ with free variables among x. On this set we define the equivalence relation

$$t(x) \sim s(x) \quad = \quad T_{\text{ag}} \vdash t(x) = s(x).$$

We define the group operations on $L_{\text{ter},x}/\!\sim$ in the obvious way. We denote by $\mathbb{Z}^{\oplus\alpha}$ the set of tuples of integers of length α that are almost always 0. The group operations on $\mathbb{Z}^{\oplus\alpha}$ are defined coordinate-wise. The following immediate proposition implies in particular that $L_{\text{ter},x}/\!\sim$ is isomorphic to $\mathbb{Z}^{\oplus\alpha}$.

8.2 Proposition Let $A \subseteq M \vDash T_{\rm ag}$. Then for every formula $\varphi(x) \in L_{\rm at}(A)$ there are $n \in \mathbb{Z}^{\oplus\alpha}$ and $c \in \langle A\rangle_M$ such that

$$\vdash \quad \varphi(x) \quad \leftrightarrow \quad \sum_{i<\alpha} n_i x_i \;=\; c$$

where $n = \langle n_i : i < \alpha\rangle$ and $x = \langle x_i : i < \alpha\rangle$.

Proof. Up to equivalence over $T_{\rm ag}$ the formula $\varphi(x)$ has the form $s(x) = t(a)$ for some parameter-free terms $s(x)$ and $t(z)$. Over Diag $\langle A\rangle_M$, we can replace $t(a)$ with a single $c \in \langle A\rangle_M$ and write $s(x)$ as the linear combination shown above. □

8.3 Definition Let $M \vDash T_{\rm ag}$. For $A \subseteq M$ and $c \in M$, we say that c is independent from A if $\langle A\rangle_M \cap \langle c\rangle_M = \{0\}$. Otherwise we say that c is dependent from A. The rank of M is the least cardinality of a subset $A \subseteq M$ such that all elements in M are dependent from A. We denote it by $\operatorname{rank}(M)$.

Note that when M is a vector space the condition $\langle A\rangle_M \cap \langle c\rangle_M = \{0\}$ is equivalent to saying that c is not a linear combination of vectors in A. Then rank M coincides with the dimension of M. In fact, what we do here for abelian groups could be easily generalized to D-modules, where D is any integral domain, and in particular to vector spaces. In practice, it is more convenient to use the following syntactic characterization of independence.

An element c of an abelian group is a torsion element if $n\,c = 0$ for some positive integer.

8.4 Proposition Let $A \subseteq M \vDash T_{\rm ag}$. Suppose that $c \in M$ is not a torsion element. Then the following are equivalent

1. c is independent from A
2. $p(x) = \text{at-tp}_M(c/A)$ is trivial (see Notation 8.1).

Proof. 1⇒2. By Proposition 8.2, formulas in $p(x)$ may be assumed to have the form $n\,x = a$ for some integer n and some $a \in \langle A\rangle_M$. As this formula is satisfied by c then $a \in \langle A\rangle_M \cap \langle c\rangle_M$. Hence $a = 0$. As c is not a torsion element, $n = 0$, therefore the equation is trivial.

2⇒1. Let $a \in \langle A\rangle_M \cap \langle c\rangle_M \neq \{0\}$. Then $n\,c = a$ for some $n \in \mathbb{Z} \smallsetminus \{0\}$. Then c satisfies the equation $nx = a$. If this equation is trivial $n = 0$ and $a = 0$. □

8.5 Remark Let $k : M \to N$ be a partial embedding and let a be an enumeration of dom k. We claim that $k \cup \{\langle b, c\rangle\} : M \to N$ is a partial embedding for every $b \in M$ and $c \in N$ that are independent from a, respectively $k\,a$. In fact, it suffices to check that $M, b, a \equiv_{\rm at} N, c, k\,a$. Suppose $\varphi(x\,;z) \in L_{\rm at}$ is such that $M \vDash \varphi(b\,;a)$. Then by independence $\varphi(x\,;a)$ is trivial, i.e.

$$T_{\rm ag} \cup \text{Diag}\langle a\rangle_M \quad\vdash\quad \varphi(x\,;a).$$

As $\langle a\rangle_M$ and $\langle k\,a\rangle_N$ are isomorphic structures

$$T_{\rm ag} \;\cup\; \text{Diag}\langle k\,a\rangle_N \quad\vdash\quad \varphi(x\,;k\,a).$$

Therefore $N \vDash \varphi(c\,;k\,a)$. This proves $M, b, a \Rrightarrow_{\rm at} N, c, k\,a$.
As the same assumptions apply to $k^{-1} : N \to M$, we also have $M, b, a \Lleftarrow_{\rm at} N, c, k\,a$.

8.2 Torsion-free abelian groups

The theory of torsion-free abelian groups extends T_{ag} with the following axioms for all positive integers n

tf $\quad nx = 0 \rightarrow x = 0.$

We denote this theory by T_{tfag}. It is not difficult to see that in a torsion-free abelian group every equation of the form $nx = a$ has at most one solution.

8.6 Proposition Let $M \vDash T_{\mathrm{tfag}}$ be uncountable. Then $\operatorname{rank} M = |M|$.

Proof. Let $A \subseteq M$ have cardinality $< |M|$. We claim that M contains some element that is independent from A. It suffices to show that the number of elements that are dependent from A is $< |M|$. If $c \in M$ is dependent from A then, by Proposition 8.7, it is a solution of some formula $L_{\mathrm{at}}(A)$. As there is no torsion, such a formula has at most one solution. Therefore the number of elements that are dependent from A is at most $|L_{\mathrm{at}}(A)|$, that is $\max\{|A|, \omega\}$. If M is uncountable the claim follows. $\Box$

8.7 Proposition Let $A \subseteq M \vDash T_{\mathrm{tfag}}$. Let $p(x) = \mathrm{at}^{\pm}\text{-}\mathrm{tp}(b/A)$, where $b \in M$. Then one of the following holds

1. b is independent from A
2. $M \vDash \varphi(b)$ for some $\varphi(x) \in L_{\mathrm{at}}(A)$ such that $\vdash \varphi(x) \rightarrow p(x)$.

Note the similarity with Example 7.15, where the independent type is $q(x)$ and the isolating formulas are the $r_i(x)$.

It is important to observe that the set A above may be infinite. This is essential to obtain Corollary 8.10, and it is one of the main differences between this example and the examples encountered in Chapter 6.

Proof. If b is dependent from A, then b satisfies a nontrivial atomic formula $\varphi(x)$ which we claim is the formula required in 2. It suffices to show that $\varphi(x)$ implies a complete $L_{\mathrm{at}^{\pm}}(A)$-type. Clearly this type must be $p(x)$. Let $\varphi(x)$ have the form $nx = a$ for some $n \in \mathbb{Z} \smallsetminus \{0\}$ and $a \in \langle A\rangle_M \smallsetminus \{0\}$. We show that for every $m \in \mathbb{Z}$ and every $c \in \langle A\rangle_M$ one of the following holds

a. $\vdash nx = a \rightarrow mx = c$

b. $\vdash nx = a \rightarrow mx \neq c$.

Suppose not for a contradiction that neither a nor b holds and fix models N_1, N_2 and some $b_i \in N_i$ such that

a′. $N_1 \vDash nb_1 = a$ and $N_1 \vDash mb_1 \neq c$

b′. $N_2 \vDash nb_2 = a$ and $N_2 \vDash mb_2 = c$.

From b′ we infer that $N_2 \vDash ma = nc$. As N_1 is torsion-free, from a' we infer that $N_1 \vDash ma \neq nc$. But $ma = nc$ is a formula with parameters in $\langle A\rangle_M$, so it should have the same truth value in all superstructures of $\langle A\rangle_M$, a contradiction. $\Box$

8.3 Divisible abelian groups

The theory of divisible abelian groups extends T_{tfag} with the following axioms for all integers $n \neq 0$

div $y \neq 0 \to \exists x\, nx = y$.

We denote this theory by T_{dag}.

8.8 Proposition Let $A \subseteq M \vDash T_{\text{tfag}}$ and let $\varphi(x) \in L_{\text{at}}(A)$, where $|x| = 1$, be consistent. Then $N \vDash \exists x\, \varphi(x)$ for every N such that $\langle A \rangle_M \subseteq N \vDash T_{\text{dag}}$.

Note that in the proposition above *consistent* means satisfied in some M' such that $\langle A \rangle_M \subseteq M' \vDash T_{\text{tfag}}$. The proposition holds more generally for all $\varphi(x) \in L_{\text{qf}}$ and also when x is a tuple of variables. This follows from Lemma 8.9, whose proof uses the proposition.

Proof. We can assume that $\varphi(x)$ has the form $nx = a$ for some $n \in \mathbb{Z}$ and some $a \in \langle A \rangle_M$. If $n = 0$, then $a = 0$ since $\varphi(x)$ is consistent, and the claim is trivial. If $n \neq 0$ then by consistency $a \neq 0$, hence a solution exist in N by axiom div. □

We are ready to prove that divisible abelian groups of infinite rank are ω-rich.

8.9 Lemma Let $M \vDash T_{\text{tfag}}$. Let $N \vDash T_{\text{dag}}$ be a model of rank $\geq \lambda$. Then for every partial isomorphism $k : M \to N$ of cardinality $< \lambda$ and every $b \in M$ there is $c \in N$ such that $k \cup \{\langle b\,;c \rangle\} : M \to N$ is a partial isomorphism.

Proof. Let a be an enumeration of $\operatorname{dom} k$ and let $p(x\,;z) = \text{at}^{\pm}\text{-tp}(b\,;a)$. The required c has to realize $p(x\,;k\,a)$. We consider two cases. If b is dependent from a, then Proposition 8.7 yields a formula $\varphi(x\,;z) \in L_{\text{at}}$ such that

i. $\vdash\ \varphi(x\,;a) \to p(x\,;a)$

ii. $\varphi(x\,;a)$ is consistent.

By isomorphism, i and ii hold with a replaced by $k\,a$. Then by Proposition 8.8 the formula $\varphi(x\,;k\,a)$ has a solution $c \in N$.

The second case, which has no analogue in Lemma 6.1, is when b is independent from a. Then by Remark 8.5 we may choose c to be any element of N independent from $k\,a$. Such an element exists because N has rank at least λ. □

Below a few important consequences of this lemma.

8.10 Corollary Work in the category of models of T_{tfag} with partial embeddings as morphisms. Then the following are equivalent

1. N is a λ-rich model
2. $N \vDash T_{\text{dag}}$ and has rank $\geq \lambda$.

In particular every uncountable $N \vDash T_{\text{dag}}$ is rich.

8.11 Corollary The theory $T_{\rm dag}$ is uncountably categorical, complete, and has quantifier elimination.

Proof. Categoricity and completeness are immediate (the category presented in the corollary above is connected). As for quantifier elimination, let $k : M \to N$ is a partial isomorphism between models of $T_{\rm dag}$. If $M \preceq M'$ and $M \preceq N'$ are elementary superstructures of uncountable cardinality then $k : M' \to N'$ is elementary by Theorem 7.11 and this suffices to conclude that $k : M \to N$ is elementary. □

8.12 Exercise Prove a converse of Proposition 8.8. Let $A \subseteq N \vDash T_{\rm ag}$ and let x be a single variable. Prove that if $N \vDash \exists x\, \varphi(x)$ for every consistent $\varphi(x) \in L_{\rm at}(A)$, then $N \vDash T_{\rm dag}$.

8.13 Exercise Prove that every model of $T_{\rm dag}$ is ω-ultrahomogeneous (independently of cardinality and rank).

8.4 Commutative rings

In this section L is the language of (unital) rings. It contains two constants 0 and 1 the unary operation $-$ and two binary operations $+$ and $\cdot$. The theory of rings contains the following axioms

a1-a4. as for abelian groups

r1. $(x{\cdot}y){\cdot}z \;=\; y{\cdot}(x{\cdot}z)$

r2. $1{\cdot}x \;=\; x{\cdot}1 \;=\; x$

r3. $(x+y){\cdot}z \;=\; x{\cdot}z + y{\cdot}z$

r4. $z{\cdot}(x+y) \;=\; z{\cdot}x + z{\cdot}y.$

All the rings we consider are commutative

c. $x{\cdot}y \;=\; y{\cdot}x.$

We denote the theory of commutative rings by $T_{\rm cr}$.

In what follows the theory $T_{\rm cr} \cup {\rm Diag}\langle A\rangle_M$, for some M clear from the context, is implicit in the sense of Notation 8.1. So it is important to remember that ${\rm Diag}\langle A\rangle_M$ is not trivial even when $A = \varnothing$. In fact, ${\rm Diag}\langle\varnothing\rangle_M$ determines the characteristic of the models.

Let $A \subseteq M \vDash T_{\rm cr}$ and let x be a tuple of variables. We write $L_{{\rm ter},x}(A)$ for the set of terms $t(x)$ with free variables among x and parameters in A. On this set we define the equivalence relation

$$t(x) \sim s(x) \quad\Leftrightarrow\quad \vdash\; t(x) = s(x).$$

On $L_{{\rm ter},x}(A)/{\sim}$ we define the ring operations in the obvious way. These make of $L_{{\rm ter},x}(A)/{\sim}$ a commutative ring. We denote by $A[x]$ the set of polynomials with variables among x and parameters in $\langle A\rangle_M$. The ring operations on $A[x]$ are defined as usual. The following proposition (which is clear) implies in particular that $L_{{\rm ter},x}(A)/{\sim}$ is isomorphic to $A[x]$. For simplicity we state it only for $|x| = 1$.

8.14 Proposition Let $A \subseteq M \vDash T_{\rm cr}$ and let x be a single variable. Then for every formula $\varphi(x) \in L_{\rm at}(A)$ there is a unique $n < \omega$ and a unique tuple $\langle a_i : i \leq n\rangle$ of elements of $\langle A\rangle_M$ such that $a_n \neq 0$ and

$$\vdash \quad \varphi(x) \quad \leftrightarrow \quad \sum_{i\leq n} a_i x^i \ = \ 0.$$

The integer n in the proposition above is called the degree of $\varphi(x)$.

8.15 Definition Let $A \subseteq M \vDash T_{\rm cr}$. An element $b \in M$ is transcendental over A if the type $p(x) = \text{at-tp}_M(b/A)$ is trivial (see Notation 3.20 and 8.1). Otherwise we say that b is algebraic over A. The transcendence degree of M is the least cardinality of a subset $A \subseteq M$ such that all the elements of M are algebraic over A.

8.16 Remark Remark 8.5 holds here with ‘independent’ replaced by ‘transcendental’ and $T_{\rm ag}$ replaced by $T_{\rm cr}$.

8.5 Integral domains

Let $a \in M \vDash T$. We say that a is a zero divisor if $a\,b = 0$ for some $b \in M \smallsetminus \{0\}$. An integral domain is a commutative ring without zero divisors. The theory of integral domains contains the axioms of commutative rings and the following

nt. $0 \neq 1$

id. $x{\cdot}y = 0 \ \rightarrow \ x = 0 \ \vee \ y = 0$.

We denote the theory of integral domains by $T_{\rm id}$.

For a prime p, we define the theory $T^p_{\rm id}$, which contains $T_{\rm id}$ and the axiom

ch$_p$. $1 + \cdots (p \text{ times}) \cdots + 1 = 0$.

The theory $T^0_{\rm id}$ contains the negation of ch$_p$ for all p. Note that all models of $T^p_{\rm id}$ have the same characteristic in the model theoretic sense defined in 3.28. In the remaining section we work in the category of models of $T_{\rm id}$ with partial embeddings as morphisms. This category consists of countably many connected components each containing all models of $T^p_{\rm id}$ for some p.

8.17 Proposition Let $M \vDash T_{\rm id}$ be uncountable. Then M has transcendence degree $|M|$.

Proof. In an integral domain every polynomial has finitely many solutions and there are $|L(A)|$ polynomials over A. □

8.18 Proposition Let $A \subseteq M \vDash T_{\rm id}$. For $b \in M$ let $p(x) = \text{at}^{\pm}\text{-tp}(b/A)$. Then one of the following holds

1. b is transcendental over A
2. $M \vDash \varphi(b)$ for some $\varphi(x) \in L_{\rm at}(A)$ such that $\vdash \ \varphi(x) \rightarrow p(x)$.

Note the similarity with Example 7.15, where the transcendental type is $q(x)$ and the isolating formulas are the $r_i(x)$.

As in Proposition 8.7, the set A may be infinite. This is essential to obtain Corollary 8.21.

Proof. Suppose b is not transcendental, i.e. it satisfies a nontrivial atomic formula. Let $\varphi(x) \in L_{\mathrm{at}}(A)$ be a nontrivial formula with minimal degree such that $\varphi(b)$. We prove that $\varphi(x)$ implies a complete $L_{\mathrm{at}^{\pm}}(A)$-type. Clearly this type must be $p(x)$. We prove that for any $\xi(x) \in L_{\mathrm{at}}(A)$ one of the following holds

1. $\vdash\ \varphi(x) \rightarrow \ \ \xi(x)$
2. $\vdash\ \varphi(x) \rightarrow \neg\xi(x)$.

Let us write $a(x) = 0$ and $a'(x) = 0$ for the formulas $\varphi(x)$ and $\xi(x)$, respectively. If $\langle A\rangle_M$ is a field, choose a polynomial $d(x)$ of maximal degree such that for some polynomials $t(x)$ and $t'(x)$ the following hold

a. $d(x)\,t(x) \ = a(x)$

a′. $d(x)\,t'(x) = a'(x)$.

If $\langle A\rangle_M$ is not a field, polynomials $d(x)$, $t(x)$ and $t'(x)$ as above exist with coefficients in the field of fractions of $\langle A\rangle_M$. Then a and a' hold up to a factor in $\langle A\rangle_M$ which we absorb in $a(x)$ and $a'(x)$.

From a we get $d(b) = 0$ or $t(b) = 0$. In the first case, as $a(x)$ has minimal degree, we conclude that $t(x)$ is constant. This implies that any zero of $a(x)$ is also a zero of $a'(x)$, that is, it implies 1.

Now suppose $t(b) = 0$. Then the minimality of the degree of $a(x)$ implies that $d(x) = d$, where d is a nonzero constant. If $\langle A\rangle_M$ is a field, apply Bézout's identity to obtain two polynomials $c(x)$ and $c'(x)$ such that $d = a(x)c(x) + a'(x)c'(x)$. Then $a(x)$ and $a'(x)$ have no common zeros, and 2 follows. If $\langle A\rangle_M$ is not a field, we use Bézout's identity in the field of fractions of $\langle A\rangle_M$ and, for some $d' \in \langle A\rangle_M \smallsetminus \{0\}$, obtain $d'd = a(x)c(x) + a'(x)c'(x)$. Then we reach the same conclusion. □

8.6 Algebraically closed fields

Let $a, b \in M \vDash T_{\mathrm{id}}$. We say that b is the inverse of a if $a{\cdot}b = 1$. A field is a commutative ring where every non-zero element has an inverse. The theory of fields contains T_{id} and the axiom

f. $\exists y\ \big[x \neq 0 \ \rightarrow\ x{\cdot}y = 1\big]$.

Fields are structures in the signature of rings: the language contains no symbol for the multiplicative inverse. So, substructures of fields are merely integral domains.

The theory of algebraically closed field, which we denote by T_{acf}, also contains the following axioms for every positive integer n

ac. $\exists x\ \big(x^n + z_{n-1}x^{n-1} + \cdots + z_1 x + z_0 \ = \ 0\big)$.

The theory T^p_{acf} is defined in analogy to T^p_{id} in the previous section.

8.19 Proposition Let $A \subseteq M \vDash T_{\mathrm{id}}$ and let $\varphi(x) \in L_{\mathrm{at}}(A)$, where $|x| = 1$, be consistent. Then $N \vDash \exists x\, \varphi(x)$ for every model $N \vDash T_{\mathrm{acf}}$.

Note that in the proposition above *consistent* means satisfied in some M' such that $\langle A\rangle_M \subseteq M' \vDash T_{\mathrm{id}}$. The claim in the proposition holds more generally for all $\varphi(x) \in$

$L_{\rm qf}$ when x is a tuple of variables. This follows from Lemma 8.20 whose proof uses the proposition.

Proof. Up to equivalence $\varphi(x)$ has the form $a_n x^n + \cdots + a_1 x + a_0 = 0$ for some $a_i \in \langle A\rangle_N$. Choose n minimal. If $n = 0$ then $a_0 = 0$ by the consistency of $\varphi(x)$ and the claim is trivial. Otherwise $a_n \neq 0$ and the claim follows from f and ac. □

8.20 Lemma Let $k : M \to N$ be a partial isomorphism of cardinality $< \lambda$, where $M \vDash T_{\rm id}$ and $N \vDash T_{\rm acf}$ has transcendence degree $\geq \lambda$. Then for every $b \in M$ there is $c \in N$ such that $k \cup \big\{\langle b, c\rangle\big\} : M \to N$ is a partial isomorphism.

The following is the same proof given for Lemma 8.9 which we repeat here for convenience.

Proof. Let a be an enumeration of $\operatorname{dom} k$ and let $p(x\,;z) = \text{at}^{\pm}\text{-tp}_M(b\,;a)$. The required c has to realize $p(x\,;k\,a)$. We consider two cases. If b is algebraic over a, then Proposition 8.18 yields a formula $\varphi(x\,;z) \in L_{\rm at}$ such that

i. $\varphi(x\,;a) \to p(x\,;a)$

ii. $\varphi(x\,;a)$ is consistent.

By isomorphism i and ii hold with a replaced by $k\,a$. Then by Proposition 8.19 the formula $\varphi(x\,;k\,a)$ has a solution in $c \in N$.

The second case, which has no analogue in Lemma 6.1, is when b is transcendental over a. Then by Remark 8.16 we may choose c to be any element of N transcendental over $k\,a$. This exists because N has transcendence degree $\geq \lambda$. □

Below a few important consequences of this lemma.

8.21 Corollary Work in the category of models of $T_{\rm id}$ with partial embeddings as morphisms. Then the following are equivalent

1. N is a λ-rich model
2. $N \vDash T_{\rm acf}$ and has transcendence degree $\geq \lambda$.

In particular every uncountable $N \vDash T_{\rm acf}$ is rich.

Proof. Implication 2 ⇒ 1 is an immediate consequence of Lemma 8.20.

In every connected component there is an $M \vDash T_{\rm acf}$ of cardinality λ and transcendence degree λ (by Proposition 8.17 when $\lambda > \omega$, by compactness for $\lambda = \omega$). As proved above, M is rich and therefore elementarily equivalent to any λ-rich model N in the same connected component. This proves 1 ⇒ 2. □

8.22 Corollary The theory $T_{\rm acf}$ has elimination of quantifiers.

Proof. Let $k : M \to N$ be a partial embedding between models of $T_{\rm acf}$. Let M' and N' be elementary superstructures of M and N respectively of sufficiently large cardinality. As M' and N' are rich, $k : M' \to N'$ is elementary by Theorem 7.11. Hence $k : M \to N$ is also elementary. □

8.23 Corollary The theories $T^p_{\rm acf}$ are complete and uncountably categorical (i.e. λ-categorical for every uncountable λ).

Proof. Two models of $T^p_{\rm acf}$ belong to the same connected component. Then, as every uncountable model of $T^p_{\rm acf}$ is rich, uncountable categoricity and completeness follow. □

8.24 Exercise Prove that every model of $T_{\rm acf}$ is ω-ultrahomogeneous (independently of cardinality and transcendence degree).

8.7 Hilbert's Nullstellensatz

Fix a tuple of variables x and a subset A of an integral domain M. In this section we are interested in formulas in $L_{\rm at}(A)$, hence we redefine the symbol of closure under logical consequence accordingly

$$\operatorname{ccl} p(x) \;=\; \Big\{\varphi(x) \in L_{\rm at}(A) \;:\; p(x) \vdash \varphi(x)\Big\}.$$

Recall that we work under the assumptions in Notation 8.1. In particular, in this section we work over the theory $T_{\rm id} \cup \operatorname{Diag}\langle A\rangle_M$, and the symbol $\vdash$ has to be interpreted accordingly.

In general, the notion of closure under logical consequence is elusive. In this respect Propositions 8.25 and 8.27 are useful, as they give a model theoretic characterization of $\operatorname{ccl} p(x)$. Corollary 8.30 gives an algebraic characterization.

8.25 Proposition Let $A \subseteq M \vDash T_{\rm id}$ and let $p(x) \subseteq L_{\rm at}(A)$. Let N be of sufficiently large cardinality and such that $\langle A\rangle_M \subseteq N \vDash T_{\rm acf}$. Then

$$\operatorname{ccl} p(x) \;=\; \Big\{\varphi(x) \in L_{\rm at}(A) \;:\; N \vDash \forall x\, \big[p(x) \rightarrow \varphi(x)\big]\Big\}.$$

The cardinality of N is sufficiently large if $|L(A)| < |N|$ and $|x| \leq |N|$; note that here x has possibly infinite length.

Proof. Only the inclusion $\supseteq$ requires a proof. Suppose $\varphi(x) \in L_{\rm at}(A)$ is such that $p(x) \wedge \neg\varphi(x)$ is consistent. Then there is a model M_1 of cardinality $< |L(A)|$ and $\leq |x|$ such that $M_1 \vDash p(a) \wedge \neg\varphi(a)$ for some $a \in M_1^x$. By Corollary 8.21, if N is large enough, there is a partial isomorphism $h : M_1 \rightarrow N$ that extends ${\rm id}_A$ and is defined on a. Therefore $N \vDash p(ha) \wedge \neg\varphi(ha)$. So $\varphi(x)$ does not belong to the set on the r.h.s. □

In the proposition above we could replace $L_{\rm at}$ by $L_{\rm at^{\pm}}$. But here we are interested in a strengthening, the Nullstellensatz, that requires positive formulas.

Hilbert's Nullstellensatz extends the validity of the proposition above to the case $A = M = N$ (assuming x finite). While Proposition 8.25 could be generalized to other theories, the Nullstellensatz rests on an exquisitely algebraic phenomenon. In fact, model theory has no general tools to deal with large sets of parameters. Algebra comes to our aid with the following lemma. Which the reader may wish to compare with Hilbert's Basis Theorem.

8.26 Lemma Let $M \vDash T_{\rm id}$. Let $p(x) \subseteq L_{\rm at}(M)$, where x is finite, be closed under logical consequence. Then $\psi(x) \vdash p(x)$, for some $\psi(x)$ conjunction of formulas in $p(x)$.

Proof. By induction on the length of x. If x is the empty tuple, the lemma is trivial. Now, let x be a finite tuple and let y be a single variable. Let $p(x,y) \subseteq L_{\rm at}(M)$ be closed under logical consequence. Write $p(x)$ for the set of formulas $\varphi(x) \in L_{\rm at}(M)$ such that $p(x,y) \vdash \varphi(x)$. Assume as induction hypothesis that there is a conjunction of formulas in $p(x)$, say $\psi(x)$, such that $\psi(x) \vdash p(x)$. We prove the lemma with x, y for x.

Let $\psi(x,y) \in L_{\rm at}(M)$ have minimal degree (in y) among the formulas in $p(x,y)$ such that $\psi(x) \nvdash \psi(x,y)$. Let n be the degree of $\psi(x,y)$, which is non zero, because $\psi(x) \nvdash \psi(x,y)$. So we may write

$$\psi(x,y) \quad = \quad \Big(t_n(x) \cdot y^n + t'(x,y) = 0\Big),$$

where $t'(x,y)$ is a polynomial of degree $< n$. We claim that

1 $\qquad \psi(x) \wedge \psi(x,y) \quad \vdash \quad p(x,y)$

Suppose not for a contradiction. Pick among the formulas in $p(x,y)$ that are a counterexample to 1, one of minimal degree, say $\varphi(x,y)$. By the minimaly of n, the degree of $\varphi(x,y)$ is $n+i$ for some $i \geq 0$. Hence we may write

$$\varphi(x,y) \quad = \quad \Big(s_{n+i}(x) \cdot y^{n+i} + s'(x,y) = 0\Big),$$

for some polynomial $s'(x,y)$ of degree $< n+i$. From $p(x,y) \vdash \varphi(x,y)$ we obtain

$$p(x,y) \quad \vdash \quad \Big(s_{n+i}(x) \cdot t_n(x) \cdot y^{n+i} + t_n(x) \cdot s'(x,y) = 0\Big).$$

Now, using that $p(x,y) \vdash \psi(x,y)$

2. $$p(x,y) \quad \vdash \quad \Big(-s_{n+i}(x) \cdot t'(x,y) \cdot y^i + t_n(x) \cdot s'(x,y) = 0\Big).$$

But the polynomial in 2 has degree $< n+i$ and this contradicts the minimality of the degree of $\varphi(x,y)$. A contradicion that proves the proposition. □

8.27 Hilbert's Nullstellensatz (model theoretic version) Let $N \vDash T_{\rm acf}$. For every type $p(x) \subseteq L_{\rm at}(N)$, where x is a finite tuple, we have

$$\operatorname{ccl} p(x) \quad = \quad \Big\{\varphi(x) \in L_{\rm at}(N) \ : \ N \vDash \forall x \, [p(x) \to \varphi(x)]\Big\}.$$

Proof. By Proposition 8.25, the equality above holds if we replace N by a sufficiently large elementary extension N', i.e.

$$= \quad \Big\{\varphi(x) \in L_{\rm at}(N) \ : \ N' \vDash \forall x \, [p(x) \to \varphi(x)]\Big\}.$$

Since $p(x)$ and $\operatorname{ccl} p(x)$ are logically equivalent, we can replace one by the other. Then, by Lemma 8.26, we can replace $\operatorname{ccl} p(x)$ by an equivalent formula $\psi(x)$

$$= \quad \Big\{\varphi(x) \in L_{\rm at}(N) \ : \ N' \vDash \forall x \, [\psi(x) \to \varphi(x)]\Big\}.$$

Now, by elementarity, we replace N back

$$= \quad \Big\{\varphi(x) \in L_{\rm at}(N) \ : \ N \vDash \forall x \, [\psi(x) \to \varphi(x)]\Big\}.$$

Finally, we replace $p(x)$ back and obtain the desired equality. □

In the rest of this section, we show how to translate the model theoretic version of the Nullstellensatz into a more common algebraic formulation.

Let $A[x]$ be the ring of polynomials with variables in x and coefficients in $\langle A\rangle_M$. We identify $L_{\rm at}(A)$ and $A[x]$ in the obvious way. A type $p(x) \subseteq L_{\rm at}(A)$ is identified with a set $\mathcal{P}_p \subseteq A[x]$. Conversely, for any subset $\mathcal{P} \subseteq A[x]$ we write $p_{\mathcal{P}}(x)$ for the associated type.

We would like to characterize ccl $p(x)$ in algebraic terms. The following is a preliminary result.

8.28 Proposition Let $A \subseteq M \vDash T_{\rm id}$ and let $p(x) \subseteq L_{\rm at}(A)$. The following are equivalent

1. $p(x)$ is a prime type
2. $\mathcal{P}_p$ is a prime ideal.

Proof. 1⇒2. Recall that, by our definition, a prime type is closed under logical consequence. Therefore to prove that $\mathcal{P}_p$ is an ideal, it suffices to note that the following entailments hold for every pair of polynomials $t(x)$ and $s(x)$

$$\begin{aligned} t(x) = 0 &\ \vdash\ s(x)t(x) = 0 \\ s(x) = t(x) = 0 &\ \vdash\ s(x) + t(x) = 0 \end{aligned}$$

Finally, to prove that $\mathcal{P}_p$ is prime, suppose that the polynomial $t(x) \cdot s(x)$ belongs to $\mathcal{P}_p$. As we are working over $T_{\rm id}$

$$t(x) \cdot s(x) = 0 \ \vdash\ t(x) = 0 \vee s(x) = 0$$

If $p(x)$ is a prime type, $p(x) \vdash t(x) = 0$ or $p(x) \vdash s(x) = 0$. As $p(x)$ is closed under logical consequence, $t(x) \in \mathcal{P}_p$ or $s(x) \in \mathcal{P}_p$.

2⇒1. First, we prove that $p(x)$ is closed under logical consequence. Equivalently, we assume that $t(x) = 0 \notin p(x)$ and prove that $p(x) \wedge t(x) \neq 0$ is consistent. As $\mathcal{P}_p$ is a prime ideal, $A[x]/\mathcal{P}_p$ is integral domain. We denote by $x + \mathcal{P}_p$ the equivalence class of the polynomial $x \in A[x]$. Then $x + \mathcal{P}_p$ satisfies exactly the formulas in $p(x)$. Hence it witnesses the consistency of $p(x) \wedge t(x) \neq 0$ in $A[x]/\mathcal{P}_p$.

We now prove that $p(x)$ is prime. Let $t_i(x)$ be polynomials such that

$$p(x) \ \vdash\ \bigvee_{i=1}^{n} t_i(x) = 0.$$

Then

$$p(x) \ \vdash\ \prod_{i=1}^{n} t_i(x) \ = \ 0$$

Since $p(x)$ is closed under logical consequence, and $\mathcal{P}_p$ is a prime ideal, $t_i(x) \in \mathcal{P}_p$ for some i. Hence $p(x)$ contains the equation $t_i(x) = 0$. By Corollary 3.19 this suffices to prove that $p(x)$ is a prime type. □

8.29 Proposition Let $A \subseteq M \vDash T_{\rm id}$ and let $p(x) \subseteq L_{\rm at}(A)$. Then the following are equivalent for every $\varphi(x) \in L_{\rm at}(A)$

1. $p(x) \vdash \varphi(x)$
2. $q(x) \vdash \varphi(x)$ for every prime type $q(x) \subseteq L_{\rm at}(A)$ containing $p(x)$.

Proof. Only 2⇒1 requires a proof. Assume that $p(x) \nvDash \varphi(x)$. Then there is an N such that $\langle A\rangle_M \subseteq N \vDash T_{\rm id}$ and $N \vDash p(a) \wedge \neg\varphi(a)$ for some $a \in N^x$. Let $q(x) =$ at-tp$_N(a/A)$. Then $q(x)$ is prime and $\varphi(x) \notin q(x)$. □

For any subset $\mathcal{P} \subseteq A[x]$ we write $\sqrt{\mathcal{P}}$ for the intersection of all prime ideals containing $\mathcal{P}$. This is called the radical ideal generated by $\mathcal{P}$. From Propositions 8.28 and 8.29 we obtain the following algebraic characterization of ccl $p(x)$.

8.30 Corollary Let $A \subseteq M \vDash T_{\rm id}$. Then $\mathcal{P}_{{\rm ccl}\, p(x)} = \sqrt{\mathcal{P}_p}$ for every $p(x) \subseteq L_{\rm at}(A)$. Equivalently, ${\rm ccl}\, p_{\mathcal{P}}(x) = p_{\sqrt{\mathcal{P}}}$ for every $\mathcal{P} \subseteq A[x]$.

We are now ready to translate the Nullstellensatz into a language more familiar to algebraists. Fix $N \vDash T_{\rm acf}$. Let $\mathcal{P} \subseteq N[x]$. The algebraic variety associated to $\mathcal{P}$, often denoted by $V(\mathcal{P})$, is the set $p_{\mathcal{P}}(N)$. Vice versa, given $A \subseteq N$, let $\mathcal{I}(A)$ be the set of polynomials in $N[x]$ that vanish at all points in A^x. Clearly, $\mathcal{I}(A)$ is an ideal of $N[x]$.

8.31 Hilbert's Nullstellensatz (standard version) Let $N \vDash T_{\rm acf}$. Let $\mathcal{P} \subseteq N[x]$ where x is a finite tuple. Then $\mathcal{I}\big(V(\mathcal{P})\big) = \sqrt{\mathcal{P}}$.

Proof. Note that $V(\mathcal{P}) = p_{\mathcal{P}}(N)$. Hence $p_{\mathcal{I}}(x)$, where $\mathcal{I} = \mathcal{I}(V(\mathcal{P}))$, is the set of formulas $\varphi(x) \in L_{\rm at}(N)$ such that $V(\mathcal{P}) \subseteq \varphi(N)$. In other words, $p_{\mathcal{I}}(x)$ is the set containing those formulas such that $N \vDash \forall x[p_{\mathcal{P}}(x) \rightarrow \varphi(x)]$. Hence from the model theoretic version of the Nullstellensatz we obtain $p_{\mathcal{I}}(x) = {\rm ccl}\, p_{\mathcal{P}}(x)$, that in turn coincides with $p_{\mathcal{I}}(x) = p_{\sqrt{\mathcal{P}}}(x)$ by the corollary above. □

Chapter 9

Saturation and homogeneity

In the first two sections we introduce saturation and homogeneity and Section 9.3 presents the notation we shall use in the following chapters when working inside a monster model.

9.1 Saturated structures

Recall that a type $p(x) \subseteq L(M)$ is finitely consistent in M if every conjunction of formulas in $p(x)$ has a solution in M. When $A \subseteq M$ we write $S_x(A)$ for the set of types whose variables are in x which are complete and finitely consistent in M. We never display M in the notation as it will always be clear from the context. When A is empty it usual to write $S_x(T)$ for $S_x(A)$ where $T = \mathrm{Th}(M)$. We write $S(A)$ for the union of $S_x(A)$ as x ranges over all tuples of variables. Similarly for $S(T)$.

The following remark will be used in the sequel without explicit reference.

9.1 Remark Let $k : M \to N$ be an elementary map and let a be an enumeration of $\mathrm{dom}\, k$. Let $p(x\,;z) \subseteq L$. If $p(x\,;a)$ is finitely consistent in M, then $p(x\,;k\,a)$ is finitely consistent in N. (We can drop *finitely* in the antecedent but not in the consequent.)

9.2 Definition Assume $|L| \le \lambda$. Let x be a single variable. We say that an infinite structure N is λ-saturated if it realizes every type $p(x)$ such that

1. $p(x) \subseteq L(A)$ for some $A \subseteq N$ of cardinality $< \lambda$;
2. $p(x)$ is finitely consistent in N.

We say that N is saturated if it is λ-saturated and $|N| = \lambda$.

We shall see that some theories have saturated models of small size. However, the existence of saturated models of arbitrary theories is problematic. The following theorem states the existence of a saturated model of cardinality λ whenever $|L| < \lambda = \lambda^{<\lambda}$. The existence of cardinals of this kind is independent of ZFC. Every inaccessible cardinal has this property and, if the generalized continuum hypothesis (GCH) holds, every successor cardinal. Both the existence of inaccessible cardinals and GCH are not generally accepted axioms.

Nevertheless, saturated models are widely used in model theory, without any worries about their existence. In fact, if consistency is an issue, they can be replaced by models that are both λ-saturated and λ-homogeneous (see next section) for some less problematic large cardinal λ. This is well known, so complications are commonly avoided by simply assuming that saturated models exist.

If T is the theory of the ring $\mathbb{Z}$, or any other sufficiently expressive theory, then one can prove that the cardinality λ of any saturated model of T is such that $\lambda = \lambda^{<\lambda}$.

As the existence of cardinals with this property has to be assumed as an extra axiom, one could simply assume the existence of saturated models and skip the proof of the following theorem.

9.3 Theorem Assume $|L| < \lambda$ where λ is such that $\lambda^{<\lambda} = \lambda$. Then every structure M of cardinality $\leq \lambda$ has a saturated elementary extension of cardinality λ.

Proof. We can assume that M has cardinality λ. We construct an elementary chain $\langle M_i : i < \lambda \rangle$ of models of cardinality λ. The chain starts with M and is the union at limit stages. Given M_i we choose as M_{i+1} any model of cardinality λ that realizes all types $p(x) \subseteq L(A)$ for all $A \subseteq M_i$ of cardinality $< \lambda$. The required M_{i+1} exists because for any given A there are at most $2^{|L(A)|} \leq \lambda^{<\lambda} = \lambda$ such types and there are $2^{<\lambda} = \lambda$ sets A.

Let N be the union of the chain. We check that N is the required extension. Let $p(x) \subseteq L(A)$ for some $A \subseteq N$ of cardinality $< \lambda$. As $\lambda^{<\lambda} = \lambda$ implies in particular that λ is a regular cardinal, $A \subseteq M_i$ for some $i < \lambda$. Then M_{i+1} realizes $p(x)$, and so does N, by elementarity. □

9.4 Remark The reader who did not read Section 7.1 may replace 2 in the theorem below with the following (and forget about $\mathcal{M}$).

2′ for every $b \in M$, every elementary map $k : M \to N$ of cardinality $< \lambda$ has an extension defined on b;

9.5 Theorem Assume $|L| \leq \lambda$. Let $\mathcal{M}$ be the category (see Section 7.1) that consists of L-structures and elementary maps between these. Then for every model N the following are equivalent

1 N is a λ-saturated structure

2 N is a λ-rich model

3 N realizes all finitely consistent types $p(z) \subseteq L(A)$, with $|z| \leq \lambda$ and $|A| < \lambda$.

Note that it is the completeness of T which makes $\mathcal{M}$ connected.

Proof. 1⇒2. Let $k : M \to N$ be an elementary map of cardinality $< \lambda$. It suffices to show that for every $b \in M$ there is a $c \in N$ such that $k \cup \{\langle b, c \rangle\} : M \to N$ is an elementary map. Let a be an enumeration of $\operatorname{dom} k$ and define $p(x\,; z) = \operatorname{tp}_M(b\,; a)$. As $p(x\,; a)$ is finitely consistent in M then $p(x\,; k\,a)$ is finitely consistent in N. The required c is any element of N such that $N \vDash p(c\,; k\,a)$. Such a c exists by saturation because $|a| < \lambda$.

2⇒3. Let $p(z)$ be as in 3. By the compactness theorem $N \preceq K \vDash p(a)$ for some model K and $a \in K^z$. By the downward Löwenheim-Skolem theorem there is a model $A, a \subseteq M \preceq K$ of cardinality $\leq \lambda$. (Here we use $|L|, |A|, |z| \leq \lambda$.) By 2, there is an elementary embedding $h : M \hookrightarrow N$ that extends id_A. (Here we use $|A| < \lambda$.) Finally, as $M \vDash p(a)$, elementarity yields $N \vDash p(h\,a)$.

3⇒1. Trivial. □

Two saturated structures of the same cardinality are isomorphic as soon as they are elementarily equivalent (i.e. as soon as $\varnothing : M \to N$ is an elementary map.) In fact, from Theorems 9.5 and 7.6 we obtain the following (reference to Theorems 7.6 may be avoided with an easy back-and-forth construction).

9.6 Corollary Every elementary map $k : M \to N$ of cardinality $< \lambda$ between saturated models of the same cardinality λ extends to an isomorphism.

As it turns out, we already have many examples of saturated structures.

9.7 Corollary The following models are ω-saturated

1 models of $T_{\rm dlo}$

2 models of $T_{\rm rg}$

3 models of $T_{\rm dag}$ with infinite rank

4 models of $T_{\rm acf}$ with infinite degree of transcendence.

Moreover, all countable models of $T_{\rm dlo}$ and $T_{\rm rg}$ and all uncountable models of $T_{\rm dag}$ or $T_{\rm acf}$ are saturated.

Proof. We proved that partial embeddings between models of these theories coincide with elementary embeddings – that is, quantifier elimination. Then saturation is proved applying Theorem 9.5 and the extension lemmas proved in Chapter 6 and 8. □

The following is a useful test for quantifier elimination.

9.8 Theorem Assume $|L| \leq \lambda$. Consider the category that consists of models of some theory T_0 and partial embeddings. Suppose λ-rich models exist and denote by T_1 their theory. Then the following are equivalent

1. every λ-saturated model of T_1 is λ-rich
2. T_1 has elimination of quantifiers.

Proof. 2⇒1. Let $N \vDash T_1$ be λ-saturated. Let $k : M \to N$ be a partial embedding of cardinality $< \lambda$ Pick $b \in M$. By elimination of quantifiers, $k : M \to N$ is an elementary map. Then reasoning as in the proof of Theorem 9.5 there is some $c \in N$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is elementary so, in particular, a partial embedding.

1⇒2. Let $k : M \to N$ be a finite partial embedding between models of T. We claim that it is an elementary map. Let $M' \succeq M$ and $N' \succeq N$ are λ-saturated models. As these are λ-rich, $k : M' \to N'$ is elementary by Theorem 7.11. □

9.9 Exercise Suppose $|L| \leq \omega$ and let M be an infinite structure. Then for every non-principal ultrafilter F on ω the structure M^ω/F is a ω_1-saturated elementary superstructure of M.

Hint: the notation is as in Chapter 4. Let $|x| = 1$ and $|z| = \omega$. It suffices to consider types of the form $p(x\,;\hat{c})$ where $p(x\,;z) = \{\varphi_i(x\,;z) : i < \omega\} \subseteq L$ and $\hat{c} \in (M^\omega)^z$. Without loss of generality we can also assume that $\varphi_{i+1}(x\,;z) \to \varphi_i(x\,;z)$, and that all formulas $\varphi_i(x\,;\hat{c})$ are consistent in M^ω.

Let $\langle X_i : i < \omega\rangle$ be a strictly decreasing chain of elements of the ultrafilter such that

$X_{i+1} \subseteq \{j : M \vDash \exists x\, \varphi_i(x, \hat{c}j)\}$. Let $\hat{a} \in M^\omega$ be such that $\varphi_i(\hat{a}j, \hat{c}j)$ holds for every $j \in X_i \smallsetminus X_{i+1}$. Then $\hat{a}$ realizes $p(x)$.

9.10 Exercise Let $L = \{<\}$ and let N be a ω_1-saturated extension of $\mathbb{Q}$. Prove that there is an embedding $f : \mathbb{R} \to N$. Is it elementary? Could it be surjective?

9.11 Exercise Let $L = \{<\}$ and let N be a saturated extension of $\mathbb{Q}$. Prove that there $2^{|N|}$ Dedekind cuts of N.

9.12 Exercise Let $\varepsilon(x, y) \subseteq L(A)$ be a type-definable equivalence relation with $< \lambda$ classes. Prove that every λ-saturated model containing A intersects every class of $\varepsilon(x, y)$.

9.2 Homogeneous structures

Definition 7.3 introduces the notions of universal and homogeneous structures in a general context. When the morphisms of the underlying category are the elementary maps, we refer to these notions as elementary homogeneity and elementary universality. However, one often omits to specify *elementary*. We repeat Definition 7.3 in this specific case.

9.13 Definition A structure N is (elementarily) λ-universal if every $M \equiv N$ of cardinality $\leq \lambda$ there is an elementary embedding $h : M \hookrightarrow N$. We say universal if it is λ-universal and of cardinality λ.

A model N is (elementarily) λ-homogeneous if every elementary map $k : N \to N$ of cardinality $< \lambda$ extends to an automorphism. As usual, N is homogeneous if it is λ-homogeneous and of cardinality λ.

As saturated structures are rich, the following theorem is an instance of Theorem 7.8.

9.14 Theorem For every structure N of cardinality $\geq |L|$ the following are equivalent

1. N is saturated
2. N is elementarily universal and homogeneous.

Given $A \subseteq N$ we denote by $\mathrm{Aut}(N/A)$ the group of A-automorphisms of N. That is the group of automorphisms that fix A point-wise. Let a be a tuple of elements of N. The orbit of a over A in N is the set

$$O_N(a/A) \quad = \quad \Big\{ f a \ : \ f \in \mathrm{Aut}(N/A) \Big\}.$$

When the model N is clear from the context we omit the subscript.

Orbits in a homogeneous structure are particularly interesting because they have a syntactical counterpart: (complete) type-definable sets. The following proposition is immediate but its importance cannot be overestimated.

9.15 Proposition Let N be a λ-homogeneous structure. Let $A \subseteq N$ have cardinality $< \lambda$ and let $a \in N^{<\lambda}$. Then $O_N(a/A) = p(N)$, where $p(x) = \mathrm{tp}_N(a/A)$.

Finally, we want to extend the equivalence in Theorem 9.14 to λ-saturated structures. For this we only need to apply Theorem 7.24.

When the morphisms of the underlying category are the elementary maps, it is usual to replace the notion of λ-universal (cf. Definition 7.3) with the following.

9.16 Definition We say that N is weakly λ-saturated if N realizes every type $p(x) \subseteq L$, where $|x| \leq \lambda$, that is finitely consistent in N.

In fact, by the Löwenheim-Skolem theorem the two notions are equivalent.

9.17 Proposition The following are equivalent

1. N weakly λ-saturated
2. N is λ-universal.

The following is an instance of Theorem 7.24 that the reader may prove directly as an exercise.

9.18 Corollary Let $|L| \leq \lambda$. The following are equivalent

1. N is λ-saturated
2. N is weakly λ-saturated and weakly λ-homogeneous.

9.19 Exercise Let M be an arbitrary structure of cardinality $\lambda \geq |L|$, a regular cardinal. Prove that M has an homogeneous elementary extension of the same cardinality.

9.20 Exercise Let M and N be elementarily homogeneous structures of the same cardinality λ. Suppose that $M \vDash \exists x\, p(x) \Leftrightarrow N \vDash \exists x\, p(x)$ for every $p(x) \subseteq L$ such that $|x| < \lambda$ Prove that the two structures are isomorphic.

9.21 Exercise Let L be a language that extends that of strict linear orders with the constants $\{c_i : i \in \omega\}$. Let T be the theory that extends T_{dlo} with the axioms $c_i < c_{i+1}$ for every $i \in \omega$. Prove that T has elimination of quantifiers and is complete (it can be deduced from what is known of T_{dlo}). Exhibit a countable saturated model and a countable model that is not homogeneous.

9.3 The monster model

In this section we present some notation and terminology frequently adopted when dealing with a complete theory T. We fix a saturated structure $\mathcal{U}$ of cardinality larger than $|L|$. We assume $\mathcal{U}$ to be large enough that among its elementary substructures we can find any model of T we might be interested in. This structure is called the monster model. We denote by κ the cardinality of $\mathcal{U}$. When appropriate, we assume κ to be inaccessible.

Some terms acquire a slightly different meaning when working inside a monster model.

truth	is always evaluated in $\mathcal{U}$, and we say that $\varphi(x)$ holds if $\mathcal{U} \vDash \forall x\, \varphi(x)$.
consistency	we say that $\varphi(x)$ is consistent if $\mathcal{U} \vDash \exists x\, \varphi(x)$.
small/large	cardinalities smaller than κ are called small.
bounded	is used in some contexts as synonym of small.
models	are elementary substructure of $\mathcal{U}$ of small cardinality they are denoted by the letters M and N.
parameters	are always in $\mathcal{U}$; the symbols A, B, C, etc. denote sets of parameters of small cardinality; calligraphic letters as $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, etc. are used for sets of arbitrary cardinality.
tuples	have length $< \kappa$ unless otherwise specified; their type is denoted by $\mathrm{tp}(a/A) = \mathrm{tp}_{\mathcal{U}}(a/A)$ and $a \equiv_A b$ abbreviates $\mathcal{U}, a \equiv_A \mathcal{U}, b$.
small types	are subsets of $L(\mathcal{U})$ of small cardinality. We may use small types in expression such as $p(x) \rightarrow \neg q(x)$, $\exists y\, p(x,y)$, etc. with the obvious meaning though these are neither first-order formulas nor types.
global types	are complete finitely consistent types over $\mathcal{U}$; the set of global types is denoted by $S(\mathcal{U})$. If $p(x)$ is a global type we write $p(x) \vdash \varphi(x)$ if $\psi(x) \rightarrow \varphi(x)$ for some conjunction $\psi(x)$ of formulas in $p(x)$.
formulas	have parameters in $\mathcal{U}$ unless otherwise specified.
definable	sets are sets of the form $\varphi(\mathcal{U})$ for some formula $\varphi(x) \in L(\mathcal{U})$; we may say A-definable if $\varphi(x) \in L(A)$.
type-definable	sets are sets of the form $p(\mathcal{U})$ for some small type $p(x)$.
orbits of tuples	under the action of $\mathrm{Aut}(\mathcal{U}/A)$ are denoted by $\mathcal{O}(a/A)$.

Let x be a tuple of variables. For any fixed $A \subseteq \mathcal{U}$ we introduce a topology on $\mathcal{U}^x$ that we call the topology induced by A or, for short, A-topology. (This is nonstandard terminology, not to be confused with the *logic* A-topology in Section **??**, which is finer when A is not a model.) The closed sets of the A-topology are those of the form $p(\mathcal{U})$ where $p(x) \subseteq L(A)$ is a type over A.

For $\varphi(x) \in L(A)$ the sets of the form $\varphi(\mathcal{U})$ are clopen in this topology (and vice versa by Proposition 9.22). They form both a base of closed sets and base of open sets, which makes these topologies *zero-dimensional*. By saturation, the topology induced by A is compact. Actually, saturation is equivalent to the compactness of all these topologies as A ranges over the sets of small cardinality.

These topologies are never T_0 as any pair of tuples $a \equiv_A b$ have exactly the same neighborhoods. Such pairs always exist by cardinality reasons. However it is immediate that the topology induced on the quotient $\mathcal{U}^x/\!\equiv_A$ is Hausdorff (this is the so-called *Kolmogorov quotient*). Indeed, this quotient corresponds to $S_x(A)$ with the topology introduced in Section 3.3.

The following proposition is an immediate consequence of compactness. When $A = B$ it says that the topology induced by A is *normal*: any two closed sets are separated by open sets. It could be called mutual normality (not a standard name) because the two closed sets belong to different topologies and the separating sets are each found in the corresponding topology.

9.22 Proposition (mutual normality) Let $p(x) \subseteq L(A)$ and $q(x) \subseteq L(B)$ be such that $p(x) \rightarrow \neg q(x)$. Then there are a conjunction $\varphi(x)$ of formulas in $p(x)$ and a conjunction $\psi(x)$ of formulas in $q(x)$ such that $\varphi(x) \rightarrow \neg\psi(x)$.

Proof. The assumptions say that $p(x) \cup q(x)$ is inconsistent (i.e. not realized in $\mathcal{U}$). Then the formulas $\varphi(x)$ and $\psi(x)$ exist by compactness (i.e. saturation). □

9.23 Remark There are many forms in which the proposition above can be applied. For instance, assuming for brevity that $p(x)$ and $q(x)$ are closed under conjunctions,

a. if $p(x) \leftrightarrow \neg q(x)$ then $p(x) \leftrightarrow \varphi(x)$ for some $\varphi(x) \in p(x)$

b. if $p(x) \leftrightarrow \psi(x)$ for some $\psi(x) \in L(\mathcal{U})$ then $p(x) \leftrightarrow \varphi(x)$ for some $\varphi(x) \in p$

c. if $p(x) \rightarrow \psi(x)$ for some $\psi(x) \in L(\mathcal{U})$ then $\varphi(x) \underset{n}{\rightarrow} \psi(x)$ for some $\varphi(x) \in p$

d. if $p(x) \rightarrow \bigvee\limits_{\psi \in \Psi} \psi(x)$, where $|\Psi| < \kappa$, then $p(x) \rightarrow \bigvee\limits_{i=1} \psi_i(x)$ for some $\psi_i \in \Psi$.

9.24 Remark A definable set has the form $\varphi(\mathcal{U}^x;b)$ for some formula $\varphi(x;z) \in L$ and some $b \in \mathcal{U}^z$. If $f \in \mathrm{Aut}(\mathcal{U})$ then

$$\begin{aligned} f[\varphi(\mathcal{U}^x;b)] &= \{fa : \varphi(a;b), \quad a \in \mathcal{U}^x\} \\ &= \{fa : \varphi(fa;fb), a \in \mathcal{U}^x\} \\ &= \varphi(\mathcal{U}^x;fb). \end{aligned}$$

Hence automorphisms act on definable sets in a very natural way. Their action on type-definable sets is similar.

We say that a set $\mathcal{D} \subseteq \mathcal{U}^x$ is invariant over A if $f[\mathcal{D}] = \mathcal{D}$ for every $f \in \mathrm{Aut}(\mathcal{U}/A)$. A formula $\varphi(x) \in L(\mathcal{U})$ is invariant if the set it defines is such. If $\mathcal{D}$ is invariant then $o(a/A) \subseteq \mathcal{D}$ for every $a \in \mathcal{D}$. By homogeneity this is equivalent to requiring that

$$q(x) \quad \rightarrow \quad x \in \mathcal{D}$$

for every $q(x) = \mathrm{tp}(a/A)$ and $a \in \mathcal{D}$.

Proposition 9.26 below is an important fact about invariant type-definable sets. It may clarify the proof to consider first the particular case of definable sets.

9.25 Proposition For every $\varphi(x) \in L(\mathcal{U})$ the following are equivalent

1. $\varphi(x)$ is equivalent to some formula $\psi(x) \in L(A)$
2. $\varphi(x)$ is invariant over A.

We give two proofs of this theorem as they are both instructive.

Proof. 1⇒2 Obvious.

2⇒1 From 2 and homogeneity we obtain

$$\varphi(x) \quad \leftrightarrow \quad \bigvee_{q(x) \in Q} q(x)$$

where Q is the set of the types in $S_x(A)$ such that $q(x) \to \varphi(x)$. By compactness, we can rewrite this equivalence

$$\varphi(x) \quad \leftrightarrow \quad \bigvee_{\vartheta(x)\in\Theta} \vartheta(x)$$

where Θ is the set of the formulas in $L(A)$ such that $\vartheta(x) \to \varphi(x)$. The latter equivalence says that $\neg\varphi(x)$ is equivalent to a type over A. Again by compactness we obtain

$$\varphi(x) \quad \leftrightarrow \quad \bigvee_{i=1}^{n} \vartheta_i(x)$$

for some formula $\vartheta_i(x) \in L(A)$. □

Second proof of Proposition 10.18. 2⇒1 Let $\varphi(x;b) \in L$ be invariant over A. Let $p(z) = \mathrm{tp}(b/A)$. As $f[\varphi(\mathcal{U}^x;b)] = \varphi(\mathcal{U}^x;fb)$ for every $f \in \mathrm{Aut}(\mathcal{U}/A)$, homogeneity and invariance yield

$$p(z) \quad \to \quad \forall x\, \big[\varphi(x\,;z) \leftrightarrow \varphi(x\,;b)\big].$$

By compactness there is a formula $\vartheta(z) \in p$ such that

$$\vartheta(z) \quad \to \quad \forall x\, \big[\varphi(x\,;z) \leftrightarrow \varphi(x\,;b)\big].$$

Hence $\varphi(x;b)$ is equivalent to the formula $\exists z\, \big[\vartheta(z) \wedge \varphi(x\,;z)\big]$, which is a formula in $L(A)$ as required. □

9.26 Proposition Let $p(x) \subseteq L(B)$. Then the following are equivalent

1. $p(x)$ is equivalent to some type $q(x) \subseteq L(A)$
2. $p(\mathcal{U}^x)$ is invariant over A.

We give two proofs of this theorem. The second one requires Proposition 9.27 below.

Proof. 1⇒2 Obvious.

2⇒1 It suffices to show that for every formula $\psi(x) \in p(x)$ there is a formula $\vartheta(x) \in L(A)$ such that $p(x) \to \vartheta(x) \to \psi(x)$. Fix $\psi(x) \in p(x)$. By invariance, any $q(x) \in S(A)$ consistent with $p(x)$ implies $p(x)$, hence

$$p(x) \quad \to \quad \bigvee_{q(x)\to\psi(x)} q(x) \quad \to \quad \psi(x)$$

where $q(x)$ above range over all types in $S_x(A)$. By compactness we can rewrite this equivalence as follows

$$p(x) \quad \to \quad \bigvee_{\vartheta(x)\to\psi(x)} \vartheta(x) \quad \to \quad \psi(x)$$

where $\vartheta(x)$ ranges over all formulas in $L(A)$. Applying mutual normality (Proposition 9.22) to the first implication we obtain a finite number of formulas $\vartheta_i(x) \in L(A)$ such that

$$p(x) \quad \to \quad \bigvee_{i=1}^{n} \vartheta_i(x) \quad \to \quad \psi(x). \qquad \square$$

The following easy proposition is very useful.

9.27 Proposition Let $p(x;z) \subseteq L(A)$. There is a type $q(x) \subseteq L(A)$ such that

$$\forall x\,\big[\exists z\, p(x;z) \;\leftrightarrow\; q(x)\big].$$

The proposition also holds when z has length κ.

Proof. It is easy to verify that the equivalence above holds with

$$q(x) \;=\; \big\{\exists z\,\varphi(x;z)\;:\;\varphi(x;z) \text{ conjunction of formulas in } p(x;z)\big\}. \qquad \square$$

Note that the proposition would not hold without saturation. For a counter example consider $\mathbb{R}$ as a structure in the language of strict orders and let $q(x,y) = \mathrm{tp}(0,1/A)$, where

$$A \;=\; \Big\{1+\frac{1}{n+1}\;:\;n\in\omega\Big\}.$$

By quantifier elimination, $0 \equiv_A 1$ but $\mathbb{R}, 1 \nvDash \exists y\, q(x,y)$. However, in any sufficiently saturated elementary extension of $\mathbb{R}$, we have $1 \vDash \exists y\, q(x,y)$.

As an application we give a second proof of the proposition above.

Second proof of Proposition 9.26. 2⇒1 Write $p(x)$ as the type $q(x;b)$ for some $q(x;z) \subseteq L$ and some $b \in \mathcal{U}^z$. Let $s(z) = \mathrm{tp}(b/A)$. By invariance and homogeneity the types $q(x;fb)$ for $f \in \mathrm{Aut}(\mathcal{U}/A)$ are all equivalent. Therefore

$$p(x) \;\leftrightarrow\; \bigvee_{f\in \mathrm{Aut}(\mathcal{U}/A)} q(x;fb)$$

$$p(x) \;\leftrightarrow\; \bigvee_{c\equiv_A b} q(x;c)$$

$$\leftrightarrow\; \exists z\,\big[s(z)\wedge q(x;z)\big].$$

Hence, by Proposition 9.27, $p(x)$ is equivalent to a type over A. $\square$

9.28 Exercise Let $\varphi(x) \in L$. Prove that the following are equivalent

1. $\varphi(x)$ is equivalent to some $\psi(x) \in L_{\mathrm{qf}}$
2. $\varphi(a) \leftrightarrow \varphi(fa)$ for every partial embedding $f : \mathcal{U} \to \mathcal{U}$ and $a \in (\mathrm{dom}\, f)^x$.

Use the result to prove Theorem 7.14 for T complete.

9.29 Exercise Prove that $\mathbb{R}, 1 \nvDash \exists y\, q(x,y)$, as claimed before Proposition 9.27.

9.30 Exercise Let $p(x) \subseteq L(A)$, with $|x| < \omega$. Prove that if $p(\mathcal{U})$ is infinite then it has cardinality κ. Show that this is not true if x has length ω.

9.31 Exercise Let $\varphi(x,y) \in L(\mathcal{U})$. Prove that if the set $\{\varphi(a,\mathcal{U}) \;:\; a \in \mathcal{U}^x\}$ is infinite then it has cardinality κ. Does the claim remain true with a type $p(x,y) \subseteq L(A)$ for $\varphi(x,y)$?

9.32 Exercise Let $\varphi(x;y) \in L(\mathcal{U})$. Prove that the following are equivalent

1. there is a sequence $\langle a_i : i\in\omega\rangle$ such that $\varphi(\mathcal{U};a_i) \subset \varphi(\mathcal{U};a_{i+1})$ for every $i<\omega$
2. there is a sequence $\langle a_i : i\in\omega\rangle$ such that $\varphi(\mathcal{U};a_{i+1}) \subset \varphi(\mathcal{U};a_i)$ for every $i<\omega$.

9.33 Exercise Let $\varphi(x;z) \in L$ and $a \in \mathcal{U}^z$. Prove that the following are equivalent

1. the type $\{\varphi(x;fa) \;:\; f \in \mathrm{Aut}(\mathcal{U})\}$ is realized (in $\mathcal{U}$)

2. there is a formula $\psi(z) \in L$ such that $\psi(a)$ and $\exists x \forall z\, [\psi(z) \to \varphi(x\,;z)]$.

9.34 Exercise Let L be a countable language containing L_{gr} and assume $\mathcal{U}$ is a group. Let $p(x) \subseteq L$ define a subgroup of $\mathcal{U}$. Prove that there are some formulas $\varphi_n(x) \in L$ such that

1. $p(x) \ \leftrightarrow\ \{\varphi_n(x) : n < \omega\}$
2. $\varphi_{n+1}(x) \wedge \varphi_{n+1}(y) \ \rightarrow\ \varphi_n(x \cdot y)$.

9.35 Exercise Let $\mathcal{U}$ be as in Exercise 9.34. Prove that the following are equivalent

1. every model M is a normal subgroup of $\mathcal{U}$
2. some ω-saturated model M is a normal subgroup of $\mathcal{U}$
3. $\mathcal{U}$ is a BFC group.

A group G is BFC (has boundedly finite conjugacy classes) if for some n the sets $\{g\,a\,g^{-1} : g \in G\}$, as a ranges over G, have at most n elements.

Chapter 10

Preservation theorems

In this chapter we present a few results dating from the 1950s that describe the relationship between syntactic and semantic properties of first-order formulas. These results characterize the classes of formulas that preserved under various sorts of morphisms. Criteria for quantifier-elimination follow from these theorems, see for instance the frequently used back-and-forth method of Corollary 10.13.

10.1 Lyndon-Robinson Lemma

We refer the reader to Exercise 9.28 for a simpler version of the main result in this section, the Lyndon-Robinson Lemma. In fact, under the additional assumption of completeness, Lemma 10.3 is essentially the same as the claim in Exercise 9.28.

However, we are interested in criteria for quantifier elimination, e.g. Corollary 10.12 below. We often need to prove quantifier elimination in order to prove completeness. Therefore any assumption of completeness would make criteria for quantifier elimination less applicable.

In this section T is a consistent theory without finite models and Δ is a set of formulas containing the formula $x = y$ closed under renaming of variables. At a first reading the reader is encouraged to assume $\Delta = L_{\mathrm{at}}$.

10.1 Definition If $\mathsf{C} \subseteq \{\forall, \exists, \neg, \vee, \wedge\}$ is a set of connectives, we write $\mathsf{C}\Delta$ for the closure of Δ with respect to all connectives in C. We may write $\Delta^{\pm}$ for $\{\neg\}\Delta$.

Recall that a Δ-morphism is a map $k : M \to N$ that preserves the truth of all formulas in Δ. It is immediate that Δ-morphism are automatically $\{\wedge\vee\}\Delta$-morphisms. Similarly, all $\{\exists\wedge\}\Delta$-morphisms are $\{\exists\wedge\vee\}\Delta$-morphisms and $\{\forall\vee\}\Delta$-morphisms are $\{\forall\wedge\vee\}\Delta$-morphisms.

Below we use the following proposition without further reference.

10.2 Proposition Fix $M \vDash T$ and $b \in M^x$. Let $q(x) = \Delta\text{-tp}_M(b)$. Then for every $\varphi(x) \in L$ the following are equivalent

1. $N \vDash \varphi(kb)$ for every Δ-morphism $k : M \to N \vDash T$ that is defined in b

1′. $N \vDash \varphi(c)$ for every $N \vDash T$ such that $N, c \Rrightarrow_\Delta M, b$

2. $T \vdash q(x) \to \varphi(x)$.

Proof. 1⇔1′. In fact, the difference is just in the notation.

2⇒1. Immediate.

1⇒2. Negate 2, then there are $N \vDash T$ and $c \in N^x$ such that $N \vDash q(c) \wedge \neg\varphi(c)$. Therefore the map $k : M \to N$, where $k = \{\langle b, c\rangle\}$, contradicts 1. Note that, as Δ

contains equality, k is indeed a function. $\square$

The following is sometimes referred to as the Lyndon-Robinson Lemma.

10.3 Lemma For every $\varphi(x) \in L$ the following are equivalent

1. $\varphi(x)$ is equivalent over T to a formula in $\{\wedge\vee\}\Delta$
2. $\varphi(x)$ is preserved by Δ-morphisms between models of T.

Proof. 1⇒2. Immediate.

2⇒1. We claim that 2 implies

$$\# \qquad T \vdash \varphi(x) \leftrightarrow \bigvee \{ p(x) \subseteq \Delta \; : \; T \vdash p(x) \rightarrow \varphi(x) \}.$$

The implication $\leftarrow$ is clear. To verify the implication $\rightarrow$, let $M \vDash T$ and let $b \in M^x$ be such that $M \vDash \varphi(b)$. From 2 it follows that $\varphi(x)$ satisfies 1 of Proposition 10.2. Therefore $T \vdash q(x) \rightarrow \varphi(x)$ for $q(x) = \Delta\text{-tp}(b)$. Hence $q(x)$ is one of the types that occur in the disjuction in # which therefore is satisfied by b.

From # and compactness we obtain

$$T \vdash \varphi(x) \leftrightarrow \bigvee \{ \psi(x) \in \{\wedge\}\Delta \; : \; T \vdash \psi(x) \rightarrow \varphi(x) \}.$$

Applying compactness again allows us to replace the infinite disjunction above with a finite one and prove 2. $\square$

10.4 Remark Note that Theorem 7.14, which we repeat for emphasis, follows immediately from the lemma. For every theory T the following are equivalent

1. T has elimination of quantifiers
2. every partial isomorphism between models of T is an elementary map.

10.5 Proposition Let N be λ-saturated and let $k : M \rightarrow N$ be a Δ-morphism of cardinality $< \lambda$. Then the following are equivalent

1. $k : M \rightarrow N$ is a $\{\exists\wedge\}\Delta$-morphism
2. for $b \in M$ there is $c \in N$ such that $k \cup \{\langle b, c\rangle\} : M \rightarrow N$ is a $\{\exists\wedge\}\Delta$-morphism
3. for $b \in M^\omega$ there is $c \in N^\omega$ such that $k \cup \{\langle b, c\rangle\} : M \rightarrow N$ is a Δ-morphism.

Proof. 1⇒2. Let a enumerate $\operatorname{dom} k$. Define $p(x\,;z) = \{\exists\wedge\}\Delta\text{-tp}_M(b\,;a)$. By 1, $p(x\,;ka)$ is finitely consistent in N. By saturation there is a $c \in N$ that realizes $p(x\,;ka)$. Therefore, $h : M \rightarrow N$ where $h = k \cup \{\langle b, c\rangle\}$, witnesses 2.

2⇒1. Iterate 2.

2⇒3. Let a enumerate $\operatorname{dom} k$ and let $|z| = |a|$. Formulas in $\{\exists\wedge\}\Delta$ with free variables among z are of the form $\exists\bar{x}\,\varphi(\bar{x}\,;z)$ where $\varphi(\bar{x}\,;z)$ is in $\{\wedge\}\Delta$ and $\bar{x}$ is of finite length. Assume $M \vDash \exists\bar{x}\,\varphi(\bar{x}\,;a)$ and let $\bar{b}$ be such that $M \vDash \varphi(\bar{b}\,;a)$. By 2 we can extend k to some Δ-morphism $h : M \rightarrow N$ defined in $\bar{b}$. Then $N \vDash \varphi(h\bar{b}\,;ha)$ and therefore $N \vDash \exists\bar{x}\,\varphi(\bar{x}\,;ka)$. $\square$

Iterating the lemma above we obtain the following.

10.6 Corollary Let N be λ-saturated and let $|M| \leq \lambda$. Let $k : M \to N$ be a Δ-morphism of cardinality $< \lambda$. Then the following are equivalent

1. $k : M \to N$ is a $\{\exists\wedge\}\Delta$-morphism
2. $k : M \to N$ extends to a total $\{\exists\wedge\}\Delta$-morphism
3. $k : M \to N$ extends to a total Δ-morphism.

The following theorem is often paraphrased as follows: a formula is existential if and only if (its truth) is preserved under extensions of structures.

10.7 Theorem For every $\varphi(x) \in L$ the following are equivalent

1. $\varphi(x)$ is equivalent over T to a formula in $\{\exists\wedge\vee\}\Delta$
2. $\varphi(x)$ is preserved by total Δ-morphisms between models of T.

Proof. 1$\Rightarrow$2. Immediate.

2$\Rightarrow$1. Negate 1. By the Lyndon-Robinson Lemma 10.3 there is a $\{\exists\wedge\}\Delta$-morphism $k : M \to N$ between models of T that does not preserve $\varphi(x)$. We can assume that N is λ-saturated for some sufficiently large λ. By Corollary 10.6 there is a total Δ-morphism $h : M \hookrightarrow N$ that extends k and contradicts 2. □

A dual version of the results above is obtained replacing total morphisms by surjective morphisms and $\{\exists\wedge\}$ by $\{\forall\vee\}$. If Δ is closed under negation, then $k : M \to N$ is a Δ-morphism if and only is $k^{-1} : N \to M$ is a Δ-morphism. In this case the dual version follows from what proved above. Without these assumptions the results need a similar but independent proof.

10.8 Proposition Let M be λ-saturated and let $k : M \to N$ be a Δ-morphism of cardinality $< \lambda$. Then the following are equivalent

1. $k : M \to N$ is a $\{\forall\vee\}\Delta$-morphism
2. for $c \in N$ there is $b \in M$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is a $\{\forall\vee\}\Delta$-morphism
3. for $c \in N^\omega$ there is $b \in M^\omega$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is a Δ-morphism.

We write $\neg\Delta$ for the set containing the negation of the formulas in Δ. Warning: do not confuse $\neg\Delta$ with $\{\neg\}\Delta$.

Proof. Left as an exercise for the reader. Hint: to prove implication 1$\Rightarrow$2 define $p(x, y) = \neg\{\forall\vee\}\Delta\text{-tp}_N(ka, c)$, where a is a tuple that enumerates $\operatorname{dom} k$. From 1 obtain that $p(a, y)$ is finitely consistent in M. Then proceed as in the proof of Proposition 10.5. □

10.9 Corollary Let M be λ-saturated and let $|N| \leq \lambda$. Let $k : M \to N$ be a Δ-morphism of cardinality $< \lambda$. Then the following are equivalent

1. $k : M \to N$ is a $\{\forall\vee\}\Delta$-morphism
2. $k : M \to N$ extends to a sujective $\{\forall\vee\}\Delta$-morphism
3. $k : M \to N$ extends to a surjective Δ-morphism.

Finally we obtain the following.

10.10 Theorem The following are equivalent

1. $\varphi(x)$ is equivalent to a formula in $\{\forall\wedge\vee\}\Delta$
2. every surjective Δ-morphism between models of T preserves $\varphi(x)$.

10.2 Quantifier elimination by back-and-forth

We say that T admits (or has) positive Δ-elimination of quantifiers if for every formula $\varphi(x)$ in $\{\exists\forall\wedge\vee\}\Delta$ there is a formula $\psi(x)$ in $\{\wedge\vee\}\Delta$ such that

$$T \quad \vdash \quad \varphi(x) \leftrightarrow \psi(x).$$

When Δ is closed under negation the attribute *positive* becomes irrelevant and will be omitted. When Δ is $L_{\mathrm{at}^\pm}$ or L_{qf}, we simply say that T admits elimination of quantifiers. This is by far the most common case.

Quantifier elimination is often used to prove that a theory is complete because it reduces it to something much simpler to prove. The following is an immediate consequence of the definition above with x replaced by the empty tuple.

10.11 Remark If T has elimination of quantifiers then the following are equivalent

1. T decides all quantifier free sentences
2. T is complete.

Hence a theory with quantifier elimination is complete if it decides the characteristic of its models, see Definition 3.27.

The following is a consequence of Lemma 10.3.

10.12 Corollary The following are equivalent

1. T has positive Δ-elimination of quantifiers
2. every Δ-morphism between models of T is an $\{\exists\wedge\}\Delta$ and a $\{\forall\vee\}\Delta$-morphism.

Proof. 1⇒2. Immediate.

2⇒1. We prove by induction of syntax that Δ-morphism preserve the truth of all formulas in $\{\exists\forall\wedge\vee\}\Delta$, this suffices by Lemma 10.3. Induction for the connectives $\vee$ and $\wedge$ is trivial. So assume as induction hypothesis that the truth of $\varphi(x,y)$ is preserved. By Lemma 10.3 $\varphi(x,y)$ is equivalent to a formula in $\{\wedge\vee\}\Delta$, hence by 2 the truth of $\exists y\, \varphi(x,y)$ and $\forall y\, \varphi(x,y)$ is preserved. □

Condition 2 of the corollary above may be difficult to verify directly. The following corollary of Proposition 10.5 and 10.8 gives a back-and-forth condition witch is easier to verify.

10.13 Corollary Let $|L| \leq \lambda$. The following are equivalent

1. T has positive Δ-elimination of quantifiers
2. for every finite Δ-morphism $k : M \to N$ between λ-saturated models of T
   a. for any $b \in M$ there is $c \in N$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is a Δ-morphism
   b. for any $c \in N$ there is $b \in M$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is a Δ-morphism

Note that when Δ is closed under negation, then $k : M \to N$ is a Δ-morphism if and only if $k^{-1} : N \to M$ is a Δ-morphism. In this case a and b are equivalent.

10.14 Exercise Let T be a complete theory without finite models in a language that consists only of unary predicates. Prove that T has elimination of quantifiers. Is the claim true if T is not complete?

10.15 Exercise Consider $\mathbb{R}$ as a structure in the language of real vector spaces expanded with the usual order relation. Prove that $\mathrm{Th}(\mathbb{R})$ has elimination of quantifiers.

10.16 Exercise Let T be the theory of discrete linear orders, that is, T extends the theory of linear orders T_{lo} (see Section 6.1) with the following two of axioms

dis↑. $\exists z \left[x < z \ \wedge \ \neg\exists y \ x < y < z\right]$
dis↓. $\exists z \left[z < x \ \wedge \ \neg\exists y \ z < y < x\right]$.

Let Δ be the set of formulas that contains (all alphabetic variants of) the formulas $x <_n y := \exists^{\geq n} z \ (x{<}z{<}y)$ and their negations, for every $n > 0$. Prove that the theory of discrete linear orders has Δ-elimination of quantifiers. Prove that the structure $\mathbb{Q} \times \mathbb{Z}$ ordered with the lexicographic order

$$(a_1, a_2) < (b_1, b_2) \quad \Leftrightarrow \quad a_1 < b_1 \ \text{ or } \ (a_1 = b_1 \ \text{ e } \ a_2 < b_2)$$

is a saturated model of T.

10.17 Exercise Let T be a consistent theory without finite models. Suppose that all completions of T are of the form $T \cup S$ for some set S of quantifier-free sentences. Prove that if all completions of T have elimination of quantifiers, so does T. Show that this fails when the completions of T have arbitrary complexity.

Note. Thought the claim follows immediately from Corollary 10.12, a direct proof by compactness is also instructive. Prove that for every formula $\varphi(x)$ there are some quantifier-free sentences σ_i and quantifier-free formulas $\psi_i(x)$ such that

$$\sigma_i \vdash \varphi(x) \leftrightarrow \psi_i(x), \qquad T \vdash \bigvee_{i=1}^{n} \sigma_i, \qquad \text{and} \qquad \sigma_i \vdash \neg\sigma_j \text{ for } i \neq j.$$

For a counter example consider the empty theory in the language with a single unary predicate.

10.18 Exercise Work in a monster model $\mathcal{U}$. Let $\varphi(x) \in L_{\mathrm{qf}}(\mathcal{U})$. Are the following equivalent?

1. $\varphi(x)$ is equivalent to some formula $\psi(x) \in L_{\mathrm{qf}}(A)$
2. $\varphi(x)$ is invariant over A.

10.3 Model-completeness

We say that T is model-complete if every embedding $h : M \hookrightarrow N$ between models of T is an elementary embedding. The terminology, introduced by Abraham Robinson, is inspired by the fact that T is model-complete if and only if $T \cup \mathrm{Diag}(M)$ is a complete theory, in the language $L(M)$, for every $M \vDash T$. See Proposition 10.21 below.

To stress positivity in the next proposition, we generalize the definition as follows. We say that T is Δ-model-complete if every total Δ-morphism $k : M \hookrightarrow N$ between models of T is a $\{\forall\exists\wedge\vee\}\Delta$-morphism.

Model-completeness is equivalent to a property akin to quantifier elimination.

10.19 Proposition The following are equivalent

1. T is Δ-model-complete
2. T has positive $\{\exists\wedge\}\Delta$-elimination of quantifiers.

Proof. 1⇒2. By 1, every formula $\{\forall\exists\wedge\vee\}\Delta$ is preserved by total Δ-morphism therefore, by Theorem 10.7, it is equivalent to a formula in $\{\exists\wedge\vee\}\Delta$.

2⇒1. Clear, because total Δ-morphism preserve formulas in $\{\exists\wedge\vee\}\Delta$. □

The theory of discrete linear orders defined in Exercise 10.16 is an example of a model-complete theory without elimination of quantifiers.

The difference between quantifier elimination and model-completeness subtle. It boils down to models of T having or not the amalgamation property, see Section 7.4.

10.20 Proposition Assume T is model-complete. Let $\mathcal{M}$ be the category that consists of models of T and partial isomorphisms. Then the following are equivalent

1. $\mathcal{M}$ has the amalgamation property
2. T has elimination of quantifiers.

Proof. 1⇒2. By Proposition 7.26 every partial morphism $k : M \to N$ extends to an embedding $g : M \hookrightarrow N'$ which, by model-completeness, is an elementary embedding. Model-completeness also implies that $N \preceq N'$. Hence $k : M \to N$ is an elementary map. This proves 2.

2⇒1. If all morphisms are elementary maps, amalgamations follows from Proposition 7.29. □

Recall that the Δ-diagram of M, which we denote by $\Delta\text{-}\mathrm{Diag}(M)$, is the set $\{\varphi(a) : \varphi(x) \in \Delta,\ a \in M^x,\ M \vDash \varphi(a)\}$. The following explains the terminology of model-completeness.

10.21 Proposition The following are equivalent

1. T is Δ-model-complete
2. if $M \vDash T$, then $T \cup \Delta\text{-}\mathrm{Diag}(M)$ entails the $\{\forall\exists\wedge\vee\}\Delta$-diagram of M.

Note that if $\Delta = L_{\mathrm{qf}}$ then 2 says that $T \cup \mathrm{Diag}(M)$ is complete.

Proof. 1⇒2. Let $M \vDash T$ and let $\varphi(x) \in L$ and $a \in M^x$ be such that $M \vDash \varphi(a)$. We show that $\varphi(a)$ holds in every model $N \vDash T \cup \Delta\text{-Diag}(M)$. Suppose not, then the map $\mathrm{id}_M : M \to N$ is a total Δ-morphism which is not a $\{\forall\exists\wedge\vee\}\Delta$-morphism. this contradicts 1.

2⇒1. Let $h : M \hookrightarrow N$ be an embedding between models of T. We show that h is a $\{\forall\exists\wedge\vee\}\Delta$-morphism. Let $\varphi(x)$ be a formula in $\{\forall\exists\wedge\vee\}\Delta$ and let $a \in M^x$ be such that $M \vDash \varphi(a)$. By 2, $\varphi(a)$ is entailed by $T \cup \Delta\text{-Diag}(M)$, hence $N \vDash \varphi(ha)$. □

10.22 Exercise Let T be a consistent theory without finite models. Suppose that, in every model $M \vDash T$, every formula $\varphi(x)$ is equivalent to some quantifier-free formula $\psi(x) \in L(M)$. Prove that T is Δ-model-complete for Δ the set of formulas of the form $\forall y\, \exists z\, \vartheta(x, y, z)$ with $\vartheta(x, y, z) \in L_{\mathrm{qf}}$.

10.23 Exercise A countable structure M is Δ-*set-homogeneous* if for every finite Δ-morphism $k : M \to M$ there is an $h \in \mathrm{Aut}(M)$ such that $h[\operatorname{dom} k] = \operatorname{rng} k$. Prove that the theory of Δ-set-homogeneous models is Δ-model-complete. Hint: prove that every prime $\{\exists\}\Delta$-type is maximal.

Chapter 11

Strongly minimal theories

In this chapter we fix a signature L, a complete theory T without finite models, and a saturated model $\mathfrak{U}$ of inaccessible cardinality κ larger than $|L|$. The notation and implicit assumptions are as in Section 9.3.

11.1 Algebraic and definable elements

Let $a \in \mathfrak{U}$ and let $\mathcal{A} \subseteq \mathfrak{U}$ be some set of parameters (of arbitrary cardinality). We say that a is algebraic over $\mathcal{A}$ if $\varphi(a) \wedge \exists^{=k} x\, \varphi(x)$ holds for a formula $\varphi(x) \in L(\mathcal{A})$ and some positive integer k. In particular, when $k = 1$ we say that a is definable over $\mathcal{A}$. We write $\mathrm{acl}(\mathcal{A})$ for the algebraic closure of $\mathcal{A}$, that is, the set of all the elements that are algebraic over $\mathcal{A}$. If $\mathcal{A} = \mathrm{acl}(\mathcal{A})$, we say that $\mathcal{A}$ is algebraically closed. The definable closure of $\mathcal{A}$ is defined similarly and is denoted by $\mathrm{dcl}(\mathcal{A})$.

Let x be a finite tuple of variables. Formulas $\varphi(x) \in L(A)$, or types $p(x) \subseteq L(A)$, with finitely many solutions are called algebraic.

11.1 Proposition For every $A \subseteq \mathfrak{U}$ and every type $p(x) \subseteq L(A)$, where x is finite, the following are equivalent

1. $\exists^{\leq n} x\; p(x)$
2. $\exists^{\leq n} x\; \varphi(x)$ for some $\varphi(x)$ which is a conjunction of formulas in $p(x)$.

Proof. Only 1⇒2 requires a proof. Let $\{a_1, \ldots, a_n\} = p(\mathfrak{U})$. Then

$$p(x) \;\rightarrow\; \bigvee_{i=1}^{n} a_i = x$$

and 2 follows by compactness. □

11.2 Theorem For every $A \subseteq \mathfrak{U}$ and every $a \in \mathfrak{U}$ the following are equivalent

1. $a \in \mathrm{dcl} A$
2. $o(a/A) = \{a\}$.

Proof. Only 2⇒1 requires a proof. Recall that $o(a/A)$ coincides with the set of realizations of $\mathrm{tp}(a/A)$. Then the theorem follows from Proposition 11.1. □

11.3 Theorem For every $A \subseteq \mathfrak{U}$ and every $a \in \mathfrak{U}$ the following are equivalent

1. $a \in \mathrm{acl} A$
2. $o(a/A)$ is finite
3. a belongs to every model containing A.

Proof. 1⇔2. This is proved as in Theorem 11.2.

1⇒3. Assume 1. Then there is a formula $\varphi(x) \subseteq L(A)$ such that $\varphi(a) \wedge \exists^{=k} x\, \varphi(x)$ for some k. By elementarity $\exists^{=k} x\, \varphi(x)$ holds in every model M containing A. Again by elementarity, the k solutions of $\varphi(x)$ in M are solutions in $\mathcal{U}$, therefore a is one of these.

3⇒2. Assume $o(a/A)$ is infinite and fix any model M containing A. By Exercise 9.30, $o(a/A)$ has cardinality κ, hence $o(a/A) \nsubseteq M$. Pick any $f \in \mathrm{Aut}(\mathcal{U}/A)$ such that $fa \notin M$. Then $a \notin f^{-1}[M]$, so $f^{-1}[M]$ is a model that contradicts 3. □

11.4 Corollary For every $A \subseteq \mathcal{U}$ and every $a \in \mathcal{U}$

1. if $a \in \mathrm{acl} A$ then $a \in \mathrm{acl} B$ for some finite $B \subseteq A$ — finite character
2. $A \subseteq \mathrm{acl} A$ — extensivity
3. if $A \subseteq B$ then $\mathrm{acl} A \subseteq \mathrm{acl} B$ — monotonicity
4. $\mathrm{acl} A = \mathrm{acl}(\mathrm{acl} A)$ — idempotency
5. $\mathrm{acl} A = \bigcap_{A \subseteq M} M$.

Properties 1-4 say that acl(-) is a closure operator with finite character.

Proof. Properties 1-3 are obvious, 4 follows from 5 which in turn follows from Theorem 11.3. □

11.5 Proposition If $f \in \mathrm{Aut}(\mathcal{U})$ then $f[\mathrm{acl} A] = \mathrm{acl}(f[A])$ for every $A \subseteq \mathcal{U}$.

Proof. We prove $f[\mathrm{acl} A] \subseteq \mathrm{acl}(f[A])$. Fix $a \in \mathrm{acl} A$ and let $\varphi(x\,;z) \in L$ and $b \in A^z$ be such that $\varphi(x\,;b)$ is algebraic formula satisfied by a. By elementarity, $\varphi(x\,;fb)$ is algebraic and satisfied by fa. Therefore fa is algebraic over $f[A]$, which proves the inclusion.

The converse inclusion is obtained by substituting f^{-1} for f and $f[A]$ for A. □

11.6 Exercise For every $a \in \mathcal{U}^x$ and $A \subseteq \mathcal{U}$, the following are equivalent

1. a is solution of some algebraic formula $\varphi(x) \in L(A)$
2. $a = a_1, \ldots, a_n$ for some $a_1, \ldots, a_n \in \mathrm{acl} A$.

11.7 Exercise Let $\varphi(z) \in L(A)$ be a consistent formula. Prove that, if $a \in \mathrm{acl}(A, b)$ for every $b \vDash \varphi(z)$, then $a \in \mathrm{acl} A$. Prove the same claim with a type $p(z) \subseteq L(A)$ for $\varphi(z)$.

11.8 Exercise Let $a \in \mathcal{U} \smallsetminus \mathrm{acl}\varnothing$. Prove that $\mathcal{U}$ is isomorphic to some $\mathcal{V} \preceq \mathcal{U}$ such that $a \notin \mathcal{V}$. Hint: let $\bar{c}$ be an enumeration of $\mathcal{U}$ and let $p(\bar{u}) = \mathrm{tp}(\bar{c})$ prove that $p(\bar{u}) \cup \{u_i \neq a : i < |\bar{u}|\}$ is realized in $\mathcal{U}$ and that any realization yields the required substructure of $\mathcal{U}$.

11.9 Exercise Let C be a finite set. Prove that if $C \cap M \neq \varnothing$ for every model M containing A, then $C \cap \mathrm{acl} A \neq \varnothing$. Hint: by induction on the cardinality of C. Suppose there is a $c \in C \smallsetminus \mathrm{acl} A$, then there is $\mathcal{V} \simeq \mathcal{U}$ such that $A \subseteq \mathcal{V} \preceq \mathcal{U}$ and $c \notin \mathcal{V}$, see Exercise 11.8. Apply the induction hypothesis to $C' = C \cap \mathcal{V}$ with $\mathcal{V}$ for $\mathcal{U}$.

11.10 Exercise Prove that for every $A \subseteq N$ there is an M such that $\mathrm{acl} A = M \cap N$. Hint: add the requirement $\mathrm{acl}(A_i) \cap N \subseteq \mathrm{acl} A$ to the construction used to prove the downward Löwenheim-Skolem theorem. You need to prove that every consistent $\varphi(x) \in L(A_i)$ has a solution a such that $\operatorname{acl}(A_i, a) \cap N \subseteq \operatorname{acl} A$. The required a has to realize the type

$$\{\varphi(x)\} \ \cup \ \Big\{\neg\big[\psi(b,x) \wedge \exists^{\leq n} y\, \psi(y,x)\big] \ : \ b \in N \smallsetminus \operatorname{acl} A, \ \ \psi(y,x) \in L(A_i), \ \ n < \omega\Big\}$$

whose consistency need to be verified.

11.11 Exercise Prove that for every $A \subseteq N$ there is an automorphism $f \in \operatorname{Aut}(\mathcal{U}/A)$ such that $\operatorname{acl} A = f[N] \cap N$. (This is a stronger version of the claim in Exercise 11.10.) Hint: let $\bar{c}$ be an enumeration of N. Let $p(\bar{x}) = \operatorname{tp}(\bar{c}/A)$. Consider the type

$$p(x) \ \cup \ \Big\{\neg\big[\psi(b,\bar{x}) \wedge \exists^{\leq n} y\, \psi(y,\bar{x})\big] \ : \ b \in N \smallsetminus \operatorname{acl} A, \ \ \psi(y,x) \in L(A), \ \ n < \omega\Big\}$$

Any $\bar{a} \vDash p(\bar{x})$ enumerates a model A-isomorphic to N.

11.12 Exercise Let $\varphi(x) \in L(\mathcal{U})$ and fix an arbitrary set A. Prove that the following are equivalent

1. $\varphi(M^x) \neq \varnothing$ for every model M containing A
2. there is a consistent formula $\psi(z_1, \ldots, z_n) \in L(A)$ such that

$$\psi(z_1, \ldots, z_n) \to \bigvee_{i=1}^{n} \varphi(z_i).$$

Hint: let $\bar{c}$ be an enumeration of M^x, where M is any model containing A. Let $p(\bar{z}) = \operatorname{tp}(\bar{c}/A)$. Note that 1 implies that $p(\bar{z}) \cup \big\{\neg\varphi(z_i) \ : \ i < |\bar{z}|\big\}$ is inconsistent.

11.13 Exercise Let $T = T_{\mathrm{rg}}$. Prove that $\operatorname{acl} A = \operatorname{dcl} A$ for every set A.

11.14 Exercise Let $p(x) \in S(A)$. Let $Q = \big\{q(x) \in S(\operatorname{acl} A) \ : \ q(x) \supseteq p(x)\big\}$. Prove that $\operatorname{Aut}(\mathcal{U}/A)$ acts on Q transitively, i.e. any two types in Q are conjugated by an automorphism in $\operatorname{Aut}(\mathcal{U}/A)$.

11.15 Exercise Let M be ω-saturated and $A \subseteq M$ finite. Suppose that every automorphism of $\mathcal{U}$ that fixes M setwise fixes also $\operatorname{acl} A$ setwise. Prove that $\operatorname{acl} A \subseteq \operatorname{acl}\varnothing$. (The interesting case is when M is countable.)

11.2 Strong minimality

Finite and cofinite sets are always (trivially) definable in every structure. We say that M is a minimal structure if all its definable subsets of arity one are finite or cofinite. Unfortunately, this notion is not elementary, i.e. it is not a property of $\operatorname{Th}(M)$. For instance $\mathbb{N}$ with only the order relation in the language is a minimal structure but none of its elementary extensions is. Hence the following definition: we say that M is a strongly minimal structure if it is minimal and all its elementary extensions are minimal.

We say that T, a consistent theory without finite models, is strongly minimal if for every formula $\varphi(x\,;z) \in L$, where x has arity one, there is an $n \in \omega$ tale che

$$T \ \vdash \ \exists^{\leq n} x\ \varphi(x\,;z) \ \vee \ \exists^{\leq n} x\ \neg\varphi(x\,;z).$$

We show that the semantic notion matches the syntactic one.

11.16 Proposition The following are equivalent

1. $\mathrm{Th}(M)$ is a strongly minimal theory
2. M is a strongly minimal structure
3. M has an elementary extension which is minimal and ω-saturated.

Proof. Implications 1⇒2⇒3 are immediate, we prove 3⇒1. Let $\varphi(x\,;z) \in L$ and let N be the elementary extension given by 3. Let $p(z) \subseteq L$ be the following type

$$p(z) \quad = \quad \Big\{\exists^{>n}x\,\varphi(x\,;z) \ \wedge\ \exists^{>n}x\,\neg\varphi(x\,;z) \quad : \quad n \in \omega\Big\}.$$

As N is minimal, $N \nvDash \exists z\,p(z)$. By ω-saturation $p(z)$ is not finitely consistent in M. Hence, for some n

$$M \quad \vDash \quad \forall z\,\Big[\exists^{\leq n}x\,\varphi(x\,;z) \ \vee\ \exists^{\leq n}x\,\neg\varphi(x\,;z)\Big].$$

which proves that $\mathrm{Th}(M)$ is strongly minimal. □

By quantifier elimination, T_{acf} and T_{dag} are strongly minimal theories.

11.17 Exercise Let T be a complete theory without finite models. Prove that the following are equivalent

1. M is minimal
2. $a \equiv_M b$ for every $a, b \in \mathfrak{U} \smallsetminus M$.

11.3 Independence and dimension

Throughout this section we assume that T is a complete strongly minimal theory.

When $a \notin \mathrm{acl}B$ we say that a is algebraically independent from B. We say that B is an algebraically independent set if every $a \in B$ is independent from $B \smallsetminus \{a\}$. Below we shall abbreviate $B \cup \{a\}$ by B, a and $B \smallsetminus \{a\}$ by $B \smallsetminus a$.

The following is a pivotal property of independence that holds in strongly minimal structures. It is called symmetry or exchange principle. For every B and every pair of elements $a, b \in \mathfrak{U} \smallsetminus \mathrm{acl}B$

$$b \in \mathrm{acl}(B, a) \quad \Leftrightarrow \quad a \in \mathrm{acl}(B, b)$$

Note that when T is the theory of vector spaces (over any fixed field) this principle is the so-called Steinitz exchange lemma.

11.18 Theorem (T strongly minimal.) Independence is symmetric. That is, if $a, b \notin \mathrm{acl}B$ then $b \in \mathrm{acl}(B, a) \Leftrightarrow a \in \mathrm{acl}(B, b)$

Proof. Suppose $b \notin \mathrm{acl}(B, a)$ and $a \in \mathrm{acl}(B, b)$. We prove that $a \in \mathrm{acl}B$. Fix a formula $\varphi(x, y) \in L(B)$ such that $\varphi(x, b)$ witnesses $a \in \mathrm{acl}(B, b)$, i.e. for some n

$$\varphi(a, b) \quad \wedge \quad \exists^{\leq n}x\,\varphi(x, b).$$

As $b \notin \mathrm{acl}(B, a)$, the formula

$$\psi(a, y) \quad = \quad \varphi(a, y) \quad \wedge \quad \exists^{\leq n}x\,\varphi(x, y).$$

is not algebraic. Therefore, by strong minimality, $\psi(a, y)$ has cofinitely many solutions. Hence every model containing B contains a solution of $\psi(a, y)$. As a is algebraic in any of these solutions, a belongs to every model containing B. Therefore, $a \in \operatorname{acl} B$ by Theorem 11.3. $\square$

We say that $B \subseteq C$ is a basis of C if B is an independent set and $C \subseteq \operatorname{acl} B$. The following theorem proves that all bases have the same cardinality, which we call the dimension of C and denote by $\dim C$. First we need the following lemma.

11.19 Lemma (T strongly minimal.) If B is an independent set and $a \notin \operatorname{acl} B$ then B, a is also an independent set.

Proof. Suppose B, a is not independent and that $a \notin \operatorname{acl} B$. Then $b \in \operatorname{acl}(B \setminus b, a)$ for some $b \in B$. As $a, b \notin \operatorname{acl}(B \setminus b)$, from symmetry we obtain $a \in \operatorname{acl}(B \setminus b, b) = \operatorname{acl} B$. Hence B is not an independent set. $\square$

11.20 Corollary (T strongly minimal.) For every $B \subseteq C$ the following are equivalent

1. B is a basis of C
2. B is a maximally independent subset of C.

Finally we prove the main theorem about basis.

11.21 Theorem (T strongly minimal.) Fix some arbitrary set C. Then

1. every independent set $B \subseteq C$ can be extended to a basis of C
2. all bases of C have the same cardinality.

Proof. By the finite character of algebraic closure, the independent set form an inductive class. Apply Zorn lemma to obtain a maximally independent subset of C containing B. By Corollary 11.20 this set is a basis of C. This proves 1.

As for 2, assume for a contradiction that $A, B \subseteq C$ are two bases of C and that $|A| < |B|$. First consider the case when B is infinite. For each $a \in A$ fix a finite set $D_a \subseteq B$ such that $a \in \operatorname{acl}(D_a)$. Let

$$D \;=\; \bigcup_{a \in A} D_a.$$

Then $A \subseteq \operatorname{acl} D$ and $|D| < |B|$. By transitivity, $C \subseteq \operatorname{acl} D$ which contradicts the independence of B.

Now we suppose that $|A| < |B| = n < \omega$. Choose B such that $|B \setminus A|$ is minimal among the bases of cardinality $\geq n$. As $|A| < |B|$, there is a $b \in B \setminus A$. Let B' be a maximal independent subset of $(B \setminus b), A$ containing $B \setminus b$. By Corollary 11.20, B' is a base of $(B \setminus b), A$, hence it is also a base of C. As $|B' \setminus A| < |B \setminus A|$, this is a contradiction. $\square$

11.22 Proposition (T strongly minimal.) Let k be an elementary map. Then $k \cup \{\langle b, c\rangle\}$ is also an elementary map for every $b \notin \operatorname{acl}(\operatorname{dom} k)$ and $c \notin \operatorname{acl}(\operatorname{rng} k)$.

Proof. Let a be an enumeration of $\operatorname{dom} k$. We need to show that $\varphi(b\,; a) \leftrightarrow \varphi(c\,; ka)$ holds for every $\varphi(x\,; z) \in L$. As k is elementary, the formulas $\varphi(x\,; a)$ and $\varphi(x\,; ka)$ are either both algebraic or both co-algebraic. As $b \notin \operatorname{acl}(a)$ and $c \notin \operatorname{acl}(ka)$, they are both false or both true respectively. So the proposition follows. $\square$

11.23 Corollary (T strongly minimal.) Every bijection between independent sets is an elementary map.

Finally we show that dimension classifies models of T.

11.24 Theorem (T strongly minimal.) Models of T with the same dimension are isomorphic.

Proof. Let A e B be bases of M and N respectively. By Corollary 11.23, any bijection between A and B is an elementary map. By Proposition 11.5, it extends to the required isomorphism between $\mathrm{acl}A = M$ and $\mathrm{acl}B = N$. □

11.25 Corollary (T strongly minimal.) Let $|L| < \lambda$. Then T is λ-categorical.

Proof. Let M have cardinality λ. Let $B \subseteq M$ be a base. Then $\lambda = |M| = |\mathrm{acl}B| = |L(B)| = \max\{|L|, |B|\}$. If $|L| < \lambda$, then $\lambda = |B|$. Therefore all models of cardinality λ are isomorphic because they all have the same dimension λ. □

11.26 Proposition (T strongly minimal.) For every model N of cardinality $\geq |L|$ the following are equivalent

1. N is saturated
2. $\dim N = |N|$.

Proof. 2⇒1. Assume 2 and let $k : M \to N$ be an elementary map of cardinality $< |N|$ and let $b \in M$. We want an extension of k defined in b. If $b \in \mathrm{acl}(\mathrm{dom}\, k)$ then the required extension exists by Proposition 11.5. Otherwise, we pick any element $c \in N \smallsetminus \mathrm{acl}(\mathrm{rng}\, k)$. Such an element exists as $|k| < \dim N = |N|$. Then $k \cup \{\langle b, c\rangle\}$ is the required extension by Proposition 11.22.

1⇒2. If $B \subseteq N$ is a basis of N the following type is not realized in N

$$p(x) \quad = \quad \Big\{\neg\varphi(x) \ : \ \varphi(x) \in L(B) \text{ is algebraic} \Big\}$$

Therefore, if N is saturated, $|B| = |N|$. □

11.27 Exercise (T strongly minimal.) Prove that every infinite algebraically closed set is a model.

11.28 Exercise (T strongly minimal, L countable.) Prove that every model is homogeneous.

11.29 Exercise (T strongly minimal.) Prove that if $\dim N = \dim M + 1$ then there is no model K such that $M \prec K \prec N$.

Chapter 12

Countable models

In this chapter L is a fix signature, T a complete theory without finite models, and $\mathcal{U}$ is a saturated model of inaccessible cardinality κ larger than $|L|$. We make no blanket assumption on the cardinality of L, but the main theorems require L to be countable. The notation and implicit assumptions are as in Sectialwaysbon 9.3.

12.1 The omitting types theorem

Troughout the chapter we work over a set of parameters A which, when necessary, we will assume to be countable. We say that the formula $\varphi(x)$ isolates the type $p(x)$ when $\varphi(x)$ is consistent and $\varphi(x) \rightarrow p(x)$. When Δ is a set of formulas, we say that Δ isolates $p(x)$ if some formula in Δ does. When $\Delta = L_x(A)$, the case we are mainly interested in, we say that $p(x)$ is isolated over A, When A is clear from the context (e.g. when $p(x)$ is presented as an type over A), we may omit reference to it. We say that a model M omits $p(x)$ if $p(x)$ is not realized in M.

All this has a topological interpretation. Work with the topology induced by on $\mathcal{U}^x$ or, better, on $\mathcal{U}^x/{\equiv_A}$ which we identify with $S_x(A)$. Then a complete type $p(x) \in S(A)$ is isolated over A if it is an isolated point of the topology. When $p(x) \subseteq L(A)$, isolated over A translates in beeing nowhere dense in the topology induced by A.

Observe that if $p(x) \subseteq L(M)$ then M realizes $p(x)$ if and only if M isolates $p(x)$. Therefore if A isolates $p(x)$, then every model containing A realizes $p(x)$. Below we prove that the converse holds when L and A are countable. This is a famous classical theorem that is called the *omitting types theorem* because it is proved by constructing a model M that omits a given non-isolated type $p(x)$.

The core of the argument lies in the following lemma.

12.1 Lemma Assume $L(A)$ is countable. Let $p(x) \subseteq L(A)$ and suppose that A does not isolate $p(x)$. Then, if $\psi(z) \in L(A)$ is consistent, $\psi(z)$ has a solution a such that A, a does not isolate $p(x)$.

Proof. We construct a sequence of formulas $\langle \psi_i(z) : i < \omega \rangle$ such that any realization a of the type $\{\psi_i(z) : i < \omega\}$ is the required solution of $\psi(z)$.

Let $\langle \xi_i(x\,;z) : i < \omega \rangle$ be an enumeration of $L_{x\,;z}(A)$. Set $\psi_0(z) = \psi(z)$ and define $\psi_{i+1}(z)$ inductively as follows:

1. if $\xi_i(x\,;z) \wedge \psi_i(z)$ is inconsistent, let $\psi_{i+1}(z) = \psi_i(z)$
2. otherwise, let $\psi_{i+1}(z) = \psi_i(z) \wedge \exists x \left[\xi_i(x\,;z) \wedge \neg\varphi(x)\right]$ for some/any $\varphi(x) \in p$ that makes $\psi_{i+1}(z)$ is consistent.

Note that 1 and 2 ensure that if $a \vDash \psi_{i+1}(z)$ then $\xi_i(x\,;a)$ does not isolate $p(x)$. Therefore the proof is complete if we can show that it is always possible to find the formula $\varphi(x)$ required in 2.

Suppose for a contradiction that no formula makes $\psi_{i+1}(z)$ consistent, that is,

$$\xi_i(x\,;z) \wedge \psi_i(z) \quad \rightarrow \quad \varphi(x)$$

for every $\varphi(x) \in p$. This immediately implies that

$$\exists z\ \big[\xi_i(x\,;z) \wedge \psi_i(z)\big] \quad \rightarrow \quad p(x),$$

that is, $p(x)$ is isolated by a formula in $L_x(A)$. This contradicts our assumption and proves the lemma. □

12.2 Omitting Types Theorem Assume $L(A)$ is countable. Then for every consistent type $p(x) \subseteq L(A)$ the following are equivalent

1. all models containing A realize $p(x)$
2. A isolates $p(x)$.

Proof. The implication 2⇒1 is clear. We prove 1⇒2. Assume that A does not isolate $p(x)$. The model M is the union of a chain $\langle A_i : i < \omega\rangle$ of countable subsets of $\mathfrak{U}$ where $A_0 = A$. Along the construction we require that A_i does not isolate $p(x)$. At the end, M will not isolate $p(x)$. Since M is a model, this is equivalent to M omitting $p(x)$.

We proceed as in the proof of the downward Löwenheim-Skolem theorem. Assume that A_i does not isolate $p(x)$. With the notation in the second proof of the Löwenheim-Skolem Theorem 2.55, at stage $i = \pi(j,k)$ apply Lemma 12.1 to find a solution a of $\varphi_k(x)$ such that $A_{i+1} = A_i, a$ does not isolate $p(x)$. □

Gerald Sacks once famously remarked: *Any fool can realize a type, but it takes a model theorist to omit one*. However, the diagonalization method in the proof of Lemma 12.1 lean towards descriptive set theory. (We invite the interested reader to compare this lemma with the Kuratowski-Ulam theorem.)

12.3 Example The following example shows that in the omitting types theorem we cannot drop the assumption that $L(A)$ is countable. Let F be the set of all bijections between two uncountable sets, X and Y. Let M be the model whose domain is domain the disjoint union of F, X and Y. The language has a ternary relation symbol for $f(x) = y$ and unary relation symbols for F, X, and Y. Let $\mathfrak{U}$ be a saturated elementary extension of M. Then $\mathfrak{U}$ is partitioned into three definable sets $\mathfrak{U}_F$, $\mathfrak{U}_X$ and $\mathfrak{U}_Y$. Each element of $\mathfrak{U}_F$ defines a bijection between $\mathfrak{U}_X$ and $\mathfrak{U}_Y$.

Note that for any two elements $a, b \in \mathfrak{U}_Y$, there is an automorphism of $\mathfrak{U}$ that fixes $\mathfrak{U}_X \cup \mathfrak{U}_Y \smallsetminus \{a, b\}$ and swaps a and b.

Now, let $Y_0 \subseteq \mathfrak{U}_Y$ be countable. Let $c \in \mathfrak{U}_Y \smallsetminus Y_0$ and let $p(y) = \mathrm{tp}(c/X, Y_0)$. We claim that $p(y)$ is realized in every model containing X, Y_0. In fact, by the remark above $p(\mathfrak{U}) = \mathfrak{U}_Y \smallsetminus Y_0$. But every model containing X also contains uncountably many elements of $\mathfrak{U}_Y$, hence it contains a conjugate of c which therefore realizes $p(y)$. We also claim that $p(y)$ is not isolated. Suppose for a contradiction there is a consistent formula $\varphi(y)$ such that $\varphi(y) \rightarrow p(y)$. Then $\varphi(y)$ has a solution in $\mathfrak{U}_Y \smallsetminus Y_0$. By the remark above, this implies that $\varphi(\mathfrak{U})$ is a cofinite subset of $\mathfrak{U}_Y$ and this contradicts $\varphi(\mathfrak{U}) \subseteq p(\mathfrak{U})$.

12.4 Exercise Let $p(x) \subseteq L(B)$ and $p_n(x) \subseteq L(A)$, for $n < \omega$, be such that

$$p(x) \quad \to \quad \bigvee_{i<\omega} p_i(x)$$

Prove that

$$p(x) \wedge \varphi(x) \quad \to \quad p_n(x)$$

for some $n < \omega$ and some formula $\varphi(x) \in L(A)$ consistent with $p(x)$.

12.5 Exercise Let $A \subseteq B$ and $p(x) \subseteq L(A)$. Suppose that $\mathrm{tp}(a/B)$ is isolated over B for every $a \vDash p(x)$. Prove that $p(x)$ is isolated over A.

12.6 Exercise Let x be a finite tuple. A set $B \subseteq \mathcal{U}^x$ is *nowhere dense* in the topology induced by A if for every consistent formula $\varphi(x) \in L(A)$ there is some consistent $\psi(x) \in L(A)$ such that $\psi(x) \to \varphi(x)$ and $\psi(B) = \varnothing$. A set $C \subseteq \mathcal{U}^x$ is *meager* if it is union of countably many nowhere dense sets.

Assume $L(A)$ is countable. Prove that for every countable model M the following are equivalent

1. C is meager in the topology topology induced by A
2. there is a model M disjoint of C

Hint: note that Lemma 12.1 holds for a countable set of non-isolated types.

12.2 Prime and atomic models

We say that M is prime over A if $A \subseteq M$ and for every N containing A there is an elementary embedding $h : M \to N$ that fixes A. When A is empty we simply say that M is prime.

There is no syntactic analogue of primeness. The closest notion, which works well for countable models in a countable language, is atomicity. For $a \in \mathcal{U}^x$ we say that a is isolated over A if the type $p(x) = \mathrm{tp}(a/A)$ is isolated. Note that this equivalent to claiming that a is an isolated point in $\mathcal{U}^x$ with respect to the A-topology defined in Section **??** We say that M is atomic over A if $A \subseteq M$ and every $a \in M^{<\omega}$ is isolated over A. When A is empty we say that M is atomic.

12.7 Proposition Let a and b be finite tuples. Then the following are equivalent

1. A isolates b, a
2. A, a isolates b and A isolates a.

Proof. Note that, if $p(x, z) = \mathrm{tp}(b, a/A)$, then $p(x, a) = \mathrm{tp}(b/A, a)$ and $\exists x\, p(x, z)$ equivalent to $\mathrm{tp}(a/A)$.

1⇒2. Let $\varphi(x, z) \in p$ be such that $\varphi(x, z) \to p(x, z)$. Then $\varphi(x, a) \to p(x, a)$ and $\exists x\, \varphi(x, z) \to \exists x\, p(x, z)$. Therefore 2 holds by the remark above.

2⇒1. Fix $\varphi(x\,; z), \psi(z) \in L(A)$ such that $\varphi(x\,; a)$ isolates $p(x\,; a)$ and $\psi(z)$ isolates $\exists x\, p(x\,; z)$. Let $\xi(x\,; z) \in p$ be arbitrary. As $\varphi(x\,; a) \to \xi(x\,; a)$, the formula $\forall x\, [\varphi(x\,; z) \to \xi(x\,; z)]$ belongs to $\mathrm{tp}(a/A)$ which, as noted above, coincides with $\exists x\, p(x, z)$. Hence $\psi(z) \to \forall x\, [\varphi(x\,; z) \to \xi(x\,; z)]$. As this holds for all $\xi(x\,; z) \in p$, we conclude that $\psi(z) \wedge \varphi(x\,; z)$ isolates $p(x, z)$. □

Implication 1⇒2 of the proposition above yields the following useful proposition.

12.8 Proposition If M is atomic over A then M is atomic over A, a for any $a \in M^{<\omega}$.

Proof. Let $b \in M^x$ be a finite tuple. Then A isolates b, a hence A, a isolates b. □

12.9 Proposition Let $k : M \to N$ be an elementary map and suppose that M is atomic over $\operatorname{dom} k$. Then for every $b \in M$ there is a $c \in N$ such that $k \cup \{\langle b, c\rangle\} : M \to N$ is elementary.

Proof. Let $p(x\,;z) = \operatorname{tp}(b\,;a)$ where a is an enumeration of $\operatorname{dom} k$. Let $\varphi(x\,;z) \in L$ be such that $\varphi(x\,;a) \to p(x\,;a)$. Note that, by elementarity, $\varphi(x\,;ka) \to p(x\,;ka)$. Hence the required c is any solution of $\varphi(x\,;ka)$ in N. □

A limiting assumption in Proposition 12.8 is that a need to be finite. Therefore the following proposition is restricted to countable models.

12.10 Proposition Any two countable models atomic over A are isomorphic.

Proof. By Propositions 12.8 and 12.9 and an easy back-and-forth argument. □

12.11 Proposition Assume $L(A)$ is countable. Then for every model M the following are equivalent

1. M is countable and atomic over A
2. M is prime over A.

Proof. 1⇒2. By Propositions 12.8 and 12.9.

2⇒1. Some countable model containing A exists, as M embeds in it, M has also to be countable. Now we prove that M is atomic over A. Suppose for a contradiction that there is some $b \in M^{<\omega}$ such that $p(x) = \operatorname{tp}(b/A)$ is not isolated. By the omitting types theorem there is a model N containing A that omits $p(x)$. Then there cannot be any A-elementary embedding of M into N. □

12.12 Proposition Assume $L(A)$ is countable. Then the following are equivalent

1. there are models atomic over A
2. every consistent $\varphi(z) \in L(A)$ has a solution that is isolated over A.

Note that 2 says that in $\mathcal{U}^z$ isolated points are dense w.r.t. the topology defined in Section 9.3.

Proof. 1⇒2. This holds by elementarity.

2⇒1. We construct by induction a sequence $\langle a_i : i < \omega\rangle$. Reasoning as in (the second proof of) the downward Löverheim-Skolem theorem we can easily ensure that $A \cup \{a_i : i < \omega\}$ is a model. To obtain an atomic model we require that $a_{\restriction i}$ is isolated over A.

Suppose $a_{\restriction i}$ has been defined and assume that some formula $\varphi(z) \in L(A)$ isolates $\mathrm{tp}(a_{\restriction i}/A)$. Let $\psi(x\,;z) \in L(A)$ be such that $\psi(x\,;a_{\restriction i})$ is consistent (we leave to the reader the details of the enumeration of such formulas). Then $\psi(x\,;z) \wedge \varphi(z)$ is also consistent and by assumption it has a solution $b\,;c$ that is isolated over A. As $a_{\restriction i} \equiv_A c$, there is an A-automorphism such that $fc = a_{\restriction i}$. Therefore $fb\,;a_{\restriction i}$ is a solution $\psi(x\,;z)$ that is also isolated over A. Then we can set $a_i = fb$. $\square$

12.3 Countable categoricity

Here we present some important characterizations of ω-categoricity. The second property below can be stated in different equivalent ways; for convenience, these equivalents are considered in a separate proposition. For the time being we introduce the following generalization (which we will prove is completely unnecessary): we say hat T is ω-categorical over A if any two countable models containing A are isomorphic over A. We say ω-categorical for ω-categorical over $\varnothing$.

12.13 Theorem (Engeler, Ryll-Nardzewsky, and Svenonius) Assume $L(A)$ is countable. The following are equivalent:

1. T is ω-categorical over A
2. every type $p(x) \subseteq L(A)$ with $|x| < \omega$ is isolated.

The set A is introduced for convenience. By 3 of Proposition 12.14 below, no theory is ω-categorical over an infinite set, and categoricity over some finite A is equivalent to categoricity over $\varnothing$ (see Exercise 12.16).

Proof. 1⇒2. This is an immediate consequence of the omitting types theorem. In fact, if $p(x)$ is a non-isolated A-type, then there are two countable models M and N containing A such that M realizes $p(x) \subseteq L(A)$ while N omits it. Then M and N cannot be isomorphic over A.

2⇒1. Observe that 2 implies that every countable model containing A is atomic over A. But, by Proposition 12.10, countable atomic models are unique up to isomorphism. $\square$

12.14 Proposition Fix a set A and a finite tuple of variables x. The following are equivalent

1. every type $p(x) \subseteq L(A)$ is isolated
2. $S_x(A)$ is finite
3. $L_x(A)$ is finite up to equivalence
4. $\mathrm{Aut}(\mathfrak{U}/A)$ induces finitely many orbits in $\mathfrak{U}^x$.

Proof. To prove the implication 1⇒2 observe that $\mathfrak{U}^x$ is the union of sets of the form $p(\mathfrak{U})$ where $p \in S_x(A)$. If these types are isolated then $\mathfrak{U}^x$ is the union of A-definable sets. By compactness this union has to be finite. To prove 2⇒1 let $p \in S_x(A)$. If $S_x(A)$ is finite, $\neg p(\mathfrak{U})$ is the union of finitely many type definable sets. A finite union of type definable sets is type definable. So $\neg p(\mathfrak{U})$ is type definable. Hence $p(\mathfrak{U})$ is isolated. We prove implication 2⇒3 observe that each formula in $L_x(A)$ is equivalent to the disjunction of the types in $S_x(A)$ that contain this formula. If $S_x(A)$ is finite, $L_x(A)$ is finite up to equivalence. Implication 3⇒2 is clear and equivalence 2⇔4 follows from the characterization of orbits as type-definable sets. □

12.15 Exercise Prove that the following are equivalent

1. T is ω-categorical
2. for every M countable and x finite, $\mathrm{Aut}(M)$ induces finitely many orbits in M^x.

12.16 Exercise Prove that the following are equivalent for every finite set A

1. T is ω-categorical
2. T is ω-categorical over A.

12.17 Exercise Prove that the following are equivalent

1. T is ω-categorical
2. there is a countable model that is both saturated and atomic.

12.18 Exercise Assume L is countable and that T is complete. Suppose that for every finite tuple x there is a model M that realizes only finitely many types in $S_x(T)$. Prove that T is ω-categorical.

12.19 Exercise Prove that if T is ω-categorical then for every $n < \omega$ there is an $m < \omega$ such that $|\mathrm{acl}A| \leq m$ for every A such that $|A| \leq n$.

12.4 Small theories

Let T be, as always in this chapter, a complete theory without finite models. We say that T is small over A if $S_x(A)$ is countable for every x of finite length. When A is empty, we simply say that T is small. The set A is introduced for convenience; in most application A is the empty set. A different term is used in another very interesting case: a theory which is small over every countable set A is said to be ω-stable. For this reason, the term 0-stable is sometimes used for small.

12.20 Proposition If T is small over A then it is small over $A \cup B$ for any finite set B.

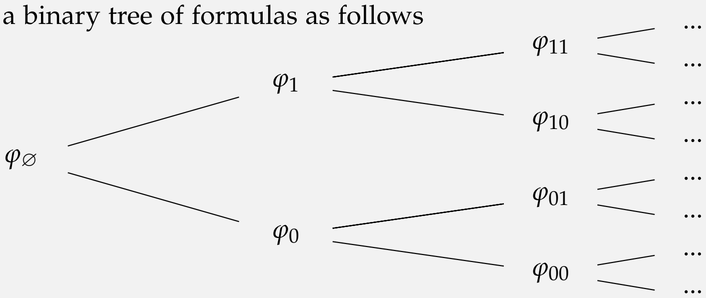
Proof. Let b be an enumeration of B. As $S_x(A, B) = \{p(x\,; b) \ :\ p \in S_{x\,;z}(A)\}$ the proposition is immediate. □

Below we identify $S_x(A)$ with $\mathcal{U}^x/{\equiv_A}$

12.21 Definition Let Δ be a set of formulas (we mainly use $\Delta = L_x(A)$ in this section). A binary tree of formulas in Δ is a sequence $\langle \varphi_s(x) \ : \ s \in 2^{<\lambda} \rangle$ of formulas in $\Delta \cup \{\top\}$ such that

1. for each $s \in 2^\lambda$ the type $p_s(x) \ = \ \{\varphi_{s\restriction i}(x) \ : \ i < \lambda\}$ is consistent
2. $p_s(x) \cup p_r(x)$ is inconsistent for any two distinct $s, r \in 2^\lambda$.

(Condition 2 is usually obtained by taking $\varphi_{s0}(x) \leftrightarrow \neg\varphi_{s1}(x)$ for every s.) We call λ the height of the tree. If the height is not specified, we assume it is ω. We may depict a binary tree of formulas as follows



where branches are consistent types and distinct branches are inconsistent.

Let $S(\Delta)$ denote the set of maximal consistent Δ-types.

12.22 Lemma Suppose Δ is countable and closed under negation. Then the following are equivalent

1. there is a binary tree of formulas in Δ
2. $|S(\Delta)| = 2^\omega$
3. $|S(\Delta)| > \omega$.

Proof. Since the implications 1⇒2⇒3 are clear, it suffices to prove 3⇒1. We assume that $S(\Delta)$ is uncountable and define a tree of formulas in Δ by induction. Begin with $\varphi_\varnothing = \top$. For $s \in 2^{<\omega}$ define

$$p_s(x) \quad = \quad \{\varphi_r(x) \ : \ r \subseteq s\}.$$

Assume inductively that $p_s(x)$ has uncountably many extensions in $S(\Delta)$. This will guarantee the consistency of the branches.

It suffices to show that there is a formula $\psi(x) \in \Delta$ such that both $p_s(x) \cup \{\psi(x)\}$ and $p_s(x) \cup \{\neg\psi(x)\}$ have uncountably many extensions in $S(\Delta)$. Then we define $\varphi_{s^\frown 0}(x) = \psi(x)$ and $\varphi_{s^\frown 1}(x) = \neg\psi(x)$.

Consider the following type that extends $p_s(x)$

$$q(x) \quad = \quad \Big\{\xi(x) \in \Delta \ : \ p_s(x) \cup \{\neg\xi(x)\} \text{ has } \leq \omega \text{ extensions in } S(\Delta)\Big\}.$$

This type is consistent, otherwise

$$p_s(x) \quad \rightarrow \quad \neg\xi_1(x) \vee \cdots \vee \neg\xi_n(x)$$

would hold for some $\xi_i(x) \in q$. This cannot happen, because $p_s(x)$ has uncountably many extensions in $S(\Delta)$, while by the definition of q each of $p_s(x) \cup \{\neg\xi_i(x)\}$ has countably many extensions.

If the formula $\psi(x)$ required above does not exist, $q(x)$ is complete, hence it belongs to $S(\Delta)$. Every type in $S(\Delta)$ that extends $p_s(x)$ and is distinct from $q(x)$ contains

$p_s(x) \cup \{\neg\xi(x)\}$ for some $\xi(x) \in q(x)$. By the definition of $q(x)$, there are countably many such types, so this contradicts the induction hypothesis. □

12.23 Proposition Suppose $L(A)$ is countable. The following are equivalent

1. T is small over A
2. there exists a countable saturated model containing A
3. there is no binary tree of formulas in $L_x(A)$ for any finite x.

Proof. 1⇒2. There is a countable model M containing A that is weakly saturated (see Proposition 9.17). There is a countable homogeneous model N containing M (see Exercise 9.18). Clearly N is also weakly saturated. Then it is saturated by Corollary 9.18.

For a direct argument: construct a countable chain of countable models M_i such that M_{i+1} realized all types in $S_x(A, B)$ for every finite $B \subseteq M_i$ and x. The union of the chain is the saturated model that proves 2.

2⇒3. Clear.

3⇒1. By Lemma 12.22. □

12.24 Proposition A small theory has atomic models over A for any countable A.

Proof. We prove that every formula in $L_x(A)$, where x is finite, has a solution isolated over A. Then it suffices to apply Proposition 12.12.

Suppose for a contradiction that $\varphi(x) \in L(A)$ is consistent but has no solution isolated over A. Then there is a formula $\psi(x) \in L(A)$ such that both $\varphi(x) \wedge \psi(x)$ and $\varphi(x) \wedge \neg\psi(x)$ are consistent, otherwise $\varphi(x)$ would imply a complete type and every solution of $\varphi(x)$ would be isolated. Fix such a $\psi(x)$. Clearly neither $\varphi(x) \wedge \psi(x)$ nor $\varphi(x) \wedge \neg\psi(x)$ have a solution isolated over A. This allows to construct a binary tree of formulas in $L_x(A)$ and prove that T is not small over A. □

12.25 Exercise Let $|x| = 1$. Prove that if $S_x(A)$ is countable for every finite set A, then T is small.

12.26 Exercise (Vaught) Prove that no complete theory has exactly 2 countable models (assume L is countable – though it is not really necessary).

Hint: suppose T has exactly two countable models. Then T is small. Therefore, there is a countable saturated model N and an atomic model $M \subseteq N$. As T is not ω-categorical, $M \not\simeq N$ and there is finite tuple a that is not isolated over $\varnothing$. Let K be an atomic model over a. Clearly $K \not\simeq M$ and, by Exercises 12.16 and 12.17, also $K \not\simeq N$.

12.5 A toy version of a theorem of Zil'ber

Recall that T is a complete theory without finite models. As an application we prove that if T is ω-categorical and strongly minimal then it is not finitely axiomatizable.

We say that T has the finite model property if for every sentence $\varphi \in L$ there is a finite substructure $A \subseteq \mathcal{U}$ such that fmp $\quad\quad \mathcal{U} \vDash \varphi \quad\Leftrightarrow\quad A \vDash \varphi$

The property is interesting to us because of the following proposition.

12.27 Proposition If T has the finite model property then it is not finitely axiomatizable.

Proof. Assume fmp and suppose for a contradiction that there is a sentence $\varphi \in L$ such that $T \vdash \varphi \vdash T$. Then $A \vDash T$ for some finite structure A. But $T \vdash \exists^{>k} x\,(x = x)$ for every k. A contradiction. □

We need the following definition. We say that $C \subseteq \mathcal{U}$ is a homogeneous set if for every finite elementary map $k : C \to C$ and every $b \in C$ there is a $c \in C$ such that $k \cup \{\langle b, c\rangle\}$ is elementary.

12.28 Proposition (T strongly minimal.) Every algebraically closed set of finite dimension is homogeneous.

Proof. Let $k : A \to A$ be elementary. Assume A is algebraically closed and pick $b \in A$. If $b \in \operatorname{acl}(\operatorname{dom} k)$, then by Proposition 11.5 there is an $f \in \operatorname{Aut}(\mathcal{U})$ that extends k and is such that $f(b) \in \operatorname{acl}(\operatorname{rng} k) \subseteq A$. Then $c = fb$ is as required. If $b \notin \operatorname{acl}(\operatorname{dom} k)$, then $\dim(\operatorname{dom} k) < \dim A$. Therefore also $\dim(\operatorname{rng} k) < \dim A$. Then we can pick any $c \in A \smallsetminus \operatorname{rng} k$. □

12.29 Lemma (T strongly minimal.) If T is ω-categorical then T has the finite model property.

Proof. We prove fmp also for formulas with parameters. Namely, we prove that for every n there is a finite structure $A \subseteq \mathcal{U}$ such that fmp holds for all sentences $\varphi \in L(A)$ such that

$\#_n \quad\quad$ number of parameters in φ $+$ number of quantifiers in φ $\leq n$.

Fix n and pick some finite algebraically closed set A such that all types $p(z) \subseteq L$ with $|z| \leq n$ have a realization in A^z. Recall that the algebraic closure is always a substructure and that, in an ω-categorical theory, finite sets have finite algebraic closure. Now we prove that $A \vDash \varphi$ for every φ as in $\#_n$.

Proceed by induction on the syntax of φ. The claim is clear for atomic formulas. Induction for Boolean connectives is straightforward. As for induction step for the existential quantifier, consider the formula $\exists x\, \varphi(x\,; a)$, where $a \in A^{<n}$ and $|x| = 1$. Implication $\Leftarrow$ of fmp follows immediately from the induction hypothesis and from the fact that, if $\exists x\, \varphi(x\,; a)$ satisfy $\#_n$, also $\varphi(b\,; a)$ satisfies it. As for $\Rightarrow$, assume that $\mathcal{U} \vDash \exists x\, \varphi(x\,; a)$. Let $b, a' \in A^{x\,z}$ be a solution of $\varphi(x\,; z)$ such that $a \equiv a'$. Such a solution exists because all types with $\leq n$ variables are realized in A. As A is a homogeneous set (apply the definition with $k = \{\langle a', a\rangle\}$), there is a $c \in A$ such that $a, c \equiv a', b$. Therefore $A \vDash \varphi(c\,; a)$ follows by induction hypothesis. □

12.30 Theorem (T strongly minimal.) If T is ω-categorical then it is not finitely axiomatizable.

12.31 Exercise Assume L is countable and let T be strongly minimal. Prove that the following are equivalent

1. T is ω-categorical
2. the algebraic closure of a finite set is finite.

Implication 1⇒2 does not require the strong minimality of T.

12.32 Exercise Let T be ω-categorical. Prove that if all algebraically closed sets are homogeneous, then T is not finitely axiomatizable.

12.6 Notes and references

An uncountable, non-isolated, complete type that cannot be omitted was produced by Gebhard Fuhrken in 1962. Example 12.3 is inspired by a post by Alex Kruckman on StackExhange [1]. I am not aware of other expositions.

Boris Zil'ber famously proved that Theorem 12.30 holds for any totally categorical theory. The same theorem has been proved independently by Cherlin, Harrington and Lachlan. Their proof uses the classification of finite simple groups. This theorem marks the birth of a subject known as *geometric stability theory* which studies in depth the geometric properties which we briefly mentioned in Chapter 11. The interested reader may consult Pillay's monograph [2]. The material in Section 12.5 comes from [2, Section 2.6]

[1] Alex Kruckman, *Counterexample to the omitting types in uncountable language* (2017), available at `https://math.stackexchange.com/q/2434851`. URL accessed 2019-02-28.

[2] Anand Pillay, *Geometric stability theory*, Oxford Logic Guides, vol. 32, 1996.

Chapter 13

Imaginaries

The description of first-order definability is simplified if we allow definable sets to be used as second-order parameters in formulas. This leads to the theory of (elimination of) *imaginaries*. The technical reason that induced Shelah to introduce imaginaries will only be clear later, see Section 18.4, but the theory is of independent interest.

In this chapter we fix a signature L, a complete theory T without finite models, and a saturated model $\mathcal{U}$ of inaccessible cardinality κ strictly larger than $|L|$. The notation and implicit assumptions are as in Section 9.3.

13.1 Many-sorted structures

A many-sorted language consists of three disjoint sets. Besides the usual L_{fun} and L_{rel}, we have a set L_{srt} whose elements are called sorts. The language also includes a (many-sorted) arity function that assigns to function and relation symbols r, f a tuple of sorts of finite positive length which we call arity.

A many-sorted structure M consists of

1. a set M_s, for each $s \in L_{\text{srt}}$
2. a function $f^M : M_{s_1} \times \cdots \times M_{s_n} \to M_{s_0}$, for each $f \in L_{\text{fun}}$ of arity $\langle s_0, \ldots, s_n \rangle$
3. a relation $r^M \subseteq M_{s_0} \times \cdots \times M_{s_n}$, for each $r \in L_{\text{rel}}$ of arity $\langle s_0, \ldots, s_n \rangle$.

For every sort s we fix a sufficiently large set of variables V_s. Now we define terms and their respective sorts by induction.

All variables are terms of their respective sort. If $t_1, \ldots, t_n$ are terms of sorts $s_1, \ldots, s_n$ and $f \in L_{\text{fun}}$ is of arity $\langle s_0, \ldots, s_n \rangle$ then $f\, t_1, \ldots, t_n$ is a term of sort s_0.

Formulas are defined as follows. If $r \in L_{\text{rel}}$ has arity $\langle s_0, \ldots, s_n \rangle$ then $r\, t_0, \ldots, t_n$ is a formula. Also, $t_1 = t_2$ is formula for every pair of terms of equal sort. All other formulas are constructed by induction using the propositional connectives $\neg$ and $\vee$ and the quantifier $\exists x$ (or any other reasonable choice of logical connectives).

Truth of formulas is defined as for one-sorted languages, except that here we require that the witness of the quantifier $\exists x$ belongs to M_s, where s is the sort of the variable x.

Models of second-order logic are arguably the most widely used examples of many-sorted structures. They may be described using a language with a sort n for every $n \in \omega$. The sort 0 is used for the first-order elements; the sort $n > 0$ is used for relations of arity n. For every $n > 0$ the language has a relation symbol $\in_n$ of arity $\langle 0^n, n \rangle$, where $0^n = 0 \overset{n \text{ times}}{\ldots\ldots} 0$. There are also arbitrarily many function and relation symbols of sort $\langle 0^n \rangle$ for any $n > 0$.

13.2 The eq-expansion

Warning: the structure $\mathcal{U}^{\text{eq}}$ and the theory T^{eq} defined below do not coincide with the standard ones introduced by Shelah. As the difference is merely cosmetic, introducing new notation would be overkill and we prefer to abuse the existing terminology. In Section 13.7 below we compare our definition with the standard one.

Given a language L, we define a many-sorted language L^{eq} which has a sort for each partitioned formula $\sigma(x\,;z) \in L$ and a sort 0 which we call the home sort. (Partitioned formulas have been introduced in Definition 1.16.) For legibility, we pretend that all formulas σ depend on the same variables. So we assume that $x\,;z$ are infinite tuples. Hence, with the notation of the previous section $L_{\text{srt}} = \{0\} \cup L_{x\,;z}$.

The home sort is also called first-order sort, the other sorts are called second-order.

Let $\mathcal{U}^{\text{eq}}$ be the many-sorted structure that has $\mathcal{U}$ as domain of the home sort. The domain for the sort $\sigma(x\,;z)$ contains the definable sets $\mathcal{A} = \sigma(\mathcal{U}^x; b)$ as b ranges over $\mathcal{U}^z$.

The language of $\mathcal{U}^{\text{eq}}$ is denoted by L^{eq}. It contains all the relations and functions of the first order language L with the same interpretation as in the one-sorted case. Moreover L^{eq} contains a relation symbol $\in_{\sigma(x\,;z)}$ for each sort $\sigma(x\,;z)$. These relation symbols have arity $\langle 0^{|x_\sigma|}, \sigma(x\,;z)\rangle$, where x_σ are the variables in x that actually occur in σ. These relations are interpreted as set membership. As there is no risk of ambiguity, in what follows we omit σ from the subscripts.

We write T^{eq} for $\text{Th}(\mathcal{U}^{\text{eq}})$. As usual L^{eq} also denotes the set of formulas constructed in this language and, if $A \subseteq \mathcal{U}^{\text{eq}}$, we write $L^{\text{eq}}(A)$ for the language and the set of formulas that use elements of A as parameters.

We write $L(A)$ for the set of formulas in $L^{\text{eq}}(A)$ that contain no second-order variables, neither free nor quantified (when $A \subseteq \mathcal{U}^{\text{eq}}$, it may contain second-order parameters).

We use the symbol $\mathcal{X}$ to denote a generic second-order variable.

It is important to note right away that this expansion of $\mathcal{U}$ is a mild one: the definable subsets of the home sort of $\mathcal{U}^{\text{eq}}$ are the same as those of $\mathcal{U}$. In particular, iterating the expansion would not yield anything new.

13.1 Proposition Let $\bar{\mathcal{X}} = \mathcal{X}_1, \dots, \mathcal{X}_n$ be a tuple of second order variables of sort $\sigma_i(x\,;z)$. Then for every formula $\varphi(u\,;\bar{\mathcal{X}}) \in L^{\text{eq}}$ there is a formula $\varphi'(u\,;\bar{z}) \in L$, where $\bar{z} = z_1, \dots, z_n$ is a tuple variables of length $|z|$, such that the following holds in $\mathcal{U}^{\text{eq}}$

$$\varphi(u\,;\bar{\mathcal{A}}) \;\leftrightarrow\; \varphi'(u\,;\bar{b})$$

for every $\bar{\mathcal{A}} = \mathcal{A}_1, \dots, \mathcal{A}_n$ and $\bar{b} = b_1, \dots, b_n$ such that $\mathcal{A}_i = \sigma(\mathcal{U}^x; b_i)$.

When $n = 0$ the proposition asserts that L^{eq}_x and L_x have the same expressive power.

Proof (sketch). By induction on syntax. When φ is atomic, we set $\varphi' = \varphi$ unless φ is of the form $t \in \mathcal{X}_i$ for some tuple of terms t or it has the form $\mathcal{X}_i = \mathcal{X}_j$. In the first case φ' is the formula $\sigma_i(t\,;z_i)$. In the second case it is the formula $\forall x\, [\sigma_i(x\,;z_i) \leftrightarrow \sigma_j(x\,;z_j)]$.

The connectives stay unchanged except for the quantifiers $\exists\mathcal{X}$, where $\mathcal{X}$ is a second-order variable, say of sort $\sigma(x\,;z)$. These quantifiers are replaced by $\exists z$. □

Proposition 13.1 implies in particular that we can always replace $\exists\mathcal{X}$ by $\exists z$ if we substitute $\sigma(t\,;z)$ for $t \in \mathcal{X}$ in the quantified formula.

13.2 Remark Proposition 13.1 should convince the reader that the move from $\mathcal{U}$ to $\mathcal{U}^{\rm eq}$ is *almost* trivial. For instance, it implies that for every $A \subseteq \mathcal{U}^{\rm eq}$, there exists a $B \subseteq \mathcal{U}$ such that $L(B)$ is at least as expressive as $L(A)$. By this we mean that every formula in $L(A)$ is equivalent to some formula in $L(B)$. The set $B \subseteq \mathcal{U}$ contains the parameters that define the definable sets in $A \subseteq \mathcal{U}^{\rm eq}$. The point of $\mathcal{U}^{\rm eq}$ is that there might not be any $B \subseteq \mathcal{U}$ such that $L(B)$ is *exactly* as expressive as $L^{\rm eq}(A)$.
For instance, suppose L contains only a binary relation which is interpreted as an equivalence relation with infinitely many infinite classes. Let $\mathcal{A}$ be an equivalence class and let $A = \{\mathcal{A}\}$. Then for $\mathcal{A}$ is definable in $L(B)$ if and only if $B \cap \mathcal{A} \neq \varnothing$. But no element of $\mathcal{A}$ is definable in $L(A)$.

If $\mathcal{V} \preceq \mathcal{U}$ we write $\mathcal{V}^{\rm eq}$ for the substructure of $\mathcal{U}^{\rm eq}$ that has $\mathcal{V}$ as domain of the home sort and the set of definable sets of the form $\sigma(\mathcal{U}^x;b)$ for some $b \in \mathcal{V}^z$ as domain of the sort $\sigma(x\,;z)$. The following proposition claims that the elementary substructures of $\mathcal{U}^{\rm eq}$ are exactly those of the form $\mathcal{V}^{\rm eq}$ for some $\mathcal{V} \preceq \mathcal{U}$.

13.3 Proposition For any structure $\mathcal{V}^\dagger$ of signature $L^{\rm eq}$, the following are equivalent

1. $\mathcal{V}^\dagger \preceq \mathcal{U}^{\rm eq}$
2. $\mathcal{V}^\dagger = \mathcal{V}^{\rm eq}$ for some $\mathcal{V} \preceq \mathcal{U}$.

Proof. Implication 2⇒1 is a direct consequence of Proposition 13.1. We prove 1⇒2. Let $\mathcal{V}$ be the domain of the home sort of $\mathcal{V}^\dagger$. It is clear that $\mathcal{V} \preceq \mathcal{U}$. Let $\mathcal{A} \in \mathcal{V}^{\rm eq}$ have sort $\sigma(x\,;z)$, say $\mathcal{A} = \sigma(\mathcal{U}^x;b)$ for some $b \in \mathcal{V}^z$. As $\exists^{=1}\mathcal{X}\,\forall x\,\big[x \in \mathcal{X} \leftrightarrow \sigma(x\,;b)\big]$ holds in $\mathcal{U}^{\rm eq}$, by elementarity it holds in $\mathcal{V}^\dagger$ and therefore $\mathcal{A} \in \mathcal{V}^\dagger$. This proves $\mathcal{V}^{\rm eq} \subseteq \mathcal{V}^\dagger$. A similar argument proves the converse inclusion. Given $\mathcal{A} \in \mathcal{V}^\dagger$ of sort $\sigma(x\,;z)$, the formula $\forall x\,\big[x \in \mathcal{A} \leftrightarrow \sigma(x\,;b)\big]$ holds in $\mathcal{V}^\dagger$ for some b in the home sort. By elementarity, $\mathcal{A} = \sigma(\mathcal{U}^x;b)$ for some $b \in \mathcal{V}$. □

13.4 Proposition Let $A \subseteq \mathcal{U}^{\rm eq}$. Then every type $p(u\,;\mathcal{X}) \subseteq L^{\rm eq}(A)$ that is finitely consistent in $\mathcal{U}^{\rm eq}$ is realized in $\mathcal{U}^{\rm eq}$. That is, $\mathcal{U}^{\rm eq}$ is saturated.

Proof. By Remark 13.2, there are some $B \subseteq \mathcal{U}$ and some $q(u\,;\mathcal{X}) \subseteq L^{\rm eq}(B)$ equivalent to $p(u\,;\mathcal{X})$. This already proves the proposition when $\mathcal{X}$ is the empty tuple. Otherwise, let $q'(u\,;z)$ be obtained by replacing every formula $\varphi(u\,;\mathcal{X})$ in $q(u\,;\mathcal{X})$ with the formula $\varphi'(u\,;z)$ given in Proposition 13.1. Then $q'(u\,;z)$ is finitely consistent in $\mathcal{U}$. Assume for clarity of notation that $\mathcal{X}$ is a single variable of sort $\sigma(x\,;z)$. If $c\,;b \vDash q'(u\,;z)$, then $c\,;\sigma(\mathcal{U}^x;b) \vDash q(u\,;\mathcal{X})$. □

Automorphisms of a many-sorted structure are defined in the obvious way: sorts are preserved and so are functions and relations. Every automorphism $f : \mathcal{U} \to \mathcal{U}$ extends to an automorphism $f : \mathcal{U}^{\rm eq} \to \mathcal{U}^{\rm eq}$ as follows. If $\mathcal{A} = \sigma(\mathcal{U}^x;b)$ we define

$f\mathcal{A} = \sigma(\mathcal{U}^x; fb) = f[\mathcal{A}]$, which clearly preserves the sort and the relation $\in$. Clearly, this extension is unique.

The homogeneity of $\mathcal{U}^{\text{eq}}$ follows by back-and-forth as in the one-sorted case.

13.5 Proposition Every elementary map $k : \mathcal{U}^{\text{eq}} \to \mathcal{U}^{\text{eq}}$ of cardinality $< \kappa$ extends to an automorphism of $\mathcal{U}^{\text{eq}}$.

13.3 The eq-definable closure

We may safely identify automorphism of $\mathcal{U}$ with automorphisms of $\mathcal{U}^{\text{eq}}$. Let $A \subseteq \mathcal{U}^{\text{eq}}$ and let a be a tuple of elements of $\mathcal{U}^{\text{eq}}$. We denote by $\text{Aut}(\mathcal{U}/A)$ the set of automorphisms (of $\mathcal{U}^{\text{eq}}$) that fix all elements of A. The symbol $\mathcal{O}(a/A)$ denotes the orbit of a over A. This has been defined in Section 9.2 and now we apply it to $\mathcal{U}^{\text{eq}}$

$$\mathcal{O}(a/A) \quad = \quad \{fa \ : \ f \in \text{Aut}(\mathcal{U}/A)\}.$$

By homogeneity, $\mathcal{O}(a/A) = p(\mathcal{U}^{\text{eq}})$ where $p(v) = \text{tp}(a/A)$. When $\mathcal{O}(a/A) = \{a\}$ we say that a is invariant over A or A-invariant, for short.

13.6 Definition Let $A \subseteq \mathcal{U}^{\text{eq}}$ and $a \in \mathcal{U}^{\text{eq}}$. When $\varphi(a) \wedge \exists^{=1} v\, \varphi(v)$ holds for some formula $\varphi(v) \in L^{\text{eq}}(A)$, we say that a is definable over A. We write $\text{dcl}^{\text{eq}}(A)$ for the set of those $a \in \mathcal{U}^{\text{eq}}$ that are definable over A. We write $\text{dcl}(A)$ for $\text{dcl}^{\text{eq}}(A) \cap \mathcal{U}$. This is the natural generalization of the notion of definability introduced in Section 11.1.

The definition above treats first- and second-order elements of $\mathcal{U}^{\text{eq}}$ uniformly. The following proposition claims that when $a \in \mathcal{U}^{\text{eq}}$ is a definable set, the notion of definability coincides with the one usually applied to sets.

13.7 Proposition Let $A \subseteq \mathcal{U}^{\text{eq}}$ and let $\mathcal{A} \in \mathcal{U}^{\text{eq}}$ have sort $\sigma(x\,; z)$. Then the following are equivalent

1. $\mathcal{A} \in \text{dcl}^{\text{eq}}(A)$
2. $\mathcal{A} = \psi(\mathcal{U}^x)$ for some $\psi(x) \in L(A)$.

Proof. Implication 2⇒1 is clear because extensionality is implicit in the definition of $\mathcal{U}^{\text{eq}}$. We prove 1⇒2. Let $\varphi(\mathcal{X}) \in L^{\text{eq}}(A)$ be a formula $\mathcal{A}$ is the unique solution of. Then 2 holds with $\exists \mathcal{X}\, [x \in \mathcal{X} \wedge \varphi(\mathcal{X})]$ for $\psi'(x)$. This $\psi'(x)$ is a formula in $L^{\text{eq}}(A)$. Proposition 13.1 yields the required formula $\psi(x) \in L(A)$. □

The saturation and homogeneity of $\mathcal{U}^{\text{eq}}$ allows us to prove the following proposition with virtually the same proof as for Theorem 11.2

13.8 Theorem For any $A \subseteq \mathcal{U}^{\text{eq}}$ and $a \in \mathcal{U}^{\text{eq}}$ the following are equivalent

1. a is invariant over A
2. $a \in \text{dcl}^{\text{eq}}(A)$.

By Proposition 13.7, Theorem 13.8 when applied to a definable set $\mathcal{A}$ gives an alternative proof of Proposition 10.18.

We conclude this section with a remark about the canonicity of the definitions of sets. The formula $\psi(x)$ in Proposition 13.7 need not be the sort $\sigma(x\,;z)$. For example, consider the theory of a binary equivalence relation $e(x\,;z)$ with two infinite classes, let $\mathcal{A}$ be one of these classes and let $A \neq \varnothing$ be such that $A \cap \mathcal{A} = \varnothing$. Then $\mathcal{A}$ is definable over A though not by some formula of the form $e(x\,;b)$ for some $b \in A$. Things change if we replace A with a model.

13.9 Proposition Let M be a model and let $\mathcal{A}$ be an element of sort $\sigma(x\,;z)$. Then the following are equivalent

1. $\mathcal{A} \in \mathrm{dcl}^{\mathrm{eq}}(M)$
2. $\mathcal{A} = \sigma(\mathcal{U}^x;b)$ for some $b \in M^z$.

In particular $M^{\mathrm{eq}} = \mathrm{dcl}^{\mathrm{eq}}(M)$.

Proof. Assume 1 and let $\psi(x) \in L(M)$ be such that $\mathcal{A} = \psi(\mathcal{U}^x)$. Such a formula exists by Proposition 13.7. Then $\exists z\, \forall x\, \big[\psi(x) \leftrightarrow \sigma(x\,;z)\big]$ holds in $\mathcal{U}$. By elementarity it holds in M, therefore $\exists z$ has a witness in M. This proves 1⇒2, the converse implication is obvious. □

13.4 The eq-algebraic closure

The following is the natural generalization of the notion introduced in Section 11.1.

13.10 Definition Let $A \subseteq \mathcal{U}^{\mathrm{eq}}$ and $a \in \mathcal{U}^{\mathrm{eq}}$. We say that a is algebraic over A if there is a formula $\varphi(v) \in L^{\mathrm{eq}}(A)$ such that $\varphi(a) \wedge \exists^{=k} v\, \varphi(v)$ holds for some and some positive integer k. We write $\mathrm{acl}^{\mathrm{eq}} A$ for the set of those $a \in \mathcal{U}^{\mathrm{eq}}$ that are algebraic over A. We write $\mathrm{acl} A$ for $\mathrm{acl}^{\mathrm{eq}} A \cap \mathcal{U}$.

The following proposition is proved with virtually the same proof as Theorem 11.3

13.11 Theorem For every $A \subseteq \mathcal{U}^{\mathrm{eq}}$ and every $a \in \mathcal{U}^{\mathrm{eq}}$ the following are equivalent

1. $\mathcal{O}(a/A)$ is finite
2. $a \in \mathrm{acl}^{\mathrm{eq}} A$
3. $a \in M^{\mathrm{eq}}$ for every model such that $A \subseteq M^{\mathrm{eq}}$.

It is worthwhile to spell out the equivalence 2⇔3 of the theorem above when a is a definable set $\mathcal{A}$. Namely, $\mathcal{A} \in \mathrm{acl}^{\mathrm{eq}} A$ if and only if $\mathcal{A}$ is definable over every model containing A.

We say finite equivalence relation for an equivalence relation with finitely many classes. A finite equivalence formula or type is a formula, respectively a type, that defines a finite equivalence relation. Theorem 13.12 below proves that sets algebraic over A are union of classes of a finite equivalence relations definable over A.

13.12 Theorem Let $A \subseteq \mathcal{U}^{\rm eq}$ and let $\mathcal{A} \in \mathcal{U}^{\rm eq}$ be an element of sort $\sigma(x\,;z)$. Then the following are equivalent

1. $\mathcal{A} \in {\rm acl}^{\rm eq}A$
2. for some finite equivalence formula $\varepsilon(x\,;y) \in L(A)$ and some $c_1,\dots,c_n \in \mathcal{U}^y$

$$x \in \mathcal{A} \;\leftrightarrow\; \bigvee_{i=1}^{n} \varepsilon(x\,;c_i).$$

Proof. 2⇒1 If $\varepsilon(x\,;y)$ has m classes, then $\mathcal{O}(\mathcal{A}/A)$ contains at most $\binom{m}{n}$ sets.

1⇒2 Let $\varphi(\mathcal{X}) \in L^{\rm eq}(A)$ be an algebraic formula that has $\mathcal{A}$ among its solutions and define

$$\varepsilon(x\,;y) \;=\; \forall\mathcal{X}\Big[\varphi(\mathcal{X}) \;\rightarrow\; \big[x \in \mathcal{X} \leftrightarrow y \in \mathcal{X}\big]\Big]$$

If $\varphi(\mathcal{X})$ has n, solutions, then $\varepsilon(x\,;y)$ has at most 2^n equivalence classes. Clearly, $\mathcal{A}$ is union of some these classes. □

13.13 Definition If $\varepsilon(a\,;b)$ holds for every finite equivalence formula $\varepsilon(x\,;y) \in L(A)$, we say that a and b have the same Shelah strong type over A. We write $a \overset{\rm Sh}{\equiv}_A b$.

By the following proposition, the Shelah strong type of a over A is ${\rm tp}(a/{\rm acl}^{\rm eq}A)$.

13.14 Theorem Let $A \subseteq \mathcal{U}^{\rm eq}$ and let $a, b \in \mathcal{U}^x$. Then the following are equivalent

1. $a \overset{\rm Sh}{\equiv}_A b$
2. $a \equiv_{{\rm acl}^{\rm eq}A} b$.

Proof. 2⇒1 Assume ¬1 and let $\varepsilon(x\,;y) \in L(A)$ be a finite equivalence formula such that $\neg\varepsilon(a\,;b)$. Let $\mathcal{D} = \varepsilon(\mathcal{U}^x;b)$, then $b \in \mathcal{D}$ and $a \notin \mathcal{D}$. As $\varepsilon(x\,;y)$ is an A-invariant finite equivalence formula, $\mathcal{D} \in {\rm acl}^{\rm eq}A$, and ¬2 follows.

1⇒2 Assume ¬2 and let $\varphi(x) \in L\big({\rm acl}^{\rm eq}A\big)$ be such that $\varphi(a) \not\leftrightarrow \varphi(b)$. Let $\mathcal{D} = \varphi(\mathcal{U}^x)$, then $\mathcal{D} \in {\rm acl}^{\rm eq}A$. Therefore, by Proposition 13.12, the set $\mathcal{D}$ is union of equivalence classes of some finite equivalence formula $\varepsilon(x\,;y) \in L(A)$. Then $\neg\varepsilon(a\,;b)$ and ¬1 follows. □

We write $\mathcal{S}(a/A)$ for the intersection of all definable sets that contain a and are algebraic over A. This is called the Shelah strong type over A (the symbol is not standard). By Theorems 13.12 and 13.14

$$\begin{aligned}
\mathcal{S}(a/A) \;&=\; \mathcal{O}(a/{\rm acl}^{\rm eq}A)\\
&=\; q(\mathcal{U}^x) && \text{where } q(x) = {\rm tp}(a/{\rm acl}^{\rm eq}A)\\
&=\; \bigcap\big\{\varepsilon(\mathcal{U}^x;a) \;:\; \varepsilon(x\,;y) \in L(A) \text{ a finite equivalence relation}\big\}\\
&=\; \bigcap\big\{\mathcal{A} \;:\; a \in \mathcal{A} \in {\rm acl}^{\rm eq}A\big\}.
\end{aligned}$$

13.15 Proposition Let $p(x) \in S(A)$ be given. Let

$$Q \quad = \quad \big\{q(x) \in S(\mathrm{acl}^{\mathrm{eq}}A) \ : \ p(x) \subseteq q(x)\big\}.$$

Then $\mathrm{Aut}(\mathcal{U}/A)$ acts transitively on Q, i.e. any two types in Q are conjugated.

Proof. By what remarked above, there is a one-to-one correspondence between Q and the set

$$Q' \quad = \quad \big\{\mathcal{S}(a/A) \ : \ a \vDash p(x)\big\}.$$

Then the proposition is immediate. □

13.16 Exercise Let $p(x) \subseteq L(A)$ and let $\varphi(x\,;y) \in L(A)$ be a formula that defines, when restricted to $p(\mathcal{U}^x)$, an equivalence relation with finitely many classes. Prove that there is a finite equivalence relation definable over A that coincides with $\varphi(x\,;y)$ on $p(\mathcal{U}^x)$.

13.17 Exercise Let $A \subseteq \mathcal{U}$ and let $\mathcal{A}$ be a definable set with finite orbit over A. Without using the eq-expansion, prove that $\mathcal{A}$ is union of classes of a finite equivalence relation definable over A.

13.18 Exercise Let T be strongly minimal and let $\varphi(x\,;z) \in L(A)$ with $|x| = 1$. For arbitrary $b \in \mathcal{U}^z$, prove that if the orbit of $\varphi(\mathcal{U}^x;b)$ over A is finite, then $\varphi(\mathcal{U}^x;b)$ is definable over $\mathrm{acl}A$. Hint: you can use Theorem 13.12.

13.19 Exercise Let M be ω-saturated. Let $A \subseteq M$ be finite. Prove M realizes every consistent $p(x) \subseteq L(\mathrm{acl}A)$, for $|x| < \omega$.

13.5 Elimination of imaginaries

For the time being, we agree that *imaginary* is just another word for definable set. Though this is not historically correct (cf. Section 13.7), it is morally true and helps to understand the terminology. The concept of elimination of imaginaries has been introduced by Poizat who also proved Theorem 13.29 below. A theory has elimination of imaginaries if for every $A \subseteq \mathcal{U}^{\mathrm{eq}}$, there is a $B \subseteq \mathcal{U}$ such that $L(B)$ and $L(A)$ have the same expressive power (i.e. they are the same up to equivalence).

13.20 Definition We say that T has elimination of imaginaries if for every definable set $\mathcal{A}$ there is a formula $\varphi(x\,;z) \in L$ such that

ei $\quad \exists^{=1}z\,\forall x\,\Big[x \in \mathcal{A} \ \leftrightarrow\ \varphi(x\,;z)\Big]$

We say that the witness of $\exists^{=1}z$ in the formula above is a canonical parameter of $\mathcal{A}$ or a canonical name for $\mathcal{A}$. A set may have different canonical parameters for different formulas $\varphi(x\,;z)$.

We say that T has weak elimination of imaginaries if

wei $\quad \exists^{=k}z\,\forall x\,\Big[x \in \mathcal{A} \ \leftrightarrow\ \varphi(x\,;z)\Big]$

for some positive integer k.

In the formulas above we allow z to be the empty string. In this case we read ei and wei omitting the quantifiers $\exists^{=1}z$, respectively $\exists^{=k}z$. Therefore $\varnothing$-definable sets have all (at least) the empty string as a canonical parameter.

To show that the notions above are well-defined properties of a theory one needs to check that they are independent of our choice of monster model. We leave this to the reader as an exercise.

We say that two tuples a and b of elements of $\mathcal{U}^{\rm eq}$ are interdefinable if $\mathrm{dcl}^{\rm eq}(a) = \mathrm{dcl}^{\rm eq}(b)$. By Theorem 13.8 this is equivalent to saying that $\mathrm{Aut}(\mathcal{U}/a) = \mathrm{Aut}(\mathcal{U}/b)$, that is, the automorphisms that fix a fix also b, and vice versa.

13.21 Theorem The following are equivalent

1. T has weak elimination of imaginaries
2. every definable set is interdefinable with a finite set
3. every definable set $\mathcal{A} \in \mathrm{dcl}^{\rm eq}(\mathrm{acl}\{\mathcal{A}\})$.

Proof. 1⇒2 Assume 1 and let $\varphi(x\,;z)$ be the formula in wei. Let $\mathcal{B}$ be the set of solutions of the formula

$$\forall x\ \Big[x \in \mathcal{A}\ \leftrightarrow\ \varphi(x\,;z)\Big].$$

Hence $\mathcal{B}$ is finite and $\mathcal{B} \in \mathrm{dcl}^{\rm eq}\{\mathcal{A}\}$. We also have $\mathcal{A} \in \mathrm{dcl}^{\rm eq}\{\mathcal{B}\}$ because $\mathcal{A}$ is definable by the formula $\exists z\,\big[z \in \mathcal{B} \wedge \varphi(x\,;z)\big]$. Therefore $\mathrm{dcl}^{\rm eq}\{\mathcal{A}\} = \mathrm{dcl}^{\rm eq}\{\mathcal{B}\}$.

2⇒3 Assume $\mathrm{dcl}^{\rm eq}\{\mathcal{A}\} = \mathrm{dcl}^{\rm eq}\{\mathcal{B}\}$ for some finite set $\mathcal{B}$. The elements of $\mathcal{B}$, say $b_1, \dots, b_n$, are the (finitely many) solutions of the formula $z \in \mathcal{B}$. Therefore $b_1, \dots, b_n \in \mathrm{acl}\{\mathcal{B}\} = \mathrm{acl}\{\mathcal{A}\}$. By the assumption there is a formula $\psi(x\,;\mathcal{B}) \in L^{\rm eq}$ that defines $\mathcal{A}$. Then $\mathcal{A}$ is also defined by a formula $\psi'(x\,;b_1, \dots, b_n)$ with parameters in $\mathrm{acl}\{\mathcal{A}\}$.

3⇒1 Assume 3. As wei holds trivially for all $\varnothing$-definable sets, we may assume $\mathcal{A} \neq \varnothing$. Let $\sigma(x\,;z)$ be such that

$$\forall x\ \Big[x \in \mathcal{A}\ \leftrightarrow\ \sigma(x\,;b)\Big].$$

for some tuple b of elements of $\mathrm{acl}\{\mathcal{A}\}$. Fix some algebraic formula $\delta(z\,;\mathcal{A})$ satisfied by b and write $\psi(z\,;\mathcal{X})$ for the formula

$$\forall x\ \Big[x \in \mathcal{X}\ \leftrightarrow\ \sigma(x\,;z)\Big]\ \wedge\ \delta(z\,;\mathcal{X}).$$

The formula $\sharp$ below is clearly satisfied by b therefore, if we can prove that it has finitely many solutions, wei follows from Proposition 13.1

$$\sharp \qquad \forall x\ \Big[x \in \mathcal{A}\ \leftrightarrow\ \sigma(x\,;z)\ \wedge\ \exists \mathcal{X}\,\psi(z\,;\mathcal{X})\Big].$$

We check that any c that satisfies $\sharp$ also satisfies $\delta(z\,;\mathcal{A})$. As $\mathcal{A}$ is non empty, $\psi(c\,;\mathcal{A}')$ holds for some $\mathcal{A}'$. By $\sharp$ and the definition of $\psi(z\,;\mathcal{X})$ we obtain $\mathcal{A}' = \mathcal{A}$ and $\delta(c\,;\mathcal{A})$. □

There are notions of elimination of imaginaries that are weaker than weak elimination. For instance, we say that T has geometric elimination of imaginaries if for every $A \subseteq \mathcal{U}$

$$\mathcal{A}\ \in\ \mathrm{acl}^{\rm eq}(\mathrm{acl}\{\mathcal{A}\})$$

This will not be applied in these notes, but see Exercises 13.24 and 13.25.

13.22 Theorem The following are equivalent

1. T has elimination of imaginaries
2. every definable set is interdefinable with a tuple of real elements
3. every definable set $\mathcal{A} \in \mathrm{dcl}^{\mathrm{eq}}(\mathrm{dcl}\{\mathcal{A}\})$.

Proof. Implications 1⇒2⇒3 are immediate. Implication 3⇒1 is identical to the homologous implication in Theorem 13.21, just substitute algebraic with definable. □

13.23 Exercise Let T have elimination of imaginaries. Let $\varphi(x\,;z) \in L(A)$ and $c \in \mathcal{U}^z$ be given. Prove that if the orbit of $\varphi(\mathcal{U}^x;c)$ over A is finite, then $\varphi(\mathcal{U}^x;c)$ is definable over $\mathrm{acl}A$.

13.24 Exercise Prove that following are equivalent for every $A \subseteq \mathcal{U}$

1. $\mathrm{acl}^{\mathrm{eq}}A = \mathrm{dcl}^{\mathrm{eq}}(\mathrm{acl}A)$ for every $A \subseteq \mathcal{U}$
2. $\mathrm{Aut}(\mathcal{U}/\mathrm{acl}^{\mathrm{eq}}A) = \mathrm{Aut}(\mathcal{U}/\mathrm{acl}A)$
3. $c \equiv_{\mathrm{acl}A} b \;\Leftrightarrow\; c \overset{\mathrm{Sh}}{\equiv}_A b$ for every $A \subseteq \mathcal{U}$ and $c, b \in \mathcal{U}^{<\omega}$.

13.25 Exercise Prove that the following are equivalent

1. T has weak elimination of imaginaries
2. T has geometric elimination of imaginaries and 1 of Exercise 13.24 holds.

13.26 Exercise Prove that the following are equivalent

1. T has weak elimination of imaginaries
2. $\mathrm{dcl}^{\mathrm{eq}}A \cap \mathrm{dcl}^{\mathrm{eq}}B = \mathrm{dcl}^{\mathrm{eq}}(A \cap B)$ for every algebraically closed sets $A, B \subseteq \mathcal{U}$.

13.6 Examples

We now consider elimination of imaginaries in two concrete theories that have been introduced in the previous chapters: algebraically closed fields and the random graph.

13.27 Lemma The following is a sufficient condition for weak elimination of imaginaries

$\sharp$ for every $A \subseteq \mathcal{U}^{\mathrm{eq}}$, every consistent $\varphi(z) \in L(A)$ has a solution in $\mathrm{acl}A$.

Proof. Let $\mathcal{A} \in \mathcal{U}^{\mathrm{eq}}$ be a definable set of sort $\sigma(x\,;z)$. Then $\forall x\,\big[x \in \mathcal{A} \leftrightarrow \sigma(x\,;z)\big]$ is consistent and, by $\sharp$ it has a solution in $\mathrm{acl}\{\mathcal{A}\}$. Hence weak elimination follows from Theorem 13.21. □

13.28 Theorem Let T be a complete, strongly minimal theory. Then, if $\mathrm{acl}\varnothing$ is infinite, T has weak elimination of imaginaries.

Proof. If $\mathrm{acl}\varnothing$ is infinite, $\mathrm{acl}A$ is a model for every A (cf. Exercise 11.27) so condition $\sharp$ of lemma 13.27 holds by elementarity and the theorem follows. □

13.29 Theorem The theories T^p_{acf} have elimination of imaginaries.

Proof. By Theorem 13.28 we know that $T^p_{\rm acf}$ has weak elimination of imaginaries. Therefore, by Theorem 13.21 it suffices to prove that every finite set $\mathcal{A}$ is interdefinable with a tuple. Let $\mathcal{A} = \{a_1,\dots,a_n\}$ where each a_i is a tuple $a_{i,1},\dots,a_{i,m}$ of elements of $\mathcal{U}$. Given $\mathcal{A}$ we define the term

$$t_{\mathcal{A}}(x\,;y) \quad = \quad \prod_{i=1}^{n}\Big(x - \sum_{k=1}^{m} a_{i,k}\, y_k\Big). \qquad\qquad \text{where } y = y_1,\dots,y_m$$

Note that (the interpretation of) the term $t_{\mathcal{A}}(x\,;y)$ is independent on particular indexing of the set $\mathcal{A}$. So, any automorphism that fixes $\mathcal{A}$, fixes the $t_{\mathcal{A}}(x\,;y)$. Now rewrite $t_{\mathcal{A}}(x\,;y)$ as a sum of monomials and let c be the tuple of coefficients of these monomials. The tuple c uniquely determines $t_{\mathcal{A}}(x\,;y)$ and vice versa. Therefore every automorphism that fixes $\mathcal{A}$ fixes c and vice versa. Hence $\mathcal{A}$ and c are interdefinable. □

The rest of this section is dedicated to the proof that the theory of the random graph has weak elimination of quantifiers.

13.30 Lemma The following is a sufficient condition for weak elimination of imaginaries

♮ $\mathcal{D} \subseteq \mathcal{U}^x$ is definable both over A and over B, then it is definable over $A \cap B$.

Proof. Let $\mathcal{D} \subseteq \mathcal{U}^x$ be a definable set, and pick a finite set $A \subseteq \mathcal{U}$ of minimal size over which $\mathcal{D}$ is definable. Now, let f be an automorphism that fixes $\mathcal{D}$. Then $\mathcal{D}$ is definable over $A \cap f[A]$, so by minimality of $|A|$ we have $A = f[A]$. Therefore the orbit under $\mathrm{Aut}(\mathcal{U}/\{\mathcal{D}\})$ of the elements of A is finite. Hence $A \subseteq \mathrm{acl}\{\mathcal{D}\}$. Therefore $\mathcal{D} \in \mathrm{dcl}^{\rm eq}(\mathrm{acl}\{\mathcal{D}\})$ The lemma follows by applying Theorem 13.21. □

13.31 Lemma Let $\mathcal{U} \vDash T_{\rm rg}$. Let a and b be tuples such that $\mathrm{rng}\, a \cap A, \mathrm{rng}\, b \cap B \subseteq A \cap B$. Assume also that $a \equiv_{A\cap B} b$. Then there is a tuple c such that $a \equiv_A c \equiv_B b$.

Proof. For simplicity assume that $\mathrm{rng}\, a \cap A = \mathrm{rng}\, b \cap B = \varnothing$. The general case follows easily.

By induction on the length of a and b. The claim is trivial if this length is 0 so, assume they have length $n+1$ and that there are $c_0,\dots,c_{n-1} \notin A, B, \mathrm{rng}\, a, \mathrm{rng}\, b$ such that $a_{\restriction n} \equiv_A c_0,\dots,c_{n-1} \equiv_B b_{\restriction n}$.

We assume that $\mathrm{rng}\, a \cap \mathrm{rng}\, b = \varnothing$ otherwise we replace b with any tuple $b' \equiv_B b$ that is disjoint of $\mathrm{rng}\, a$.

Now, let $A' = A \cup \{a_0,\dots,a_{n-1}\}$ and $B' = B \cup \{b_0,\dots,b_{n-1}\}$. By the assumption above $A' \cap B' = A \cap B$.

Let c_n be a vertex the satisfies the following formulas

1a $\quad \displaystyle\bigwedge_{a' \in A'} \Big[r(x,a') \;\leftrightarrow\; r(a_n,a')\Big]$

1b $\quad \displaystyle\bigwedge_{b' \in B'} \Big[r(x,b') \;\leftrightarrow\; r(b_n,b')\Big]$

2a $\quad \displaystyle\bigwedge_{i<n} \Big[r(x,c_i) \;\leftrightarrow\; r(a_n,a_i)\Big]$

2b $\displaystyle\bigwedge_{i<n}\Big[r(x,c_i)\ \leftrightarrow\ r(b_n,b_i)\Big]$

Note that 1a and 1b are mutually consistent by the assumption $a \equiv_{A\cap B} b$. By the same assumption 2a and 2b. As $c_0,\dots,c_{n-1} \notin A, B, \operatorname{rng} a, \operatorname{rng} b$ the four formulas above are mutually consistent.

Clearly, any c_n not in $A, B, \operatorname{rng} a, \operatorname{rng} b$ proves the lemma. □

13.32 Lemma The theory $T_{\rm rg}$ has weak elimination of imaginaries.

Proof. We prove ♮ of Lemma 13.30. Suppose not and pick $a, b \in \mathcal{U}^x$ such that $a \equiv_{A\cap B} b$ and $a \in \mathcal{D} \not\leftrightarrow b \in \mathcal{D}$.

Note that $\mathcal{D}$ is definable over $f[A]$ for every $f \in \operatorname{Aut}(\mathcal{U}/B)$ and the same holds swapping A and B. Therefore we can replace A and B with suitable $f[A]$ and $g[B]$ that are disjoint of $\operatorname{rng} a$ and $\operatorname{rng} b$, respectively (for simplicity we are assuming that a and b do not contain elements in $A \cap B$). Clearly $f[A] \cap g[B] = A \cap B$.

By Lemma 13.31, there is c such that $a \equiv_{f[A]} c \equiv_{g[B]} b$. This contradicts the fact that $\mathcal{D}$ is definable both over $f[A]$ and over $g[B]$. □

13.33 Exercise Prove that the theory of the random graph has not elimination of imaginaries.

13.7 Imaginaries: the true story

The point of the expansion to $\mathcal{U}^{\rm eq}$ is to add a canonical parameter for each definable set. In fact, in $\mathcal{U}^{\rm eq}$ every definable subset of $\mathcal{U}^z$ is the canonical parameter of itself. This allows us to deal with theories without elimination of imaginaries in the most straightforward way.

The expansion to $\mathcal{U}^{\rm eq}$ that was originally introduced by Shelah (and still used everywhere else) is slightly different from the one introduced here. For a given set $\mathcal{A} = \sigma(\mathcal{U}^x; b)$ Shelah considers the equivalence relation defined by the formula

$$\varepsilon(z\,;z') \;=\; \forall x\Big[\sigma(x\,;z)\ \leftrightarrow\ \sigma(x\,;z')\Big].$$

The equivalence class of b in the relation $\varepsilon(z\,;z')$ is what Shelah uses as canonical parameter of the set $\mathcal{A}$.

Shelah's $\mathcal{U}^{\rm eq}$ has a sort for each $\varnothing$-definable equivalence relation $\varepsilon(z\,;z')$. The domain of the sort $\varepsilon(z\,;z')$ contains the classes of the equivalence relation defined by $\varepsilon(z\,;z')$. These equivalence classes are called imaginaries. Shelah's $L^{\rm eq}$ contains functions that map tuples in the home sort to their equivalence class.

13.8 Uniform elimination of imaginaries

Sometimes in the literature the notion of elimination of imaginaries is conflated with that of uniform elimination of imaginaries, see e.g. [3, Definition 8.4.2]. Theorem 13.35 below shows that the difference is immaterial. This section is more technical and could be skipped at a first reading.

Let us rephrase the definition of elimination of imaginaries: for every formula $\varphi(x\,;u) \in L$ and every tuple $c \in \mathcal{U}^u$ there is a formula $\sigma(x\,;z) \in L$ such that

$$\exists^{=1} z\ \forall x\ \Big[\varphi(x\,;c)\ \leftrightarrow\ \sigma(x\,;z)\Big].$$

A priori, the formula $\sigma(x\,;z)$ may depend on c in a very wild manner. We say that T has uniform elimination of imaginaries if for every $\varphi(x\,;u)$ there are a formula $\sigma(x\,;z)$ and a formula $\rho(z)$ such that

uei $$\forall u\ \exists^{=1} z\ \Big[\rho(z)\ \wedge\ \forall x\ \big[\varphi(x\,;u)\ \leftrightarrow\ \sigma(x\,;z)\big]\Big].$$

The role of the formula $\rho(z)$ above is mysterious. It is clarified by the following propositions. In fact, uniform elimination of imaginaries is equivalent to a very natural property which in words says: every definable equivalence relation is the kernel of a definable function. Recall that the kernel of the function f is the relation $fa = fb$.

Uniform elimination of imaginaries is convenient when dealing with interpretations of structure inside other structures. Let us consider a simple concrete example. Suppose $\mathcal{U}$ is a group and let $\mathcal{H}$ be a definable normal subgroup of $\mathcal{U}$. The elements of the quotient structure $\mathcal{U}/\mathcal{H}$ are equivalence classes of a definable equivalence relation. If there is uniform elimination of imaginaries we can identify $\mathcal{U}/\mathcal{H}$ with an actual definable subset of $\mathcal{G}$ (the range of the function f above). Moreover, the group operation of $\mathcal{G}$ are definable functions. As working in $\mathcal{U}/\mathcal{H}$ may be notationally cumbersome, $\mathcal{G}$ may offer a convenient alternative.

13.34 Proposition The following are equivalent

1. T has uniform elimination of imaginaries
2. for every $\varphi(x\,;u)$ such that $\forall u\ \exists x\ \varphi(x\,;u)$ there is a formula $\sigma(x\,;z)$ such that
$$\forall u\ \exists^{=1} z\ \forall x\ \Big[\varphi(x\,;u)\ \leftrightarrow\ \sigma(x\,;z)\Big];$$
3. for every equivalence formula $\varepsilon(x\,;u) \in L$ there is given by $\sigma(x\,;z) \in L$ such that
$$\forall u\ \exists^{=1} z\ \forall x\ \Big[\varepsilon(x\,;u)\ \leftrightarrow\ \sigma(x\,;z)\Big].$$

Note that the definable function $fu = z$ mentioned above is defined by the formula

$$\vartheta(u\,;z)\ =\ \forall x\ \Big[\varepsilon(x\,;u)\ \leftrightarrow\ \sigma(x\,;z)\Big].$$

Proof. 1⇒2. If $\forall u\ \exists x\ \varphi(x\,;u)$ we can rewrite uei as

$$\forall u\ \exists^{=1} z\ \forall x\ \Big[\varphi(x\,;u)\ \leftrightarrow\ \rho(z) \wedge \sigma(x\,;z)\Big].$$

2⇒3. Clear.

3⇒1. Apply 3 to the equivalence formula

$$\varepsilon(u\,;v)\ =\ \forall x\ \Big[\varphi(x\,;u)\ \leftrightarrow\ \varphi(x\,;v)\Big]$$

Let $\vartheta(u\,;z)$ be defined as above. The reader may check that uei holds substituting for $\delta(z)$ the formula $\exists u\ \vartheta(u\,;z)$ and and for $\sigma(x\,;z)$ the formula $\exists u\ \big[\varphi(x\,;u) \wedge \vartheta(u\,;z)\big]$. □

By the following theorem, uniformity comes almost for free.

13.35 Theorem The following are equivalent

1. T has uniform elimination of imaginaries
2. $\mathrm{dcl}\varnothing$ contains at least two elements and T has elimination of imaginaries.

Proof. 1⇒2. Let $\varphi(x,u)$ be the formula $u_1 = u_2$. From 1 we obtain a formula $\sigma(x\,;z) \in L$ such that

$$\forall u_1, u_2\, \exists^{=1} z \Big[\rho(z) \ \wedge\ \forall x\ \big[u_1 = u_2 \ \leftrightarrow\ \sigma(x\,;z)\big]\Big].$$

Therefore the formulas $\exists^{=1} z\ \big[\rho(z) \wedge \forall x\, \sigma(x\,;z)\big]$ and $\exists^{=1} z\ \big[\rho(z) \wedge \forall x \neg\sigma(x\,;z)\big]$ are both true. The witnesses of $\exists^{=1} z$ in these two formulas are two distinct elements of $\mathrm{dcl}\varnothing$.

2⇒1. Assume 2 and fix a formula formula $\varphi(x\,;u)$ such that $\varphi(x,a)$ is consistent for every $a \in \mathcal{U}^{|u|}$. We prove 2 of Proposition 13.34.

Let $p(u)$ be the type that contains the formulas

$$\neg\, \exists^{=1} z\, \forall x \Big[\varphi(x\,;u) \ \leftrightarrow\ \sigma(x;z)\Big],$$

where $\sigma(x;z)$ ranges over all formulas in L. By elimination of imaginaries $p(u)$ is not consistent. Therefore, by compactness, there are some formulas $\sigma_i(x,z)$ such that

$$\sharp \qquad \forall u \bigvee_{i=0}^{n} \exists^{=1} z\, \forall x \Big[\varphi(x\,;u) \ \leftrightarrow\ \sigma_i(x\,;z)\Big].$$

To prove the theorem we need to move the disjunction in front of the $\sigma_i(x\,;z)$.

We can assume that if $\sigma_i(x\,;b) \leftrightarrow \sigma'_i(x\,;b')$ for some $\langle b,i\rangle \neq \langle b',i'\rangle$ then $\sigma_i(x\,;b)$ is inconsistent. Otherwise we can substitute the formula $\sigma_i(x\,;z)$ with

$$\sigma_i(x\,;z) \ \wedge\ \bigwedge_{j\leq i} \neg\exists y \neq b\, \forall x \Big[\sigma_j(x\,;y) \leftrightarrow \sigma_i(x\,;z)\Big].$$

As $\varphi(x,a)$ is consistent for every a, the substitution does not break the validity of $\sharp$.

Fix some distinct $\varnothing$-definable tuples $d_0,\ldots,d_n$ of the same length (these are easy to obtain from two $\varnothing$-definable elements). We claim that from $\sharp$ it follows that

$$\forall u\ \exists^{=1} z, y\, \forall x \Big[\varphi(x\,;u) \ \leftrightarrow\ \bigvee_{i=1}^{n} \Big[\sigma_i(x\,;z) \ \wedge\ y = d_i\Big]\Big].$$

(The tuple z, y plays the role of z.) We fix some a and check that the formula below has a unique solution

$$\flat \qquad \forall x \Big[\varphi(x\,;a) \ \leftrightarrow\ \bigvee_{i=1}^{n} \Big[\sigma_i(x\,;z) \ \wedge\ y = d_i\Big]\Big].$$

Existence follows immediately from $\sharp$. As for uniqueness, note that if b, d_i and $b', d_{i'}$ are two distinct solution of $\flat$ then $\sigma_i(x\,;b) \leftrightarrow \sigma_{i'}(x\,;b')$ for some $\langle b,i\rangle \neq \langle b',i'\rangle$. By what assumed on $\sigma_i(x\,;z)$, we obtain that $\varphi(x\,;a)$ is inconsistent. A contradiction which proves the theorem. □

Chapter 14

Invariant sets

In this chapter, L is a signature, T is a complete theory without finite models, and $\mathcal{U}$ is a saturated model of inaccessible cardinality κ strictly larger than $|L|$. We use the same notation and make the same implicit assumptions as in Section 9.3.

14.1 Invariant sets and types

Let $\mathcal{D} \subseteq \mathcal{U}^z$, where z is a tuple of length $< \kappa$. We say that $\mathcal{D}$ is invariant over A, or A-invariant for sort, if it is fixed setwise by all A-automorphisms. That is, $f[\mathcal{D}] = \mathcal{D}$ for every automorphism $f \in \mathrm{Aut}(\mathcal{U}/A)$ or, yet in other words, if

is1. $\qquad a \in \mathcal{D} \leftrightarrow fa \in \mathcal{D} \qquad$ for every $a \in \mathcal{U}^z$ and every $f \in \mathrm{Aut}(\mathcal{U}/A)$,

which, by homogeneity, is equivalent to,

is2. $\qquad a \in \mathcal{D} \leftrightarrow b \in \mathcal{D} \qquad$ for all $a, b \in \mathcal{U}^z$ such that $a \equiv_A b$.

The latter condition yields the following bound on the number of invariant sets.

14.1 Proposition Let $\lambda = |L_z(A)|$. There are at most 2^{2^λ} sets $\mathcal{D} \subseteq \mathcal{U}^z$ that are invariant over A.

Proof. By is2, sets that are invariant over A are unions of equivalence classes of the relation $\equiv_A$, that is, unions of sets of the form $p(\mathcal{U})$ where $p(z) \in S(A)$. Then the number of A-invariant sets is at most $2^{|S_z(A)|}$. Clearly $|S_z(A)| \leq 2^\lambda$. □

A formula is invariant if the set it defines is invariant. The same for small types.

For non-small types the definition of invariant type is less straightforward. One is tempted to require that the formulas in the type are invariant – but this is too strong a demand as it implies that the type is small. Alternatively, one could require that the set defined by the type is invariant – but this is an empty requirement for types that are not realized in $\mathcal{U}$.

First we introduce some notation. Let $\varphi(x\,;z) \in L(\mathcal{U})$. A φ-formula is a Boolean combination of formulas of the form $\varphi(x\,;b)$ for some $b \in \mathcal{U}^z$. A φ-type is a set of φ-formulas. We denote by $S_\varphi(\mathcal{A})$ the set of complete φ-types that contain φ-formulas that are invariant over $\mathcal{A}$. Typically, $\mathcal{A}$ is either $\mathcal{U}$ or some small set $A \subseteq \mathcal{U}$. Types in $S_\varphi(\mathcal{U})$ are called global φ-types.

Let $p(x) \subseteq L(\mathcal{U})$. For $f \in \mathrm{Aut}(\mathcal{U})$, we write $fp(x)$ for the type containing the formulas $\varphi(x\,;fa)$ for every $\varphi(x\,;a) \in p$, where $\varphi(x\,;z) \in L$. We say that $p(x)$ is invariant over A, or A-invariant for short, if for every $f \in \mathrm{Aut}(\mathcal{U}/A)$

it1. $\qquad p(x) \vdash fp(x)$

Equivalently, if for every $\varphi(x\,;z) \in L(A)$, every $a \in \mathcal{U}^z$, and every $f \in \mathrm{Aut}(\mathcal{U}/A)$

it1′. $p(x) \vdash \varphi(x\,;a) \;\Leftrightarrow\; p(x) \vdash \varphi(x\,;fa)$.

Let $p(x) \in S(\mathfrak{U})$ be a global type. For every formula $\varphi(x\,;z) \in L(\mathfrak{U})$ we define

$$\mathcal{D}_{p,\varphi} \;=\; \Big\{a \in \mathfrak{U}^z \;:\; p(x) \vdash \varphi(x\,;a)\Big\}.$$

A global φ-type $p(x) \in S_\varphi(\mathfrak{U})$ can be identified with the set $\mathcal{D}_{p,\varphi}$. Therefore a global type $p(x) \in S(\mathfrak{U})$ can be identified with the collection of the sets $\mathcal{D}_{p,\varphi}$, where $\varphi(x\,;z)$ ranges over L.

Hence $p(x)$ is invariant exactly when all the sets $\mathcal{D}_{p,\varphi}$ are. That is, $p(x)$ is invariant over A if for all $\varphi(x\,;z) \in L(A)$ and $a, b \in \mathfrak{U}^z$

it2. $a \equiv_A b \;\Rightarrow\; \Big(p(x) \vdash \varphi(x\,;a) \;\Leftrightarrow\; p(x) \vdash \varphi(x\,;b)\Big)$

If $p(x) \in S_\varphi(\mathfrak{U})$ it2 is equivalent to requiring that for every $a, b \in \mathfrak{U}^z$

it2′. $a \equiv_A b \;\Rightarrow\; p(x) \;\vdash\; \varphi(x\,;a) \leftrightarrow \varphi(x\,;b)$.

Finally, note the following useful characterization of it2′

it3. $a \equiv_A b \;\Rightarrow\; \varphi(c\,;a) \leftrightarrow \varphi(c\,;b)$

for every $a, b \in \mathfrak{U}^z$ and $c \vDash p_{\restriction A,a,b}$. When $p(x) \in S(\mathfrak{U})$, this can be rephrased as

it3′. $a \equiv_A b \;\Rightarrow\; a \equiv_{A,c} b$.

14.2 Exercise Let $\varphi(x\,;z) \in L(A)$. Let $M \supseteq A$ be a sufficiently saturated model. Let $p(x) \in S_\varphi(M)$ be invariant over A in the following sense: $p(x) \vdash \varphi(x\,;b) \leftrightarrow \varphi(x\,;b')$ for every $b \equiv_A b'$ in M^z. Prove that there is the unique global φ-type that extends $p(x)$ and is invariant over A. Prove that such a type do not depends on the choice of M. Find a bound on the number of global types that are invariant over A.

14.2 Heirs and coheirs

The easiest way to obtain types that are invariant over A is via types that are finitely satisfiable in A. We say that a type $p(x)$ is finitely satisfiable in A if every conjunction of formulas in $p(x)$ has a solution in A^x.

14.3 Proposition Every $p(x) \in S(\mathfrak{U})$ that is finitely satisfiable in A is invariant over A.

Proof. Suppose not. By it2′ and the completeness of $p(x)$ there are $a \equiv_A b$ and $\varphi(x\,;z) \in L$ such that $p(x) \vdash \varphi(x\,;a) \not\leftrightarrow \varphi(x\,;b)$. Then the finite satisfiability of $p(x)$ contradicts $a \equiv_A b$. □

14.4 Proposition Every type $q(x) \subseteq L(\mathfrak{U})$ that is finitely satisfiable in A has an extension to a global type that is finitely satisfiable in A.

Proof. Let $p(x) \subseteq L(\mathfrak{U})$ be maximal among the types that contain $q(x)$ and are finitely satisfiable in A. We prove that $p(x)$ is complete. If for a contradiction $p(x)$ contains neither $\psi(x)$ nor $\neg\psi(x)$, then neither $p(x) \cup \{\psi(x)\}$ nor $p(x) \cup \{\neg\psi(x)\}$ are finitely satisfiable in A. This contradicts the finite satisfiability of $p(x)$. □

In most cases we work with types that are finitely satisfiable over a model. The reason is explained by the next proposition, which is clear by elementarity.

14.5 Remark Every consistent type over a model is finitely satisfiable in that model, that is, whenever $p(x) \subseteq L(M)$ is consistent, $p(x)$ is finitely satisfiable in M.

14.6 Definition A type $p(x) \subseteq L(\mathfrak{U})$ that is finitely satisfiable in A is said to be a coheir of $p_{\restriction A}(x)$.

In many cases it is convenient to work with elements instead of types. We introduce the following notation to express that $\mathrm{tp}(a/A, b)$ is finitely satisfied in A.

14.7 Definition For every $a \in \mathfrak{U}^x$ and $b \in \mathfrak{U}^z$ we define

⚠
$$a \mathop{\smile\!\!\!\!\!\!|}_A b \quad \Leftrightarrow \quad \varphi(A^x, b) \neq \varnothing \text{ for all } \varphi(x\,;z) \in L(A) \text{ such that } \varphi(a\,;b).$$

We say that $\mathrm{tp}(a/A, b)$ is a coheir of $\mathrm{tp}(a/A)$ or, equivalently, that $\mathrm{tp}(b/A, a)$ is an heir of $\mathrm{tp}(b/A)$. We define the type

$$x \mathop{\smile\!\!\!\!\!\!|}_A b \quad = \quad \Big\{\neg\varphi(x\,;b) \;:\; \varphi(x\,;z) \in L(A) \text{ and } \varphi(A^x; b) = \varnothing\Big\}.$$

We write $a \equiv_A x \mathop{\smile\!\!\!\!\!\!|}_A b$ for the union of the types $x \mathop{\smile\!\!\!\!\!\!|}_A b$ and $\mathrm{tp}(a/A)$. By 4 in Lemma 14.9 below, this is a consistent type over A, b.

The tuples a realizing $x \mathop{\smile\!\!\!\!\!\!|}_A b$ are exactly those such that $a \mathop{\smile\!\!\!\!\!\!|}_A b$. Note that $\mathrm{tp}(a/A, b)$ is a coheir of $\mathrm{tp}(a/A)$ according to Definition 14.6, so the terminology is consistent.

We may think of $a \mathop{\smile\!\!\!\!\!\!|}_A b$ as saying that a is independent from b over A. This is a strong notion of independence. In general it is not symmetric, that is, $b \mathop{\smile\!\!\!\!\!\!|}_A a$ is not equivalent to $a \mathop{\smile\!\!\!\!\!\!|}_A b$. In Chapter 18 we will see that symmetry is equivalent to stability.

14.8 Remark In general there is no type that says $x \mathop{\smile\!\!\!\!\!\!|}_A z$, see Exercise 14.20. However, for every $b \in \mathfrak{U}^z$ there is a type that says $x \mathop{\smile\!\!\!\!\!\!|}_A z \equiv_A b$, namely

$$p(x\,;z) \quad \Leftrightarrow \quad \big\{\neg\varphi(x\,;z) \;:\; \varphi(A^x\,;b) = \varnothing,\;\; \varphi(x\,;z) \in L(A)\big\} \;\cup\; q(z)$$

where $q(z) = \mathrm{tp}(b/A)$.

We will use the following easy lemma without explicit reference.

14.9 Lemma The following properties hold for all A, a, b, and c

1. $a \mathop{\smile\!\!\!\!\!\!|}_A b \;\Rightarrow\; fa \mathop{\smile\!\!\!\!\!\!|}_A fb$ for every $f \in \mathrm{Aut}(\mathfrak{U}/A)$ *invariance*
2. $a \mathop{\smile\!\!\!\!\!\!|}_A b \;\Leftrightarrow\; a_0 \mathop{\smile\!\!\!\!\!\!|}_A b_0$ for all finite $a_0 \subseteq a$ and $b_0 \subseteq b$ *finite character*
3. $a \mathop{\smile\!\!\!\!\!\!|}_{A,c} b$ and $c \mathop{\smile\!\!\!\!\!\!|}_A b \;\Rightarrow\; a, c \mathop{\smile\!\!\!\!\!\!|}_A b$ *transitivity*
4. $a \mathop{\smile\!\!\!\!\!\!|}_A b \;\Rightarrow$ there exists $a' \equiv_{A,b} a$ such that $a' \mathop{\smile\!\!\!\!\!\!|}_A b, c$ *coheir extension*
5. $a \mathop{\smile\!\!\!\!\!\!|}_A b_1, b_2$ and $b_1 \equiv_A b_2 \;\Rightarrow\; b_1 \equiv_{A,a} b_2$ *non-splitting*

Proof. Properties 1-3 follow immediately from Definition 14.7. We prove 4. Assume $a \mathop{\smile\!\!\!\!\!\!|}_A b$, that is, $\mathrm{tp}(a/A, b)$ is finitely satisfiable in A. By Proposition 14.4 $\mathrm{tp}(a/A, b)$ extends to a global type $p(x)$ that is finitely satisfiable in A. Then any $a' \models p_{\restriction A,b,c}(x)$ proves the lemma. The proof of 5 is left to the reader. □

14.10 Proposition Let $a \mathop{\smile\!\!\!\!|}_A b$ then there is $\mathcal{V} \preceq \mathcal{U}$ that is isomorphic to $\mathcal{U}$ over A, b and such that $a \mathop{\smile\!\!\!\!|}_A \mathcal{V}$.

Proof. Let c be an enumeration of $\mathcal{U}$. Let $p(w, z) = \mathrm{tp}(c, b/A)$. We can take $\mathcal{V}$ to be the structure enumerated by the tuple that realizes symultaneously $p(w, b)$ and the following type

$$\Big\{\neg\varphi(a, w) \quad : \quad \varphi(x, w) \in L(A), \ \ \varphi(A^x, c) = \varnothing\Big\}.$$

We only need to prove finite consistency. If inconsistent, for some $\vartheta(w, z) \in p$ and some $\varphi_i(x, w)$ in the above set

$$\vartheta(w, b) \quad \rightarrow \quad \bigvee_{i=1}^{n} \varphi_i(a, w).$$

As $a \mathop{\smile\!\!\!\!|}_A b$, we can replace a by some $a' \in A^x$. Finally, substituting c for w, we obtain that $\varphi_i(a, c)$ holds for some i, a contradiction $\square$

When A is a model, the above proposition has a dual version.

14.11 Proposition Let $a \mathop{\smile\!\!\!\!|}_M b$ then there is $\mathcal{V} \preceq \mathcal{U}$ that is isomorphic to $\mathcal{U}$ over M, a and such that $\mathcal{V} \mathop{\smile\!\!\!\!|}_A b$.

Proof. Let c be an enumeration of $\mathcal{U}$. Let $p(w, x) = \mathrm{tp}(c, a/M)$. We can take $\mathcal{V}$ to be the structure enumerated by the tuple that realizes symultaneously $p(w, a)$ and the type $w \mathop{\smile\!\!\!\!|}_M b$, that is,

$$\Big\{\neg\varphi(w, b) \quad : \quad \varphi(w, z) \in L(M), \ \ \varphi(M^w, b) = \varnothing\Big\}.$$

We only need to prove finite consistency. If inconsistent, for some $\vartheta(w, x) \in p$ and some $\varphi_i(w, b)$ in the above set

$$\vartheta(w, a) \quad \rightarrow \quad \bigvee_{i=1}^{n} \varphi_i(w, b).$$

As $a \mathop{\smile\!\!\!\!|}_M b$, we can replace a by some $a' \in M^x$. We can also require that $\exists w\, \vartheta(w, a')$. Then, by elementarity, $\vartheta(c', a')$ holds for some $c' \in M^w$. Therefore, $\varphi_i(c', b)$ holds for some i, a contradiction. $\square$

As a matter of fact, on the left-side of the $\mathop{\smile\!\!\!\!|}_M$-relation we can require a stronger form of saturation.

14.12 Proposition Let $C \mathop{\smile\!\!\!\!|}_M b$. There is $\mathcal{W} \mathop{\smile\!\!\!\!|}_M b$ such that $C \cup M \subseteq \mathcal{W} \preceq \mathcal{U}$, and $\mathcal{W}$ realizes every type $p(x) \subseteq L(B, b)$, for any $B \subseteq \mathcal{W}$, that is finitely satisfied in $\mathcal{W}$.

Proof. Construct inductively a chain of sets $C_i \mathop{\smile\!\!\!\!|}_M b$, for $i < \kappa$. The chain starts with $C_0 = C \cup M$, and is the union at limit stages. At successor stages we define $C_{i+1} = C_i \cup \{c_i\}$ for some suitable $c_i \mathop{\smile\!\!\!\!|}_{C_i} b$. By 3 in Lemma 14.9, this ensures $C_{i+1} \mathop{\smile\!\!\!\!|}_M b$. We choose the sequence $\langle c_i : i < \kappa\rangle$ so that for every n every complete type $p(x) \in S(C_n, b)$ that is finitely satisfiable in C_n is eventually realized by some

c_i. This ensure that $\mathcal{W}$, the union of the chain, realizes all the required types. To check that $\mathcal{W} \preceq \mathcal{U}$ note that, by Proposition 14.11, every consistent formula in $L(\mathcal{W})$ has a solution in some $\mathcal{V} \downarrow_M b$. □

14.13 Definition We say that the type $x \downarrow_A b$ is stationary if for every $a \in \mathcal{U}^x$ the type $a \equiv_A x \downarrow_A b$ is complete over A, b. We say that $\downarrow_A$ is stationary if $x \downarrow_A b$ is stationary for every $b \in \mathcal{U}^z$ and any finite tuples of variables x and z. We say n-stationary if we limit the request to a tuple x of length n.

A first application of stationarity is given in Section 15.3. Stationarity is deeply concted with definability as illustrated by the following proposition. This phenmenon will receive the due attention in Section 18.7.

14.14 Proposition The following are equivalent

1. $x \downarrow_M b$ is stationary
2. for every $\varphi(x\,;z) \in L(M)$ there is a $\psi(x) \in L(M)$ such that $\varphi(M^x\,;b) = \psi(M^x)$.

Proof. 2⇒1. Let $b \in \mathcal{U}^z$ and $a_1, a_2 \in \mathcal{U}^x$ be such that $a_i \downarrow_M b$ and $a_1 \equiv_M a_2$. We claim that $a_1 \equiv_{M,b} a_2$. We need to prove that $\varphi(a_1\,;b) \leftrightarrow \varphi(a_2\,;b)$ for every $\varphi(x\,;z) \in L(M)$. Let $\psi(x) \in L(M)$ be such that $\varphi(M^x;b) = \psi(M^x)$. From $a_i \downarrow_M b$ we obtain that $\varphi(a_i\,;b) \leftrightarrow \psi(a_i)$. Finally, the claim follows because $a_1 \equiv_M a_2$.

1⇒2. Let $\mathcal{W}$ be as in Proposition 14.12. Then the set $\varphi(\mathcal{W}^x\,;b)$ is invariant under $\mathrm{Aut}(\mathcal{W}/M)$. As $\mathcal{W}$ is saturated in the language expanded with predicate for $\varphi(\mathcal{W}^x\,;b)$, invariance implies that $\varphi(\mathcal{W}^x\,;b)$ is definable over M. □

14.15 Remark There are theories where the stationarity of $\downarrow_M$ holds for some particular M. Indeed, if every subset of M^n is M-definable then $\downarrow_M$ is n-stationary by Proposition 14.14. This will help in the proof of Theorem 15.18. For a natural example, let $T = T_{\mathrm{dlo}}$ and let $M \subseteq \mathcal{U}$ have the order type of $\mathbb{R}$. By quantifier elimination every definable subset of $\mathcal{U}$ is a union of finitely many intervals. By Dedekind completeness, the trace on M of any interval of $\mathcal{U}$ coincides with that of an M-definable interval. Therefore $\downarrow_M$ is 1-stationary.

14.16 Exercise Let T be strongly minimal. Let $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^z$, where x and z are finite tuples. Prove that the following are equivalent

1. $a \downarrow_M b$
2. $\dim(b/M) = \dim(b/M, a)$
3. $b \downarrow_M a$.

14.17 Exercise Let $a \downarrow_M b$ then there is $\mathcal{V} \preceq \mathcal{U}$ that is isomorphic to $\mathcal{U}$ over M, a and such that $\mathcal{V} \downarrow_M b$. Prove that we can also require that every type over $\mathcal{V}, b$ with $< \kappa$ parameters that is finitely satisfiable in $\mathcal{V}$ is realized in $\mathcal{V}$.

14.18 Exercise Let $a \downarrow_M b$. Prove that for every c there is $b' \equiv_{M,a} b$ such that $a, c \downarrow_M b'$.

14.19 Exercise Let $N \succeq M$ be saturated and of cardinality $> |M|$. Let $a_i \downarrow_M N, b$, for $i = 1, 2$. Prove that if $a_1 \equiv_N a_2$ then $a_1 \equiv_{N,b} a_2$.

14.20 Exercise Show that in general there is no type $p(x\,;z) \subseteq L(M)$ such that

$$a \mathop{\smile\!\!\!\!\!\!\downarrow}_{M} b \quad\Leftrightarrow\quad a\,;b \vDash p(x\,;z).$$

Hint: consider the theory of dense linear orders without endpoints.

14.3 Morley sequences and indiscernibles

In what follows α is some ordinal $\leq \kappa$, typically ω, and x is a tuple of variables of length $< \kappa$. Let $p(x) \in S(\mathfrak{U})$ be a global type. We say that $\bar{c} = \langle c_i : i < \alpha \rangle$ is a Morley sequence of $p(x)$ over A if for every $i < \alpha$

Ms. $\qquad c_i \quad\vDash\quad p_{\restriction A, c_{\restriction i}}(x).$

When $p(x)$ is finitely satisfiable in A, we say that $\bar{c}$ is a coheir sequence of $p(x)$ over A. When we say that $\bar{c}$ is a coheir sequence over A (with no explicit reference to a global type), we mean that *there is* a type $p(x) \in S(\mathfrak{U})$ that is finitely satisfiable in A such that $\bar{c}$ is a coheir sequence of $p(x)$.

The following is a convenient characterization of coheir sequences.

14.21 Lemma The following are equivalent

1. $\bar{c} = \langle c_n : n < \omega \rangle$ is a coheir sequence over M
2. $c_n \mathop{\smile\!\!\!\!\!\!\downarrow}_M c_{\restriction n}$ and $c_{n+1} \equiv_{M, c_{\restriction n}} c_n$ for every $n < \omega$.

Proof. 1$\Rightarrow$2. Assume 1 and let $p(x) \in S(\mathfrak{U})$ be a global type that is finitely satisfiable in M and such that $c_i \vDash p_{\restriction M, c_{\restriction i}}(x)$. The requirement $c_{n+1} \equiv_{M, c_{\restriction n}} c_n$ is clear. Now, suppose $\varphi(c_{n+1})$ for some $\varphi(x) \in L(M, c_{\restriction n+1})$. Then $\varphi(x)$ belongs to $p(x)$, so $\varphi(\mathfrak{U}^x) \cap M^x \neq \varnothing$ because $p(x)$ is finitely satisfiable in M. This proves $c_n \mathop{\smile\!\!\!\!\!\!\downarrow}_M c_{\restriction n}$.

2$\Rightarrow$1. Let $q(x) = \{\varphi(x) \in L(M, \bar{c})\ :\ \varphi(c_n)$ holds for cofinitely many $n\}$. We claim that $q(x)$ is finitely satisfiable in M. Let $\varphi(x\,;z) \in L(M)$ be such that $\varphi(x\,;c_{\restriction n}) \in q$. By the definition of $q(x)$, the formula $\varphi(c_m\,;c_{\restriction n})$ holds for all sufficiently large m. Hence, from 2 we infer $c_m \mathop{\smile\!\!\!\!\!\!\downarrow}_M c_{\restriction n}$ and conclude that $\varphi(x\,;c_{\restriction n})$ is satisfied in M.

Let $p(x)$ be any global extension of $q(x)$ finitely saisfied in M. We prove that $\bar{c}$ is a Morley sequence of $p(x)$ over M. By 2 either $c_m \vDash p_{\restriction M, c_{\restriction n}}(x)$ for all $m \geq n$ or $c_m \nvDash p_{\restriction M, c_{\restriction n}}(x)$ for all $m \geq n$. As $p(x)$ extends $q(x)$, the latter cannot occur. $\square$

Let $(I, <_I)$ be a linear order. A function $\bar{a} : I \to \mathfrak{U}^x$ is said to be an *I*-sequence, or simply a sequence when I is clear. We will often introduce an I-sequence as $\bar{a} = \langle a_i : i \in I \rangle$.

If $I_0 \subseteq I$ we call $a_{\restriction I_0}$ a subsequence of $\bar{a}$. The subsets $I_0 \subseteq I$ that are well-ordered by $<_I$, in particular the finite ones, are especially relevant. When I_0 has order type α, an ordinal, we identify $a_{\restriction I_0}$ with a tuple of length α.

Recall that $I^{(n)}$ denotes that the set of n-subsets of I, i.e. the subsets of I of cardinality n. The notation

$$\binom{I}{n} \quad=\quad I^{(n)}$$

is also common.

14.22 Definition Let $(I, <_I)$ be an infinite linear order and let $\bar{a}$ be an I-sequence. We say that $\bar{a}$ is a sequence of indiscernibles over A or, an A-indiscernible sequence, if $a_{\restriction I_0} \equiv_A a_{\restriction I_1}$ for every $I_0, I_1 \in I^{(n)}$ and $n < \omega$.

The indiscernibility condition can be formulated in a number of equivalent ways. For example, we can require that, for every formula $\varphi(x_1, \ldots, x_n) \in L(A)$ and every pair of tuples in I^n such that $i_0 < \cdots < i_n$ and $j_0 < \cdots < j_n$,

$$\varphi(a_{i_0}, \ldots, a_{i_n}) \quad \leftrightarrow \quad \varphi(a_{j_0}, \ldots, a_{j_n})$$

Alternatively, we can simply say that for all $i_0, \ldots, i_n \in I$ the type $\operatorname{tp}(a_{i_0}, \ldots, a_{i_n}/A)$ only depends on the order type of $i_0, \ldots, i_n$.

14.23 Proposition Let $p(x) \in S(\mathfrak{U})$ be a global A-invariant type and let $\bar{c} = \langle c_i : i < \alpha \rangle$ be a Morley sequence of $p(x)$ over A. Then $\bar{c}$ is an A-indiscernibles sequence.

Proof. We prove by induction on $n < \omega$ that

$$\sharp \qquad c_{\restriction n} \;\equiv_A\; c_{\restriction I_0} \quad \text{for every } I_0 \subseteq \alpha \text{ of cardinality } n.$$

For $n = 0$ the claim is trivial. We assume inductively that $\sharp$ above is true and prove that

$$c_{\restriction n}, c_n \;\equiv_A\; c_{\restriction I_0}, c_i \quad \text{for every } I_0 < i < \alpha.$$

As $\bar{c}$ is Morley sequence, $c_n \equiv_{A, c_{\restriction n}} c_i$ whenever $n < i$. Hence we can equivalently prove that

$$c_{\restriction n}, c_i \;\equiv_A\; c_{\restriction I_0}, c_i,$$

which is equivalent to

$$c_{\restriction n} \equiv_{A, c_i} c_{\restriction I_0}.$$

The latter holds by induction hypothesis $\sharp$ and the invariance of $p(x)$ as formulated in it3 of Section 14.1. □

14.4 Product of types

We will often need to combine two global invariant types $p(x)$ and $q(z)$ into a single global invariant type $r(x, z)$ that extends both $p(x)$ and $q(z)$.

14.24 Fact Let $p(x), q(z) \in S(\mathfrak{U})$ be global types invariant over A. Let M be a sufficiently saturated model containing A. Then there is a unique A-invariant global type $r(x, z) \in S(\mathfrak{U})$ containing $p(x) \cup q(z)$ and the formulas $\varphi(x\,;z) \in L(M)$ such that $\varphi(x\,;b) \in p$, for some (equivalently, for every) $b \vDash q(z){\restriction}M$. This type is independent of the choice of M.

Proof. First, note that the following type is consistent and contains both $p(x)$ and $q(z)$ (this is straightforward to check)

$$s(x, z) \quad = \quad \{\varphi(x\,;z) \in L(M) \;:\; \varphi(x\,;b) \in p \text{ for all } b \vDash q(z){\restriction}M\}.$$

We claim that $s(x, z)$ is complete over M. Suppose not. Then there is a formula $\varphi(x\,;z) \in L(M)$ neither $\varphi(x\,;z)$ nor $\neg\varphi(x\,;z)$ belongs to $s(x, z)$. Then for some

$b, b' \models q(z){\restriction}M$ such that $\neg\varphi(x\,;b) \in p$ and $\varphi(x\,;b') \in p$. As $q(z)$ is complete, $b \equiv_M b'$. This contradicts the invariance of $p(x)$ over M and proves the claim.

Finally, we claim that $s(x,z)$ is invariant in the sense of Exercise 14.2. Suppose not. Then $\psi(x\,;z\,;c), \neg\psi(x\,;z\,;c') \in s$ for some $\psi(x\,;z\,;y) \in L$ and some $c, c' \in M^y$ such that $c \equiv_A c'$. Then $\psi(x\,;b\,;c), \neg\psi(x\,;b\,;c') \in p$ for some $b \models q(z){\restriction}M$. Let $f \in \operatorname{Aut}(\mathcal{U}/A)$ be such that $f[M] = M$ and $f\,c = c'$. This exists by the saturation of M. By the invariance of $p(x)$ over A, it follows that $\psi(x\,;f\,b\,;c') \in p$. By the invariance of $q(z)$ over A, we have $f\big(q(z){\restriction}M\big) = q(z){\restriction}M$, so $f\,b \models q(z){\restriction}M$. In particular $b \equiv_M f\,b$. Then $\psi(x\,;f\,b\,;c'), \neg\psi(x\,;b\,;c') \in p$ contradicts the invariance of $p(x)$ over M and proves the claim.

Now, by Exercise 14.2, there is a unique global type $r(x,z)$ that extends $s(x,z)$ and is invariant over A. Again by Exercise 14.2, $r(x,z)$ extends both $p(x)$ and $q(z)$. □

14.25 Definition Let $p(x)$ and $q(z)$ be global types invariant over A. The type $r(x,z)$ defined in the above fact is called the product of $p(x)$ and $q(z)$ and is denoted by $p(x) \otimes q(z)$.

14.26 Fact Let $p_i(x_i)$, for $i = 1,2,3$, be global types invariant over A. Then

$$\Big(p_1(x_1) \otimes p_2(x_2)\Big) \otimes p_3(x_3) \quad = \quad p_1(x_1) \otimes \Big(p_2(x_2) \otimes p_3(x_3)\Big).$$

Let $p(x)$ be a global type invariant over A. Let $\bar{x} = \langle x_i : i < \omega\rangle$ be a tuple of copies of x. Define $p^{(0)}(x_0) = p(x_0)$ and inductively

$$p^n(x_0,\dots,x_n) \quad = \quad p(x_n) \otimes p^{(n-1)}(x_0,\dots,x_{n-1}).$$

Finally set

$$p^{(\omega)}(\bar{x}) \quad = \quad \bigcup_{n<\omega} p^{(n)}(x_0,\dots,x_{n-1}).$$

14.27 Exercise Let $p(x)$ be a global type invariant over A. Prove that any $\bar{a} = \langle a_i : i < \omega\rangle$ that realizes $p^{(\omega)}(\bar{x})$ is a Morley sequence of $p(x)$ over A.

Chapter 15

Ramsey theory

In Section 15.1 we prove Ramsey's theorem and in Section 15.2 we present its major application in model theory: the Ehrenfeucht-Mostowski construction of indiscernibles.

In the remaining sections we prove two important results of Ramsey theory. These results will not be used elsewhere in these notes. Our only purpose is to illustrate a concrete (relatively speaking) application of the notion of coheir.

15.1 Ramsey's theorem from coheir sequences

In this chapter we are interested in finite partitions. We may represent the partition of a set X into k subsets with a map $f : X \to (k]$. The elements of $(k] = \{1, \ldots, k\}$ are also called colors, and the partition a coloring, or k-coloring, of X. We say that $Y \subseteq X$ is monochromatic if $f_{\restriction Y}$ is constant on Y.

Let M be an arbitrary infinite set. Fix $n, k < \omega$ and fix a coloring f of the set of all n-subsets of M, also known as the complete n-uniform hypergraph with vertex set M,

$$f \quad : \quad \binom{M}{n} = M^{(n)} \quad \to \quad (k].$$

We say that $H \subseteq M$ is a monochromatic set if f is constant on $H^{(n)}$. In combinatorics, monochromatic sets are also called homogeneous sets.

The following is a very famous theorem which we prove here in an unusual (and overly convoluted) way. Our purpose is to illustrate the method of proof of Theorem 15.18. In the latter the model theoretic overhead really pays off.

15.1 Ramsey Theorem Let M be an infinite set and fix some positive integers n and k. Fix an arbitrary k-coloring of $M^{(n)}$. Then there is an infinite monochromatic set $H \subseteq M$.

Proof. Let L be a language that contains k relation symbols $r_1, \ldots, r_k$ of arity n. Given a k-coloring f we define a structure with domain M. The interpretation the relation symbol r_i is

$$r_i^M \quad = \quad \Big\{ a \in M^n \ : \ |\operatorname{rng}(a)| = n \ \text{ and } \ f\big(\operatorname{rng}(a)\big) = i \Big\}.$$

We may assume that M is an elementary substructure of some large saturated model $\mathcal{U}$. Let $q(x)$, where $|x| = 1$, be the type that says that x is not in M. Let $p(x) \in S(\mathcal{U}^x)$ be an extension of $q(x)$ that is finitely satisfied in M. Finally, let $\bar{c} = \langle c_i : i < \omega \rangle$ be a coheir sequence of $p(x)$ over M.

There is a first-order sentence saying that the formulas $r_i(x_1, \ldots, x_n)$ are a coloring of $M^{(n)}$. Then by elementarity the same holds in $\mathcal{U}$. By indiscernibility, all tuples of n distinct elements of $\bar{c}$ have the same color, say 1.

The theorem is proved if we can find in M a sequence $\bar{a} = \langle a_i : i < \omega\rangle$ with color 1. We construct $a_{\restriction i}$ by induction on i as follows.

Assume as induction hypothesis that the subsequences of length n of $a_{\restriction i}, c_{\restriction n}$ have all color 1. Our goal is to find $a_i \in M$ such that the same property holds for $a_{\restriction i}, a_i, c_{\restriction n}$. By the indiscernibility of $\bar{c}$, the property holds for $a_{\restriction i}, c_{\restriction n}, c_n$, and this can be written by a formula $\varphi(a_{\restriction i}, c_{\restriction n}, c_n)$. As $\bar{c}$ is a coheir sequence, by Lemma 14.21 we can find $a_i \in M$ such that $\varphi(a_{\restriction i}, c_{\restriction n}, a_i)$. So, as the order is irrelevant, $a_{\restriction i}, a_i, c_{\restriction n}$ satisfies the induction hypothesis. □

15.2 Exercise Let M be an arbitrary model. Let $\varphi(x,y) \in L(M)$, where $|x| = |y| = 1$. Prove that there is a sequence $\langle a_i : i < \omega\rangle$ in M such that

$$\varphi(a_i, a_j) \leftrightarrow \varphi(a_h, a_k)$$

for every $i < j < \omega$ and $h < k < \omega$.

15.3 Exercise Let M be a graph with the property that for every finite $A \subseteq M$ there is a $c \in M$ such that $A \subseteq r(c, M)$. The edges of M are colored with infinitely many colors but every clique contains at most k colors, for some fixed $k \in \omega$. Prove that M has an infinite monochromatic complete subgraph.

15.2 The Ehrenfeucht-Mostowski theorem

Let $I, <_I$ be an infinite linear order. To minimize notation, we assume throughout the section that I has no largest element. It is immediate how to tweak the definition below so that it makes sense without this assumption.

Let $\bar{a}$ be an I-sequence. Fix a tuple of variables $\bar{x} = \langle x_i : i < \omega\rangle$. We write $p(\bar{x}) = \text{EM-tp}(\bar{a}/A)$ and say that $p(\bar{x})$ is the Ehrenfeucht-Mostowski type of $\bar{a}$ over A if

$$p(\bar{x}) \quad = \quad \Big\{\varphi(\bar{x}) \in L(A) \ : \ \varphi(a_{\restriction I_0}) \text{ holds for every } I_0 \in I^{(\omega)}\Big\}.$$

Where $I^{(\omega)}$ is the set of subsets of I ordered as ω. As $\bar{x}$ is of order-type ω, the expression $\varphi(a_{\restriction I_0})$ is sound.

Note that if $\bar{a}$ is A-indiscernible then $\text{EM-tp}(\bar{a}/A)$ is a complete type, and vice versa.

15.4 Remark Let $I, <_I$ and $J, <_J$ be infinite linear orders with no largest element. Let $\bar{c} = \langle c_i : i \in I\rangle$ be a sequence of A-indiscernibles. By compactness, there is a sequence of A-indiscernibles $\bar{a} = \langle a_j : j \in J\rangle$ such that $\text{EM-tp}(\bar{a}/A) = \text{EM-tp}(\bar{c}/A)$.

Finally, we can state the Ehrenfeucht-Mostowski theorem.

15.5 Ehrenfeucht-Mostowski Theorem Let $I, <_I$ be an infinite linear order with no largest element. Then for every sequence $\bar{a} = \langle a_i : i \in I\rangle$ there is an A-indiscernible sequence $\bar{c} = \langle c_i : i < \omega\rangle$ such that $\text{EM-tp}(\bar{a}/A) \subseteq \text{tp}(\bar{c}/A)$.

Proof. Let $q(\bar{x}) = \text{EM-tp}(\bar{a}/A)$. Any realization of the following type is a sequence of A-indiscernibles $\bar{c}$ as required by the theorem.

$$q(\bar{x}) \quad \cup \quad \Big\{\varphi(x_{\restriction I_0}) \leftrightarrow \varphi(x_{\restriction J_0}) \ : \ \varphi(\bar{x}) \in L(A), \ I_0, J_0 \in \omega^{(\omega)}\Big\}.$$

We will prove that any finite subset of the type above is realized by a subsequence of $\bar{a}$. First note that $q(\bar{x})$ is realized by $a_{\restriction H}$ for every $H \subseteq I^{(\omega)}$. Then we only need to pay attention to the set on the right.

We prove that for k and n arbitrarily large and every $\varphi_1(\bar{x}), \dots, \varphi_k(\bar{x}) \in L(A)$ there is an infinite $H \subseteq \omega$ such that $a_{\restriction H}$ realizes

$$\Big\{\varphi_i(x_{\restriction I_0}) \leftrightarrow \varphi_i(x_{\restriction J_0}) \ : \ I_0, J_0 \in \omega^{(\omega)} \text{ and } i \in (k]\Big\}.$$

Let n be larger than any i such that x_i occurs in $\varphi_1(\bar{x}), \dots, \varphi_k(\bar{x})$. Consider the subsets of $(k]$ as colors and let f be the coloring of $I^{(n)}$ that maps I_0 to the set $\{i \ : \ \varphi_i(a_{\restriction I_0})\}$. By the Ramsey theorem, there is some infinite monochromatic set $H \subseteq I$ with the order type of ω. Hence

$$\Big\{\varphi_i(a_{\restriction I_0}) \leftrightarrow \varphi_i(a_{\restriction J_0}) \ : \ I_0, J_0 \in H^{(n)} \text{ and } i \in (k]\Big\}.$$

As H has order type ω, it is immediate that $a_{\restriction H}$ reøalizes # as required. □

15.6 Proposition Let $\bar{a} = \langle a_i : i \in I\rangle$ be a sequence of A-indiscernibles. Then $\bar{a}$ is indiscernible over some model M containing A.

Proof. Fix a model M containing A. By Theorem 15.5 and the remark there is an I-sequence of M-indiscernibles $\bar{c}$ such that $\text{EM-tp}(\bar{a}/M) \subseteq \text{EM-tp}(\bar{c}/M)$. As $\bar{a}$ is an A-indiscernible sequence $\bar{a} \equiv_A \bar{c}$. Therefore $h\,\bar{c} = \bar{a}$ for some $h \in \text{Aut}(\mathcal{U}/A)$. Hence $\bar{a}$ is indiscernible over $h[M]$. □

15.7 Exercise Let $I, <_I$ and $J, <_J$ be infinite linear orders with no largest element. Prove that for every sequence $\bar{a} = \langle a_i : i \in I\rangle$ there is an A-indiscernible sequence $\bar{c} = \langle c_i : i \in J\rangle$ such that $\text{EM-tp}(\bar{a}/A) \subseteq \text{EM-tp}(\bar{c}/A)$.

15.8 Exercise Let $\bar{a} = \langle a_i : i \in I\rangle$ be an A-indiscernible sequence with no largest element and let J with $|J| \le \kappa$ be a linear order extending I. Prove that there is an A-indiscernible sequence that extends $\bar{a}$. I.e. some $\bar{c} = \langle c_i : i \in J\rangle$ such that $c_{\restriction I} = \bar{a}$.

15.9 Exercise Let $p(x) \in S(\mathcal{U})$ be a global type invariant over A. Let $a, b \vDash p_{\restriction A}(x)$. Prove that there is a sequence $\bar{c} = \langle c_i : i < \omega\rangle$ such that $a, \bar{c}$ and $b, \bar{c}$ are both sequences of A-indiscernibles.

15.10 Exercise Let $\langle c_i : i < \omega\rangle$ be an indiscernible sequence. Prove that there is an indiscernible sequence $\langle d_i : i < \omega\rangle$ such that $d_0, d_1 = c_1, c_0$.

15.11 Exercise Let $\langle \mathcal{D}_i : i < \omega\rangle$ be an A-indiscernible sequence in $\mathcal{U}^{\text{eq}}$. Prove that there is a formula $\varphi(x\,;z) \in L$ and an A-indiscernible sequence $\langle b_i : i < \omega\rangle$ in $\mathcal{U}^z$ such that $\mathcal{D}_i = \varphi(\mathcal{U}\,;b_i)$.

15.3 Idempotent orbits in semigroups

In this and the following sections we focus on semigroups definable in a first-order structure. For a lighter notation, we assume that the semigroup operation $\cdot$ is among the symbols of L and that the whole monster model is a semigroup $\mathcal{S} = \mathcal{U}$.

In this section we work over a given model $S \preceq \mathcal{S}$. For any two sets $\mathcal{A}, \mathcal{B} \subseteq \mathcal{S}$ we define

$$\mathcal{A} * \mathcal{B} \quad = \quad \Big\{ a{\cdot}b \ :\ a \in \mathcal{A},\ b \in \mathcal{B} \text{ and } a \mathop{\downarrow}_{S} b \Big\}$$

In this and the next section we abbreviate $\mathcal{O}(a/S)$, the orbit of a under $\mathrm{Aut}(\mathcal{U}/S)$, by $(a)_S$. When ajacent to $*$, we write a for $(a)_S$. E.g. we write $\mathcal{A} * b$ for $\mathcal{A} * (b)_S$ and $a * b$ for $(a)_S * b$.

15.12 Proposition If $\mathcal{A}$ is type definable over S then so is $\mathcal{A} * b$ for any b.

Proof. Let $p(x\,;z)$ the type in Remark 14.8. Then the set $\mathcal{A} * b$ is defined by the type $q(y) = \exists x, z\ \big[y = x{\cdot}z \ \wedge\ x \in \mathcal{A} \ \wedge\ p(x\,;z)\big]$. □

From the invariance of $\downarrow_S$ it follows immediately that $f[\mathcal{A} * \mathcal{B}] = f[\mathcal{A}] * f[\mathcal{B}]$ for every $f \in \mathrm{Aut}(\mathcal{U}/S)$. Therefore, when $\mathcal{A}$ and $\mathcal{B}$ are invariant over S, also $\mathcal{A} * \mathcal{B}$ is invariant over S. Below we are mostly concerned with sets that are type-definable, therefore invariant, over S.

15.13 Proposition For every S-invariant sets $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$

$$\mathcal{A} * (\mathcal{B} * \mathcal{C}) \quad \subseteq \quad (\mathcal{A} * \mathcal{B}) * \mathcal{C}$$

Proof. Let $a{\cdot}b{\cdot}c$ be an arbitrary element of the l.h.s. where $a \downarrow_S b{\cdot}c$ and $b \downarrow_S c$. By extension (Lemma 14.9), there exists a' such that $a \equiv_{S,b\cdot c} a' \downarrow_S b{\cdot}c, b, c$. By transitivity (again Lemma 14.9), $a'{\cdot}b \downarrow_S c$. Therefore $a'{\cdot}b{\cdot}c$ belongs to the r.h.s. Finally, as $a' \equiv_{S,b\cdot c} a$, also $a{\cdot}b{\cdot}c$ belongs to the r.h.s. by S-invariance. □

We show that, under the assumption of stationarity, $*$ is an operation on $\mathcal{U}^x/{\equiv_S}$.

15.14 Proposition Assume $\downarrow_S$ is stationary, see Definition 14.13. Fix $a \downarrow_S b$ arbitrarily. Then $a'{\cdot}b' \equiv_S a{\cdot}b$ for every $a' \equiv_S a$ and $b' \equiv_S b$ such that $a' \downarrow_S b'$. Or, in other words, $(a{\cdot}b)_S = a * b$.

Proof. By invariance we can assume $b = b'$. By stationarity, $a \equiv_{S,b} a'$. Hence $a{\cdot}b \equiv_S a'{\cdot}b$. □

15.15 Corollary (associativity) Assume $\downarrow_S$ is stationary. Then for every S-invariant sets $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$.

$$\mathcal{A} * (\mathcal{B} * \mathcal{C}) \quad = \quad (\mathcal{A} * \mathcal{B}) * \mathcal{C}$$

Proof. We can assume that $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ are S-orbits. Say of a, b, and c respectively. As we are working over a model, we can assume that $a \downarrow_S b{\cdot}c$ and $b \downarrow_S c$. By Proposition 15.14 the set on the l.h.s. equals $(a{\cdot}b{\cdot}c)_S$. By a similar argument the set on the r.h.s. equals $(a'{\cdot}b'{\cdot}c')_S$ for some elements a', b', and c'. Proposition 15.13 proves that inclusion $\subseteq$ holds in general. But inclusion between orbits amounts to equality. □

Let $\mathcal{A}$ be a nonempty set. When $\mathcal{A} * \mathcal{A} \subseteq \mathcal{A}$, we say that $\mathcal{A}$ is idempotent. We say that $a \in \mathcal{S}$ is idempotent if it has idempotent orbit. Note that if $\mathcal{A}$ is idempotent and $\mathcal{B} \subseteq \mathcal{A}$, then also $\mathcal{A} * \mathcal{B}$ is idempotent.

15.16 Lemma Assume $\downfree_S$ is stationary. If $\mathcal{A}$ is minimal among the idempotent sets that are type-definable over S, then $\mathcal{A} = (b)_S$ for $b \in \mathcal{A}$.

Proof. Pick arbitrarily some $b \in \mathcal{A}$. The set $\mathcal{A} * b$ is contained in $\mathcal{A}$, idempotent and type-definable over S by Proposition 15.12. Therefore by minimality $\mathcal{A} * b = \mathcal{A}$. Let $\mathcal{A}' \subseteq \mathcal{A}$ contain those a such that $a * b = (b)_S$. This set is nonempty because $b \in \mathcal{A} * b$. It is easy to verify that $\mathcal{A}'$ is type-definable over S, b. As it is clearly invariant over S, it is type-definable over S. By associativity, it is idempotent. Hence, by minimality, $\mathcal{A}' = \mathcal{A}$. Then $b \in \mathcal{A}'$, which implies $b * b = (b)_S$. That is, b is idempotent. Finally, by minimality, $\mathcal{A} = (b)_S$. □

15.17 Corollary Under the same assumptions of the lemma above, every type-definable idempotent set contains an idempotent element.

15.4 Hindman's theorem

In this section we merge the theory of idempotents presented in Section 15.3 with the proof of Ramsey's theorem to obtain Hindman's theorem in a straightforward way.

Let $\bar{a}$ be a tuple of elements of $\mathcal{U}^x$ of length $\leq \omega$. We write $\operatorname{fp} \bar{a}$ for the set of finite products of elements of $\bar{a}$ taken in increasing order. Namely,

$$\operatorname{fp} \bar{a} \quad = \quad \Big\{ a_{i_0} \cdot \dots \cdot a_{i_k} \;\; : \;\; i_0 < \dots < i_k < |\bar{a}|, \;\; k < |\bar{a}| \Big\}.$$

Let $\prec$ be a relation on $\mathcal{U}$. Let $\mathcal{A}, \mathcal{C} \subseteq \mathcal{U}$. We say that $\mathcal{A}$ is $\prec$-covered by $\mathcal{C}$ if for every $a_1, \dots, a_n \in \mathcal{A}$ there are infinitely many $c \in \mathcal{C}$ such that $a_i \prec c$ for all i. When $\mathcal{A} = \mathcal{C}$ we simply sat that $\mathcal{A}$ is $\prec$-covered. We say that $\mathcal{A}$ is $\prec$-closed if $a \prec b \prec c$ implies $a \prec b \cdot c$ for all $a, b, c \in \mathcal{A}$. A $\prec$-chain in $\mathcal{U}$ is a tuple $\bar{a} \in \mathcal{U}^{\leq \omega}$ such that $a_i \prec a_{i+1}$.

The requirements on $\prec$ are hardly restrictive. For example, on a free semigroup we can take the preorder relation given by the length of the words. Or, on any semigroup S, we could take the trivial relation S^2 –the theorem below would remain nontrivial.

15.18 Hindman's Theorem Let $\prec$ be a relation on a semigroup S. Assume that S is $\prec$-closed and $\prec$-covered. Then for every finite coloring of S there is an infinite $\prec$-chain $\bar{a}$ such that $\operatorname{fp} \bar{a}$ is monochromatic.

This implies that every commutative semigroup S has an infinite monochromatic subset closed under finite sums of distinct elements (order S arbitrarily).

Our proof follows closely the proof of Ramsey's theorem 15.1. The novelty is all in Lemma 15.16.

Proof. We interpret S as a structure in a language that extends the natural language of semigroups with a symbol for $\prec$ and one for each subset of S. Let $\mathcal{S}$ be a saturated elementary superstructure of S. As observed in Remark 14.15, the language makes $\downfree_S$ trivially stationary.

We write $\mathcal{S}'$ for the type-definable set $\{x : S \prec x\}$, which is nonempty because S is $\prec$-covered. We claim that $\mathcal{S}'$ is idempotent. In fact, if $a, b \in \mathcal{S}'$ then, as $S \prec a, b$ and $a \mathop{\smile\!\!\!\!\!\!\;|}_S b$, we must have that $a \prec b$. Therefore, from the $\prec$-closure of $\prec$ we infer $a{\cdot}b \in \mathcal{S}'$.

Let g_0 be an idempotent element of $\mathcal{S}'$ as given by Corollary 15.17. Let g_1 realize $g_0 \equiv_S x \mathop{\smile\!\!\!\!\!\!\;|}_S g_0$. Assume for definiteness that g_0 has color 1. Now, we define $\bar{a} \in S^{\omega}$ inductively such that all elements of fp $\bar{a}$ have color 1.

Assume inductively that we have $a_{\restriction i} \in S^i$ such that all elements of $\mathrm{fp}(a_{\restriction i}, g_0)$ have color 1. Our goal is to find a_i such that the same property holds for $\mathrm{fp}(a_{\restriction i+1}, g_0)$.

Note that $\mathrm{fp}(a_{\restriction i}, g_1, g_0)$ has color 1. In fact, $g_1 \equiv_S g_0 \equiv_S g_1{\cdot}g_0$. Now, $g_1 \mathop{\smile\!\!\!\!\!\!\;|}_S g_0$ yields some $a_i \in S$ such that $\mathrm{fp}(a_{\restriction i+1}, g_0)$ has color 1. □

15.19 Exercise The following claim is too strong to be true. For every finite coloring of $\mathbb{N}^{(2)}$ there is an infinite sequence $\bar{a} = \langle a_i : i < \omega \rangle$ in $\mathbb{N}$ such that $\mathrm{fp}(\bar{a})^{(2)}$ is monochromatic. Replace $\mathrm{fp}(\bar{a})^{(2)}$ with a (slightly) smaller set that makes the claim true.

15.5 The Hales-Jewett Theorem

The Hales-Jewett Theorem is a purely combinatorial statement that implies the van der Waerden Theorem.

Throughout this section we assume that $\mathop{\smile\!\!\!\!\!\!\;|}_S$ is stationary.

We work with the same notation as Section 15.3. We need a few definitions. Let $\mathcal{C} \subseteq \mathcal{S}$ be an idempotent set. If $\varnothing \neq \mathcal{A} \subseteq \mathcal{C}$ is such that $\mathcal{C} * \mathcal{A} \subseteq \mathcal{A}$, we say that $\mathcal{A}$ is a left ideal of $\mathcal{C}$. When $\mathcal{C}$ is clear we omit reference to it. A minimal left ideal is one that does not properly contain any other left ideal. The following fact is immediate.

15.20 Fact The following are equivalent for any $\mathcal{M}$ that is a left ideal of $\mathcal{C}$

1. $\mathcal{M}$ is minimal
2. $\mathcal{M} = \mathcal{M} * b$ for every $b \in \mathcal{M}$
3. $\mathcal{M} = \mathcal{C} * b$ for every $b \in \mathcal{M}$

Moreover, when b is not in $\mathcal{M}$, we have the following.

15.21 Fact Let $\mathcal{M}$ be a minimal left ideal of $\mathcal{C}$. Then $\mathcal{M} * b$ is a minimal left ideal for any $b \in \mathcal{C}$.

Proof. Clearly, $\mathcal{M} * b$ is a left ideal. Let $a * b$, for some $a \in \mathcal{M}$, be an arbitrary element of $\mathcal{M} * b$. By Fact 15.20, it sufficies to prove that $\mathcal{M} * b * a * b = \mathcal{M} * b$. This holds, in fact $b * a \in \mathcal{M}$, hence $\mathcal{M} * b * a = \mathcal{M}$ by the minimality of $\mathcal{M}$. □

Now we prove the existence of minimal left ideals.

15.22 Fact Let $\mathcal{C}$ be a type-definable idempotent set. Then $\mathcal{C}$ contains a minimal left ideal which is type-definable.

Proof. Construct inductively a descending chain of type-definable left ideals $\mathcal{M}_i$ where $\mathcal{M}_0 = \mathcal{C}$ and $\mathcal{M}_{i+1} = \mathcal{M}_i * b_i$ for some $b_i \in \mathcal{M}_i$. By compactness the chain stabilizes at some ordinal yielding some type-definable left ideal $\mathcal{M}_\alpha$ which is minimal as it satisfies 2 in the above fact. □

It is also interesting to note that we can define, in the obvious way, *right ideals* and even *two-sided ideals.* But note the proof of the above fact would not apply because, in general, $b * \mathcal{C}$ is not type-definable. However for two-sided ideals we have the following remarkable fact (which is not needed in the sequel).

15.23 Fact Let $\mathcal{C}$ be a type-definable idempotent set. Define

$$\mathcal{K} \quad = \quad \bigcup \{\mathcal{M} \ : \ \mathcal{M} \text{ a minimal left ideal of } \mathcal{C}\}$$

Then $\mathcal{K}$ is a minimal two-sided ideal.

Proof. Clearly, $\mathcal{K}$ is a left ideal. To prove that $\mathcal{K}$ is also a right ideal note that, by Fact 15.21, for every $a \in \mathcal{C}$

$$\mathcal{K} * a \quad = \quad \bigcup \{\mathcal{M} * a \ : \ \mathcal{M} \text{ a minimal left ideal of } \mathcal{C}\} \quad \subseteq \quad \mathcal{K}.$$

As for minimality, it sufficies to prove that any two-sided ideal $\mathcal{H}$ contains any minimal left ideal $\mathcal{M}$. Note that $\mathcal{H} \cap \mathcal{M}$ is not empty in fact, as $\mathcal{H}$ is in particular a right ideal $\mathcal{H} * \mathcal{M} \subseteq \mathcal{H} \cap \mathcal{M}$. As $\mathcal{H} \cap \mathcal{M}$ is in particular a left ideal and it is contained in $\mathcal{M}$, by minimality, it coincides with $\mathcal{M}$. □

Note that every left ideal is an idempotent set and therefore contains idempotent elements.

15.24 Lemma Let $\mathcal{C}$ be a type-definable idempotent set. Let $\mathcal{M}$ be a minimal left ideal of $\mathcal{C}$. Let $u, v \in \mathcal{M}$ be two idempotents. Then the followings hold (to avoid notation overload, we confuse the elements of $\mathcal{C}$ with their orbits)

1. $a * u = a$ for every $a \in \mathcal{M}$
2. $*$ is a group operation on $u * \mathcal{M}$ with identity u
3. $a \mapsto v * a$ define a group isomorphism between $u * \mathcal{M}$ and $v * \mathcal{M}$
4. if u and v are distinct (i.e. have distict orbits), then $u * \mathcal{M}$ and $v * \mathcal{M}$ are disjoint
5. $\mathcal{M}$ is covered by sets of the form $u * \mathcal{M}$ as $u \in \mathcal{M}$ ranges over the idempotents
6. if $b \in \mathcal{C}$ is an idempotent, then $b * \mathcal{M} * b$ contains an idempotent.

Proof. 1. By minimality $\mathcal{M} * u = \mathcal{M}$. Therefore $a' * u = a$ for some $a' \in \mathcal{M}$. Then $a * u = a' * u * u = a$.

2. By 1 and idempotency, u is an identity of $u * \mathcal{M}$. It suffices to prove the existence of a left inverse (in fact, by algebra, this is also a right inverse). Pick any $a \in u * \mathcal{M}$. From $\mathcal{M} * a = \mathcal{M}$ we obtain that $u * \mathcal{M} * a = u * \mathcal{M}$. Then $u * a' * a = u$ for some $a' \in \mathcal{M}$. This proves that $u * a'$ is the left inverse of a.

3. By 1, the map $a \mapsto v * a$ is a semigroup homomorphism. By 1 and 2, its inverse is $a \mapsto u * a$. So, $a \mapsto v * a$ is a group isomomorphism.

4. Suppose there is an $a \in u * \mathcal{M} \cap v * \mathcal{M}$. Let $a' \in \mathcal{M}$ be such that $a * a' = u$. Then $u \in v * \mathcal{M} * a' = v * \mathcal{M}$. Then u is an identity in the group $v * \mathcal{M}$.

5. Let $a \in \mathcal{M}$. As $\mathcal{M} * a = \mathcal{M}$, the set $\{c \in \mathcal{M} : c * a = a\}$ is nonempty. As this is a type-definable idempotent set, it contains an idempotent element u. Then $a \in u * \mathcal{M}$.

6. As $\mathcal{M} * b$ is a minimal left ideal by Fact 15.21, it contains an idempotent element $u \in \mathcal{M} * b$. It is immediate to verify that $b * u$ is the required idempotent. □

The following is a technical lemma that is required in the proof of the main theorem.

15.25 Proposition Let $\sigma : \mathcal{S} \to \mathcal{S}$ be a semigroup homomorphism definable over M. Then

1. $\sigma\big[(a)_S\big] = (\sigma\, a)_S$
2. $\sigma\big[a * b\big] = \sigma\, a * \sigma\, b$.

Proof. 1. As $a \equiv_S a'$ implies $\sigma\, a \equiv_S \sigma\, a'$, inclusion $\subseteq$ is clear. For the converse, note that the type $\exists y\, \big[\sigma\, y = x \wedge y \equiv_S a\big]$ is trivially realized by $\sigma\, a$. By invariance it is equivalent to a type over S. Therefore it is realized by all elements of $(\sigma\, a)_S$. Hence all elements of $(\sigma\, a)_S$ are the image of some element in $(a)_S$.

2. We have to prove the following equality

$$\Big\{\sigma(a'\cdot b') \;:\; a \equiv_S a' \underset{S}{\downarrow} b' \equiv_S b\Big\} \quad = \quad \Big\{a'\cdot b' \;:\; \sigma\, a \equiv_S a' \underset{S}{\downarrow} b' \equiv_S \sigma\, b\Big\}.$$

It suffices to prove one inclusion because by Proposition 15.14 both sides are orbits. We prove $\subseteq$. Note that $a' \downarrow_S b'$ implies $\sigma\, a' \downarrow_S \sigma\, b'$. Hence the set on l.h.s. is contained in the following

$$\Big\{\sigma(a'\cdot b') \;:\; \sigma\, a' \downarrow_S \sigma\, b',\; a' \equiv_S a,\; b' \equiv_S b\Big\}$$

which is in turn contained in the set on the r.h.s. □

The following theorem is an elegant rephrasing of the celebrated Hales-Jewett Theorem. Let S be a semigroup. A subsemigroup C is nice if it has the property that if $a\cdot b \in C$ then $a, b \in C$.

15.26 Hales-Jewett Theorem (Koppelberg's formulation) Let S be a semigroup with an infinite nice subsemigroup C. Let Σ be a finite set of retractions of S onto C, that is, homomorphisms $\sigma : S \to C$ such that $\sigma_{\restriction C} = \mathrm{id}_C$. Then, for every finite coloring of C, there is an $a \in S \smallsetminus C$ such that $\{\sigma\, a : \sigma \in \Sigma\}$ is monochromatic.

Proof. Let $\mathcal{S}$ be a saturated elementary extension of S in a language that expands the language of semigroups with a symbol for all subsets of S. As observed in Remark 14.15, this makes $\downarrow_S$ stationary. The language contains also symbols for the retractions $\sigma : S \to C$. Let $\mathcal{C}$ be the interpretation of C in $\mathcal{S}$. Then $\sigma : \mathcal{S} \to \mathcal{C}$.

By nicety, $\mathcal{S} \smallsetminus \mathcal{C}$ is a left ideal of $\mathcal{S}$ (as a matter of fact, a two-sided ideal). Let $\mathcal{M} \subseteq \mathcal{S} \smallsetminus \mathcal{C}$ be minimal a left ideal of $\mathcal{S}$. Let $\mathcal{N} \subseteq \mathcal{C}$ be a minimal left ideal of $\mathcal{C}$. Let $v \in \mathcal{N}$ be an idempotent. By Proposition 15.25, for every retraction σ, we have that $\sigma(\mathcal{M}) * v$ is a left ideal contained in $\mathcal{N}$. Therefore they all coincide with $\mathcal{N}$. Let $u \in v * \mathcal{M} * v$ be an idempotent. Then $\sigma(u)$ is an idempotent of

$v * \mathcal{N}$. As $v * \mathcal{N}$ contains only one idempotent orbit, $\sigma(u) \equiv_S v$ for every σ. In particular $\{\sigma(u) : \sigma \in \Sigma\}$ is monochromatic. As Σ is finite , the theorem follows by elementarity. □

Finally we derive the classical Hales-Jewett Theorem from the above algebraic version.

15.27 Hales-Jewett Theorem (classical formulation) Let C be an infinite semigroup. Let $F \subseteq C$ be a finite set. Then, given any finite coloring of C, there is a non-constant term $t(x)$ in the language of semigroups with parameters in C such that the set $\{t(a)^C : a \in F\}$ is monochromatic (x is a single variable and $t(a)^C$ is the evaluation of $t(a)$ in C).

Proof. Let S be the set of the terms n the language of semigroups with parameters in C where at most the variable x occur. Then S is a semigroup under the natural operation and C is a nice subgroup. For every $a \in C$ the map σ_a that takes $t(x)$ to $t(a)^C$. These are a retraction of S onto C. Hence we can apply Theorem 15.26. □

We also derive a famous consequence of the Hales-Jewett Theorem. An *arithmetical progression* of length k is a sequence of natural numbers $\langle n{+}m{\cdot}i : i < k\rangle$ for some $n, m \in \mathbb{N}$.

15.28 van der Waerden's Theorem For every finite coloring of $\mathbb{N}$ and for every $k < \omega$, there is a monochromatic arithmetical progression of length k.

Proof. Let $C = \mathbb{N}$ and let S be the semigroup of terms in the language of semigroups with parameters in $\mathbb{N}$ where at most the variable x occur. That is terms of the form $n{+}m{\cdot}x$ for some $n, m \in \mathbb{N}$. Then $\mathbb{N}$ is a nice subsemigroup of S. For $i < k$ let σ_i be the retraction that takes $n{+}m{\cdot}x \in S$ to $n{+}m{\cdot}i \in \mathbb{N}$. Apply Theorem 15.26 to obtain some $n, m \in \mathbb{N}$ such that the set $\{n{+}m{\cdot}i \ : \ i < k\}$ is monochromatic. □

15.29 Exercise Let $\mathcal{M}$ be a type-definable minimal left ideal of $\mathcal{C}$. Let $u \in \mathcal{M}$ be an idempotent. Prove that $u * \mathcal{C}$ is a minimal right ideal of $\mathcal{C}$.

15.30 Exercise Let $\mathcal{M}$ and $\mathcal{N}$ be type-definable minimal left ideals of $\mathcal{C}$. Let $u \in \mathcal{M}$ and $v \in \mathcal{N}$ be idempotents. Prove that $u * \mathcal{M}$ and $v * \mathcal{N}$ are isomorphic groups. The isomorphism is given by the map $a \mapsto v * a$ with inverse $a \mapsto u * a$ works if we can assume that u and v are such that $u * v = u$ and $v * u = v$.

15.31 Exercise For every finite coloring of $\mathbb{N}$ and every $k < \omega$, there is an infinite sequence $\bar{a} = \langle a_i(x) : i < \omega\rangle$ of nonconstant univariate polynomials with coefficients in $\mathbb{N}$ such that $\{a(i) : a(x) \in \operatorname{fp}(\bar{a}),\ i < k\}$ is monochromatic. Can we require that the $a_i(x)$ have increasing degrees?

15.32 Exercise Prove that

$$\bigcup\{\mathcal{M} \ : \ \mathcal{M} \text{ a minimal left ideal of } \mathcal{C}\} = \bigcup\{\mathcal{M} \ : \ \mathcal{M} \text{ a minimal right ideal of } \mathcal{C}\}.$$

15.6 Notes and references

The first application of the algebraic structure of βS (the Stone-Čech compactification of a semigroup S) to Ramsey Theory is the celebrated Galvin-Glazer proof of Hindman's theorem. Here we have used saturated models in place of Stone-Čech compactification. The idea to replace the semigroup βS with types has been pioneered by Ludomir Newelski in the study of applications of topological dynamics to model theory. Our exposition follows [1].

The original proof of the Hales-Jewett Theorem by Alfred Hales and Robert Jewett is combinatorial. An alternative proof, also combinatorial, has been given by Saharon Shelah. The formulation with retractions is from [2].

[1] Eugenio Colla and Domenico Zambella, *Ramsey's coheirs*, J. Symb. Log. **87** (2022), no. 1, 377–391. arXiv:1901.04363.

[2] Sabine Koppelberg, *The Hales-Jewett theorem via retractions*, Proceedings of the 18th Summer Conference on Topology and its Applications, 2004, pp. 595–601.

Chapter 16

Lascar invariant sets

In this chapter we fix a signature L, a complete theory T without finite models, and a saturated model $\mathcal{U}$ of inaccessible cardinality κ strictly larger than $|L|$. The notation and implicit assumptions are as in Section 9.3.

16.1 Expansions

This section is only marginally required in the present chapter so it can be postponed with minor consequences.

We will find it convenient to expand the language L with a predicate for a given $\mathcal{D} \subseteq \mathcal{U}^z$. We denote by $\langle \mathcal{U}, \mathcal{D} \rangle$ the corresponding expansion of $\mathcal{U}$. Generally, we write $L(\mathcal{X})$ for the expanded language but, when the intended interpretation of $\mathcal{X}$ is only going to be $\mathcal{D}$, we may write $L(\mathcal{D})$ and abbreviate $\langle \mathcal{U} ; \mathcal{D} \rangle \vDash \varphi(\mathcal{X})$ as $\varphi(\mathcal{D})$.

16.1 Remark The definitions above are straightforward when z finite tuple. When z is an infinite tuple the intuition stays the same but a more involved definition is required. In fact, first-order logic does not allow predicates with infinite arity. We think of $L(\mathcal{X})$ for a two sorted language. The *home sort*, denoted by 0, and the z-*sort*, denoted by z. The expansion $\langle \mathcal{U}, \mathcal{D} \rangle$ has domain $\mathcal{U}$ for the home sort, and $\mathcal{U}^z$ for the z-sort. Besides the symbols of L, there is a function symbol π_i for every $i < |z|$ which is interpreted as the projection to the i-coordinate. These functions have sort $\langle z, 0 \rangle$ (see Section 13.1 for the notation). There is also a predicate of sort $\langle z \rangle$ interpreted as $\mathcal{D}$.

What said above is adapted to define the expansion $L(\mathcal{X}_i : i < \lambda)$, where $\mathcal{X}_i$ are predicates of sort z_i. Again, when the sets $\mathcal{D}_i \subseteq \mathcal{U}^{z_i}$ are the only intended interpretation of $\mathcal{X}_i$, we write $L(\mathcal{D}_i : i < \lambda)$.

16.2 Definition If $\mathcal{C}, \mathcal{D} \subseteq \mathcal{U}^z$ we abbreviate $\langle \mathcal{U}, \mathcal{C} \rangle \equiv_A \langle \mathcal{U}, \mathcal{D} \rangle$ as $\mathcal{C} \equiv_A \mathcal{D}$. We also say that $\mathcal{D}$ is saturated if so is the model $\langle \mathcal{U} ; \mathcal{D} \rangle$.

16.3 Remark For every $\mathcal{D} \subseteq \mathcal{U}^z$ there is a saturated $\mathcal{C} \equiv \mathcal{D}$. In fact, it suffices to find a saturated model $\langle \mathcal{U}', \mathcal{D}' \rangle \equiv \langle \mathcal{U}, \mathcal{D} \rangle$ of cardinality κ. By saturation, there is an isomorphism $f : \mathcal{U}' \to \mathcal{U}$. Therefore $f[\mathcal{D}']$ is the required set $\mathcal{C}$.

16.4 Proposition If $\mathcal{D} \subseteq \mathcal{U}^z$ is invariant over A then every A-indiscernible sequence is indiscernible in the language $L(A ; \mathcal{D})$.

Proof. Let $\bar{c} = \langle c_i : i \in I \rangle$ be an A-indiscernible sequence. For every $I_0, I_1 \in I^{[n]}$

there is an $f \in \mathrm{Aut}(\mathcal{U}/A)$ such that $fc_{\restriction I_0} = c_{\restriction I_1}$, Hence

$$\varphi(c_{\restriction I_0}\,;\mathcal{D}) \quad\leftrightarrow\quad \varphi(fc_{\restriction I_0}\,;f[\mathcal{D}]) \quad\leftrightarrow\quad \varphi(c_{\restriction I_1}\,;\mathcal{D}).$$ □

16.5 Exercise Prove that if $\mathcal{C} \subseteq \mathcal{U}^z$ is type-definable over B then $\equiv_{A,B}$ implies $\equiv_{A\,;\mathcal{C}}$.

16.6 Exercise Prove that if $\mathcal{D} \subseteq \mathcal{U}^z$ is saturated and invariant over A than it is definable over A.

16.2 Lascar strong types

Let $\mathcal{D} \subseteq \mathcal{U}^z$, where z is a tuple of length $< \kappa$. The orbit of $\mathcal{D}$ over A is the set

$$o(\mathcal{D}/A) \quad=\quad \Big\{f[\mathcal{D}] \;:\; f \in \mathrm{Aut}(\mathcal{U}/A)\Big\}$$

So, $\mathcal{D}$ is invariant over A when $o(\mathcal{D}/A) = \big\{\mathcal{D}\big\}$. We say that $\mathcal{D}$ is Lascar invariant over A if it is invariant over every model $M \supseteq A$. Recall that this means that if $a \equiv_M c$ for some model M containing A then $a \in \mathcal{D} \leftrightarrow c \in \mathcal{D}$.

16.7 Proposition Let $\lambda = |L_z(A)|$. There are at most 2^{2^λ} sets $\mathcal{D} \subseteq \mathcal{U}^z$ that are Lascar invariant over A.

Proof. Let N be a model containing A of cardinality $\leq \lambda$. Every set that is Lascar invariant over A is invariant over N. As $|L_z(N)| = \lambda$ the bound follows from Proposition 14.1. □

16.8 Theorem Let $\lambda = |L_z(A)|$. For every $\mathcal{D}$ and every $A \subseteq M$ the following are equivalent

1. $\mathcal{D}$ is Lascar invariant over A
2. every set in $o(\mathcal{D}/A)$ is M-invariant
3. $o(\mathcal{D}/A)$ has cardinality $\leq 2^{2^\lambda}$
4. $o(\mathcal{D}/A)$ has cardinality $< \kappa$
5. $c_0 \in \mathcal{D} \leftrightarrow c_1 \in \mathcal{D}$ for every A-indiscernible sequence $\langle c_i : i < \omega\rangle$.

Proof. 1⇒2. This implication is clear because all sets in $o(\mathcal{D}/A)$ are Lascar invariant over A.

2⇒3. When $|M| \leq |L(A)|$ the implication follows from the bounds discussed in Section 14.1. We temporary add this assumption on M. Once the proof of the proposition is completed, is easily seen to be redundant (by 4 and Proposition 16.7).

3⇒4. This implication holds because κ is a strong limit cardinal.

4⇒5. Assume ¬5. Then we can find an A-indiscernible sequence $\langle c_i : i < \kappa\rangle$ such that $c_0 \in \mathcal{D} \not\leftrightarrow c_1 \in \mathcal{D}$. Define

$$E(u\,;v) \quad\Leftrightarrow\quad u \in \mathcal{C} \leftrightarrow v \in \mathcal{C} \quad \text{for every } \mathcal{C} \in o(\mathcal{D}/A).$$

Then $E(u\,;v)$ is an A-invariant equivalence relation. As $\neg E(c_0\,;c_1)$, by Proposition 16.4, indiscernibility over A implies that $\neg E(c_i\,,c_j)$ for every $i < j < \kappa$. Then $E(u\,;v)$ has κ equivalence classes. As κ is inaccessible, this implies ¬4.

5⇒1. Fix any $a \equiv_N b$ where $A \subseteq N$. It suffices to prove that $a \in \mathcal{D} \leftrightarrow b \in \mathcal{D}$. Let $p(z) \in S(\mathcal{U})$ be a global coheir of $\operatorname{tp}(a/N) = \operatorname{tp}(b/N)$. Let $\bar{c} = \langle c_i : i < \omega \rangle$ be a Morley sequence of $p(z)$ over N, a, b. Then both $a, \bar{c}$ and $b, \bar{c}$ are A-indiscernible sequences. Therefore, from 5 we obtain $a \in \mathcal{D} \leftrightarrow c_0 \in \mathcal{D} \leftrightarrow b \in \mathcal{D}$. □

For definable sets Lascar invariance reduces to definablity over the algebraic closure.

16.9 Corollary For every definable set $\mathcal{D}$ the following are equivalent

1. $\mathcal{D}$ is Lascar invariant over A
2. $\mathcal{D}$ is definable over every model containing A
3. $\mathcal{D} \in \operatorname{acl}^{\mathrm{eq}} A$.

The following corollary easily follows from Theorem 16.8 and Proposition 16.4. As an exercise the reader may prove it using Proposition 15.6.

16.10 Corollary The following are equivalent

1. $\mathcal{D}$ is Lascar invariant over A
2. every sequence of A-indiscernibles is $L(A\,;\mathcal{D})$-indiscernible.

⚠ Given a tuple $a \in \mathcal{U}^z$, we write $\mathcal{L}(a/A)$ for the intersection of all sets containing a that are Lascar invariant over A. Clearly $\mathcal{L}(a/A)$ is Lascar invariant over A. The symbol $\mathcal{L}(a/A)$ is not standard.

16.11 Definition We write $a \stackrel{\mathrm{L}}{\equiv}_A b$ and say that a and b have the same Lascar strong type over A if the equivalence $a \in \mathcal{D} \leftrightarrow b \in \mathcal{D}$ holds for every set $\mathcal{D}$ that is Lascar invariant over A or, in other words, if $\mathcal{L}(a/A) = \mathcal{L}(b/A)$. The notation $\text{L-stp}(a/A) = \text{L-stp}(b/A)$ and $a\,E_{L/A}b$ are also common in the the literature.

16.12 Proposition The relation $\stackrel{\mathrm{L}}{\equiv}_A$ is the finest equivalence relation with $< \kappa$ classes that is invariant over A.

Proof. Clearly $\stackrel{\mathrm{L}}{\equiv}_A$ is an equivalence relation invariant over A. Each equivalence class is Lascar invariant over A, hence the number of equivalence classes is bounded by the number of Lascar invariant sets over A. To see that $\stackrel{\mathrm{L}}{\equiv}_A$ is the finest of such equivalences. Suppose $\mathcal{D}$ is an equivalence class of an A-invariant equivalence relation with $< \kappa$ classes. Then $o(\mathcal{D}/A)$ has also cardinality $< \kappa$. Then $\mathcal{D}$ is Lascar invariant and as such it is union of classes of the relation $\stackrel{\mathrm{L}}{\equiv}_A$. □

Let $p(x) \in S(\mathcal{U})$ be global type. We say that p is Lascar invariant over A if the sets $\mathcal{D}_{p,\varphi}$, for $\varphi(x\,;z) \in L$, are all Lascar invariant over A. The sets $\mathcal{D}_{p,\varphi}$ are defined in Section 14.1.

16.13 Proposition Let $p(x) \in S(\mathcal{U})$. Then the following are equivalent

1. $p(x)$ is Lascar invariant over A
2. every A-indiscernible sequence $\bar{c}$ is indiscernible over A, a for every $a \vDash p_{\restriction A,\bar{c}}(x)$
3. every A-indiscernible sequence $\bar{c}$ is indiscernible over a for every $a \vDash p_{\restriction \bar{c}}(x)$.

For convenience the tuples c_i have length $|z| = \omega$.

Proof. 1⇒2. Assume 1 and fix an A-indiscernible sequence $\bar{c} = \langle c_i : i < \omega \rangle$ and some $a \vDash p_{\restriction \bar{c}}$. We need to prove that for every formula $\varphi(x\,;z_1,\dots,z_n) \in L(A)$

$$\varphi(a\,;c_{\restriction I_0}) \quad \leftrightarrow \quad \varphi(a\,;c_{I_1}).$$

holds for every $I_0, I_1 \subseteq \omega$ of cardinality n. Without loss of generality we can assume that $I_0 < I_1$. Define I_n for $n > 1$ arbitrarily such that $I_n < I_{n+1}$. Then the sequence $c'_n = c_{\restriction I_n}$ is a sequence of A-indiscernibles and $c'_0 \in \mathcal{D}_{p,\varphi} \not\leftrightarrow c'_1 \in \mathcal{D}_{p,\varphi}$. By Theorem 16.8 this contradicts 1.

2⇒3 is trivial.

3⇒1. If $p(x)$ is not Lascar invariant over A then $c_0 \in \mathcal{D}_{p,\varphi} \not\leftrightarrow c_1 \in \mathcal{D}_{p,\varphi}$ for some A-indiscernible sequence $\bar{c} = \langle c_i : i < \omega \rangle$ and some $\varphi(x\,;z) \in L$. Then $p(x)$ contains the formula $\varphi(x\,;c_0) \not\leftrightarrow \varphi(x\,;c_1)$. Hence, $\bar{c}$ is not indiscernible over any realization of $p(x)_{\restriction c_0,c_1}$. □

16.3 Coheirs over sets

In this section we consider a property that implies the existence of global Lascar invariant types and has simple syntactic characterization. It is a natural generalization to a set of the notion of finite satisfiability over a model that we introduced in Section 14.2.

In Section 14.2 finite satisfiability over a model has been used to prove that every consistent type over a model has a global extension to a type that is invariant over that model. Ideally, we would like to prove here a similar existence property for Lascar invariance over a set. But this is not possible in general. In fact, in Example 16.17, we present a theory that has no global Lascar invariant type.

⚠ A global type $p(x) \in S(\mathcal{U})$ that is finitely satisfiable in every model containing A, is called a Lascar coheir of $p_{\restriction A}(x)$. Note that when A is a model we obtain the same notion defined in Section 14.2.

16.14 Proposition Let $\varphi(x\,;z) \in L$. Every type $p(x) \in S_\varphi(\mathcal{U})$ that is finitely satisfiable in every $M \supseteq A$ is Lascar invariant over A.

Proof. Let $\langle c_i : i < \omega \rangle$ be an A-indiscernible sequence. Suppose for a contradiction that $c_0 \in \mathcal{D}_{p,\varphi} \not\leftrightarrow c_1 \in \mathcal{D}_{p,\varphi}$ Then $p(x) \vdash \varphi(x\,;c_0) \not\leftrightarrow \varphi(x\,;c_1)$. Let $M \supseteq A$ be such that $\langle c_i : i < \omega \rangle$ is indiscernible over M. As $p(x)$ is finitely satisfiable in M, for some $a \in M^x$ we have $\varphi(a\,;c_0) \not\leftrightarrow \varphi(a\,;c_1)$. This contradicts indiscernibility over M. □

We say that A is a extension base for Lascar coheirs if every type that is finitely satisfiable in every $M \supseteq A$ has an extension to a global type with the same property. The terminology is not standard.

16.15 Proposition The following are equivalent

1. A is a extension base for Lascar coheirs
2. for every $\psi(x), \varphi(x) \in L(\mathcal{U})$, if $\psi(x) \vee \varphi(x)$ is finitely satisfied in every $M \supseteq A$, then $\psi(x)$ or $\varphi(x)$ is finitely satisfied in every $M \supseteq A$.

Proof. 1⇒2. Clear.

2⇒1. Assume 2 and repeat the proof of Proposition 14.4. □

Trivially, every model is an extension base for Lascar coheirs. The following example shows that some theories may have no other extension bases.

16.16 Example Consider T_{dlo}. Suppose A is not a model. We show that A is not an extension base for Lascar coheirs. As A is not a model, there are $a < c$ in A such that either $A \cap (a, c) = \varnothing$ or $a \leq A$ or $A \leq c$. In the first case let $b \in (a, c)$. Then every model containing A intersects $(a, b) \cup (b, c)$, but there are both models that omit (a, b) and models that omit (b, c). In the second case let $b < a$ and consider $(-\infty, b) \cup (b, a)$. The third case is symmetric.

In Chapter 18 we will see that for some important class of theories, *stable theories*, all sets are extension bases for Lascar coheirs.

16.17 Example We present a theory without global types that are Lascar invariant over the empty set. Let T be the theory of $\mathbb{Q}$ with the cyclic order, i.e. with a ternary predicate $\mathrm{cyc}(a, b, c)$ that holds if

$$(a < b < c) \ \vee \ (c < a < b) \ \vee \ (b < c < a).$$

The picture to keep in mind is that of a circle, where $\mathrm{cyc}(a, b, c)$ holds if we encounter a, b, c in that order when going clockwise.

We will implicitly use that T has elimination of quantifiers. Let $p(x) \in S(\mathcal{U})$. We claim that $p(x)$ is not Lascar invariant over the empty set. Suppose it is, for a contradiction. We can find $a < b$ in $\mathcal{U}$ such that either $\mathrm{cyc}(a, x, b)$ or $\mathrm{cyc}(a, b, x)$ is in $p(x)$. We show that both these possibilities are contradictory.

1. If $p(x)$ contains $\mathrm{cyc}(a, x, b)$, let $M < a$ be a model and pick c such that $b < c$. Then $a, b \equiv_M b, c$, hence $\mathrm{cyc}(b, x, c)$ is also in $p(x)$. A contradiction.
2. If $p(x)$ contains $\mathrm{cyc}(a, b, x)$, let $a < M < b$ and pick a' such that $a < a' < M$. As $a, a' \equiv_M b, a$, also $\mathrm{cyc}(b, a, x)$ is in $p(x)$. A contradiction.

16.18 Exercise Let A be a extension base for Lascar coheirs. Prove that for every $b \in \mathcal{U}^z$ there is a structure $\mathcal{V} \preceq \mathcal{U}$ isomorphic to $\mathcal{U}$ over A such that $\mathcal{V} \mathrel{\smile\!\!\!\!\!\!|}_M b$ for every $A \subseteq M \preceq \mathcal{V}$.

16.19 Exercise Let A be a extension base for Lascar coheirs. Prove that for every $a \in \mathcal{U}^x$ there is a structure $\mathcal{V} \preceq \mathcal{U}$ isomorphic to $\mathcal{U}$ over A such that $a \mathrel{\smile\!\!\!\!\!\!|}_M \mathcal{V}$ for every $A \subseteq M \preceq \mathcal{V}$.

16.20 Exercise Let $\varphi(x\,; z) \in L$. Let $p(x) \in S(\mathcal{U})$ be finitely satisfied in every $M \supseteq A$. Prove that if $\mathcal{D}_{p,\varphi}$ is saturated, see definition 16.2, then it is definable and belongs to $\mathrm{acl}^{\mathrm{eq}} A$.

16.4 The Lascar graph

Here we study Lascar strong types from a different viewpoint. The Lascar graph over A has an arc between all pairs $a, b \in \mathfrak{U}^x$ such that $a \equiv_M b$ for some model M containing A. We write $d_A(a,b)$ for the distance between a and b in the Lascar graph over A. Let us spell this out: $d_A(a,b) \leq n$ if there is a sequence $a_0, \dots, a_n$ such that $a = a_0$, $b = a_n$, and $a_i \equiv_{M_i} a_{i+1}$ for some models M_i containing A. We write $d_A(a,b) < \infty$ if a and b are in the same connected component of the Lascar graph over A.

16.21 Proposition For every $a \in \mathfrak{U}^x$

$$\mathcal{L}(a/A) \quad = \quad \big\{c \ : \ d_A(a,c) < \infty\big\}.$$

Proof. To prove inclusion $\supseteq$ it suffices to show that every Lascar A-invariant set containing a contains the set on the r.h.s. Let $\mathcal{D}$ be Lascar A-invariant, and let $b \in \mathcal{D}$. Then $\mathcal{D}$ contains also every c such that $b \equiv_M c$ for some model M containing A. That is, $\mathcal{D}$ contains every c such that $d_A(b,c) \leq 1$. It follows that $\mathcal{D}$ contains every c such that $d_A(a,c) < \infty$.

To prove inclusion $\subseteq$ we prove the set on the r.h.s. is Lascar A-invariant. Suppose the sequence $a_0, \dots, a_n$, where $a_0 = a$ and $a_n = c$, witnesses $d_A(a,c) \leq n$ and suppose that $c \equiv_M b$ for some M containing A, then the sequence $a_0, \dots, a_n, b$ witnesses $d_A(a,b) \leq n+1$. □

We write $\mathrm{Aut}^{\mathrm{f}}(\mathfrak{U}/A)$ for the subgroup of $\mathrm{Aut}(\mathfrak{U}/A)$ that is generated by the automorphisms that fix point-wise some model M containing A. (The "f" in the symbol stands for *fort*, the French word for *strong*.) It is easy to verify that $\mathrm{Aut}^{\mathrm{f}}(\mathfrak{U}/A)$ is a normal subgroup of $\mathrm{Aut}(\mathfrak{U}/A)$. The following is a corollary of Proposition 16.21.

16.22 Corollary The following are equivalent

1. $a \overset{\mathrm{L}}{\equiv}_A b$
2. $a = fb$ for some $f \in \mathrm{Aut}^{\mathrm{f}}(\mathfrak{U}/A)$.

It may not be immediately obvious that the relation $d_A(x,y) \leq n$ is type-definable.

16.23 Proposition There is a type $p_n(x,y) \subseteq L(A)$ equivalent to $d_A(x,y) \leq n$.

Proof. It suffices to prove the proposition with $n = 1$. Let $\lambda = |L(A)|$ and fix a tuple of distinct variables $w = \langle w_i : i < \lambda \rangle$, then $p_1(x,y) \;=\; \exists w\, p(w,x,y)$ where

$$p(w,x,y) \quad = \quad q(w) \;\cup\; \Big\{\varphi(x,w) \leftrightarrow \varphi(y,w) \ : \ \varphi(x,w) \in L(A)\Big\}$$

and $q(w) \subseteq L(A)$ is a consistent type with the property that all its realizations enumerate a model containing A.

It remains to verify that such a type exist. Let $\langle \psi_i(z, w_{\restriction i}) : i < \lambda \rangle$ be an enumeration of the formulas in $L_{z,w}(A)$, where z is a single variable. Let

$$q(w) \quad = \quad \Big\{\exists z\, \psi_i(z, w_{\restriction i}) \;\rightarrow\; \psi_i(w_i, w_{\restriction i}) \ : \ i < \lambda\Big\}.$$

Any realization of $q(w)$ satisfy the Tarski-Vaught test therefore it enumerates a model containing A. Vice versa it is clear that we can realize $q(w)$ in any model containing A. □

16.24 Exercise Prove that the equivalence relation $a \stackrel{\text{L}}{\equiv}_A b$ is the transitive closure of the relation: there is a sequence $\langle c_i : i < \omega \rangle$ indiscernible over A such that $c_0 = a$ and $c_1 = b$. Hint: use Exercise 15.10 to prove symmetry, then reason as in Proposition 16.21.

16.25 Exercise Let $G = \{f \in \mathrm{Aut}(\mathcal{U}/A) : f\mathcal{D} = \mathcal{D}$ for every $\mathcal{D}$ with $o(\mathcal{D}/A)$ bounded$\}$. Prove that $G = \mathrm{Aut}^{\mathrm{f}}(\mathcal{U}/A)$.

Chapter 17

Five notions of largeness

Chapter under revision.

The notions intriduced in the first sections of this chapter are independent of model theory. In particular, the set $\mathcal{X}$ mentioned below does not live inside any first order structure. The instantiation of these notions to the context of the model theory is presented in Section 17.6, and will be applied in Section 18.5.

From Section 17.6 we fall back to usual setting: L is a signature, T is a complete theory without finite models, and $\mathcal{U}$ is a saturated model of inaccessible cardinality κ strictly larger than $|L|$; we use the same notation and make the same implicit assumptions as in Section 9.3.

17.1 The dual perspective on invariance

We set the stage in great (admitedly, somewhat overblown) generality.

Let G be a semigroup that acts on Γ, a Boolean algebra of subsets of $\mathcal{X}$. We assume that the action of G commutes with Boolean connectives. Such a Boolean algebra is called a G-algebra. We write $\cdot$ both for the operation of G and the action of G on Γ.

The G-algebra Γ is fixed throughout this section. The sets $\mathcal{A}, \mathcal{B}, \mathcal{C}, \ldots$ always denote elements of Γ. We also pick a filter π of subsets of $\mathcal{X}$. We encourage the reader to assume $\pi = \{\mathcal{X}\}$ on a first reading. In the following both Γ and π are mostly left implicit.

If $\sigma \subseteq \Gamma$, we say that σ is finitely consistent with π to mean that $\pi \cup \sigma$ generates a proper filter of subsets of $\mathcal{X}$. We say that σ covers π to mean that $\cup\sigma$ is in π.

We say that a set $\mathcal{A}$ is syndetic under the action of G , or G-syndetic for short, if finitely many G-translates of $\mathcal{A}$ cover π; we say n-G-syndetic if $\leq n$ translates suffices. Dually, we say that $\mathcal{A}$ is thick under the action of G, or G-thick for short, if any finitely many G-translates of $\mathcal{A}$ is finitely consistent with π; we say n-G-thick when the request is limited to $\leq n$ translates.

A set $\sigma \subseteq \Gamma$ is G-syndetic (respectively, G-thick) if every intersection of sets in σ is such.

When more than one filter π is at play, we say that $\mathcal{A}$ is G-syndetic (respectively, G-thick) relative to π.

17.1 Notation When $C \subseteq G$ we write $C\cdot\mathcal{A}$ for $\{h\cdot\mathcal{A} : h \in C\}$.

Therefore, $\mathcal{A}$ is G-syndetic if there is a finite $C \subseteq G$ such that $\cup C\cdot\mathcal{A}$ is in π; and $\mathcal{A}$ is G-thick if for every finite $C \subseteq G$ the set $\cap C\cdot\mathcal{A}$ is finitely consistent with π.

In this chapter many proofs require some juggling with negations as epitomized by the following fact.

17.2 Fact The following are equivalent

1. $\mathcal{A}$ is not G-syndetic
2. $\neg\mathcal{A}$ is G-thick (throughout the chapter $\neg\mathcal{A}$ denotes the complement in $\mathcal{X}$).

Proof. Spelling out the definitions, 1 and 2 are, respectively, equivalent to

1′. there is no finite $C \subseteq G$ such that $\cup C{\cdot}\mathcal{A}$ covers π.

2′. $\cap C{\cdot}\neg\mathcal{A}$ finitely consistent with π for every finite $C \subseteq G$.

Then the equivalence is evident. □

The following dual pair of corollaries are linked to important properties discussed in Section 17.4.

17.3 Corollary The following are equivalent

1. every G-syndedic set is G-thick
2. $\mathcal{A}$ and $\neg\mathcal{A}$ are not both G-syndetic.

17.4 Corollary The following are equivalent

1. every G-thick set is G-syndetic
2. $\mathcal{A}$ and $\neg\mathcal{A}$ are not both G-thick.

We write Σ_G for the collection of G-syndetic sets.

We write $S_\pi(\Gamma)$ for the set of ultrafilters of the Bolen algebra Γ that are finitely consistent with π. Note that when $\Sigma_G \cup \{\mathcal{D}\}$ is finitely consistent with π, then $\neg\mathcal{D}$ is not G-syndetic. Then $\mathcal{D}$ is G-thick. We prove that the converse implication holds if we replace $\mathcal{D}$ with an ultrafilter.

There is also a natural action of G on $S_\pi(\Gamma)$ defined by $g{\cdot}p = \{g{\cdot}\mathcal{D} : \mathcal{D} \in p\}$. We say that $p \in S_\pi(\Gamma)$ is G-invariant if $g{\cdot}p = p$ for every $g \in G$.

17.5 Theorem For every $p \in S_\pi(\Gamma)$ the following are equivalent

1. p is G-invariant
2. $\Sigma_G \subseteq p$
3. p is G-thick.

Proof. 1⇒2. Let $\mathcal{D}$ be G-syndetic. Pick $C \subseteq G$ be finite such that $C{\cdot}\mathcal{D}$ covers π. Then p is finitely consistent with $\cup C{\cdot}\mathcal{D}$. Therefore, by maximality, $h{\cdot}\mathcal{D} \in p$ for some $h \in C$. Finally, by invariance, $\mathcal{D} \in p$.

2⇒3. From 2 it follows that if conjunctions of formulas in p are finitely consistent with Σ_G. Therefore thickness follows from the above remark.

3⇒1. Assume $g{\cdot}p \neq p$. If $\mathcal{D} \in p$ and $g{\cdot}\mathcal{D} \notin p$ for some $\mathcal{D}$ then, by maximality, $(\mathcal{D} \cap \neg g{\cdot}\mathcal{D}) \in p$. Clearly $\mathcal{D} \cap \neg g{\cdot}\mathcal{D}$ is not 2-G-thick as it is infinitely consistent with its g-translate. If instead $\mathcal{D} \notin p$ and $g{\cdot}\mathcal{D} \in p$ then, by the same argument, $(\neg\mathcal{D} \cap g{\cdot}\mathcal{D}) \in p$ is not G-thick. □

The following immediate consequence is worth highlighting

17.6 Corollary If a G-thick $p \in S_\pi(\Gamma)$ exists then

1. Σ_G is finitely consistent with π.
2. there are G-thick $p \in S_\pi(\Gamma)$.

The following theorem gives a necessary and sufficient condition for the existence of global G-invariant ultrafilters. Ideally, we would like that every G-thick filter extends to a G-thick ultrafilter. Unfortunately this is not true in general (it is a strong assumption which is discussed in Section 17.4).

A set $\mathcal{D}$ is G-wide if every finite cover of $\mathcal{D}$ by sets in Γ contains a set that is G-thick. A filter is G-wide if every formula in the filter is G-wide.

17.7 Theorem For every set $\mathcal{D} \in \Gamma$ the following are equivalent

1. $\Sigma_G \cup \{\mathcal{D}\}$ is finitely consistent
2. $\mathcal{D} \in p$ for some G-thick $p \in S_\pi(\Gamma)$
3. $\mathcal{D}$ is G-wide.

Proof. 1⇒2. Pick any $p \in S_\pi(\Gamma)$ that extends $\Sigma_G \cup \{\mathcal{D}\}$. Then p is G-thick by Theorem 17.5.

2⇒1. By Theorem 17.5.

2⇒3. Let $\mathcal{C}_1, \ldots, \mathcal{C}_n$ cover $\mathcal{D}$. Pick p as in 2. By completeness, $\mathcal{C}_i \in p$ for some i. Therefore, $\mathcal{C}_i$ is G-thick.

3⇒2. Let p be a maximal G-wide subset of Γ containing $\mathcal{D}$. We claim that p is a G-thick ultrafilter. First we prove completeness. Suppose for a contradiction that $\mathcal{D}, \neg\mathcal{D} \notin p$. By maximality there are some $\mathcal{C}$, that is an intersection of sets in p, and some sets $\mathcal{C}_1, \ldots, \mathcal{C}_n$ that cover both $\mathcal{C} \cap \mathcal{D}$ and $\mathcal{C} \smallsetminus \mathcal{D}$ and such that no $\mathcal{C}_i$ is G-thick. As $\mathcal{C}_1, \ldots, \mathcal{C}_n$ cover $\mathcal{C}$ this is a contradiction. It is only left to show that p is G-thick. This follows from completeness and Theorem 17.5. □

17.8 Exercise The following characterization of thick sets is also useful (and is sometimes taken as the definition). The following are equivalent

1. $\mathcal{A}$ is G-thick
2. $\mathcal{A} \cap \mathcal{B} \neq \varnothing$ for every set $\mathcal{B}$ that is G-syndetic.

17.9 Exercise Prove that for every $\mathcal{D}$ the following are equivalent

1. $\mathcal{D}$ is G-thick
2. some $p \in S_\pi(\Gamma)$ contains $\{x \in g{\cdot}\mathcal{D} \ : \ g \in G\}$.

17.10 Exercise Let $p \in S_\pi(\Gamma)$. Prove that the following are equivalent

1. p is G-invariant
2. $\mathcal{D} \in p$ for every 2-G-syndetic definable set $\mathcal{D}$
3. p is 2-G-thick.

17.11 Exercise Prove that the following are equivalent

1. $\mathcal{D} \in p$ for some G-invariant $p \in S_\pi(\Gamma)$.
2. every finite cover of $\mathcal{D}$ by sets in Γ contains a 2-G-thick set.

17.12 Exercise Give an example of a thick set that is not wide. Hint: find inspiration in Example 16.17.

17.2 Notable subsemigroups

Unfortunately, syndeticity is not preserved under intersection. In particular Σ_G is not a G-syndetic set, and it may even be infinitely consistent. Then the following notion is relevant. Recall that an underlying proper filter π is implicit in all what follows.

17.13 Definition

$$Q_G \quad = \quad \{q \subseteq \Sigma_G \; : \; q \text{ maximally } G\text{-syndetic}\}.$$

In other words, the sets in Q_G are maximal among the subsets of Σ_G that are closed under conjunction.

It is easy to see that Q_G is closed under the action of G. We write $\mathrm{Stab}(q)$ for the stabilizer of $q \subseteq \Sigma_G$ in G, that is, the subsemigroup $\{g \in G : g{\cdot}q = q\}$. We write $\mathrm{Stab}(\mathcal{D})$ with a similar meaning. Finally we define

$$G^1 \quad = \quad \bigcap_{q \in Q_G} \mathrm{Stab}(q).$$

It is easy to verify that $G^1 \trianglelefteq G$.

17.14 Proposition

$$G^1 \quad = \quad \{g \in G \; : \; \mathcal{D} \cap g{\cdot}\mathcal{D} \in \Sigma_G \text{ for every } \mathcal{D} \in \Sigma_G\}.$$

Proof. $\subseteq$. Pick any $k \in G^1$ and $\mathcal{D} \in \Sigma_G$. Let $q \in Q_G$ be a filter containing $\mathcal{D}$. From the G^1-invariance of q we obtain that $k{\cdot}\mathcal{D} \in q$. Then $\mathcal{D} \cap k{\cdot}\mathcal{D} \in q$, hence $\mathcal{D} \cap k{\cdot}\mathcal{D}$ is G-syndetic.

$\supseteq$. Pick any $g \notin G^1$. Then $q \neq g{\cdot}q$ for some $q \in Q_G$. Let $\mathcal{D} \in q$ such that $g{\cdot}\mathcal{D} \notin q$. By maximality, $\mathcal{C} \cap g{\cdot}\mathcal{D}$ is not G-syndetic for some $\mathcal{C} \in q$. As q is closed under conjunction, we can assume $\mathcal{D} = \mathcal{C}$, then g does not belong to the set on the r.h.s. □

17.15 Theorem Any finite conjunction of formulas in Σ_{G^1} is G-syndetic. In particular Σ_{G^1} is finitely consistent.

Proof. Notice that from Proposition 17.14 it easily follows that for every $\mathcal{D} \in \Sigma_G$ and every finite $F \subseteq G^1$ the set $\cap F{\cdot}\mathcal{D}$ is G-syndetic.

Let $\mathcal{D}_1, \ldots, \mathcal{D}_n \in \Sigma_{G^1}$. Assume inductively that $\mathcal{D}_1 \cap \cdots \cap \mathcal{D}_{n-1}$ is G-syndetic. Let $F \subseteq G^1$ be such that $\cup F{\cdot}\mathcal{D}_n$ covers π. Then

$$\begin{aligned}
\cup F{\cdot}[\mathcal{D}_1 \cap \cdots \cap \mathcal{D}_n] \quad &\supseteq \quad \cap F{\cdot}[\mathcal{D}_1 \cap \cdots \cap \mathcal{D}_{n-1}] \;\cap\; \cup F{\cdot}\mathcal{D}_n \\
&= \quad \cap F{\cdot}[\mathcal{D}_1 \cap \cdots \cap \mathcal{D}_{n-1}].
\end{aligned}$$

This last set is G-syndetic by the inductive hypothesis and what remarked above. The G-syndeticity of $\mathcal{D}_1 \cap \cdots \cap \mathcal{D}_n$ follows. □

Unfortunately, we are unable to conclude that the intersection of G^1-syndetic sets is G^1-syndetic.

From Theorems 17.7 and 17.15 it follows that G^1-syndetic sets are G^1-wide. But we can do better. First, we remark a useful consequence of normality.

17.16 Remark Assume $H \trianglelefteq G$. For every $\mathcal{D}$ and every $g \in G$

$$\mathcal{D} \text{ is } H\text{-foo} \Leftrightarrow g{\cdot}\mathcal{D} \text{ is } H\text{-foo},$$

where *foo* can be replaced by *syndetic, invariant, thick, wide.* In particular, Σ_H is G-invariant.

Recall that when Σ_H is finitely consistent with π then H-syndetic sets are H-wide, see Theorem 17.7. As it happens, under the assumption of normality, this can be strengthened as follows.

17.17 Proposition Assume $H \trianglelefteq G$ and that Σ_H is finitely consistent with π. Then every G-syndetic $\mathcal{D}$ is H-wide. In particular every $q \in Q_G$ is G^1-wide.

Proof. Let $p \in S_\pi(\Gamma)$ be finitely consistent with Σ_H. As $\mathcal{D}$ is G-syndetic, by completeness $g{\cdot}\mathcal{D} \in p$ for some $g \in G$. As p is H-thick, by Remark 17.16 also $g{\cdot}\mathcal{D}$ is H-thick. Then $g{\cdot}\mathcal{D}$ is finitely consistent with Σ_H and the proposition follows from Theorem 17.7. □

17.3 Strong syndeticity

A set $\mathcal{D}$ is strongly G-syndetic if for every finite $F \subseteq G$ the set $\cap F{\cdot}\mathcal{D}$ is G-syndetic (recall that $F{\cdot}\mathcal{D}$ stands for $\{h{\cdot}\mathcal{D} : h \in F\}$). Dually, we say that $\mathcal{D}$ is weakly G-thick if for some finite $F \subseteq G$ the set $\cup F{\cdot}\mathcal{D}$ is thick. Again, the same properties is attributed to a collection of sets if every finite intersection has the property.

⚠ In topological dynamic, strong syndedic sets are called *thickly syndetic* and weak thickness is called *piecewise syndetic.* Newelski in [2] says *weak generic* for weakly thick. These terminologies defy my intuition.

17.18 Lemma The intersection of two strongly G-syndetic sets is strongly G-syndetic.

Proof. Let $\mathcal{D}$ and $\mathcal{C}$ be strongly G-syndetic and let $C \subseteq G$ be an arbitrary finite set. It suffices to prove that $\mathcal{B} = \cap C{\cdot}(\mathcal{C} \cap \mathcal{D})$ is G-syndetic. Clearly $\mathcal{B} = \mathcal{C}' \cap \mathcal{D}'$, where $\mathcal{C}' = \cap C{\cdot}\mathcal{C}$ and $\mathcal{D}' = \cap C{\cdot}\mathcal{D}$. Note that $\mathcal{C}'$ and $\mathcal{D}'$ are both strongly G-syndetic. In particular $\cup F{\cdot}\mathcal{D}'$ covers π for some finite $F \subseteq G$. Note that

$$\begin{aligned} \cup F{\cdot}\mathcal{B} &= \cup F{\cdot}\big[\mathcal{C}' \cap \mathcal{D}'\big] \\ &\supseteq \big(\cap F{\cdot}\mathcal{C}'\big) \cap \big(\cup F{\cdot}\mathcal{D}'\big) \\ &= \cap F{\cdot}\mathcal{C}' \end{aligned}$$

As $\mathcal{C}'$ is strongly G-syndetic, $\cap F{\cdot}\mathcal{C}'$ is G-syndetic. Therefore $\cup F{\cdot}\mathcal{B}$ is also G-syndetic. The G-syndeticity of $\mathcal{B}$ follows. □

Define the following collection of sets

$$ {}^{\mathrm{s}}\Sigma_G \;=\; \{\mathcal{D} : \ \mathcal{D} \text{ is strongly } G\text{-syndetic}\}. $$

Note that Theorem 17.15 shows that $\Sigma_{G^1} \subseteq {}^{\mathrm{s}}\Sigma_G$.

17.19 Corollary Then ${}^{\mathrm{s}}\Sigma_G$ is finitely consistent, strongly G-syndetic, and G-invariant. Moreover, ${}^{\mathrm{s}}\Sigma_G$ is the maximal collection of sets that is G-syndetic and G-invariant. We also have

$$ {}^{\mathrm{s}}\Sigma_G \;=\; \bigcap_{q \in Q_G} q. $$

Proof. Strong G-syndeticity is an immediate consequence of Lemma 17.18. Finite consistency is a consequence of syndeticity. Finally, G-invariance is clear because any translate of a strongly G-syndetic formula is also strongly G-syndetic. □

The following should be compared with Theorem 17.7.

17.20 Corollary For every $\mathcal{D}$ the following are equivalent

1. ${}^{\mathrm{s}}\Sigma_G \cup \{\mathcal{D}\}$ is finitely consistent
2. $\mathcal{D}$ is weakly G-thick.

Proof. 1⇒2. If ${}^{\mathrm{s}}\Sigma_G \cup \{\mathcal{D}\}$ is finitely consistent, then $\neg\mathcal{D}$ is strongly G-syndetic. From Fact 17.2, we obtain that $\neg\mathcal{D}$ not being strongly G-syndetic is equivalent to $\mathcal{D}$ being weakly G-thick.

2⇒1. If ${}^{\mathrm{s}}\Sigma_G \cup \{\mathcal{D}\}$ is not finitely consistent, then $\neg\mathcal{D}$ is strongly G-syndetic. From Fact 17.2, $\mathcal{D}$ is not weakly G-thick. □

The following theorem asserts that the property of weak thickness is partition regular.

17.21 Theorem If $\mathcal{C} \cup \mathcal{B}$ is weakly G-thick then $\mathcal{B}$ or $\mathcal{C}$ is weakly G-thick.

Proof. As ${}^{\mathrm{s}}\Sigma_G$ is closed under conjunction. If $\mathcal{C} \cup \mathcal{B}$ is finitely consistent with ${}^{\mathrm{s}}\Sigma_G$ then so is one of the two sets. □

17.22 Exercise Prove that the following are equivalent

1. $\mathcal{D}$ is weakly G-thick
2. $\mathcal{D} = \mathcal{C} \cap \mathcal{B}$ for some G-syndetic set $\mathcal{B}$ and some G-thick set $\mathcal{C}$
3. there is a non G-syndetic set $\mathcal{C}$ such that $\mathcal{D} \cup \mathcal{C}$ is G-syndetic.

17.23 Exercise Prove that G-syndetic sets are weakly G-thick.

17.4 A tamer landscape

Under suitable assumptions some notions introduced in this chapter coalesce, and we are left with a tamer landscape. We will see an example in Theorem 18.29.

17.24 Theorem The following are equivalent

1. G-thick sets are G-wide
2. G-syndetic sets are closed under intersection
3. G-syndetic sets are strongly G-syndetic
4. weakly G-thick sets are G-thick.

Proof. Clearly 2⇔3⇔4.

1⇒2. Let $\mathcal{C}$ and $\mathcal{D}$ be G-syndetic sets. Suppose for a contradiction that $\mathcal{C} \cap \mathcal{D}$ is not G-syndetic. Then $\neg(\mathcal{C} \cap \mathcal{D})$ is G-thick. By 1 and Theorem 17.7 there is a G-invariant $p \in S_\pi(\Gamma)$ such that $\mathcal{C} \cap \mathcal{D} \notin p$. By completeness either $\mathcal{C} \notin p$ or $\mathcal{D} \notin p$. This is a contradiction because by Theorem 17.5 $\mathcal{C} \in p$ and $\mathcal{D} \in p$.

4⇒1. By Theorem 17.21 □

It is convenient to prove one last characterization of the above phenomenon. This is used in the next section.

17.25 Proposition The equivalent conditions in Theorem 17.24 are also equivalent to

5. $$\Sigma_G \;=\; \bigcap \Big\{ p \in S_\pi(\Gamma) \;:\; p \text{ is } G\text{-invariant} \Big\}.$$

Note that $\subseteq$ is always true, therefore 5 amounts to claiming that the type on the r.h.s. is G-syndetic.

Proof. Assume 1 of Theorem 17.24. It suffices to prove $\supseteq$, because the converse inclusion is always true. Let $\mathcal{D}$ be non G-syndetic. Then $\neg\mathcal{D}$ is G-thick and therefore G-wide. Then some G-invariant $p \in S_\pi(\Gamma)$ entails $\neg\mathcal{D}$. Therefore $\mathcal{D}$ does not belong to the type on the r.h.s.

Vice versa, assume 5 and note that set on the r.h.s. is closed under conjunction. Then 2 of Theorem 17.24 immediately follows. □

We say that π is G-stationary if there is a unique G-invariant $p \in S_\pi(\Gamma)$.

The conditions in Theorem 17.24 together with stationarity, produce the complete collapse of all notions of largeness introduced in this chapter.

17.26 Fact Let π be G-stationary. Assume that the equivalent conditions in Theorem 17.24 hold. Then the following are equivalent

1. $\mathcal{D}$ is G-syndetic
2. $\mathcal{D}$ is G-wide (equivalently, G-thick).

Proof. 1⇒2. As the consistency of Σ_G follows immediately from the assumption, G-syndetic sets are G-thick. (Stationarity is not required in this direction.)

2⇒1. If $\mathcal{D}$ was not G-syndetic then $\neg\mathcal{D}$ would be G-thick and, by the assumption, G-wide. This contradicts G-stationarity. □

17.5 Topology

Note that a topological space $\mathcal{X}$ is Hausdorff if and only if the diagonal of $\mathcal{X}^2$ is closed in the product topology. Let E be the intersection of all closed equivalence relations. Then $\mathcal{X}/E$ endowed with the quotient topology (i.e. a set is open if its pull back is open) is Hausdorff. By the above observation every continuous map from $\mathcal{X}$ into a Hausdorff space factor through the projection of $\mathcal{X}$ onto $\mathcal{X}/E$.

17.6 Group action on definable sets

We now focus on a particular G-algebra Γ that consists of definable sets.

Let $\Delta \subseteq L_{x\,z}(\mathcal{U})$ be a small set of formulas. By Δ^{B} we denote the set of formulas $\vartheta(x\,;\bar{z})$, where $\bar{z} = z_1,\dots,z_n$ is a tuple of variables of the same sort as z, that are Boolean combinations of formulas $\varphi(x\,;z_i)$ for some $\varphi(x\,;z) \in \Delta$. Let $\mathcal{Z} \subseteq \mathcal{U}^z$. We write $\Delta^{\mathrm{B}}(\mathcal{Z})$ for the set of formulas $\vartheta(x\,;\bar{c})$ for some $\vartheta(x\,;\bar{z}) \in \Delta^{\mathrm{B}}$ and $\bar{c} \in \mathcal{Z}^n$. In this section $\mathcal{X} \subseteq \mathcal{U}^x$ and Γ contains the subsets of $\mathcal{X}$ that are $\Delta^{\mathrm{B}}(\mathcal{Z})$-definable.

Assume that G acts on $\mathcal{Z}$. Then G acts on Γ as follows. When $g \in G$ and $\mathcal{D} = \vartheta(\mathcal{X};\bar{c})$ then $g\cdot\mathcal{D}$ is the set $\vartheta(\mathcal{X};g\cdot\bar{c})$.

17.27 Example (random graph) We present a simple example. Let $\mathcal{U} = \mathcal{Z} = \mathcal{U}^z = \mathcal{X} = \mathcal{U}^x$ be a saturated model of the of theory of the random graph.

Set $\pi = \{\mathcal{X}\}$. Let $G = \mathrm{Aut}(\mathcal{U})$ act naturally on $\mathcal{Z}$. We prove that the equivalent conditions in Theorem 17.24 hold.

There are two G-invariant global types $p_i(x)$. The first says that x is adjacent to all vertices, the second says that x is adjacent to none. The type on the r.h.s. of (5) in Proposition 17.25 is equivalent to the disjunction of $p_1(x)$ and $p_2(x)$. This is in turn equivalent to the type containing the formulas

$$\varphi_n(x,a) \quad = \quad r(x,a_1) \leftrightarrow r(x,a_2) \leftrightarrow \cdots\cdots \leftrightarrow r(x,a_n)$$

for every tuple of vertices $a = a_1,\dots,a_n$. Therefore we only need to prove that these formulas are G-syndetic.

17.28 Theorem The above formula $\varphi_n(x,a)$ is G-syndetic for all $n \geq 2$ and all $a \in \mathcal{Z}^n$.

Proof. Let $b^1,\dots,b^n$ be tuples of length n with disjoint ranges that have the same type as a (over $\varnothing$). Let C be the set of tuples $c = c_1,\dots,c_n$ such that $c_i \in \mathrm{range}(b^i)$. Finally when we pick the b^i, we do so in such a way that every tuple $c \in C$ has the same type as a.

We claim that the following disjunction is a tautology

$$\bigvee_{c\in C} \varphi(x,c) \;\vee\; \bigvee_{i=1}^{n} \varphi(x,b^i)$$

If $\varphi(x,b^i)$ holds for some i, then we are done. Otherwise, for every i we have some k such that $r(x,b^i_k)$ holds (as well as some h such that $r(x,b^i_h)$ does not). Let $c_i = b^i_k$ for some k such that $r(x,b^i_k)$ holds. Then $\varphi(x,c)$ holds. □

17.7 Definable groups

Section under major revision.

In this section $\mathcal{X}$ is a type-definable subset of $\mathcal{U}^x$. We assume that $G \subseteq \mathcal{Z} \subseteq \mathcal{U}^z$, and we require that the semigroup operation is among the function symbols of L. We also require that every formula $\varphi(z) \in L(G, \mathcal{X})$ that is satisfied in $\mathcal{Z}$ is satisfied in G. So, if we view G and $\mathcal{Z}$ as structures in their own right (in the proper language), then $G \preceq \mathcal{Z}$.

We pick some $\Delta \subseteq L_{x\,z}(\mathcal{U})$ such that $\varphi(x\,;g\cdot z) \in \Delta$ for every $\varphi(x\,;z) \in \Delta$ and $g \in G$. The set Γ contains the subsets of $\mathcal{X}$ that are $\Delta^{\mathrm{B}}(\mathcal{Z}'')$-definable. The action of G on $\mathcal{Z}$, and by extension on Γ, is given by the semigroup operation.

When $g \in \mathcal{Z}$ and $\mathcal{D} \in \Gamma$ we write $g \mathop{\smile\!\!\!\!\!\mid}_G \mathcal{D}$ if for every $\mathcal{D}_1, \ldots, \mathcal{D}_n \in \Gamma$ there is an $h \in G$ such that $h\cdot\mathcal{D}$ and $g\cdot\mathcal{D}$ generate the same Boolean algebra over $\mathcal{D}_1, \ldots, \mathcal{D}_n$.

$$\mathcal{C} \diamond \mathcal{D} \quad = \quad \Big\{ g\cdot a \ : \ g \in \mathcal{C},\ a \in \mathcal{D},\ g \mathop{\smile\!\!\!\!\!\mid}_G a \Big\}$$

We write $\mathcal{C} \diamond a$ for $\mathcal{C} \diamond \{a\}$.

When $P \subseteq S_\pi(\Gamma)$, we write $\mathrm{cl}(P)$ for the closure of P in the topology generated by sets $[\mathcal{D}]$ for $\mathcal{D} \in \Gamma$. That is $\mathrm{cl}(P) = \bigcap \big\{ \mathcal{C} \in \Gamma \ : \ P \subseteq [\mathcal{C}] \big\}$.

17.29 Fact For every $p \in S_\pi(\Gamma)$

$$\mathrm{cl}_\Gamma(G\cdot p) \quad = \quad \mathcal{Z} \diamond p$$

Proof. $\supseteq$. Let $q \in \mathcal{Z} \diamond p$, say $q = g\cdot p$ for some $g \mathop{\smile\!\!\!\!\!\mid}_G p$. Pick some arbitrary $\mathcal{D} \in \bigcap G\cdot p$ and suppose for a contradiction that $\mathcal{D} \notin g\cdot p$. Then $g \mathop{\smile\!\!\!\!\!\mid}_G p$ implies $g\cdot a \in \mathcal{D}$. This proves that $b \in \mathrm{cl}_\Gamma(G\cdot a)$.

$\subseteq$. Let $b \in \mathrm{cl}_\Gamma(G\cdot a)$ be given and define $p(x) = \mathrm{tp}(b/G)$. It suffices to prove that $z \mathop{\smile\!\!\!\!\!\mid}_G a$ is finitely consistent with $p(z\cdot a)$. In fact, if $g \mathop{\smile\!\!\!\!\!\mid}_G a$, then $g\cdot a \in \mathcal{Z} \diamond a$. And, as $\mathcal{Z} \diamond a$ is G-invariant, if $p(g\cdot a)$ then $b \in \mathcal{Z} \diamond a$. Now, to prove concistency, assume for a contradiction that $z \mathop{\smile\!\!\!\!\!\mid}_G a \rightarrow \neg\varphi(z\cdot a)$ for some $\varphi(x) \in p$. As any $c \in Z$ realizes $z \mathop{\smile\!\!\!\!\!\mid}_G a$, no element of $Z\cdot a$ satisfies $\varphi(x)$. This is a contradiction by 2 above. □

Let $\mathcal{D} \subseteq \mathcal{X}$ be an M-definable set. As Z is small, $\mathcal{D}$ is Z-thick if and only if $\bigcap Z\cdot\mathcal{D}$ is finitely consistent.

17.30 Fact Let $\mathcal{D} \subseteq \mathcal{X}$ be definable over M. Then the following are equivalent

1. $\mathcal{D}$ is Z-thick
2. $\mathcal{Z} \diamond a \subseteq \mathcal{D}$ for some $a \in \mathcal{D}$.

Proof. 1⇒2. Pick $a \in \bigcap Z\cdot\mathcal{D}$. Suppose for a contradiction that $g\cdot a \notin \mathcal{D}$ for some $g \in \mathcal{Z}$ such that $g \mathop{\smile\!\!\!\!\!\mid}_M a$. Then $c\cdot a \notin \mathcal{D}$ for some $c \in Z$. This contradicts the choice of a.

2⇒1. It suffices to note that $c \cdot \mathcal{Z} \diamond a = \mathcal{Z} \diamond a$ for every $c \in Z$. □

The following is the dual version of the above fact.

17.31 Fact Let $\mathcal{D} \subseteq \mathcal{X}$ be definable over M. Then the following are equivalent

1. $\mathcal{D}$ is Z-syndetic
2. $(\mathcal{Z} \diamond a) \cap \mathcal{D} \neq \varnothing$ for every $a \in \mathcal{D}$.

We say that $\mathcal{C} \subseteq \mathcal{X}$ is $\mathcal{Z}\diamond$-invariant if $\mathcal{Z} \diamond \mathcal{C} \subseteq \mathcal{C}$. The following fact is easy.

17.32 Fact Every (M-invariant?) minimal $\mathcal{Z}\diamond$-invariant subset of $\mathcal{X}$ is of the form $\mathcal{Z} \diamond a$ for some $a \in \mathcal{X}$ so, in particular, it is type-definable over M.

We say that $a \in \mathcal{X}$ is minimal if $\mathcal{Z} \diamond a$ is a minimal $\mathcal{Z}\diamond$-invariant subset of $\mathcal{X}$.

17.33 Theorem Let $\mathcal{D} \subseteq \mathcal{X}$ be definable over M. Then the following are equivalent

1. $\mathcal{D}$ is weakly Z-thick
2. $\mathcal{Z} \diamond a \subseteq \mathcal{D}$ for some minimal $a \in \mathcal{D}$.

Proof. 2⇒1. Let $b \in \mathcal{Z} \diamond a$. Then $b \in \mathcal{Z} \diamond \mathcal{D}$. Therefore $b \in c \cdot \mathcal{D}$ for some $c \in Z$. This proves that $\mathcal{Z} \diamond a \subseteq \bigcup Z \cdot \mathcal{D}$. By compactness $\mathcal{Z} \diamond a \subseteq \bigcup C \cdot \mathcal{D}$ for some finite $C \subseteq Z$. By Fact 17.30, $\mathcal{D}$ is weakly Z-thick.

1⇒2. Let $C \subseteq Z$ be a finite set such that $\bigcup C \cdot \mathcal{D}$ is Z-thick. By Fact 17.30, there is $a \in \bigcup C \cdot \mathcal{D}$ such that $\mathcal{Z} \diamond a \subseteq \bigcup C \cdot \mathcal{D}$. Let $a' \in \mathcal{Z} \diamond a$ be minimal. As $a' \in c \cdot \mathcal{D}$ for some $c \in C$, we conclude that $c^{-1} \cdot a' \in \mathcal{D}$. Then $c^{-1} \cdot a'$ is the minimal element required by 2. □

17.8 Notes and references

In Example **??** we prove a theorem of Newelski's [4]. The original proof is rather long and complex. A simplified proof (also due, reportedly, to Newelski) appears in [5, Section 3.3] and [1, Chapter 9]. The proof here is a streamlined and generalized version of the latter – inspired by [6].

[1] Enrique Casanovas, *Simple theories and hyperimaginaries*, Lecture Notes in Logic, vol. 39, Cambridge University Press, 2011.

[2] Ludomir Newelski, *Topological dynamics of definable group actions*, J. Symbolic Logic (2009).

[3] Ehud Hrushovski, *Stable group theory and approximate subgroups*, J. Amer. Math. Soc. **25** (2012), no. 1, 189–243.

[4] Ludomir Newelski, *The diameter of a Lascar strong type*, Fund. Math. (2003).

[5] Rodrigo Peláez, *About the Lascar group*, PhD Thesis, Universitat de Barcelona, Departament de Lógica, História i Filosofia de la Ciéncia, 2008.

[6] Domenico Zambella, *On the diameter of Lascar strong types after Ludomir Newelski*, A tribute to Albert Visser, Coll. Publ., [London], 2016, arXiv:1605.00218.

Chapter 18

Stability

⚠ Chapter under major revision

In this chapter we fix a signature L, a complete theory T without finite models, and a saturated model $\mathcal{U}$ of inaccessible cardinality $\kappa > |L|$. The notation and implicit assumptions are as in Section 9.3.

18.1 Externally definable sets

Let $p(x) \subseteq L(\mathcal{U})$ be a finitely consistent type. Recall from Section 14.1 that for every formula $\varphi(x\,;z) \in L$ we define

$$\mathcal{D}_{p,\varphi} \quad = \quad \Big\{a \in \mathcal{U}^z \ : \ \varphi(x\,;a) \in p\Big\}.$$

We say that $\mathcal{D}$ is externally definable if $\mathcal{D} = \mathcal{D}_{p,\varphi}$ for some type $p(x) \in S_\varphi(\mathcal{U})$.

Equivalently, a set $\mathcal{D}$ is externally definable if it is the trace over $\mathcal{U}^z$ of a set which is definable in some elementary extension of $\mathcal{U}$. More precisely, $\mathcal{D} = \varphi({}^*b\,;{}^*\mathcal{U}^z) \cap \mathcal{U}^z$ where ${}^*\mathcal{U} \succeq \mathcal{U}$ and ${}^*b \in {}^*\mathcal{U}^x$. This explains the terminology.

We prefer to deal with external definability in a different, though equivalent, way.

18.1 Definition We say that $\mathcal{D}$ is approximated by the formula $\varphi(x\,;z)$ if for every finite $B \subseteq \mathcal{U}^z$ there is a tuple $a \in \mathcal{U}^x$ such that $\varphi(a\,;B) = \mathcal{D} \cap B$. We call $\varphi(x\,;z)$ the sort of $\mathcal{D}$. If in addition $\varphi(a\,;\mathcal{U}^z) \subseteq \mathcal{D}$, we say that $\mathcal{D}$ is approximated from below. Equivalently, we say that $\mathcal{D}$ is approximated from below if for every finite $B \subseteq \mathcal{D}$ there is a tuple $a \in \mathcal{U}^x$ such that $B \subseteq \varphi(a\,;\mathcal{U}^z) \subseteq \mathcal{D}$. The dual notion of approximation from above is defined as expected (and coincides with $\neg\mathcal{D}$ being approximated by $\neg\varphi(x\,;z)$ from below).

The following proposition is clear by compactness.

18.2 Proposition For every $\mathcal{D}$ the following are equivalent

1. $\mathcal{D}$ is approximated by $\varphi(x\,;z)$
2. $\mathcal{D}$ is externally definable by $\varphi(x\,;z)$.

The rest of this section is only required in Chapter 19.

Approximability from below is an adaptation to our context of the notion of *having an honest definition* in [1].

18.3 Example Every definable set is trivially approximable. Sets may be approximable by different formulas. For instance, if $T = T_{\text{dlo}}$, then $\mathcal{D} = \{z \in \mathcal{U} : a \leq z \leq b\}$ is approximable by the formula $x_1 < z < x_2$ though not definable from below nor from above by $x_1 < z < x_2$.

Now, let $T = T_{\rm rg}$. Then every $\mathcal{D} \subseteq \mathcal{U}$ is approximable and, when $\mathcal{D}$ has small infinite cardinality, it is approximable from above but not from below.

In Definition 18.1, the sort $\varphi(x\,;z)$ is fixed (otherwise any set would be approximable) but we can drop this requirement if we allow B to have infinite cardinality.

18.4 Proposition For every $\mathcal{D}$ the following are equivalent

1. $\mathcal{D}$ is approximable
2. if $C \subseteq \mathcal{U}^z$ has cardinality $\le |L|$, then $\psi(C) = \mathcal{D} \cap C$ for some $\psi(z) \in L(\mathcal{U})$.

Proof. To prove 2⇒1 negate 1 For each formula $\psi(x\,;z) \in L$ choose a finite set B such that $\psi(b\,;B) \neq \mathcal{D} \cap B$ for every $b \in \mathcal{U}^x$. Let C be the union of all these finite sets. Then C witnesses the failure of 2. □

From the following easy observation of Chernikov and Simon [1] we obtain an interesting (and mysterious) quantifier elimination result originally due to Shelah, see Corollary 19.6 below.

18.5 Proposition Let $\mathcal{C} \subseteq \mathcal{U}^{y\,z}$ be approximated from below by the formula $\varphi(x\,;y\,z)$. Then $\mathcal{D} = \big\{z : \exists y\ \big(y\,z \in \mathcal{C}\big)\big\}$ is approximated from below by $\exists y\,\varphi(x\,;y\,z)$.

Proof. Let $B \subseteq \mathcal{U}$ be finite. We want $a \in \mathcal{U}^x$ such that

a. $\quad \exists y\ \big(y\,b \in \mathcal{C}\big) \ \leftrightarrow\ \exists y\,\varphi(a\,;y\,b) \quad$ for every $b \in B$

b. $\quad \forall z\ \Big[\exists y\,\varphi(a\,;y\,z) \ \rightarrow\ \exists y\ \big(y\,z \in \mathcal{C}\big)\Big]$

Let $D \subseteq \mathcal{U}^y$ be a finite set such that

c. $\quad \exists y \in D\ \big(y\,b \in \mathcal{C}\big) \ \leftrightarrow\ \exists y\ \big(y\,b \in \mathcal{C}\big) \quad$ for every $b \in B$

As $\mathcal{C}$ is approximable from below, there is an a such that

a′. $\quad d\,b \in \mathcal{C} \ \leftrightarrow\ \varphi(a\,;d\,b) \quad$ for every $d\,b \in D, B$

b′. $\quad \forall y\,z\ \Big[\varphi(a\,;y\,z) \ \rightarrow\ y\,z \in \mathcal{C}\Big]$

We obtain b from b′ simply by logic. Implication $\rightarrow$ in a follows from a′ and c. Implication $\leftarrow$ follows from b. □

18.6 Corollary If $p(x) \in S(\mathcal{U})$ is honestly definable then the family of sets externally definable by $p(x)$ is closed under quantifiers and Boolean combinations.

Proof. The sets externally definable by $p(x)$ are always closed under Boolean operations. By the proposition above, they are closed under quantifiers. □

18.2 Ladders and definability

Let $\pi \subseteq \mathcal{U}^x \times \mathcal{U}^z$ be a binary relation (which need not be definable in any sense). We denote by $\pi(x\,;z)$ the predicate associated to π. We say that $\langle a_i\,;b_i : i < \alpha\rangle$ is a ladder of length α for $\pi(x\,;z)$ if for every $i < j < \alpha$

$$\pi(a_i\,;b_j) \quad\wedge\quad \neg\pi(a_j\,;b_i).$$

We say that $\pi(x\,;z)$ is unstable if it admits ladder sequences of arbitrary finite length; stable otherwise. If $\pi(x\,;z)$ is unstable we may also sat that it has the order property.

We will mainly deal with relations that are definable but it noteworthy that a few results in this section do not need this assumption.

When $\pi(x\,;z)$ is a formula, compactness allows to simplify the definition of stability as in the following fact.

18.7 Fact The following are equivalent for every $\pi(x\,;z) \in L(\mathcal{U})$

1. $\pi(x\,;z)$ is stable
2. $\pi(x\,;z)$ admits no ladder of length ω.

Sometimes 2 in the above fact is taken as definition of stability. The following is another useful characterization.

18.8 Theorem Let $\pi(x\,;z) \in L(A)$. Then the following are equivalent

1. $\pi(x\,;z)$ is stable
2. $\pi(a_0\,;b_1) \to \pi(a_1\,;b_0)$ for every A-indiscernible sequence $\langle a_i\,;b_i : i < \omega\rangle$.

Proof. 2⇒1. Suppose that the ladder $\langle a_i\,;b_i : i < \omega\rangle$ witnessee the failure of 2 in Fact 18.7. Let $\langle a'_i\,;b'_i : i < \omega\rangle$ be a sequence of A-indiscernibles with the same EM-type as $\langle a_i\,;b_i : i < \omega\rangle$. By A-invariance $\pi(a'_0\,;b'_1)$ and $\neg\pi(a'_1\,;b'_0)$.

2⇒1. Immediate by indiscernibility. □

18.9 Lemma Boolean combinations of stable relations are stable. Moreover, if $\pi(x\,;z)$ then $\pi^{-1}(x\,;z)$ and $\pi(x,x'\,;z,z')$ are also stable (in the latter x' and z' are dummy variables).

Proof. The second claim is immediate. To prove the first, it suffices to consider conjunction and negation. Negation is immediate, in fact if $\langle a_i\,;b_i : i < n\rangle$ is a ladder for $\pi(x\,;z)$ then $\langle a_{n-i}\,;b_{n-i} : i < n\rangle$ is a ladder for $\neg\pi(x\,;z)$. As for conjunction, note that when π is invariant over A it easily follows from Theorem 18.8. In general we reason as follows.

Assume $\pi_1(x\,;z) \wedge \pi_2(x\,;z)$ is unstable and pick a ladder sequence $\langle a_i\,;b_i : i < m\rangle$, where m is specified below. Let $H_1, H_2 \subseteq m^{(2)}$ contain those pairs $j < i$ such that $\neg\pi_1(a_i\,;b_j)$, respectively $\neg\pi_2(a_i\,;b_j)$. Then $H_1 \cup H_2 = m^{(2)}$. If m is large enough, by the (finite) Ramsey Theorem there is a set H of size n such that $H^{(2)}$ is either contained in H_1 or in H_2. Suppose the first for definiteness. Let $H = \{h_1, \dots, h_n\}$ then $\langle a_{h_i}\,;b_{h_i} : i < n\rangle$ is a ladder sequence for $\pi_1(x\,;z)$. □

The following theorem claims what is arguably one of the most important properties of stable formulas: any set that is externally definable by a stable formula is definable (by a positive Boolean combination of the formula).

18.10 Theorem Any $\mathcal{D} \subseteq \mathcal{U}^z$ approximated by a stable formula is definable. In general, if $\pi(x\,;z)$ is a stable relation that approximates $\mathcal{D}$ then there are $a_{i,j} \in \mathcal{U}^x$ such that

$$z \in \mathcal{D} \quad \leftrightarrow \quad \bigvee_{i=1}^{n} \bigwedge_{j=1}^{m} \pi(a_{i,j}\,;z)$$

Theorem 18.18 proves that the converse holds for formulas: if every set $\mathcal{D}$ approximated by the formula $\varphi(x\,;z)$ is definable then $\varphi(x\,;z)$ is stable.

Proof. The theorem follows immediately from the two lemmas below. □

18.11 Lemma If $\mathcal{D}$ is approximated from below by a stable relation $\pi(x\,;z)$ then

$$z \in \mathcal{D} \quad \leftrightarrow \quad \bigvee_{i=0}^{n} \pi(a_i\,;z)$$

for some $a_0,\ldots,a_n \in \mathcal{U}^x$.

Proof. The elements $a_0,\ldots,a_n$ are defined recursively together with some auxiliary elements $b_0,\ldots,b_{n-1} \in \mathcal{D}$.

Suppose $b_0,\ldots,b_{n-1}$ have been defined (this assumption is empty if $n = 0$). We first define a_n, then b_n. Choose $a_n \in \mathcal{U}^x$ such that $b_0,\ldots,b_{n-1} \in \pi(a_n\,;\mathcal{U}^z) \subseteq \mathcal{D}$. This is possible because $\mathcal{D}$ is approximated from below. Now, if possible, choose b_n such that

$$b_n \quad \in \quad \mathcal{D} \smallsetminus \bigcup_{i=0}^{n} \pi(a_i\,;\mathcal{U}^z).$$

Then $\langle a_i\,;b_i : i \leq n\rangle$ is a ladder sequence. By stability, for some n, the tuple b_n does not exist. This yields the required $a_0,\ldots,a_n$. □

18.12 Lemma If $\mathcal{D}$ is approximated by a stable formula $\pi(x\,;z)$. Then, for some m, the formula

$$\psi(x_0,\ldots,x_m\,;z) \quad = \quad \bigwedge_{j=0}^{m} \pi(x_j\,;z)$$

approximates $\mathcal{D}$ from below.

Proof. Let m be such that there is no ladder sequence for $\pi(x\,;z)$ of length greater than m. Let $C \subseteq \mathcal{D}$ be finite. We prove that there are some $a_0,\ldots,a_m$ such that $C \subseteq \psi(a_0,\ldots,a_m\,;\mathcal{U}^z) \subseteq \mathcal{D}$. As in the proof above, we define by recursion a ladder sequence for $\pi(x\,;z)$. Suppose that $a_0,\ldots,a_{n-1}$ and $b_0,\ldots,b_{n-1} \notin \mathcal{D}$ have been defined. We first define a_n, then b_n. Choose $a_n \in \mathcal{U}^x$ such that

$$C \;\subseteq\; \pi(a_n\,;\mathcal{U}^z) \;\subseteq\; \mathcal{U}^z \smallsetminus \{b_0,\ldots,b_{n-1}\}.$$

This a_n exists, because $\mathcal{D}$ is approximated by $\pi(x\,;z)$. (Apply Definition 18.1 with any B such that $C \cup \{b_0,\ldots,b_{n-1}\} \subseteq B$.) Then, if possible, let b_n such that

$$b_n \quad \in \quad \bigcap_{i=0}^{n} \pi(a_i,\mathcal{U}^z) \smallsetminus \mathcal{D}$$

This procedure has to stop at some $n \leq m$. Therefore the required parameters are $a_1,\ldots,a_n = a_{n+1} = \cdots = a_m$. □

By Lemma 18.9 the formula $\psi(x_0,\dots,x_m\,;z)$ is stable therefore this concludes the proof of Theorem 18.10.

18.13 Remark Theorem 18.10 is often cast in the following apparently more general form. Let $\mathcal{X} \subseteq \mathcal{U}^x$ and $\mathcal{Z} \subseteq \mathcal{U}^z$. We say that the relation $\pi(x\,;z)$ is stable between the sets $\mathcal{X}$ and $\mathcal{Z}$ if for some n no ladder of length n exists with $a_i \in \mathcal{X}$ and $b_i \in \mathcal{Z}$.
Let $\mathcal{D}$ be approximable by $\pi(x\,;z) \wedge x \in \mathcal{X} \wedge z \in \mathcal{Z}$. Then there are $a_{1,1},\dots,a_{n,m} \in \mathcal{X}$ such that for every $b \in \mathcal{Z}$

$$b \in \mathcal{D} \quad\leftrightarrow\quad \bigvee_{i=1}^{n} \bigwedge_{j=1}^{m} \pi(a_{i,j}\,;b).$$

Note also that, when $\mathcal{X}$ and $\mathcal{Z}$ are type-definable, Fact 18.7, Theorem 18.8, and Lemma 18.9 generalize in the obvious way.

18.14 Exercise Prove that a relation $\pi(x\,;z)$ is instable if and only if for every $n < \omega$ there is a set B and a sequence $\langle a_i : i < n\rangle$ such that $\pi(a_i\,;B) \subset \pi(a_{i+1}\,;B)$ for all $i < n$.

18.15 Exercise Find some unstable $p(x\,;z) \subseteq L(A)$ that admits no ladder of infinite length. Hint: work in a dense linear order, and consider the type containing the formulas $x < y \ \vee\ y \notin \{a_0,\dots,a_n\}$ for some suitable $\langle a_i : i < n\rangle$.

18.16 Exercise Let $\varphi(x,y) \in L$, where $|x| = |y| = 1$. Suppose there is an infinite set $A \subseteq \mathcal{U}$ such that $\varphi(a,b) \not\leftrightarrow \varphi(b,a)$ for every two distinct $a,b \in A$. Prove that $\varphi(x\,;y)$ is unstable.

18.17 Exercise Prove that the following are equivalent for every relation $\pi(x\,;z)$

1. $\pi(x\,;z)$ is instable
2. there are arbitrary long finite sequences $\langle a_i\,;b_i : i < n\rangle$ such that for every $i,j < n$
$$i < j \quad\Leftrightarrow\quad \pi(a_i\,;b_j)$$
3. there are arbitrary long finite sequences $\langle a_i\,;b_i : i < n\rangle$ such that for every $i,j < n$
$$j \leq i \quad\Leftrightarrow\quad \pi(a_i\,;b_j)$$

18.3 Stability and the number of types

The following proposition highlights the connection between the stability of the formula $\varphi(x\,;z)$ and the cardinality of $S_\varphi(\mathcal{U})$, equivalently the number of sets externally definable by $\varphi(x\,;z)$.

18.18 Theorem The following are equivalent

1. $\varphi(x\,;z)$ is stable
2. every subset of $\mathcal{U}^z$ that is externally definable by $\varphi(x\,;z)$ is definable
3. there are $\leq \kappa$ subsets of $\mathcal{U}^z$ that are externally definable by $\varphi(x\,;z)$
4. there are $< 2^\kappa$ subsets of $\mathcal{U}^z$ that are externally definable by $\varphi(x\,;z)$.

Proof. 1⇒2 Clear by Proposition 18.2 and Theorem 18.10.

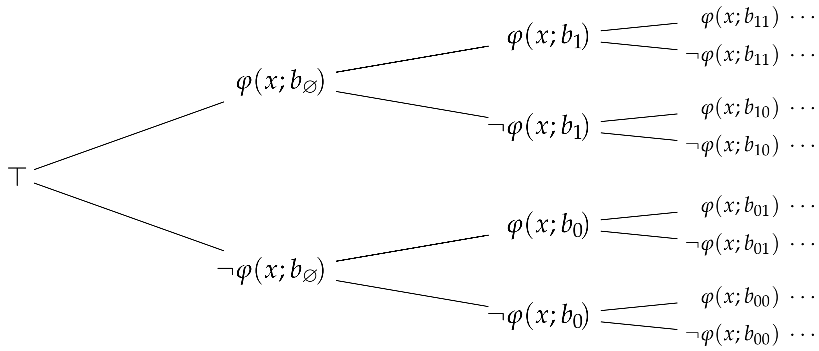
2⇒3⇒4 Obvious.

4⇒1 Suppose that $\varphi(x\,;z)$ is not stable. By compactness there is a ladder sequence $\langle a_i\,; b_i : i \in I\rangle$ where $I, <_I$ a dense linear order of cardinality κ with 2^κ cuts. Here, a *cut* is a subset $c \subseteq I$ that is closed downward. For every such $c \subseteq I$ we pick a global type

$$p_c(x) \quad \supseteq \quad \{\varphi(x\,;b_i) \leftrightarrow i \in c \;:\; i \in I\}.$$

Clearly the sets $\mathcal{D}_{p_c,\varphi}$ are all distinct. □

Binary trees of formulas have been introduced in Definition 12.21. Here we restrict to trees of a particular form. Namely, $\langle \psi_s : s \in 2^{<\omega}\rangle$ where $\psi_\varnothing = \top$ and for $s \in 2^{<\omega}$ and $i \in 2$ we have $\psi_{s\frown 0}(x) = \neg\varphi(x\,;b_s)$ and $\psi_{s\frown 1}(x) = \varphi(x\,;b_s)$. If we define $\varphi^0 = \neg\varphi$ and $\varphi^1 = \varphi$ the condition of consistency becomes for every $s \in 2^\omega$ the type $\{\varphi^{s_n}(x\,;b_{s\restriction n}) \;:\; n < \omega\}$.



When a tree of this form exists, we say that $\varphi(x\,;z)$ has the binary tree property.

18.19 Theorem The following are equivalent

1. $\varphi(x\,;z)$ is unstable
2. $\varphi(x\,;z)$ has the binary tree property.

Proof. 1⇒2. The argument is the same as in the proof of Lemma 12.22. Assume 1. By Theorem 18.18 there are 2^κ sets externally definable by $\varphi(x\,;z)$. Then there is $b_\varnothing \in \mathcal{U}^z$ such that there are 2^κ sets $\mathcal{D}$ externally definable by $\varphi(x\,;z)$ and such that $b_\varnothing \in \mathcal{D}$ and 2^κ sets such that $b_\varnothing \notin \mathcal{D}$.

Assume inductively that $b : 2^n \to \mathcal{U}^z$ is such that for all $s \in 2^n$ and $r \in 2^{n+1}$ there are 2^κ sets $\mathcal{D}$ externally definable by $\varphi(x\,;z)$ and such that $b_{s\restriction i} \in \mathcal{D} \leftrightarrow r(i) = 1$. Reasoning as above we can extend b to a map $b' : 2^{n+1} \to \mathcal{U}^z$ with the same property.

2⇒1. From 2, by compactness, there is a binary tree of height κ. Hence there are 2^κ sets that are externally definable by $\varphi(x\,;z)$. Therefore, by Theorem 18.18, $\varphi(x\,;z)$ is not stable. □

18.20 Corollary The following are equivalent

1. $\varphi(x\,;z)$ is a stable formula
2. $\big|S_\varphi(A^z)\big| \leq |A|$ for all countable sets A
3. $\big|S_\varphi(A^z)\big| < 2^{|A|}$ for all countable sets A.

Proof. It follows immediately from Lemma 12.22 and Theorem 18.19. □

We use binary trees to prove the following fundamental lemma (attributed by Harnik and Harrington [2] to Martin Ziegler).

18.21 Lemma Let $\varphi(x\,;z), \psi(x\,;z) \in L(A)$. Let $b \in \mathfrak{U}^z$. Assume that $\varphi(x\,;z)$ is stable and that $\varphi(x\,;b) \vee \psi(x\,;b)$ is finitely satisfied in every $M \supseteq A$. Then $\varphi(x\,;b)$ or $\psi(x\,;b)$ is finitely satisfied in every $M \supseteq A$.

Proof. Negate the theorem. Then there are two models containing A, the first omitting $\varphi(x\,;b)$, the second omitting $\psi(x\,;b)$. It is easy to see that we can expand these models to two substructures $\mathfrak{U}_0$ and $\mathfrak{U}_1$ that are A-isomorphic to $\mathfrak{U}$. Then $\varphi(\mathfrak{U}_0^x\,;b) = \psi(\mathfrak{U}_1^x\,;b) = \varnothing$.

Let $f_0, f_1 : \mathfrak{U} \to \mathfrak{U}_i$ be the A-isomorphisms mentioned above. Set $f_\varnothing = \mathrm{id}_{\mathfrak{U}}$, then define inductively $f_{s^\frown 0} = f_0 \circ f_s$ and $f_{s^\frown 1} = f_1 \circ f_s$, for every $s \in 2^{<\omega}$. We write $\mathfrak{U}_s = f_s[\mathfrak{U}]$ and $b_s = f_s\, b$. Then, we obtain

$$\varphi(\mathfrak{U}^x_{s^\frown 0}\,;b_s) \quad = \quad \psi(\mathfrak{U}^x_{s^\frown 1}\,;b_s) \quad = \quad \varnothing$$

We prove that the branches of the tree $\langle \varphi(x\,;b_s) \;:\; s \in 2^{<\omega}\rangle$ are consistent. By Theorem 18.19, this contradicts the stability of $\varphi(x\,;z)$. Assume provisionally that there is a consistent formula $\vartheta(x) \in L(A)$ such that

$$\vartheta(x) \quad \to \quad \varphi(x\,;b) \vee \psi(x\,;b)$$

Let $a \in \vartheta(\mathfrak{U}^x)$ be arbitrary. Let $s \in 2^n$ be given. We prove that $a_s = f_s\, a$ witnesses the consistency of $\{\varphi^{s_i}(x\,;b_{s\restriction i}) \,:\, i < n\}$, where φ^0 and φ^1 stands for $\neg\varphi$, respectively φ.

Clearly $a_s \in \vartheta(\mathfrak{U}^x_s)$, and, as $\mathfrak{U}_s \preceq \mathfrak{U}_{s\restriction i}$, we also have that $a_s \in \vartheta(\mathfrak{U}^x_{s\restriction i})$. We prove that $\varphi^{s_i}(a_s\,;b_{s\restriction i})$. When $s_i = 0$, from # we obtain

$$\mathfrak{U}^x_{s\restriction i^\frown 0} \quad \subseteq \quad \neg\varphi(\mathfrak{U}^x_{s\restriction i^\frown 0}\,;b_{s\restriction i})$$

Then $\varphi^0(a_s\,;b_{s\restriction i})$ follows because $\mathfrak{U}^x_s \subseteq \mathfrak{U}^x_{s\restriction i^\frown 0}$. Similarly, when $s_i = 1$ we obtain

$$\mathfrak{U}^x_{s\restriction i^\frown 1} \quad \subseteq \quad \neg\psi(\mathfrak{U}^x_{s\restriction i^\frown 1}\,;b_{s\restriction i})$$

Then from ##

$$\vartheta(\mathfrak{U}^x_{s\restriction i^\frown 1}) \quad \subseteq \quad \varphi(\mathfrak{U}^x_{s\restriction i^\frown 1}\,;b_{s\restriction i})$$

and $\varphi^1(a_s\,;b_{s\restriction i})$ follows.

We are left with proving that the provisional assumption is redundant. By Exercise 11.12 there is a formula $\vartheta(\bar{x}) \in L(A)$ such that

$$\vartheta(\bar{x}) \quad \to \quad \varphi'(\bar{x}\,;b) \vee \psi'(\bar{x}\,;b)$$

where $\bar{x} = x_1, \ldots, x_n$ and

$$\varphi'(\bar{x}\,;b) \quad = \quad \bigvee_{i=1}^{n} \varphi(x_i\,;b)$$

$$\psi'(\bar{x}\,;b) \quad = \quad \bigvee_{i=1}^{n} \psi(x_i\,;b)$$

Note that $\varphi'(\bar{x}\,;z)$ is a stable formula. Therefore we can apply what proved above to the formulas $\varphi'(\bar{x}\,;b)$ and $\psi'(\bar{x}\,;b)$ and note that these formulas are satisfied in $M^{\bar{x}}$ if and only if $\varphi(x\,;b)$, respectively $\psi(x\,;b)$, are satisfied in M^x. $\square$

18.4 Symmetry and stationarity

The following corollary of Theorem 18.21 is of fundamental importance. It implies, in particular the existence of Lascar invariant types.

18.22 Corollary Let $\varphi(x\,;z) \in L(A)$ be stable. Let $q(x) \subseteq L(\mathcal{U})$ be finitely satisfiable in every $M \supseteq A$. Then there is a type $p(x) \in S_\varphi(\mathcal{U})$ such that $q(x) \cup p(x)$ is finitely satisfiable in every $M \supseteq A$.

Proof. Let $p(x) \subseteq L_\varphi(\mathcal{U})$ be maximal such that $q(x) \cup p(x)$ is finitely satisfiable in every $M \supseteq A$. We prove that $p(x)$ is complete. Suppose not than there are a conjunction $\alpha(x)$ of formulas in $q(x)$, a conjunction $\vartheta(x\,;b_1,\ldots,b_n)$ of formulas in $p(x)$, and a formula $\varphi(x\,;b)$ such that

$$\alpha(x) \wedge \vartheta(x\,;b_1,\ldots,b_n) \wedge \varphi(x\,;b) \quad \text{and} \quad \alpha(x) \wedge \vartheta(x\,;b_1,\ldots,b_n) \wedge \neg\varphi(x\,;b)$$

are not satisfied in every $M \supseteq A$. The disjunction of the two formulas is satisfied in every $M \supseteq A$ by the definition of $p(x)$. Moreover the formulas obtained replacing $b, b_1, \ldots, b_n$ by $z, z_1, \ldots, z_n$ are stable. This contradicts Theorem 18.21 $\square$

18.23 Theorem (Symmetry) Let $\varphi(x\,;z) \in L(M)$ be stable. Let $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^z$ be such that $a \mathop{\smile\!\!\!\!\!\!\downarrow}_M b$. Then

$$\varphi(a\,;b) \quad\Rightarrow\quad \varphi(a\,;M^z) \neq \varnothing.$$

Note that, when all formulas in $L_{x\,z}(M)$ are stable, the theorem says

$$a \mathop{\smile\!\!\!\!\!\!\downarrow}_M b \quad\Rightarrow\quad b \mathop{\smile\!\!\!\!\!\!\downarrow}_M a.$$

Proof. Assume $\varphi(a\,;b)$. Let $\mathcal{V} \preceq \mathcal{U}$ be isomorphic to $\mathcal{U}$ over M and such that $b \in \mathcal{V}^z$ and $a \mathop{\smile\!\!\!\!\!\!\downarrow}_M \mathcal{V}$. Such $\mathcal{V}$ exists by Proposition 14.10. By stability, there is $\psi(z) \in L(\mathcal{V})$ such that $\psi(\mathcal{V}^z) = \varphi(a, \mathcal{V}^z)$. Recall that by Lemma 14.9 (non-splitting) if $b' \equiv_M b''$ are in $\mathcal{V}$ then $b' \equiv_{M,a} b''$. Then $\psi(\mathcal{V}^z)$ is invariant under $\mathrm{Aut}(\mathcal{V}/M)$, so we can assume that $\psi(z) \in L(M)$. Therefore $\psi(z)$ is satisfied in M and so is $\varphi(a, z)$. $\square$

We deduce a version of Harrington's mysterious Lemma cf. [3, Lemma 8.3.4].

18.24 Corollary Let $\varphi(x\,;z) \in L(M)$ be stable. Let $p(x) \in S_\varphi(\mathcal{U})$ and $q(z) \in S_{\varphi^{\mathrm{op}}}(\mathcal{U}^x)$ be invariant over M. Then for every $a \vDash p(x){\restriction}M$ and $b \vDash q(z){\restriction}M$

$$a \in \mathcal{D}_{q,\varphi^{\mathrm{op}}} \quad\leftrightarrow\quad b \in \mathcal{D}_{p,\varphi}.$$

Proof. Note that we can assume $a \mathop{\smile\!\!\!\!\!\!\downarrow}_M b$. Then, from we obtain immediately

$$\varphi(a;b) \quad\leftrightarrow\quad a \in \mathcal{D}_{q,\varphi^{\mathrm{op}}}.$$

In fact, as $\mathcal{D}_{q,\varphi^{\mathrm{op}}}$ is definable over M, the contrary would contradict $b \vDash q(z){\restriction}M$. The corollary follows that we can prove

1. $$\varphi(a\,;b) \quad\leftrightarrow\quad b \in \mathcal{D}_{p,\varphi}$$

Note that 1 is a stable formula. Hence from its negation and Theorem 18.23 we obtain that $\varphi(a\,;b') \not\leftrightarrow b' \in \mathcal{D}_{p,\varphi}$ holds for some $b' \in M^z$. This would contradict $a \vDash p(x){\restriction}M$. $\square$

Now we show that under the assumption of stability Lascar invariance reduces to a tamer form of invariance.

18.25 Proposition Let $\varphi(x\,;z) \in L(A)$ be stable. Then for every $p(x) \in S_\varphi(\mathcal{U})$ the following are equivalent

1. $\mathcal{D}_{p,\varphi} \in \mathrm{acl^{eq}}A$
2. $p(x)$ is finitely satisfied in every $M \supseteq A$
3. $p(x)$ is Lascar invariant over A
4. $p(x)$ is invariant over $\mathrm{acl^{eq}}A$.

Proof. 2⇒3⇒1 and 4⇔1 are clear.

1⇒2 Let $M \supseteq A$ be given. Let $\psi(x)$ be a conjunction of formulas in $p(x)$, we prove that $\psi(x)$ is satisfied in M. For simplicity we assume that $\psi(x)$ and the form $\varphi(x\,;b)$. The same argument extends easily because $\mathrm{acl^{eq}}A$ is closed under Boolean combinations and Cartesian product.

Pick $a \vDash p(x) \restriction M$. Then $\varphi(a\,;M^z) = \mathcal{D}_{p,\varphi} \cap M^z$. We can assume $b \downarrow_M a$. We claim that $\varphi(a\,;b) \leftrightarrow b \in \mathcal{D}_{p,\varphi}$. Otherwise, as by 1 the formula $\varphi(x\,;z) \not\leftrightarrow z \in \mathcal{D}_{p,\varphi}$ is $L(M)$, using $b \downarrow_M a$ we would contradict $\varphi(a\,;M^z) = \mathcal{D}_{p,\varphi} \cap M^z$. As $b \in \mathcal{D}_{p,\varphi}$ is true, so is $\varphi(a\,;b)$. By symmetry $\varphi(M^x;b) \neq \varnothing$. □

A type $q(x) \subseteq L(\mathcal{U})$ is stationary if there is a unique global type $p(x) \in S_\varphi(\mathcal{U})$ that is Lascar invariant over A. The formula $\varphi(x\,;z)$ and the set A are to be inferred from the context. The following proposition says that in a stable theory with elimination of imaginaries types over algebraically closed sets are stationary.

18.26 Theorem Let $\varphi(x\,;z) \in L(A)$ be stable. Then every $q(x) \in S_\varphi(\mathrm{acl^{eq}}A)$ is stationary.

Recall that $S_\varphi(\mathrm{acl^{eq}}A)$ is the set of maximal types that contains Boolean combinations of formulas of the form $\varphi(x\,;b)$ for any $b \in \mathcal{U}^z$ that are invariant over $\mathrm{acl^{eq}}A$.

Proof. Existence follows from Corollary 18.22 and Proposition 18.25. To prove uniqueness, pick two types $p_i(x) \in S_\varphi(\mathcal{U})$, for $i = 1,2$, that are finitely consistent with $q(x)$ and invariant over $\mathrm{acl^{eq}}A$ (equivalently, Lascar invariant over A). It suffices to prove that $\mathcal{D}_{p_1,\varphi} = \mathcal{D}_{p_2,\varphi}$. Let $b \in \mathcal{U}^z$ be given.

There exists a type $t(z) \in S_{\varphi^{\mathrm{op}}}(\mathcal{U})$ that is finitely satisfied in every model $M \supseteq A$ and is consistent with $\mathrm{tp}(b/\mathrm{acl^{eq}}A)$. By invariance $\mathcal{D}_{p_i,\varphi}$ and $\mathcal{D}_{t,\varphi^{\mathrm{op}}}$ are both in $\mathrm{acl^{eq}}A$. Let $M \supseteq A$. Let $a_i \vDash p_i(x){\restriction}M$ and $b' \vDash t(z){\restriction}M$. By Corollary 18.24, $a_i \in \mathcal{D}_{t,\varphi^{\mathrm{op}}} \leftrightarrow b' \in \mathcal{D}_{p_i,\varphi}$. Note that the formula $x \in \mathcal{D}_{t,\varphi^{\mathrm{op}}}$ is among those decided by $q(x)$. Then $a_1 \in \mathcal{D}_{t,\varphi^{\mathrm{op}}} \leftrightarrow a_2 \in \mathcal{D}_{t,\varphi^{\mathrm{op}}}$. Therefore $b' \in \mathcal{D}_{p_1,\varphi} \leftrightarrow b' \in \mathcal{D}_{p_2,\varphi}$. As $\mathcal{D}_{p_i,\varphi}$ are both in $\mathrm{acl^{eq}}A$, we also have $b \in \mathcal{D}_{p_1,\varphi} \leftrightarrow b \in \mathcal{D}_{p_2,\varphi}$. □

The following theorem presents a property of definability that seems to go in the opposite direction that what discussed in Section 18.2. It has an application in the proof of Theorem 18.29.

18.27 Theorem Let $\varphi(x;z) \in L(M)$ be stable. Let $b \in \mathcal{U}^z$ be arbitrary. Then, for some $b_i \equiv_M b$, where $i = 1, \dots, n$, there is a positive Boolean combination of the formulas $\varphi(x;b_i)$ which is equivalent to a formula in $L(M)$.

Proof. Let $\mathcal{V} \preceq \mathcal{U}$ be isomorphic to $\mathcal{U}$ over M and such that $b \mathop{\smile\!\!\!\!\!\!\!\;\;\,|}_M \mathcal{V}$. Such a $\mathcal{V}$ exists by Proposition 14.10. By Theorem 18.10 with $\mathcal{V}$ for $\mathcal{U}$ and the role of the two sorts reversed, there are some $b_i \equiv_M b$ in $\mathcal{V}$, such that $\varphi(\mathcal{V}^x;b)$ is a positive Boolean combination of the sets $\varphi(\mathcal{V}^x;b_i)$. Let $\vartheta(x;\bar{b})$, where $\bar{b} = b_1, \dots, b_n$, be the formula that defines such a Boolean combination. By non-splitting, i.e. reasoning as in Theorem 18.23, there is a formula $\psi(x) \in L(M)$ such that $\psi(\mathcal{V}^x) = \varphi(\mathcal{V}^x;b)$. Finally, as $\bar{b}$ is in $\mathcal{V}$, by elementarity we obtain that $\psi(\mathcal{U}^x) = \vartheta(\mathcal{U}^x;\bar{b})$. □

A more general version of the theorem holds, but it requires a different proof. It is not necessary in the following.

18.28 Theorem Let $\varphi(x;z) \in L(A)$ be stable. Let $b \in \mathcal{U}^z$ be arbitrary. Then, for some $b_i \equiv_{\mathrm{acl}^{\mathrm{eq}}A} b$, where $i = 1, \dots, n$, there is a positive Boolean combination of the formulas $\varphi(x;b_i)$ which is equivalent to a formula in $L(A)$.

Proof. Let $q(z) = \mathrm{tp}(b/\mathrm{acl}^{\mathrm{eq}}A)$. By Corollary 18.22 there is a type $p(z) \in S_\varphi(\mathcal{U})$ that is finitely consistent with $q(z)$ and invariant over $\mathrm{acl}^{\mathrm{eq}}A$. By Theorem 18.10, with the role of the two sorts reversed, there are some $b_1, \dots, b_n \models q(z)$ such that $\mathcal{D}_{p,\varphi^{\mathrm{op}}}$ is equivalent to a positive Boolean combination of the sets $\varphi(\mathcal{U}^x;b_i)$. Let $\vartheta(x;b_1,\dots,b_n)$ be the formula that defines such a Boolean combination. As $\mathcal{D}_{p,\varphi^{\mathrm{op}}} \in \mathrm{acl}^{\mathrm{eq}}A$ by Proposition 18.25, this proves the theorem when $A = \mathrm{acl}^{\mathrm{eq}}A$.

For the general case, reason as follows. Let

$$\bigcup_{f \in \mathrm{Aut}(\mathcal{U}/A)} f\mathcal{D}_{p,\varphi^{\mathrm{op}}} \quad = \quad \bigvee_{f \in \mathrm{Aut}(\mathcal{U}/A)} \vartheta(\mathcal{U}^x; fb_1, \dots, fb_n).$$

As $\mathcal{D}_{p,\varphi^{\mathrm{op}}} \in \mathrm{acl}^{\mathrm{eq}}A$, this is a finite union, respectively disjunction. The set on the l.h.s. is invariant (hence definable) over A. The formula on the r.h.s. is the required positive Boolean combination. □

18.5 The action of the Lascar group on stable formulas

We adopt the notation and terminology of Chapter 17, specifically of Section 17.6. In this section $\Delta \subseteq L_{x\,z}(A)$ is always a collection of stable formulas. We set $\mathcal{Z} = \mathcal{U}^z$, and let Γ be the collection of $\Delta^{\mathrm{B}}(\mathcal{Z})$-definable sets.

We are interested in the action of the groups $\mathrm{Aut}(\mathcal{U}/A)$ and $\mathrm{Aut}(\mathcal{U}/\mathrm{acl}^{\mathrm{eq}}A)$, which we abbreviate to G and H, respectively. Clearly $H \trianglelefteq G$.

The following theorem shows that the equivalent conditions of Theorem 17.24 hold for suitable choice of π.

18.29 Theorem Let $\Delta \subseteq L_{xz}(A)$ be stable. Let $\pi(x) \subseteq L(\mathcal{U}^{\mathrm{eq}})$ be finitely satisfiable in every $M \supseteq A$. Then for every $\mathcal{D} \in \Gamma$ the following are equivalent

1. $\mathcal{D}$ is H-thick
2. $\mathcal{D}$ is H-wide.

Proof. We only need to prove 1⇒2 because the converse is trivial. As Boolean combinations of stable formulas are stable, for simplicity we assume that $\mathcal{D}$ is defined by the formula $\varphi(x\,;b)$.

It suffices to prove that $\pi(x) \cup \{\varphi(x\,;b)\}$ is finitely satisfied in every $M \supseteq A$. Then from Corollary 18.22 we obtain a type $p(x) \in S_\varphi(\mathcal{U})$ containing $\varphi(x\,;b)$ such that $\pi(x) \cup p(x)$ is finitely satisfied in every model $M \supseteq A$. Then $p(x)$ is H-invariant, and thertefore $\mathcal{D}$ is H-wide by Theorems 17.5 and 17.7.

Let $M \supseteq A$ be given. By invariance we can assume that $\pi(x) \subseteq L(M)$. Let $\sigma(x) \in \pi$, we claim that $\sigma(x) \wedge \varphi(x\,;b)$ is satisfied in M. Let $\vartheta(x\,;\bar{b})$ be the formula in the proof of Theorem 18.27 which is equivalent to a formula $\psi(x) \in L(M)$. As $\vartheta(x\,;\bar{b})$ is a positive Boolean combination of $\mathrm{Aut}(\mathcal{U}/M)$-translations of $\mathcal{D}$ and $\mathcal{D}$ is H-thick, $\vartheta(x\,;\bar{b})$ is satisfied in $\sigma(\mathcal{U}^x)$.

Then also $\psi(x)$ is satisfied in $\sigma(\mathcal{U}^x)$ and, by elementarity in $\sigma(M^x)$. Uning again the positivity of $\vartheta(x\,;\bar{b})$ we conclude that $\sigma(x) \wedge \varphi(x\,;b_i)$ is satisfied in M for some i. The claim follows because $b_i \equiv_M b$. □

Now, putting together the above theorem, Theorems 18.26, and Fact 17.26 we obtain the following

18.30 Corollary With the same notation as Theorem 18.29, assume $\pi(x) \in S_\Delta(\mathrm{acl}^{\mathrm{eq}}A)$. Then for every $\mathcal{D}$, the following are equivalent

1. $\mathcal{D}$ is H-syndetic
2. $\mathcal{D}$ is H-thick.

In other words, either $\mathcal{D}$ or $\neg\mathcal{D}$ is H-syndetic.

Stationarity and definability of stable types allows to strengthen Theorem 13.14. This is known as Shelah's *finite equivalence relation theorem.* (A major difference is that below in 5 we can use a single finite equivalence relation for the relation $\overset{\mathrm{Sh}}{\equiv}$.)

18.31 Theorem Let $\Delta \subseteq L_{x\,z}(A)$ be a finite set of stable formulas. Let $\pi(x) \in S_\Delta(A)$. Define

$$P = \{p(x) \in S_\Delta(\mathcal{Z}) \ : \ \text{consistent with } \pi(x) \text{ and } H\text{-invariant}\}.$$
$$Q = \{q(x) \in S_\Delta(\mathrm{acl}^{\mathrm{eq}}A) \ : \ \text{consistent with } \pi(x)\}.$$

Then the following hold

1. concistency is a one-to-one correspondence between P and Q
2. G acts transitively on P
3. P is finite
4. the types in P are G-syndetic
5. there is a finite equivalence relation $\varepsilon(x\,;y) \in L(A)$ such that for every $c \in \mathcal{U}^x$ there is a unique $p(x) \in P$ consistent with $\varepsilon(x\,;c)$.

Proof. 1. Clear, as the types in Q are stationary by Theorem 18.26.

2. By Proposition 13.15, $\mathrm{Aut}(\mathcal{U}/A)$ acts transitively on Q. Then by 1 it also acts transitively on P.

3. We may identify a type $p(x) \in S_\varphi(\mathcal{U})$ with the set $\mathcal{D}_{p,\varphi}$. Therefore types in $S_\Delta(\mathcal{U})$ are identified with tuples of such sets. The orbit over A of $p(x) \in P$ corresponds to the orbit over A of a tuple of sets of the form $\mathcal{D}_{p,\varphi}$. As $p(x) \in P$ is invariant over $\mathrm{acl}^{\mathrm{eq}}A$, the sets $\mathcal{D}_{p,\varphi}$ have a finite orbit over A. Therefore by 2, P is finite.

4. By 2 and 3.

5. By Theorem 13.14 and 3. □

18.32 Proposition Let $\Delta \subseteq L_{x\,z}(A)$ be a finite set of stable formulas. Let $\pi(x) \in S(A)$. Then for every $\mathcal{D}$ the following are equivalent

1. $\mathcal{D}$ is G-syndetic
2. $\mathcal{D}$ is H-wide.

Proof. 1⇒2. By stability, there exist a type $p(x) \in S_\Delta(\mathcal{U})$ that is H-invariant. Then $\Sigma_H(x)$ is consistent and Proposition 17.17 applies.

2⇒1. By 2, there is an H-invariant type $p(x) \in S_\Delta(\mathcal{U})$ consistent with $x \in \mathcal{D}$. Then $p(x) \in P$ of Theorem 18.31 and, therefore it is G-syndetic. □

18.33 Theorem Let $\Delta \subseteq L_{x\,z}(A)$ be a finite set of stable formulas. Let Q_G be as in Definition 17.13 and P be as in Theorem 18.31. Let $\pi(x) \in S(A)$. Then $Q_G = P$.

Proof. ⊇. Let $p(x) \in P$. All formulas in $p(x)$ are H-wide, and therefore G-syndetic by Proposition 18.32. As $p(x)$ complete, $p(x) \in Q_G$.

⊆. Let $q(x) \in Q_G$. By Theorem 18.32, $q(x)$ is H-wide, then $q(x)$ is consistent with some $p(x) \in P$. Then, by the converse inclusion, $q(x) = p(x)$. □

18.6 Stable groups

⚠ Section to be overhauled.

18.7 Stable theories

⚠ Section under revision.

We say that T is a stable theory if $\Delta = L_{x\,z}$, where $|x| = |z| = \omega$, is stable.

18.34 Corollary If T is stable, the following are equivalent

1. $a \overset{\mathrm{L}}{\equiv}_A b$, see Definition 16.11
2. $a \overset{\mathrm{Sh}}{\equiv}_A b$, see Definition 13.13

Proof. 1⇒2. This is left as an exercise to the reader – stability is not required.

2⇒1. Assume 2 which, by Theorem 13.14, is equivalent to $a \equiv_{\mathrm{acl}^{\mathrm{eq}}A} b$. Write $q(x)$ for $\mathrm{tp}(a/\mathrm{acl}^{\mathrm{eq}}A) = \mathrm{tp}(b/\mathrm{acl}^{\mathrm{eq}}A)$. Let $p(x) \in S(\mathcal{U}^{\mathrm{eq}})$ be the unique global type that is invariant over $\mathrm{acl}^{\mathrm{eq}}A$ and extends $q(x)$ which we obtain from Theorem 18.26. Let $\bar{c} = \langle c_i : i < \omega\rangle$ be such that $c_i \vDash p(x){\restriction}\mathrm{acl}^{\mathrm{eq}}A,\ a,\ b,\ c_{\restriction i}$. Then $a, \bar{c}$ and $b, \bar{c}$ are A-indiscernible sequences, which proves 1, see Exercise 16.24. □

18.35 Theorem (Pierre Simon) If every formula $\varphi(x\,;y) \in L(\mathcal{U})$, where $|x| = |y| = 1$, is stable then T is stable.

Proof. Suppose $\varphi(x\,;y,z) \in L(\mathcal{U})$ is not stable. We prove that there is a formula $\psi(x\,;y) \in L(\mathcal{U})$ that is not stable. Let $\langle a_i\,;b_i,c_i \;:\; i \in \mathbb{Q}\rangle$ is a sequence of indiscernibles such that

$$i < j \quad\Leftrightarrow\quad \varphi(a_i\,;b_j,c_j) \qquad\text{for all } i,j \in \mathbb{Q}.$$

Let $\mathbb{Q}^* = \mathbb{Q} \smallsetminus \{0\}$. Assume first that the sequence $\langle a_i \;:\; i \in \mathbb{Q}^*\rangle$ is indiscernible over c_0. Then for every $k \in \mathbb{Q}^*$ the type below is consistent

$$p_k(y) \quad=\quad \{\varphi(a_i\;y,c_0) \leftrightarrow i < k \;\;:\; i \in \mathbb{Q}^* \smallsetminus \{k\}\}.$$

In fact, by indiscernibility, b_0 witnesses the consistency of all finite subsets of $p_k(y)$. Let $b'_k \vDash p_k(y)$. Then

$$i < k \quad\Leftrightarrow\quad \varphi(a_i\,;b'_k,c_0) \qquad\text{for all } i,j \in \mathbb{Q}^*, i \neq j$$

From this the instability of $\psi(x\,;y) = \varphi(x\,;y,c_0)$ follows easily.

Now, assume instead that $\langle a_i \;:\; i \in \mathbb{Q}^*\rangle$ is not indiscernible over c_0. Note that the sequences $\langle a_i \;:\; i < 0\rangle$ and $\langle a_i \;:\; i > 0\rangle$ are mutually indiscernible over c_0. Then there is a maximal n such that

$$a_{\restriction\{-1,\ldots,-n\}} \;\equiv_{c_0}\; a_{\restriction\{1,\ldots,n\}}.$$

Let $A = a \restriction \{\pm 1, \ldots, \pm n\}$. By maximality, $a_i \not\equiv_{c_0,A} a_j$ for every $-1 < i < 0 < j < 1$. Let $\psi(x\;y)$ be a formula such that $\psi(a_i\,;c_0)$ and $\neg\psi(a_j\,;c_0)$. We claim that for every $k \in (-1,0) \cup (0,1)$ the type below is consistent

$$q_k(z) \quad=\quad \{\psi(a_i\;y) \leftrightarrow i < k \;\;:\; i \in (-1,0) \cup (0,1)\}.$$

In fact as $\langle a_i \;:\; i \in (-1,0) \cup (0,1)\rangle$ is indiscernible over A, all finite subsets of $q_k(z)$ are realized by c_0. Finally let $c'_k \vDash p_k(z)$. Then $\langle a_i, c'_i \;:\; i \in (-1,0) \cup (0,1)\rangle$ witness the instability of $\psi(x\;z)$. □

18.36 Exercise Prove that if every formula $\varphi(x\,;z) \in L$, where $|x| = 1$, is stable then T is stable.

Proof. Let $\varphi(x, y\,; z) \in L$ not stable. Then there are indiscernibles $\langle a_i, b_i\,; c_i \;:\; i \in \mathbb{Q}\rangle$ such that □

18.37 Exercise A sequence $\langle a_i : i < \omega\rangle$ is totally indiscernible if $a_1, \ldots, a_n \equiv a_{i_1}, \ldots, a_{i_n}$ for every distinct $i_1, \ldots, i_n$. Prove that the following are equivalent

1. T is stable
2. every indiscernible sequence is totally indiscernible.

18.38 Exercise Prove that the following are equivalent

1. T is unstable
2. there is an infinite set $A \subseteq \mathcal{U}^n$ and a formula $\psi(x\,; y)$, with $|x| = |y| = n$ such that A is linearly ordered by the relation $a < b \leftrightarrow \psi(a\,; b)$.

18.39 Exercise Prove that strongly minimal theories are stable.

18.8 Notes and references

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Chapter 19

Vapnik-Chervonenkis theory

⚠ Chapter under revision.

In this chapter we fix a signature L, a complete theory T without finite models, and a saturated model $\mathcal{U}$ of inaccessible cardinality $\kappa > |L|$. The notation and implicit assumptions are as in Section 9.3.

19.1 Vapnik-Chervonenkis dimension

We say that the formula $\varphi(x\,;z) \in L$ has Vapnik-Chervonenkis dimension n if this is the largest finite cardinatity of a set $B \subseteq \mathcal{U}^z$ such that $|S_\varphi(B)| = 2^n$. If such n does not exist, we say that we say that $\varphi(x\,;z)$ has infinite VC-dimension.

Note that the condition $|S_\varphi(B)| = 2^n$ is equivalent to saying that every subset of B is the trace of some definable set of sort $\varphi(x\,;z)$.

For instance, the formula $x_1 < z < x_2$ in $T_{\rm dlo}$ has VC-dimension 2.

Arguing by compactness we obtain the following proposition whose proof is left as an exercise for the reader.

19.1 Proposition The following are equivalent

1. $\varphi(x\,;z) \in L$ has finite VC-dimension
2. there is no infinite set $B \subseteq \mathcal{U}^z$ such that every subset of B is the trace of some definable set of sort $\varphi(x\,;z)$.

From the proposition above and Proposition 18.19 below it follows that all stable formulas have finite VC-dimension.

We say that the sequence of sentences $\langle \varphi_i : i < \omega \rangle$ converges if the truth value of φ_i is eventually constant.

19.2 Lemma The following are equivalent

1. $\varphi(x\,;z) \in L$ has finite VC-dimension
2. $\langle \varphi(a\,;b_i) : i < \omega \rangle$ converges for anyany indiscernible sequence $\langle b_i : i < \omega \rangle$.

Proof. 1⇒2 Negate 2 and let $n < \omega$. It suffices to prove that for every $I \subseteq n$ the formula $\psi_I(x\,;b_0,\dots,b_{n-1})$ that says

$$\varphi(x\,;b_i) \quad \Leftrightarrow \quad i \in I$$

is consistent. If there is a a such that the truth value of $\langle \varphi(a\,;b_i) : i < \omega \rangle$ oscillates at least n times, then we can find $k_0 < \dots < k_{n-1}$ such that

$$\varphi(a\,;b_{k_i}) \quad \Leftrightarrow \quad i \in I.$$

Then the formula $\psi_I(x\,;b_{k_0},\dots,b_{k_{n-1}})$ is consistent. Therefore, by indiscernibility, also the formula $\psi_I(x\,;b_0,\dots,b_{n-1})$ is consistent.

2⇒1 Negate 1 and let $\langle c_i : i < \omega\rangle$ be an infinite sequence that is shattered by $\varphi(x\,;z)$. Let $\langle b_i : i < \omega\rangle$ be an indiscernible sequence that models the EM-type of $\langle c_i : i < \omega\rangle$. Then $\langle b_i : i < \omega\rangle$ satisfies $\exists x\, \psi_{I\restriction n}(x\,;z_0,\dots,z_{n-1})$ for all n. Let $I \subseteq \omega$ be the set of even integers. By compactness there is a a such that

$$\varphi(a\,;b_i) \quad\Leftrightarrow\quad i \in I.$$

This proves $\neg$2. □

In the next section we need the following corollary.

19.3 Corollary If $\mathcal{C} \subseteq \mathcal{U}^z$ is a set approximable by a formula with finite VC-dimension, then $\langle b_i \in \mathcal{C} : i < \omega\rangle$ converges for any indiscernible sequence $\langle b_i : i < \omega\rangle$.

19.2 Honest definitions

In this section we present a beautiful theorem by Chernikov and Simon and their alternative proof of a famous quantifier elimination result by Shelah.

We write $\neg^n$ for $\neg \overset{n \text{ times}}{\dots\dots} \neg$. We abbreviate $\neg^n(\cdot \in \cdot)$ as $\notin^n$.

Saturated sets have been defined in Definition 16.2.

19.4 Lemma Let $\mathcal{C}$ be saturated set approximable by a formula with finite VC-dimension and let A be a set of parameters. Then every global A-invariant type $p(z)$ contains a formula $\psi(z)$ such that either $\psi(\mathcal{U}^z) \subseteq \mathcal{C}$ or $\psi(\mathcal{U}^z) \subseteq \neg\mathcal{C}$. Moreover, we can require that $\psi(z) \in L(N)$, for any sufficiently saturated model N.

Proof. By Corollary 19.3, there is no infinite sequence $\langle b_i : i < \omega\rangle$

$$b_i \quad\vDash\quad p_{A,b\restriction i}(z) \;\cup\; \{z \notin^i \mathcal{C}\}$$

Let n be the largest integer such that there is a sequence $\langle b_i : i < n\rangle$ that satisfies the condition above. Then

$$p_{A,b\restriction n}(z) \quad\rightarrow\quad z \notin^{n+1} \mathcal{C}$$

and the first claim of the lemma follows by compactness.

As for the second claim note that we can pick $b_i \in N^z$ as soon as $A \subseteq N$ and $\langle N, \mathcal{C}\rangle$ is $|A|^+$-saturated. □

19.5 Corollary Let $\mathcal{C}$ be a set approximable by a formula with finite VC-dimension and let A be a set of parameters. Then there is a definable set $\mathcal{D} \supseteq A^z$ such that $\mathcal{D} \cap \mathcal{C}$ is definable. In particular, $\mathcal{C}$ is approximable from below.

Proof. Let M be a model containing A. Let c enumerate some $|M|^+$-saturated model containing M. For every $b \mathbin{\downarrow_M} c$ the type $\mathrm{tp}(b/c)$ extends to a global coheir over M. By the lemma above, there is a formula $\psi_b(z) \in \mathrm{tp}(b/c)$ such that either $\psi_b(\mathcal{U}^z) \subseteq \mathcal{C}$ or $\psi_b(\mathcal{U}^z) \subseteq \neg\mathcal{C}$, depending on whether $b \in \mathcal{C}$ or $b \notin \mathcal{C}$. Hence

$$z \mathop{\downarrow}_M c \quad\rightarrow\quad \bigvee\Big\{\psi_b(z) \;:\; b \mathop{\downarrow}_M c\Big\}.$$

By compactness,

$$z \mathop{\downarrow}_{M} c \quad \rightarrow \quad \bigvee_{i=1}^{n} \psi_{b_i}(z).$$

Again by compactness, there is a formula $\varphi(z)$ such that

$$\varphi(z) \quad \rightarrow \quad \bigvee_{i=1}^{n} \psi_{b_i}(z).$$

Let $\mathcal{D} = \varphi(\mathcal{U}^z)$. Let $\psi(z)$ is the disjunction of those $\psi_{b_i}(z)$ such that $b_i \in \mathcal{C}$. Then $\mathcal{D} \cap \mathcal{C}$ is defined by $\varphi(z) \wedge \psi(z)$.

As $\mathcal{C} =_A \mathcal{D} \cap \mathcal{C}$, we obtain in particular that $\mathcal{C}$ is approximable from below, see Lemma 18.4. □

When all formulas have finite VC-dimension, we say that the theory T has the non-independence property or, for short, that T is nip.

Let $\langle \mathcal{D}_i : i < \lambda \rangle$ be the collection of all subsets of $\mathcal{U}$, of arbitrary finite arity, that are externally definable. The expansion of $\mathcal{U}$ to the language $L(\mathcal{X}_i : i < \lambda)$ is called the Shelah expansion of $\mathcal{U}$ and is denoted by $\mathcal{U}^{\mathrm{Sh}}$.

From Corollary 19.5 and Proposition 18.5 we obtain the following.

19.6 Corollary If T is nip then $\mathcal{U}^{\mathrm{Sh}}$ has L-elimination of quantifiers. (I.e. every formula is Boolean combination of formulas in L and formulas of the form $z \in \mathcal{D}_i$.)

19.7 Exercise (T is nip.) Let $\langle a_i : i < \omega \rangle$ be a sequence of indiscernibles over A. Let

$$p(x) \quad = \quad \{\varphi(x) \in L(\mathcal{U}) \ : \ \varphi(a_i) \text{ holds for almost all } i\}$$

Prove that $p(x)$ is complete.

19.3 Notes and references

[1] Artem Chernikov and Pierre Simon, *Externally definable sets and dependent pairs*, Israel J. Math. **194** (2013), no. 1, 409–425. ArXiv:1007.4468.